



(12) **United States Patent**
Ash et al.

(10) **Patent No.:** **US 12,406,234 B2**
(45) **Date of Patent:** ***Sep. 2, 2025**

(54) **MULTI-SERVICE BUSINESS PLATFORM SYSTEM HAVING EVENT SYSTEMS AND METHODS**

G06Q 30/01 (2013.01); *G06Q 30/0201* (2013.01); *G06Q 30/0203* (2013.01); *G06Q 30/0246* (2013.01); *G06Q 30/0255* (2013.01);
(Continued)

(71) Applicant: **HUBSPOT, INC.**, Cambridge, MA (US)

(58) **Field of Classification Search**
None
See application file for complete search history.

(72) Inventors: **Bryan Ash**, Arlington, MA (US); **Mohit Dilawari**, Belmont, MA (US)

(73) Assignee: **HUBSPOT, INC.**, Cambridge, MA (US)

(56) **References Cited**

U.S. PATENT DOCUMENTS

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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(Continued)

This patent is subject to a terminal disclaimer.

Primary Examiner — Elijah S. Harper

(21) Appl. No.: **18/900,996**

(74) Attorney, Agent, or Firm — Cooper Legal Group, LLC

(22) Filed: **Sep. 30, 2024**

(65) **Prior Publication Data**

US 2025/0021947 A1 Jan. 16, 2025

Related U.S. Application Data

(63) Continuation of application No. 17/654,519, filed on Mar. 11, 2022, now Pat. No. 12,154,079.
(Continued)

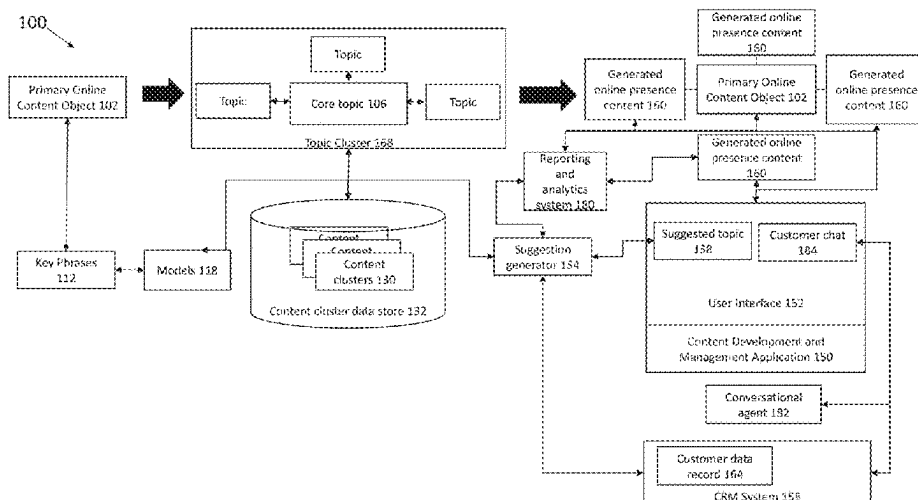
(57) **ABSTRACT**

The disclosure is directed to various ways of improving the functioning of computer systems, information networks, data stores, search engine systems and methods, and other advantages. Among other things, provided herein are methods, systems, components, processes, modules, blocks, circuits, sub-systems, articles, and other elements (collectively referred to in some cases as the “platform” or the “system”) that collectively enable, in one or more datastores (e.g., where each datastore may include one or more databases) and systems. A system and method for maintaining unified events such as maintaining an ontology data store and an event datastore. The event datastore may store event record instances of one or more different types of event records. The system and method may identify one or more types of event records that track a specific event type based on a specific event type of a new event instance.

(51) **Int. Cl.**
G06F 16/95 (2019.01)
G06F 16/951 (2019.01)
G06N 5/02 (2023.01)
G06Q 10/0631 (2023.01)
G06Q 10/0633 (2023.01)
(Continued)

(52) **U.S. Cl.**
CPC *G06Q 20/02* (2013.01); *G06F 16/951* (2019.01); *G06N 5/02* (2013.01); *G06Q 10/06315* (2013.01); *G06Q 10/0633* (2013.01);

15 Claims, 81 Drawing Sheets



Page 2

* cited by examiner

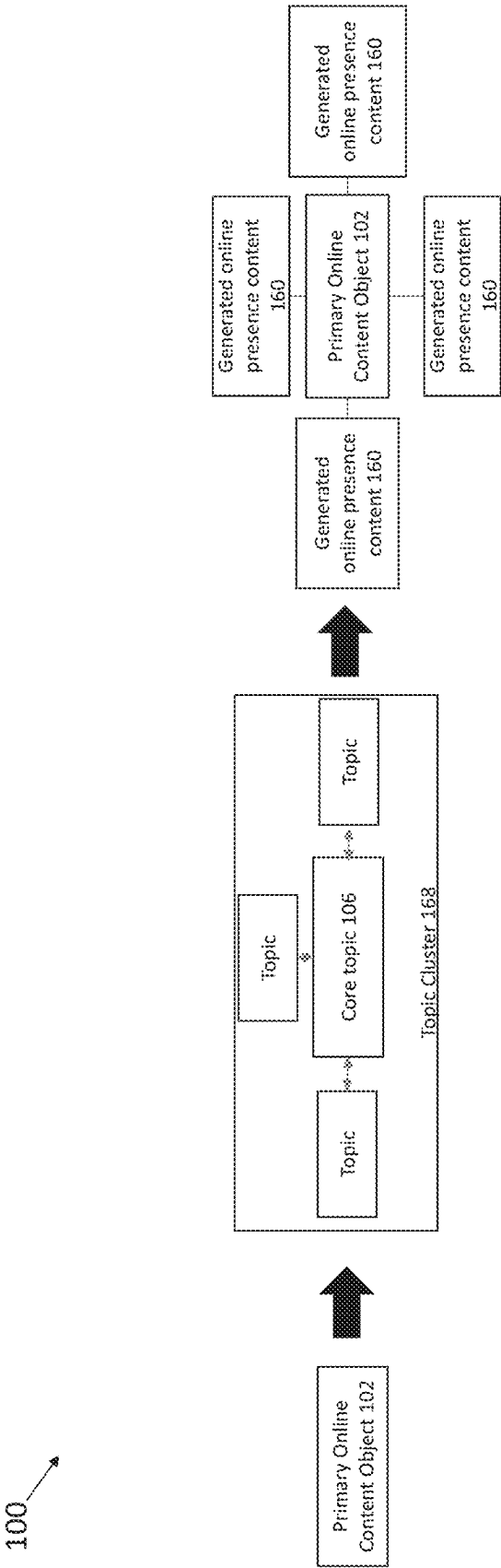


FIG. 1

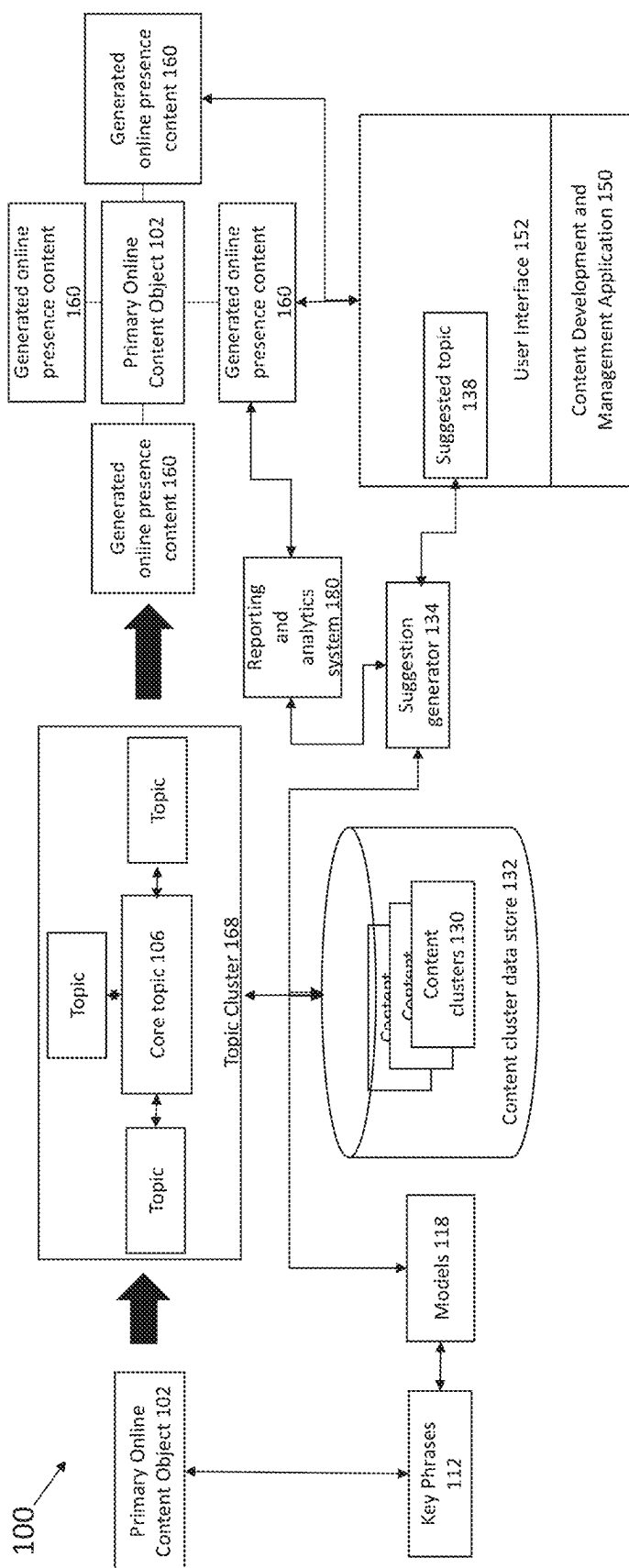


FIG. 2

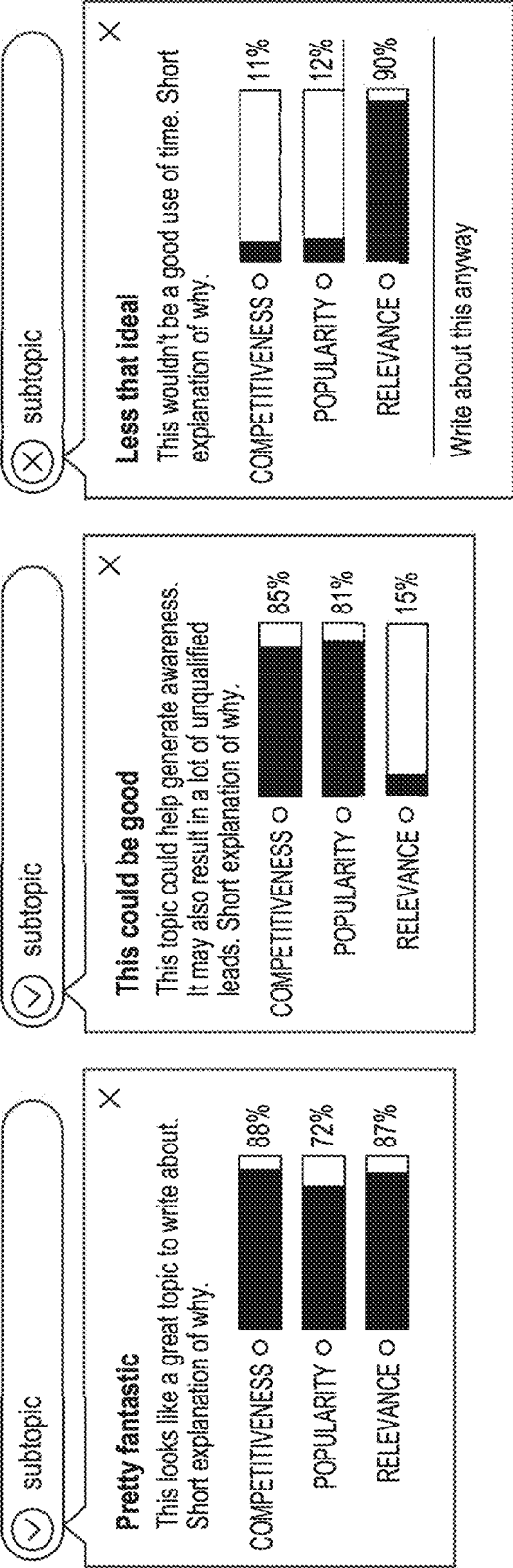


FIG. 3

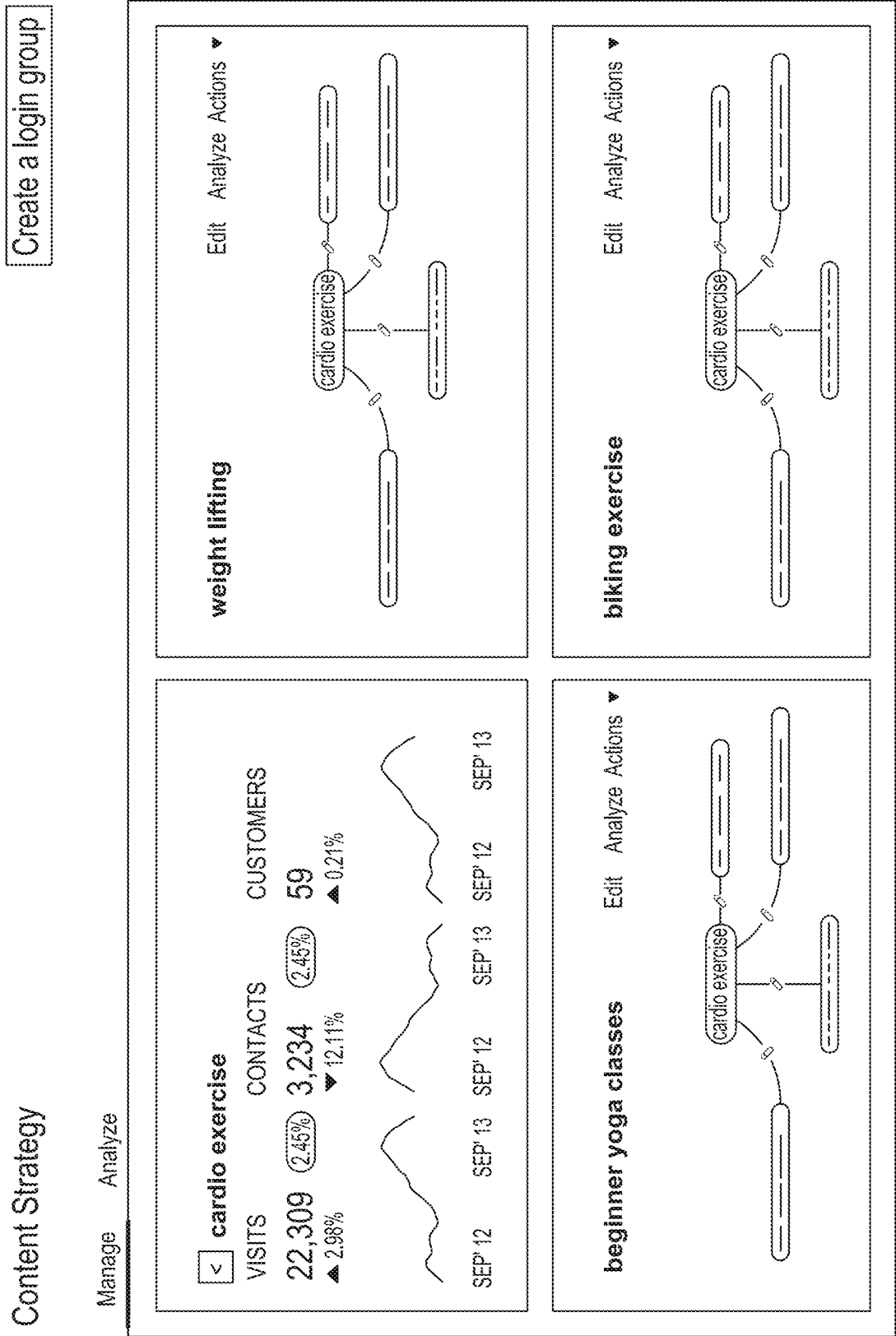


FIG. 4

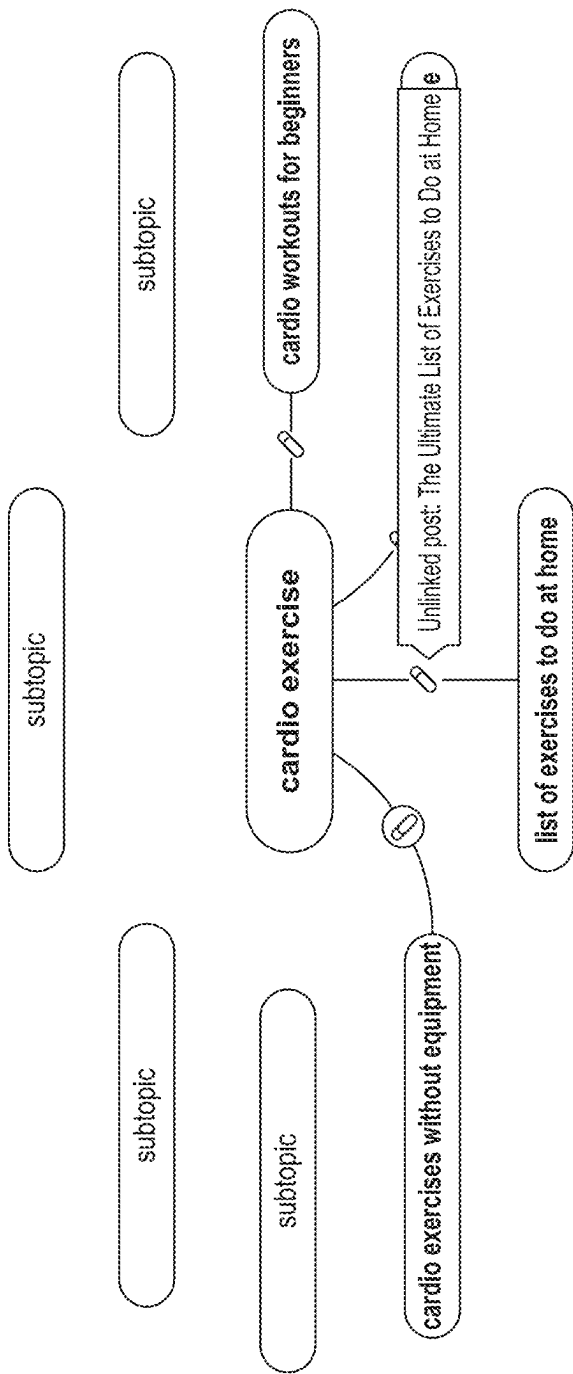
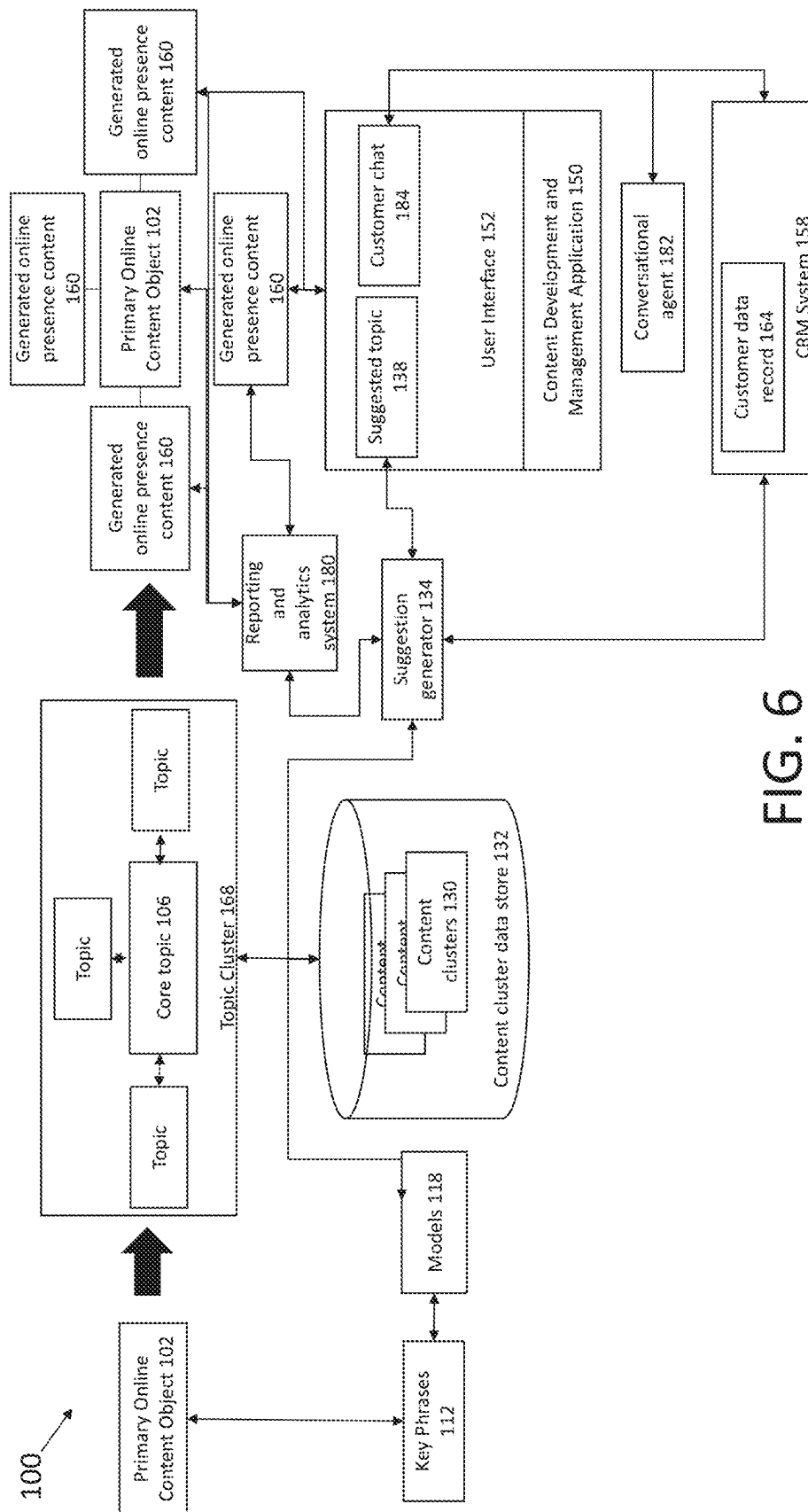
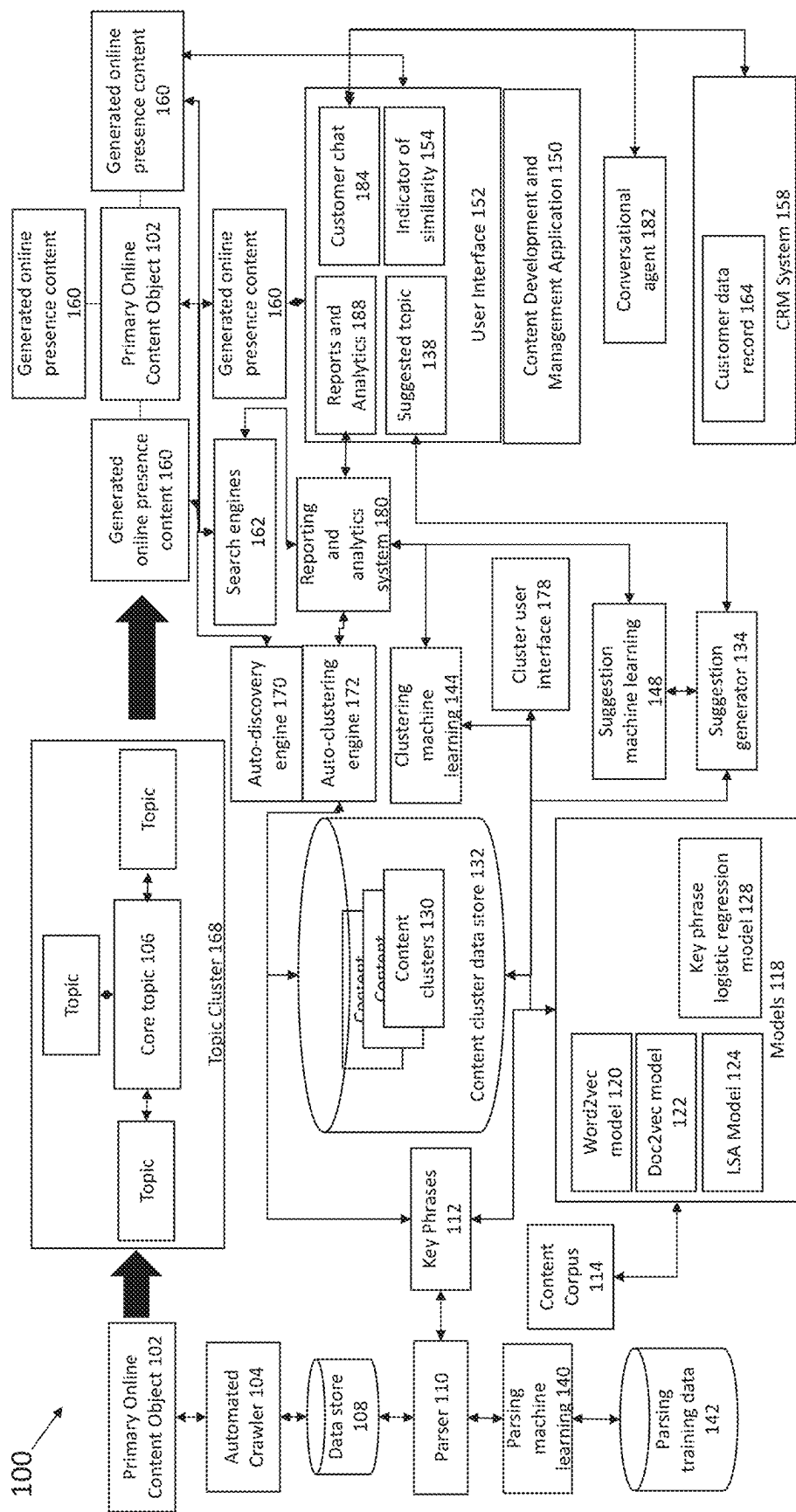
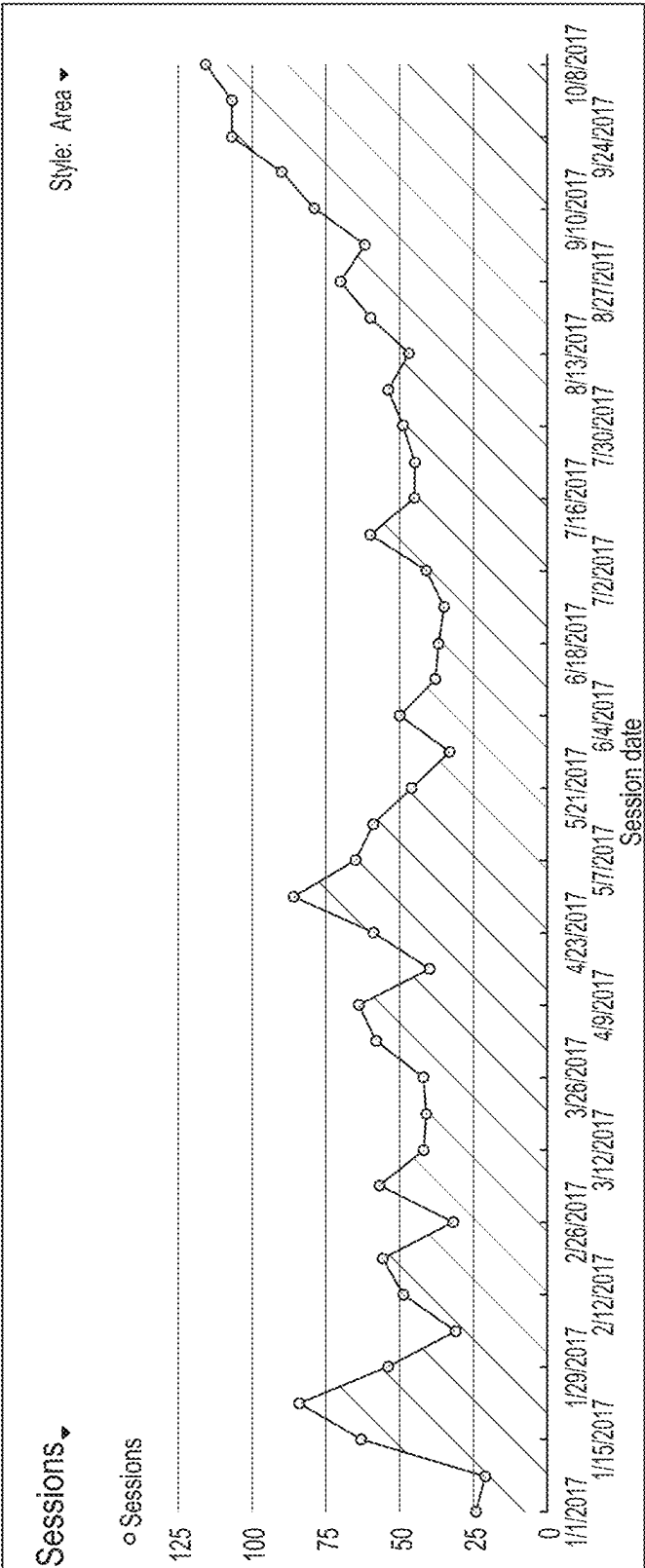


FIG. 5







Edit columns

TOPIC CLUSTER	SESSIONS	CONTACT CONVERSATION RATE	NEW CONTACTS	CUSTOMER CONVERSATION RATE	BOUNCE RATE	AVG. SESSION LENGTH
☐ Marketing Funnel	1,186	1.35%	16	-	89.21%	a minute
☐ Website Redesign	514	1.75%	9	11.11%	82.3%	a minute

FIG. 8

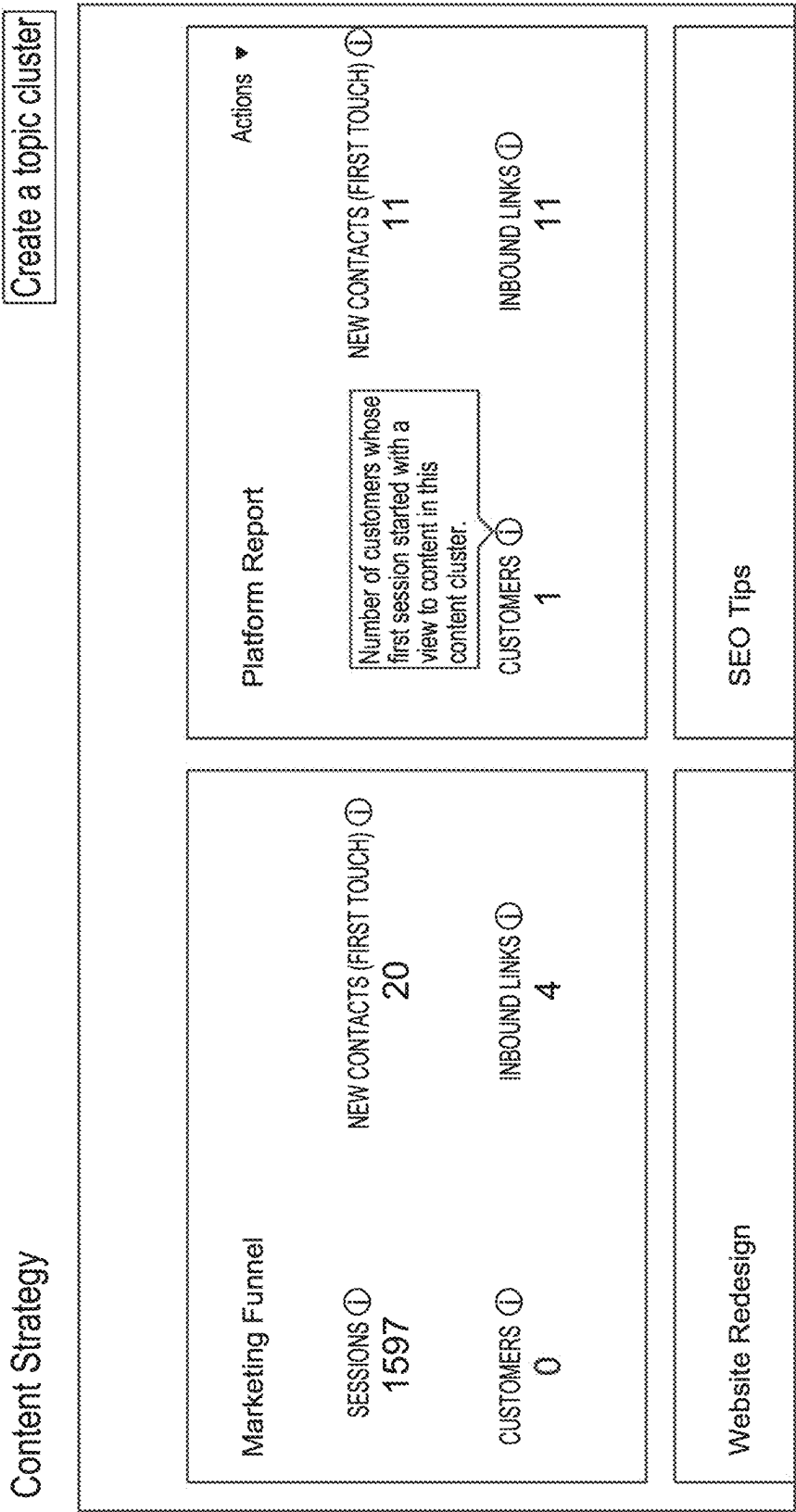


FIG. 9

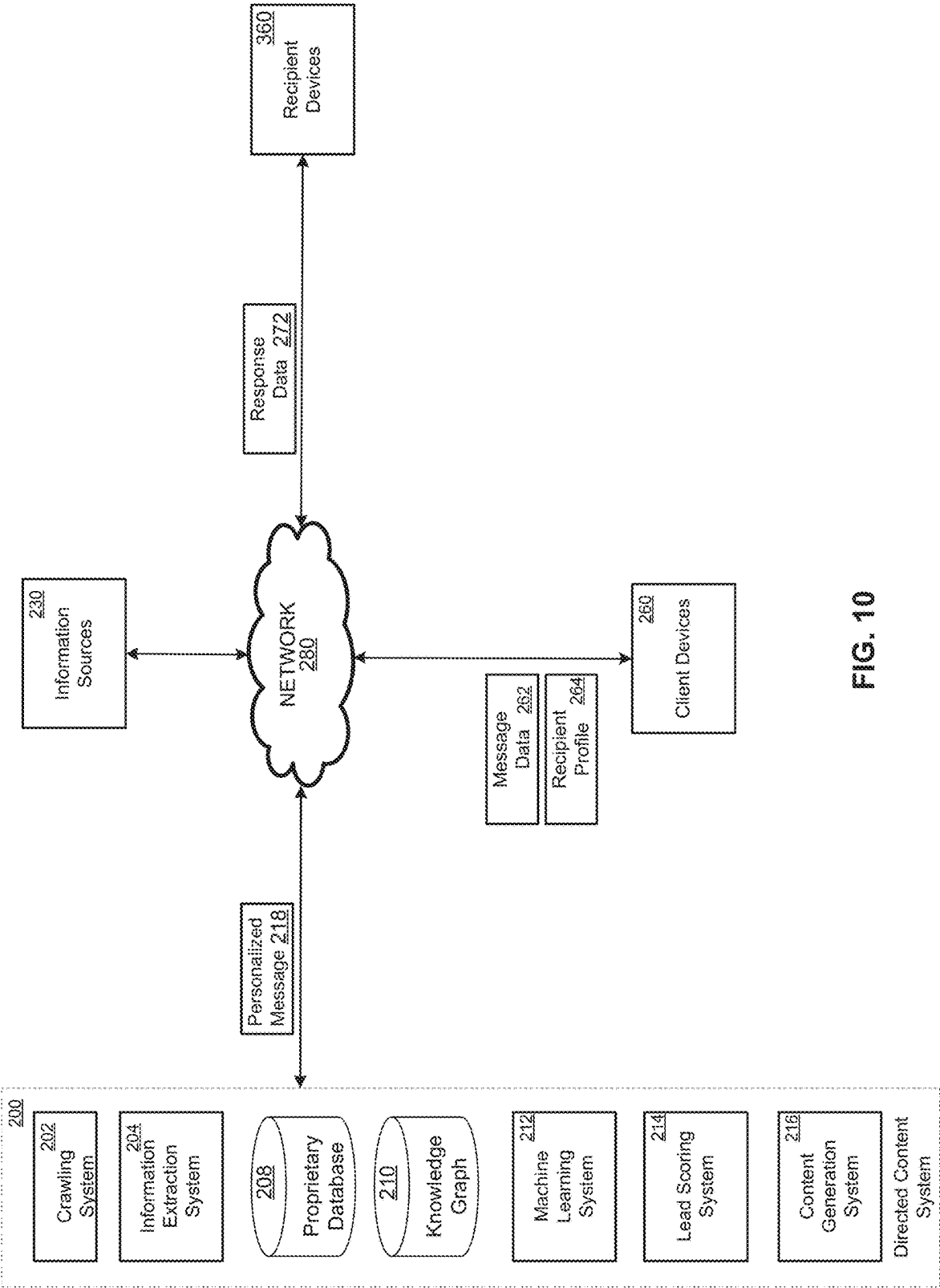
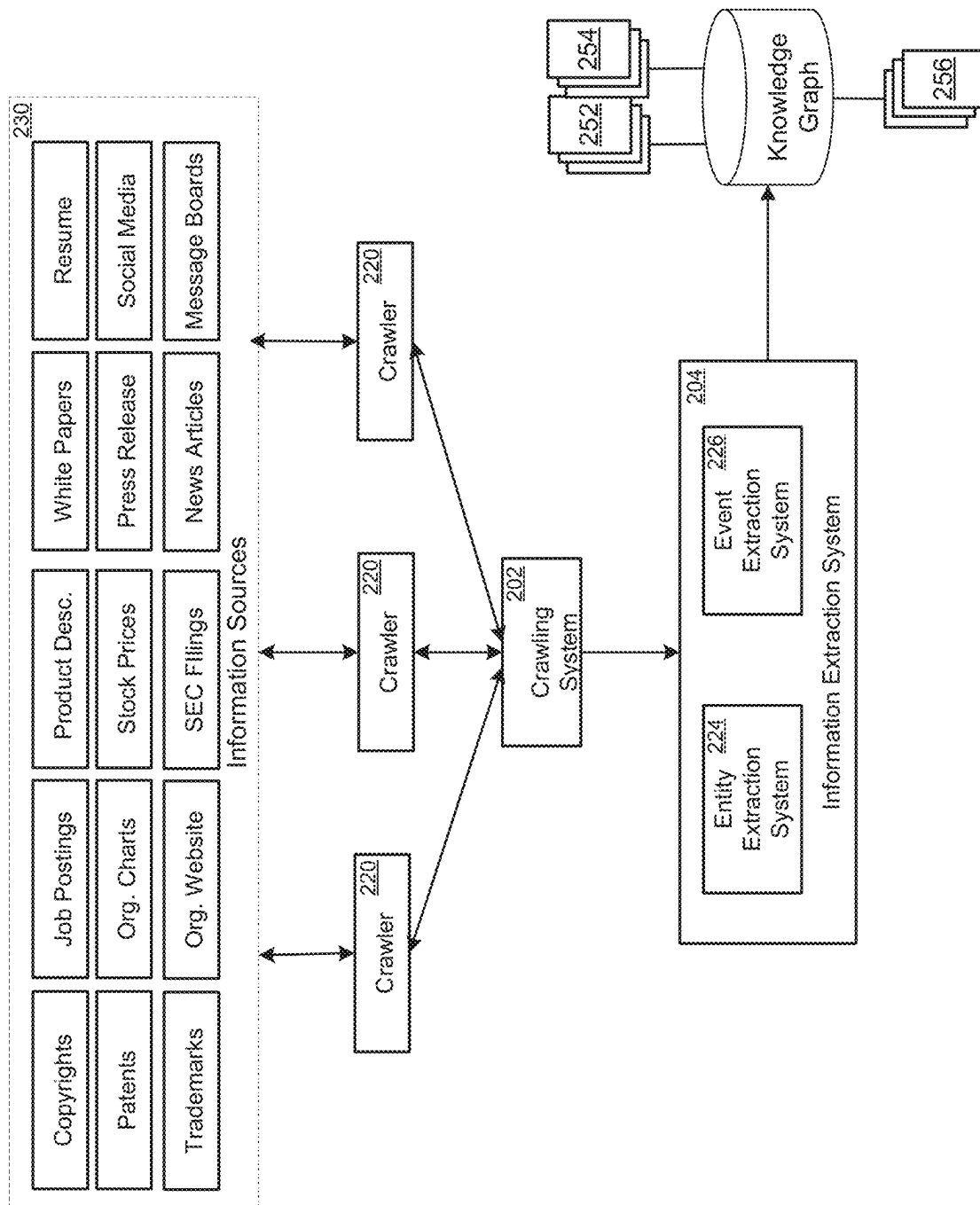


FIG. 10



FFOL

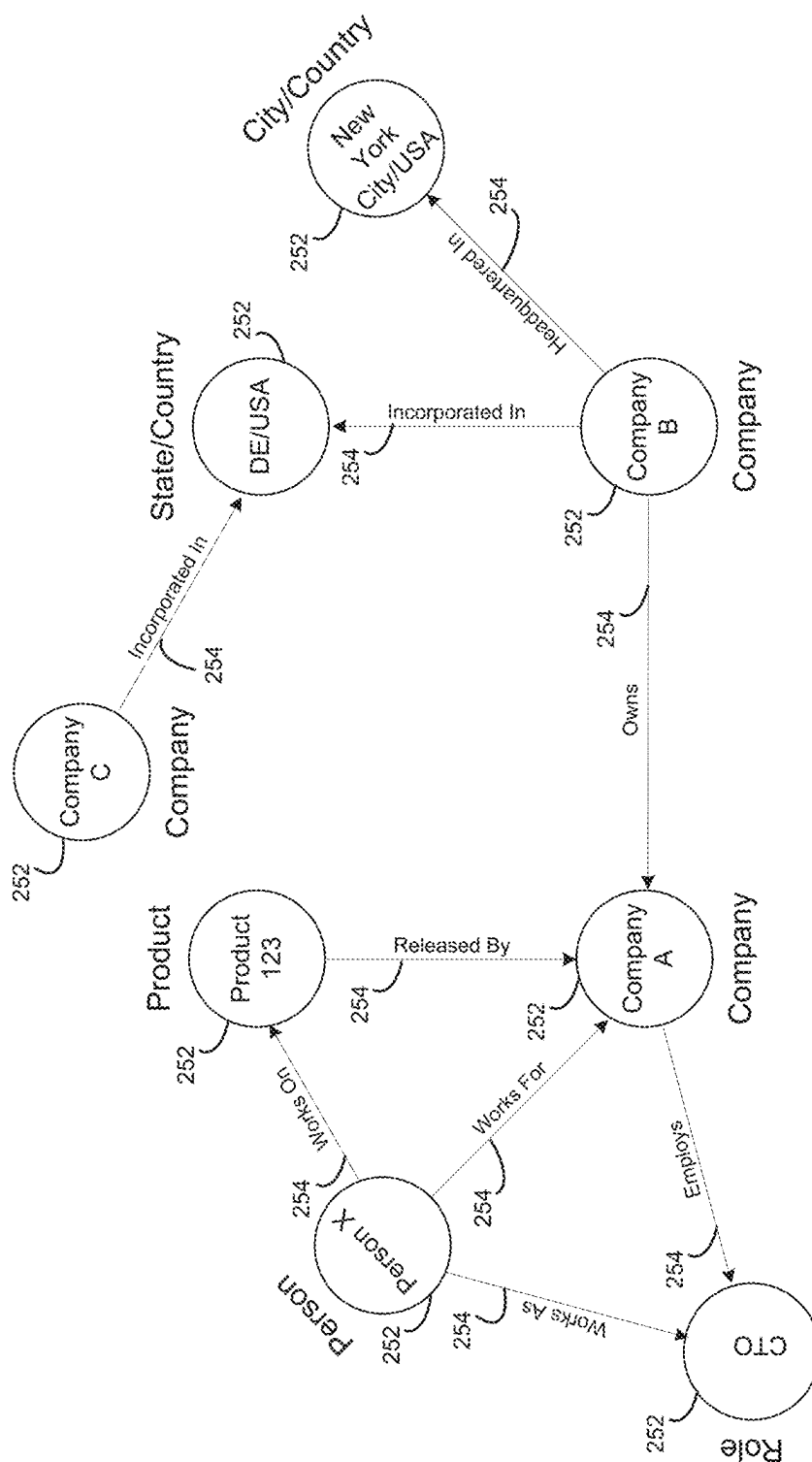


FIG. 12

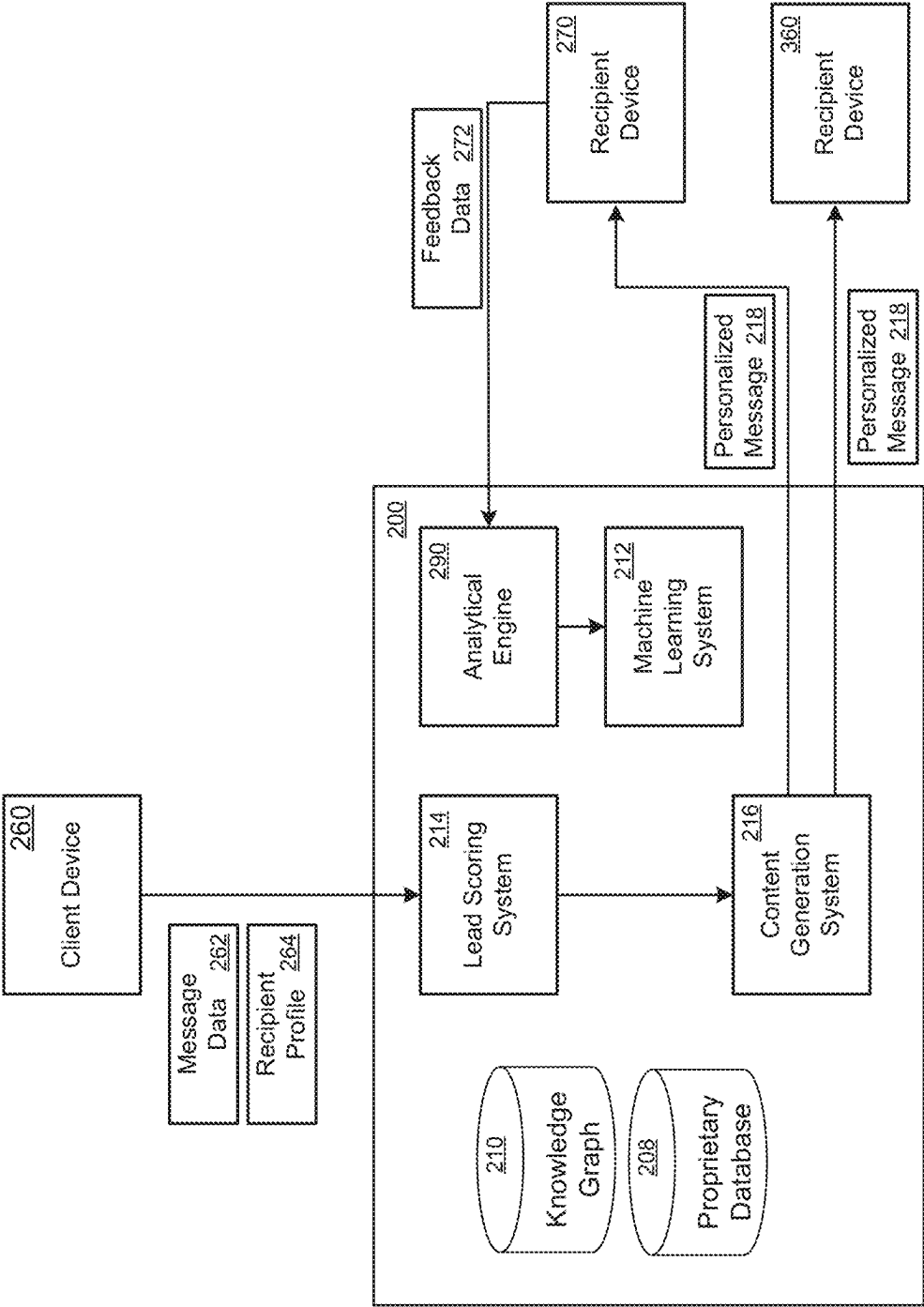


FIG. 13

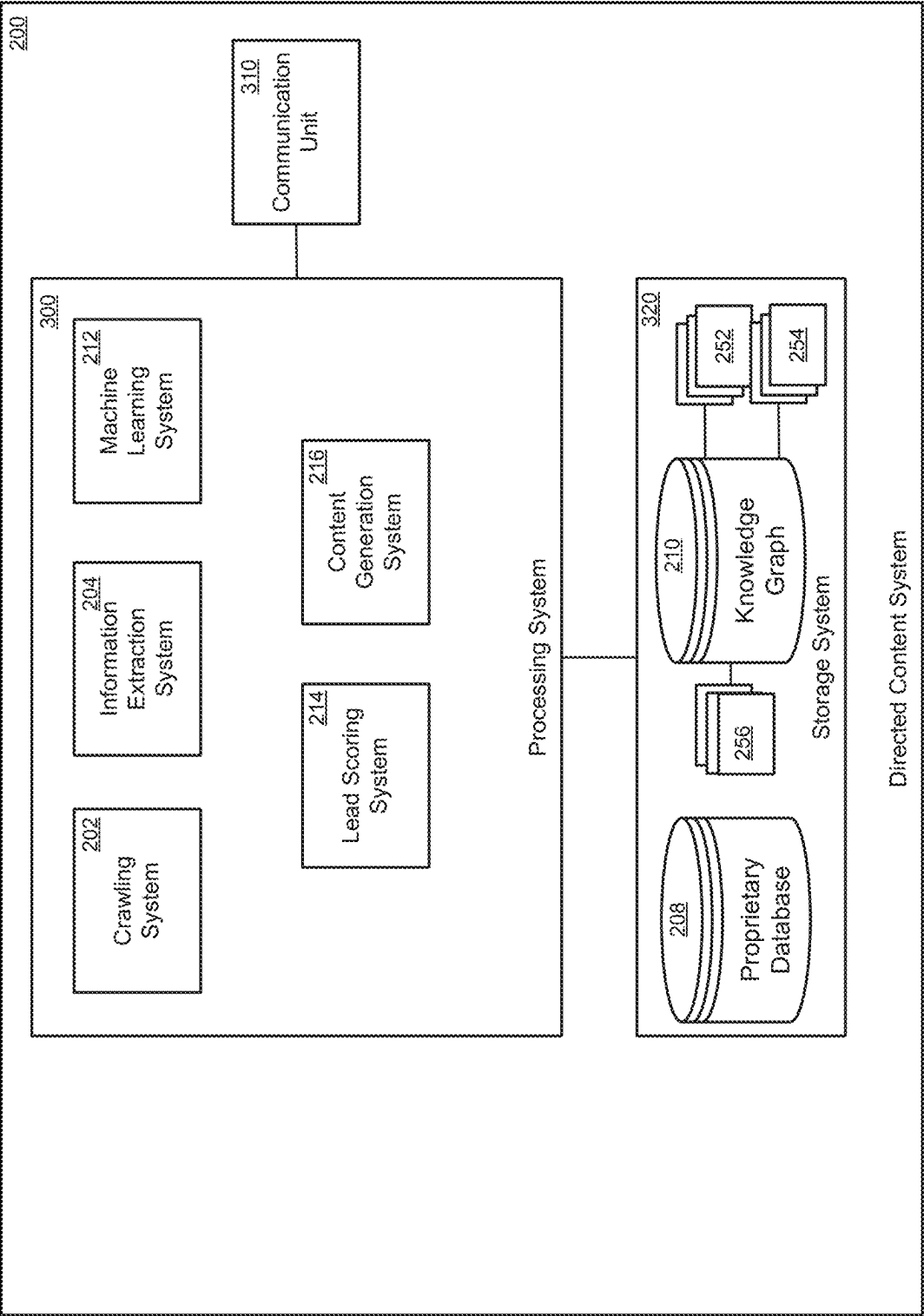


FIG. 14

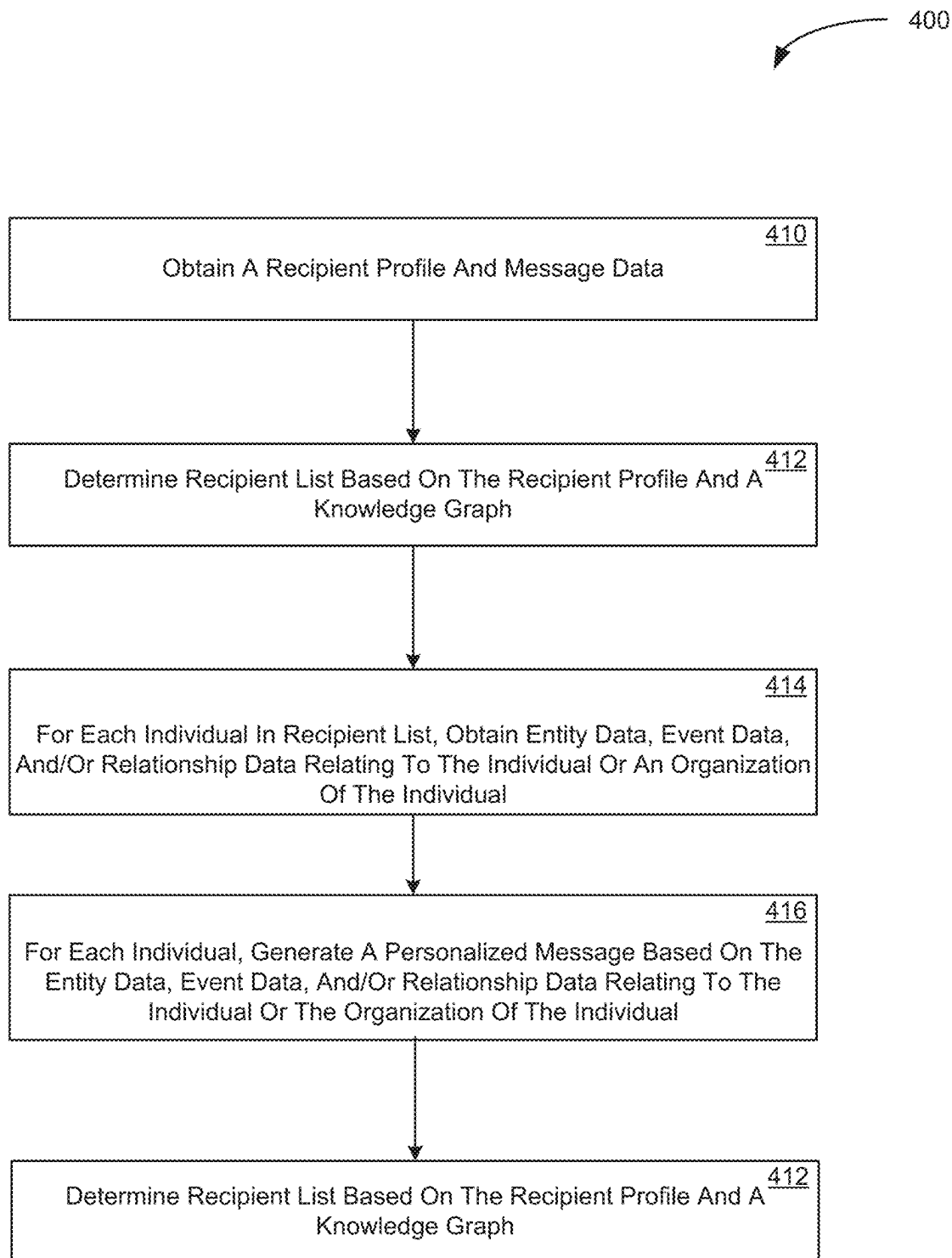


FIG. 15

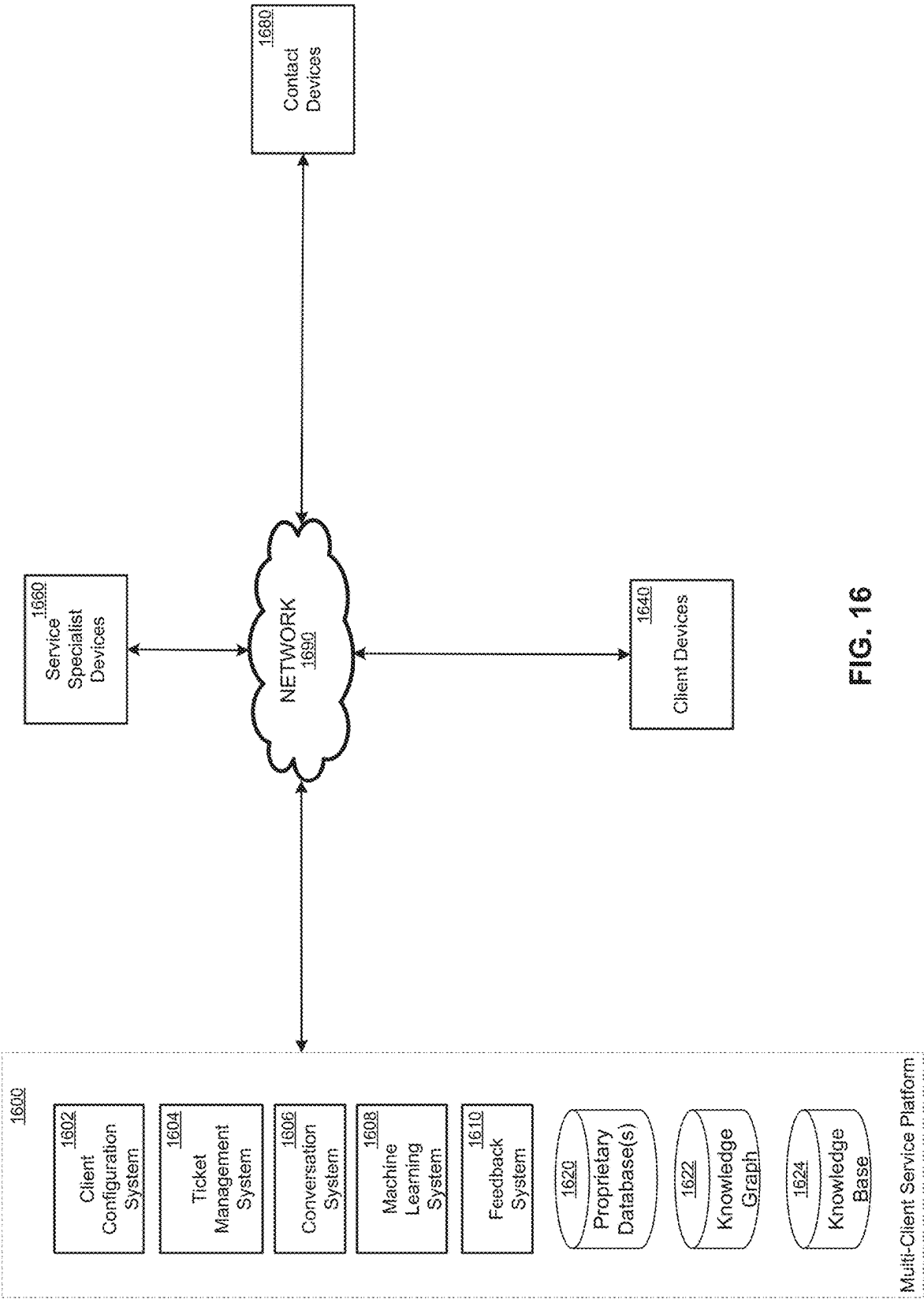


FIG. 16

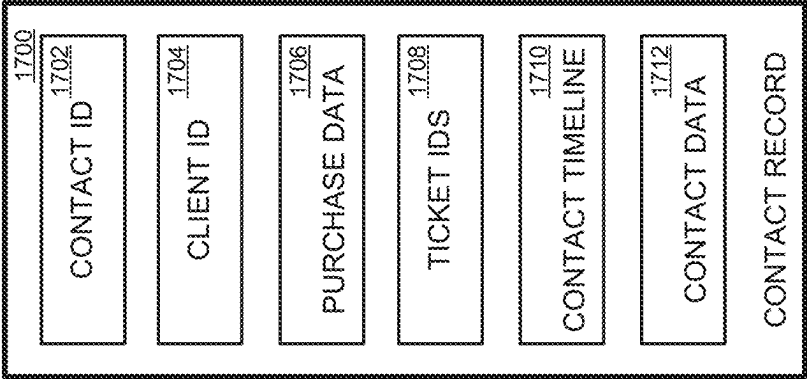


FIG. 17A

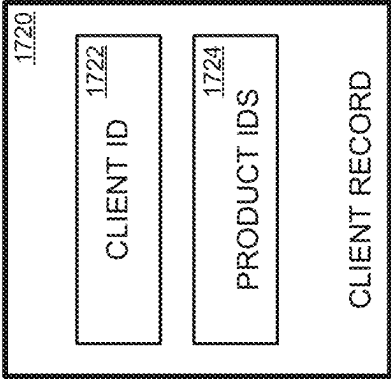


FIG. 17B

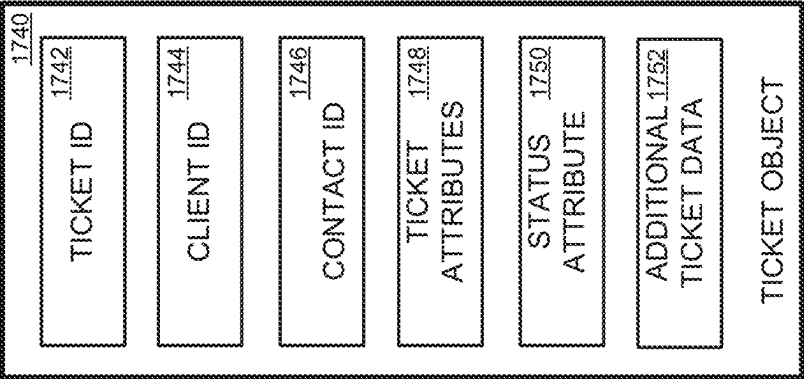


FIG. 17C

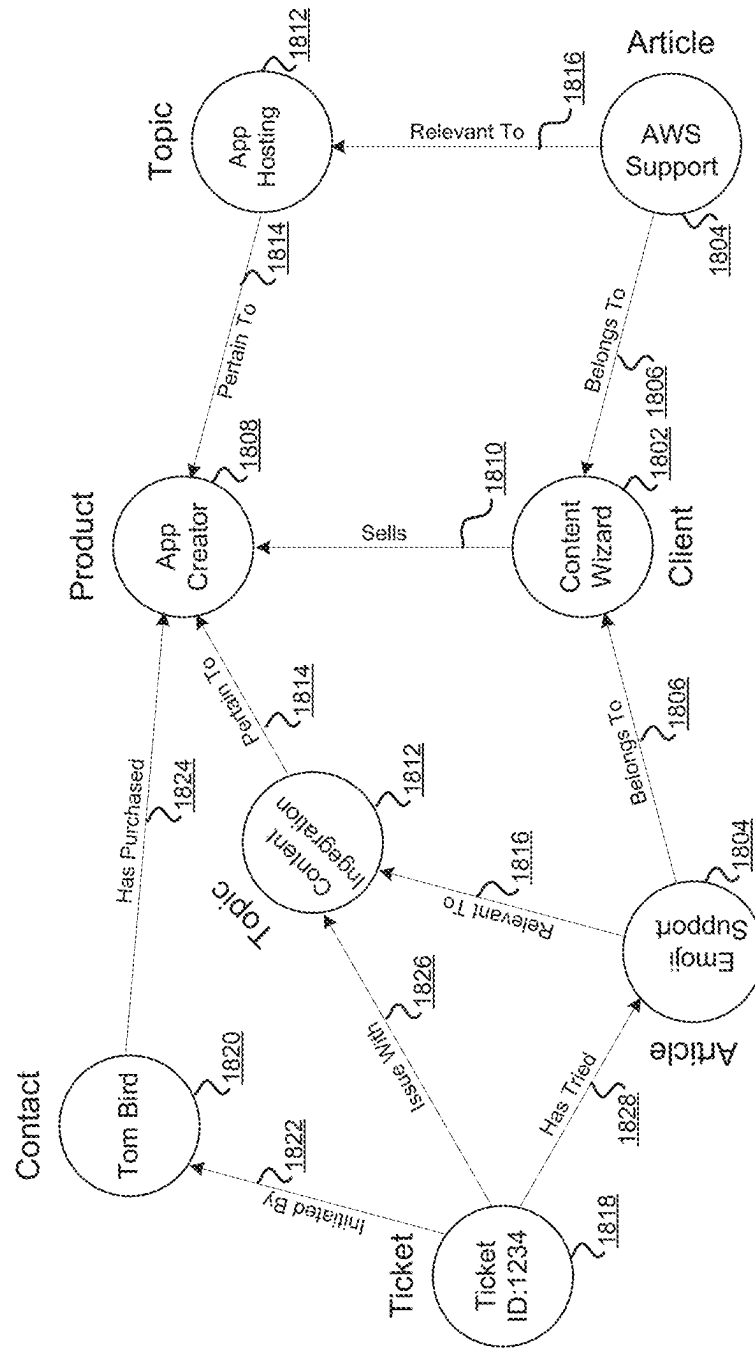


FIG. 18

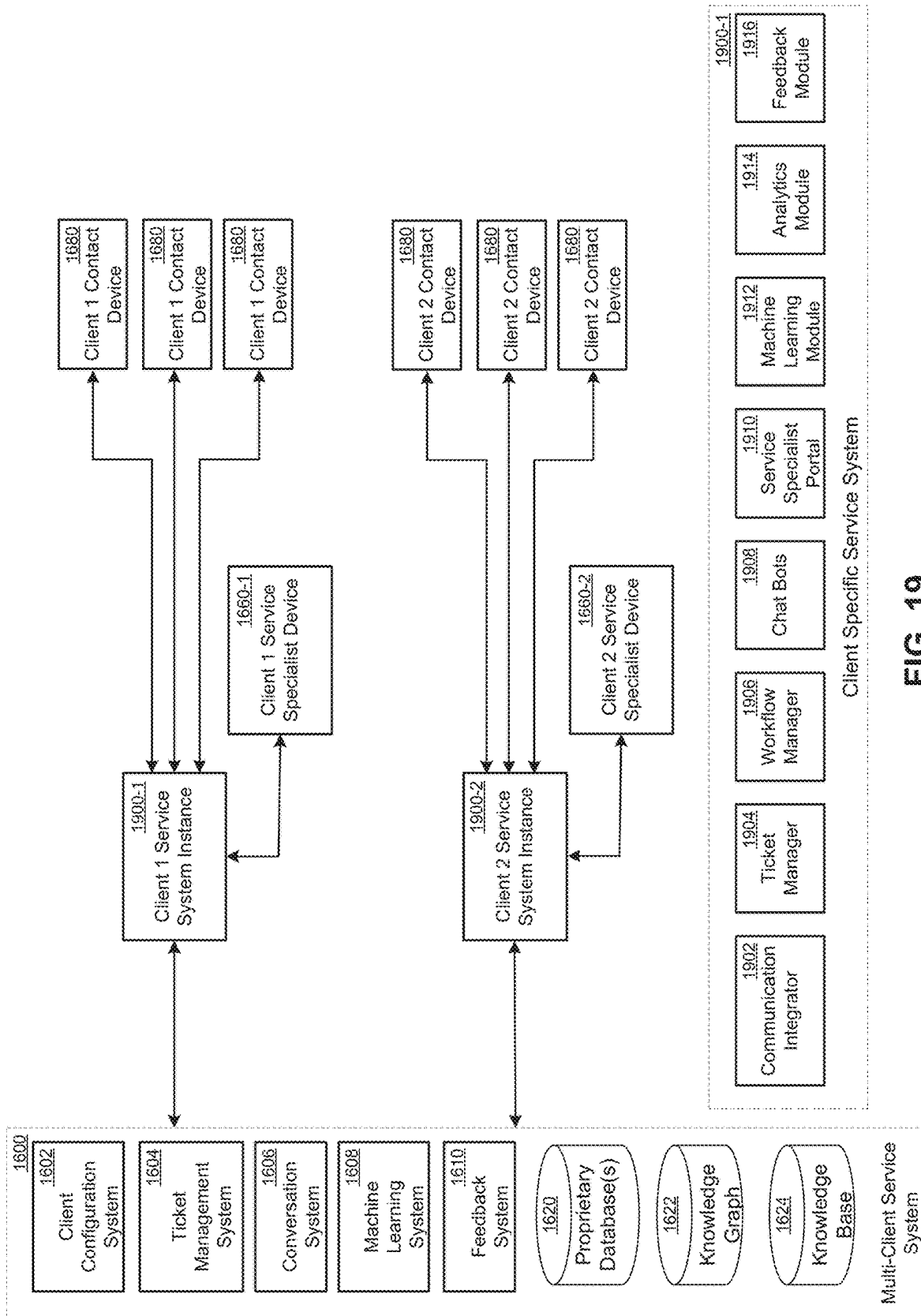


FIG. 19

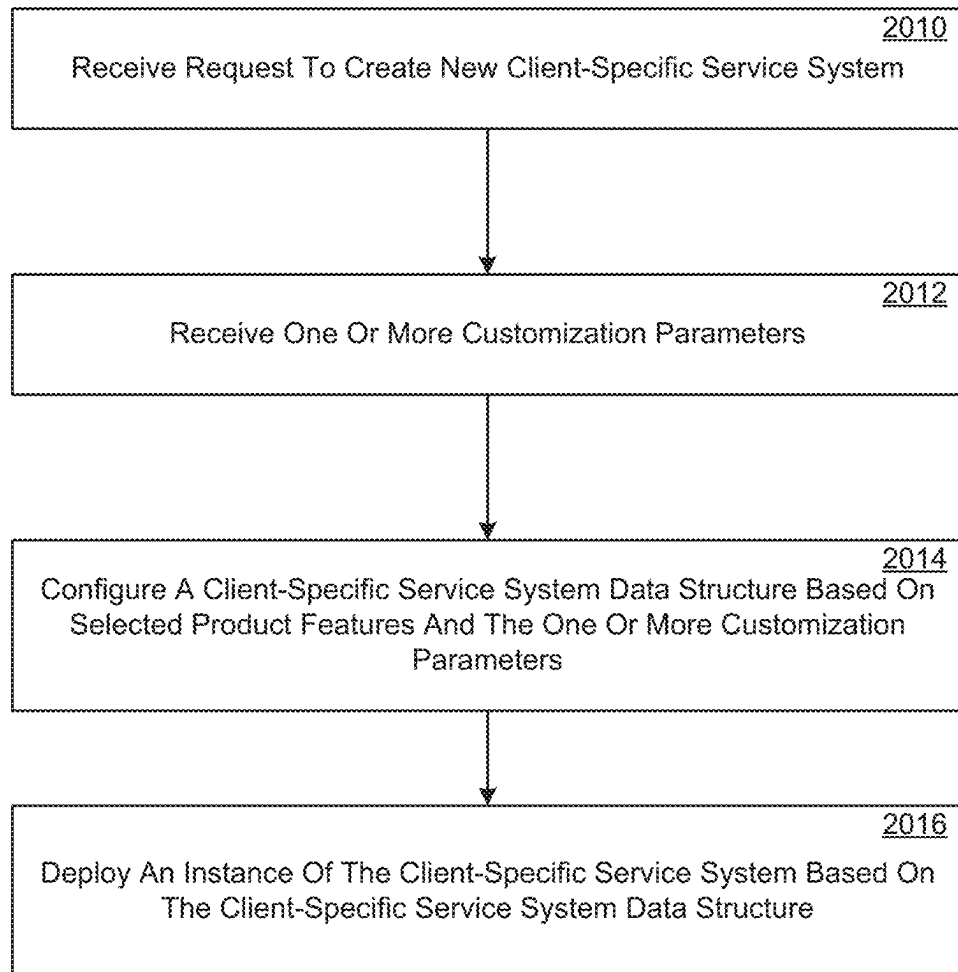


FIG. 20

2100

Dashboard Contacts Marketing Customer Sales Automation Reporting Search

Help with landing page

Email case queue ✓ — ✓ — ✓ — ○ — ○ — Create Date 09/13/2017

▼ About this case Actions

Case status
Email received

Close Date
09/13/2017

HubSpot Owner
Sophie Higga (shigga@hubspot.com)

Last Activity Date
09/18/2017 7:39 PM EDT

Create Date
09/13/2017

Case details
Jenna was having an issue styling the form submit button in the way that she wanted to. I sent her a knowledge doc walking through the styling issues

Send more

Case origin
Email

View Customize Properties

▼ Contacts

Jenna Glat
jglat@hubspot.com

Add another contact

▼ Company Actions

HubSpot, Inc.
http://biglytics.net

Name %
HubSpot, Inc.

Company Domain Name %
biglytics.net

Industry

New note Email Call Log activity Create task Schedule

Start typing to leave a note...

September 2017 Filter timeline (6/7)

Deal moved from Email received to Waiting on customer
September 18th at 8:06 pm

You sent a tracked email to Jenna Glat Glat (jglat@hubspot.com)
September 18th at 7:38 pm Actions

Status: Opened Opens: 1 Clicks: 0 Details

Re: New case "Help with landing page"

Hi Jenna,
That's great! I'm glad everything is sorted now.

You left a note
September 18th at 7:38 pm Edit Actions

Jenna was having an issue styling the form submit button in the way that she wanted to. I sent her a knowledge doc walking through the styling issues and detailed how she can make the changes she wants to.

You sent an email
September 17th at 3:36 pm Edit Actions

Re: New case "Help with landing page"

That worked! Thanks so much. I can get the landing page out on line now.

Expand

Help

FIG. 21



FIG. 22

2300



Dashboard

Contacts

Marketing

Customer

Sales

Automation

Reporting

Search

loggly.com

Hub ID: 2450743

Table

Board

Search for a ticket

Create ticket

Customize

Tickets

All tickets

All saved filters

All tickets

27 tickets

+ Add filter

Options

NEW	WAITING ON CUSTOMER	EMAIL RECEIVED	DEV ISSUE FIXED	CLOSED
Update account name Create date: September 30, 2017 PC	Help with landing page Create date: September 13, 2017 PC	Page performance Create date: September 28, 2017 PC	Error message? Create date: September 10, 2017 PC	How do I change the filter Create date: September 10, 2017 PC
Wrong number on dashboard? Create date: September 30, 2017 PC	Activity excluded based on IP Create date: September 29, 2017 PC		Emails not sending Create date: September 10, 2017 PC	Export has incorrect data Create date: September 1, 2017 PC
Feature request Create date: September 30, 2017 PC				Custom module Create date: September 8, 2017 PC
A/B testing Create date: September 30, 2017 PC				Report building Create date: August 26, 2017 PC

FIG. 23

2400

Customer

Dashboard

Contacts

Feedback

Settings

Search

Customer feedback

Customer Loyalty

Support Experience

Support agent performance

Trends

All feedback submissions

Detractors (0)

Passives (2)

Promoters (10)

Archived (0)

Filter

Show text only submissions

USER	RATING
I loved getting an in-depth demo of each tool, ...	9 Promoter
It was great to come together and get a look a...	10 Promoter
Hugely valuable to hear the full story from mar...	10 Promoter
	9 Promoter
It was great to hear from everyone on progress...	8 Passive

NPS

Health

Details

9 Promoter

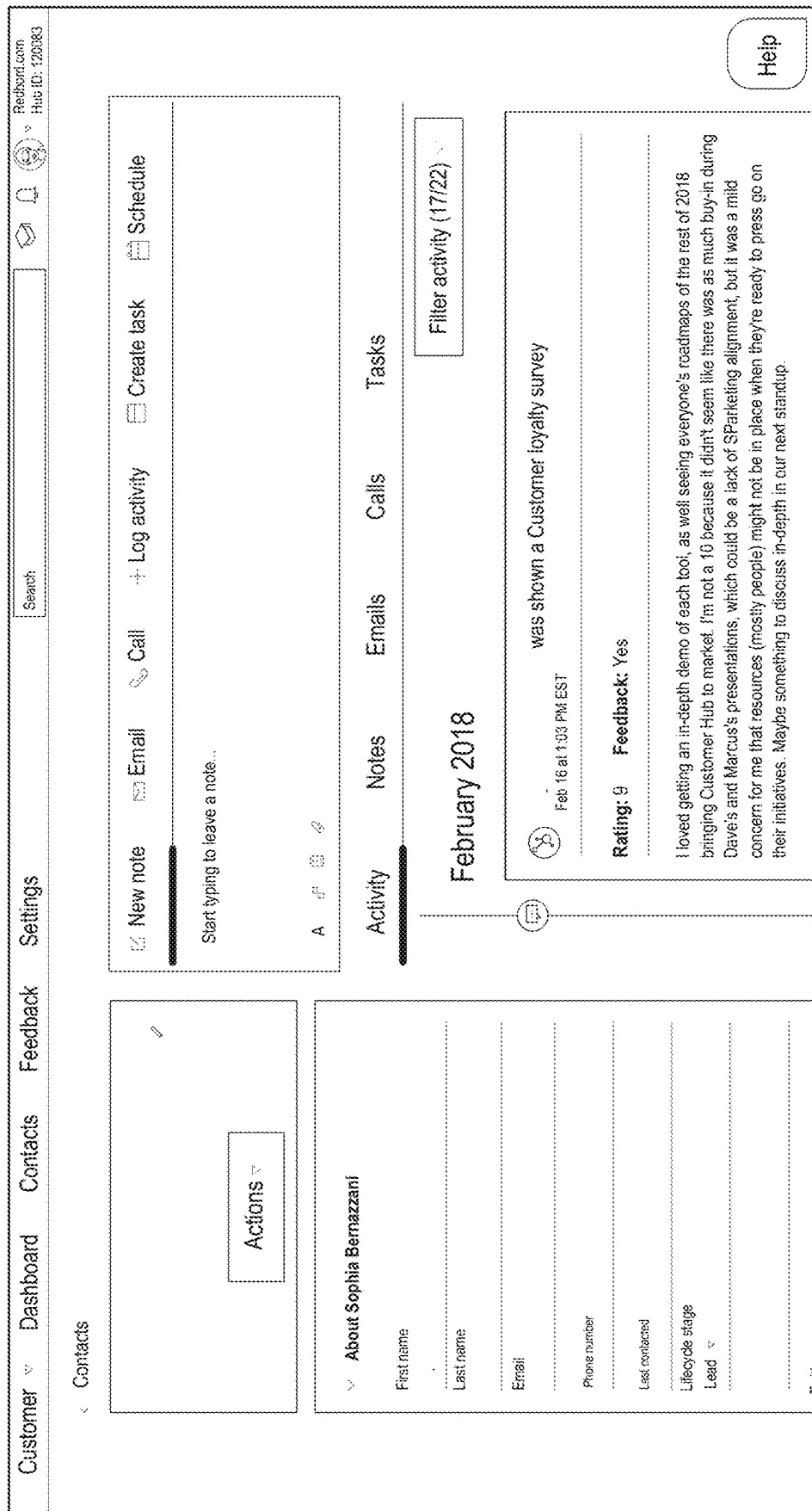
Can you help us by expanding on your score?

I loved getting an in-depth demo of each tool, as well as seeing everyone's roadmaps of the rest of 2018 bringing Customer Hub to market. I'm not a 10 because it didn't seem like there was as much buy-in during Dave's and Marcus's presentations, which could be a lack of SParketing alignment, but it was a mild concern for me that resources (mostly people) might not be in place when they're ready to press go on their initiatives. Maybe something to discuss in-depth in our next standup.

4 days

View profile

FIG. 24



25
G.
E

2600
↙



Actions ▾

⚠ Some of this contact's activity has been excluded based on their IP address. [Learn more](#)

▾ About

First name

Last name

Email

Phone number

Last contacted

Lifecycle stage
Customer

Lead status

[View all properties](#) [View property history](#)

▾ Tickets (5)

When can i have a pony? x

Can i have a pony? x

☒ New note ☒ Email ☒ Call ☒ Log activity ☒ Create task ☒ Schedule

Start typing to leave a note...

A    

Activity Notes Emails Calls Tasks

May 2018 [Filter activity \(17/22\) ▾](#)

No events matching current filters for May 2018

April 2018

  as submitted feedback on a customer loyalty survey
Apr 6 at 9:46 AM GMT+1

Rating: 2 Feedback: Yes

This is simply hogwash. Your support team has been ignoring me for 2 weeks!

  as sent the marketing email **Would you recommend us?**
Apr 4 at 5:23 PM GMT+1

☒ SENT ☒ DELIVERED ☒ OPENED ☒ CLICKED Clicks: 1 [Details](#)

  The lifecycle stage for  as changed to "Customer"
Apr 4 at 1:49 PM GMT+1

  as created from Offline Sources
Apr 4 at 1:49 PM GMT+1

FIG. 26

2700

Delivery

Customize

Follow-up

Thank you

Recipients

Options

Preview

Summary

Choose recipients ⓘ

Decide who gets the survey, based on where they are in the customer lifecycle

CloneDelete

The contact property Lifecycle stage is equal to Customer

AND

OR

OR

EXISTING CONTACTS

When to send

×

Send the survey while the purchase is fresh in customers' minds, but make sure you give them enough time to use your product or service too.

existing contacts:

all contacts who already meet these criteria.

Send the first survey: ⓘ

30

days after meeting the required criteria

Send recurring surveys: ⓘ

Every 3 months (recommended)

FIG. 27

2700

Delivery

Customize

Follow-up

Thank you

Recipients

Options

Preview

Summary

Choose recipients ⓘ

Decide who in the customer base to send to

Who to send to ⓘ

Don't be too selective. For a true representation of how customers are feeling, it's a good idea to send the survey to everyone.

The contact property Lifecycle stage is equal to Customer ⓘ

AND

OR

EXISTING CONTACTS

835

Enroll existing contacts: ☒ Enroll contacts who already meet these criteria.

Send the first survey: ⓘ

30 days after meeting the required criteria

Send recurring surveys: ⓘ

Every 3 months (recommended) ▼

FIG. 28



Customer Loyalty

Use this survey to track how likely your customers are to recommend your product.

Understand your customers' loyalty over time.

Use an **Net Promoter Score survey (NPS)*** to track how likely your customers are to recommend your product.

Preview

How it works

...

From: Biglytics

Subject: Hey Johnny, would you recommend us?

Share your feedback on Biglytics



Your opinion matters.

Please take a moment to rate us from 0-10 below.

How likely are you to recommend Biglytics to a friend or colleague?

0

1

2

3

4

5

6

7

8

9

10

☐

☐

☐

☐

☐

☐

☐

☐

☐

☐

☐

Not at all likely

Extremely likely

FIG. 29

2900

Delivery

Customize

Follow-up

Thank you

Recipients

Options

Preview

Summary

Customize your follow-up question

A good follow-up NPS question aims to discover why customers gave you the score they did.

Play around with the words to inject your brand's personality, but don't complicate it. A clear, simple question will lead to a higher response rate.

0-5 Detractors	7-5 Passive	9-10 Promoters
----------------	-------------	----------------

Follow-up question

Can you tell us more about your experience, so we can get it right the next time?

How likely are you to recommend to a friend or colleague?

0

1

2

3

4

5

6

7

8

9

10

☐
☐
☐
☒
☐
☐
☐
☐
☐
☐
☐

Not at all likely

Extremely likely

Can you tell us more about your experience, so we can get it right the next time?

Submit

150

This survey is a service from

Delivered by

Customer Hub

69

36

2900

Delivery

Display

Customize

Follow-up

Thank you

Recipients

Options

Preview

Summary

Choose where the survey will display on your website.

Select a type

Slide-in box left

Slide-in box right

Dropdown banner

How likely are you to recommend to a friend or colleague?

0

1

2

3

4

5

6

7

8

9

10

Not at all likely

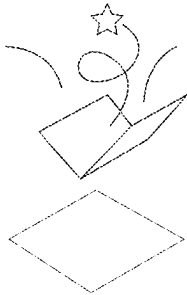
Extremely likely

FIG. 31

2900



Share your feedback on



Your opinion matters.

Please take a moment to rate our services
from 0-10 below.

How likely are you to recommend
to a friend or colleague?

0	1	2	3	4	5
☐	☐	☐	☒	☐	☐
6	7	8	9	10	
☐	☐	☐	☐	☐	

0 - Not at all likely

10 - Extremely likely

FIG. 32

2900

Delivery

Customize

Follow-up

Thank you

Recipients

Options

Preview

Summary

Choose recipients ①

Decide who gets the survey, based on where they are in the customer lifecycle.

CloneDelete

The contact property Lifecycle stage is equal to Customer.

AND

OR

OR

EXISTING CONTACTS

How often to send

If you change your offering often we recommend a quarterly check in with customers.

For a product or service that doesn't change often, or if it takes a while to make improvements after listening to feedback, we recommend waiting longer.

For a once off product or service, there's no need to send a recurring survey.

Send

30

Send recurring surveys: ①

Every 3 months (recommended)

Existing contacts:

contacts who already meet these criteria

FIG. 33

2900 ↗

Delivery

Customize

Follow-up

Thank you

Recipients

Options

Preview

Summary

Choose recipients ⓘ

Decide who gets the survey, based on where they are in the customer lifecycle.

CloneDelete

The contact property Lifecycle stage is equal to Customer. ⓘ

and

The contact property Close date is known. ⓘ

AND

OR

EXISTING CONTACTS

0

Enroll existing contacts:

☐ Enroll contacts who already meet these criteria.

Send the first survey: ⓘ

30

days after meeting the required criteria

Send recurring surveys: ⓘ

Every 3 months (recommended) ▾

FIG. 34

2900

Delivery

Customize

Follow-up

Thank you

Recipients

Options

Preview

Summary

Choose recipients ⓘ

Decide who gets the survey, based on where they are in the customer lifecycle.

< Back

Contact properties

Please select a property ▾

Search properties 🔍

Broadcast Clicks

Last email name

Emails opened

Lead status

Total revenue

Last name

Postal code

Close date

IP City

FIG. 35

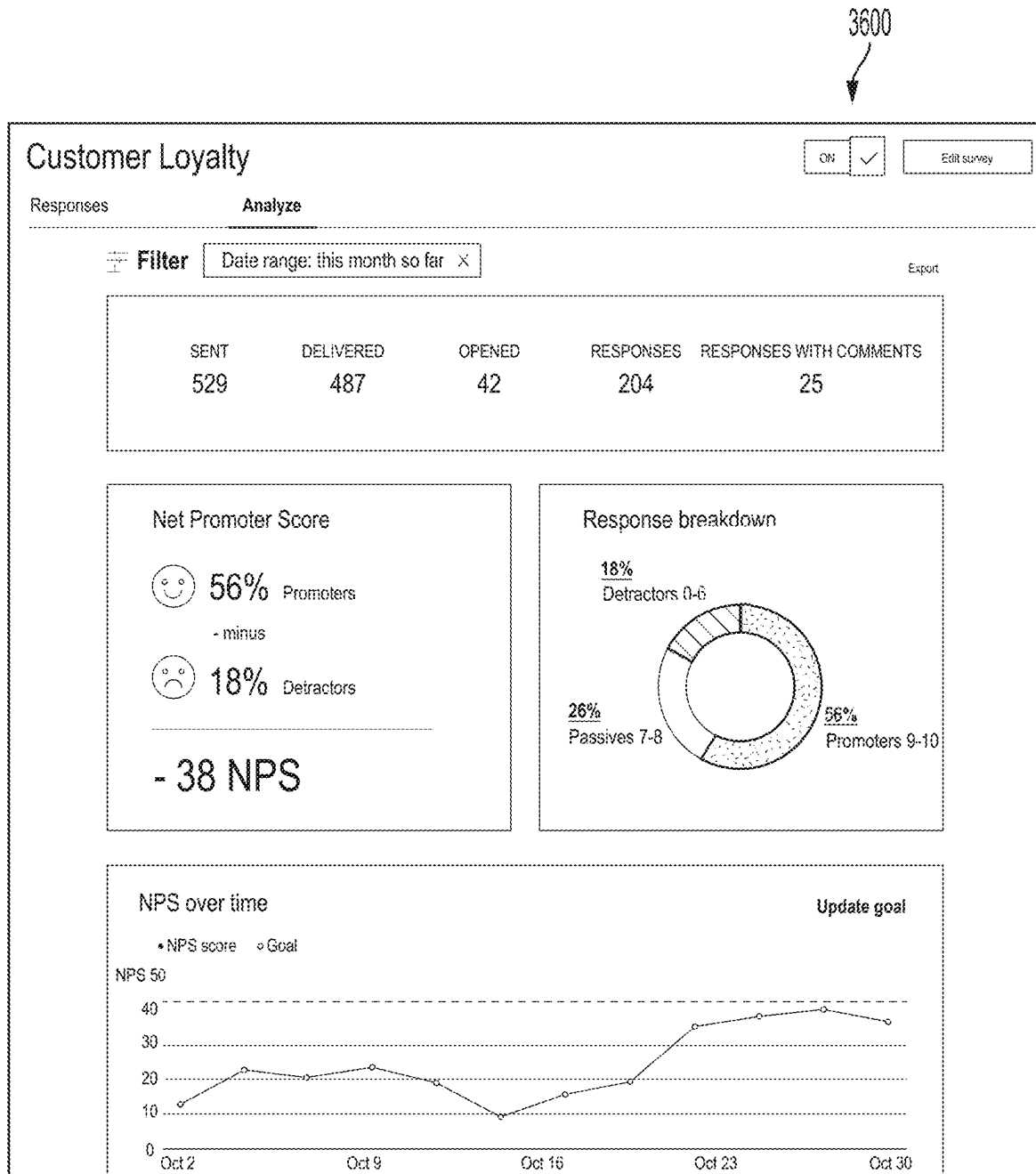


FIG. 36

3700

Settings

Account Defaults

Marketing

Properties

Sales

Customer

Feedback

Knowledge

User & Teams

Knowledge Base

General

Customization

Knowledge base not live

You must connect your domain before you can publish your knowledge base to your customers

Domain

Knowledge Base Domain

http://hs-sites.com/

Connect domain

Knowledge Base root URL

http://hs-sites.com/

knowledge

Language

By default your knowledge base will inherit the language of your domain. You can change the language here and it will only apply to your knowledge base.

Knowledge base language

English

Preview knowledge base

FIG. 37

3800

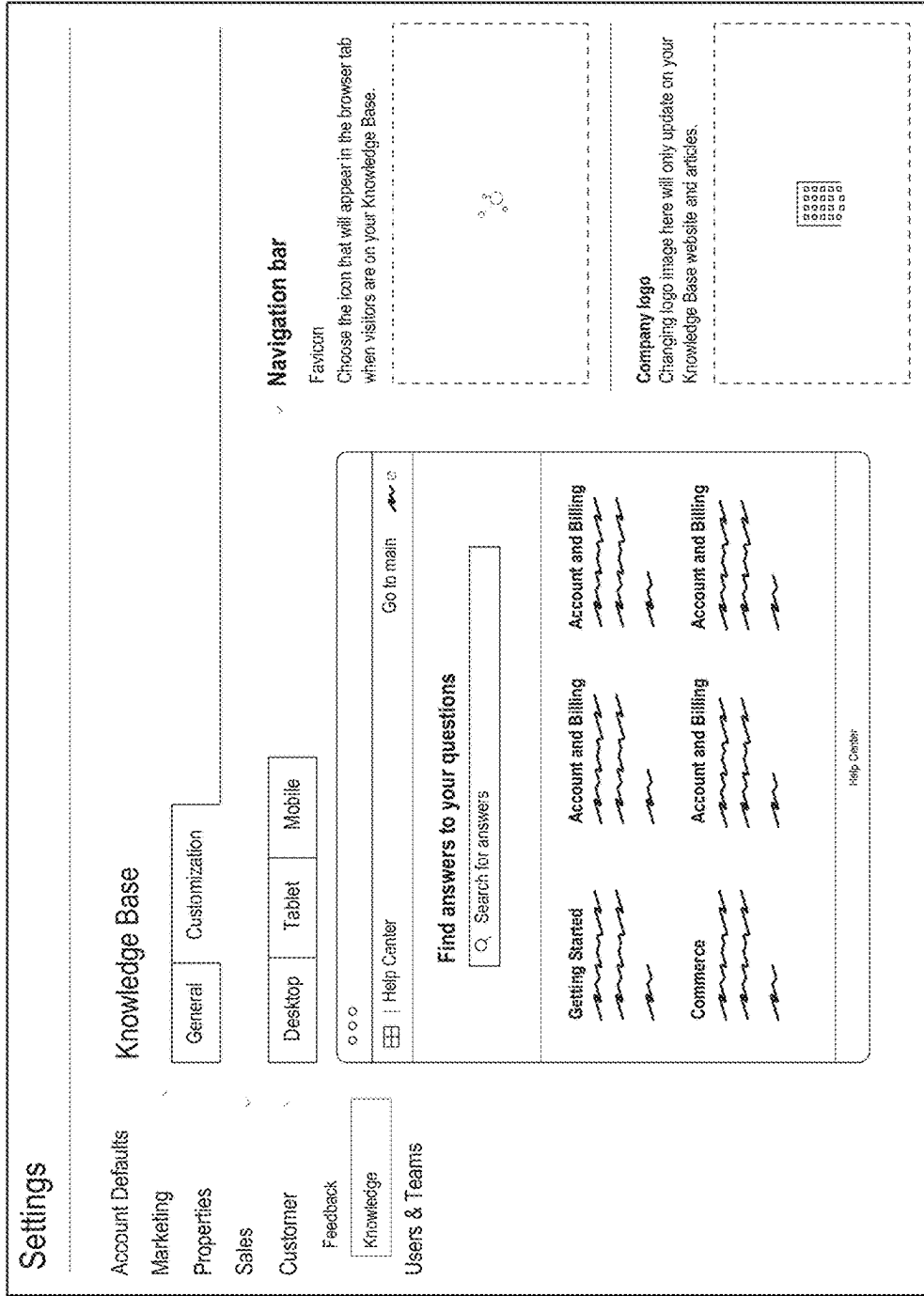


FIG. 38

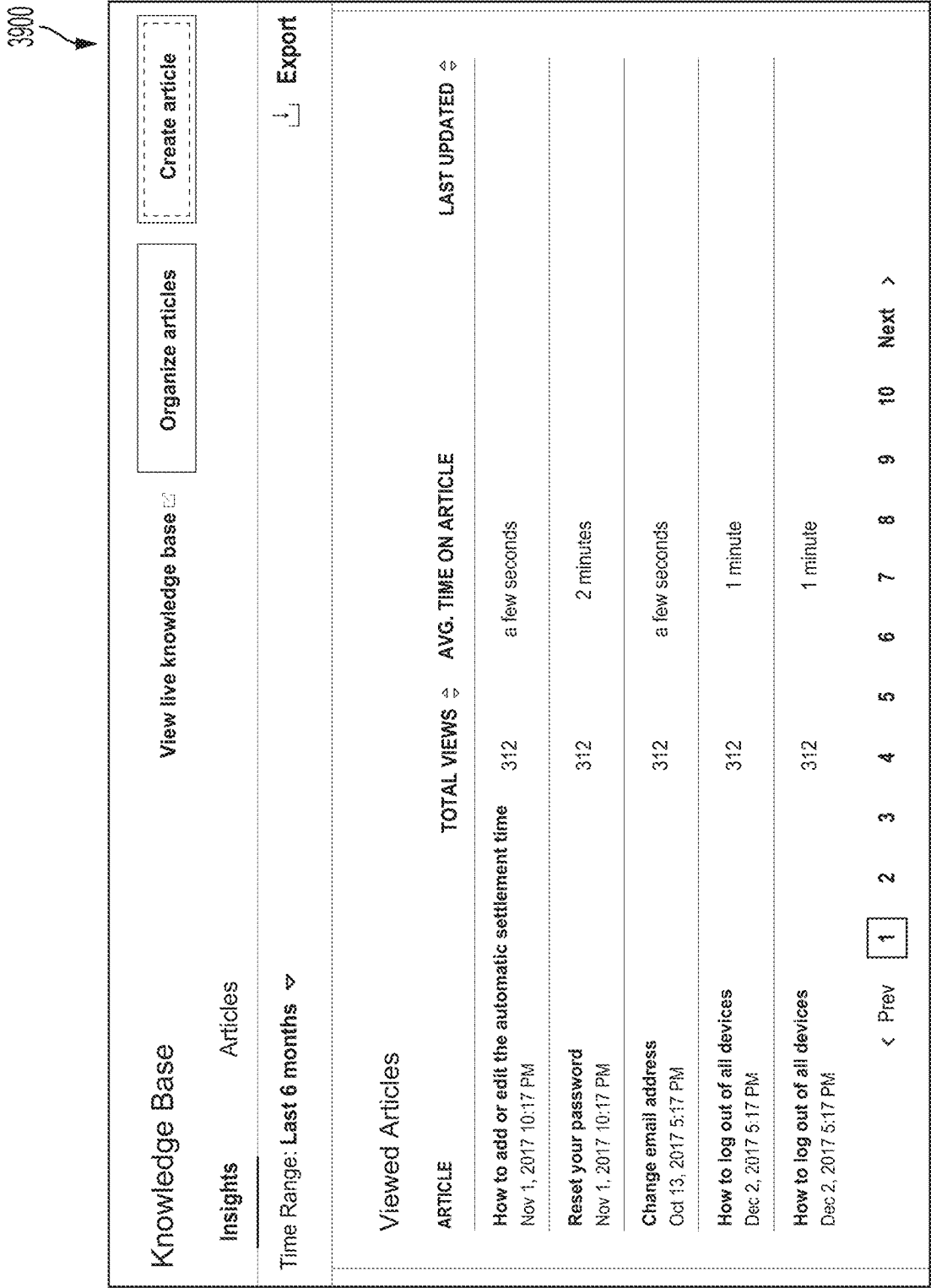


FIG. 39

4000

Settings

Account Defaults

Analytics & Reporting

Marketing

Properties

Sales

Service

Something

Something

Tickets

Something

Something

Something

Something

< Back to ticket settings

Support Ticket Pipeline

Configure

Automation

Enable automation for this pipeline

Trigger the actions below when a ticket moves into the associated ticket stage.

When ticket reaches...	When ticket reaches...	When ticket reaches...	When ticket reaches...
NEW	WAITING ON CONTACT	WAITING ON US	CLOSED
+ Create workflow	+ Create workflow	+ Create workflow	+ Create workflow

Search settings

FIG. 40

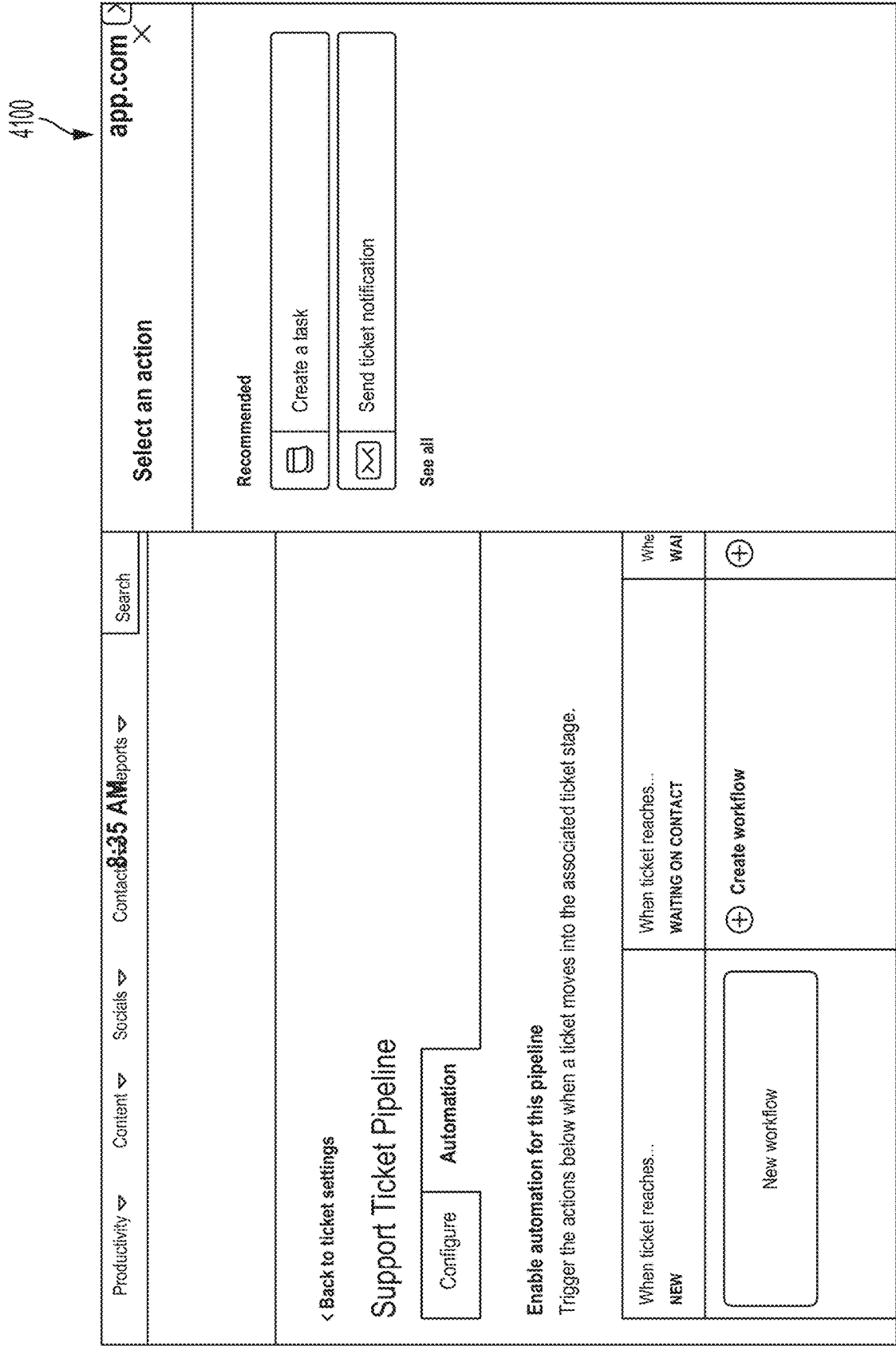


FIG. 41

4200

Support Ticket Pipeline

Configure

Automation

When ticket reaches...
NEW

New workflow

When ticket reaches...
WAITING ON CONTACT

⊕

Create workflow

When ticket reaches...
WAITING ON CONTACT

When ticket reaches...
NEW

New workflow

When ticket reaches...
WAITING ON CONTACT

⊕

Create workflow

When ticket reaches...
WAITING ON CONTACT

When ticket reaches...
NEW

New workflow

When ticket reaches...
WAITING ON CONTACT

⊕

Create workflow

When ticket reaches...
WAITING ON CONTACT

Select an action

Recommended

Create a task

Send ticket notification

See less

Search actions

Building blocks

Add delay

Productivity

Create a task

Send internal email

Send internal SMS

Marketing

Send marketing email

FIG. 42

4300

Productivity ▾

Contact ▾

Socials ▾

8:35 AM

Search

Send ticket notification

app.com

Back to ticket settings

Support Ticket Pipeline

Configure

Automation

Enable automation for this pipeline

Trigger the actions below when a ticket moves into the associated ticket stage.

When ticket reaches...	When ticket reaches...	When...
NEW	WAITING ON CONTACT	WAITING ON CONTACT
New workflow	⊕ Create workflow	⊕

Base template

Ticket received

Customize

Subject: Ticket ID

Your ticket has been received

Your request has been received and is being reviewed by our support staff. To add additional comments, reply to this email.

You can update the styling of these emails in your settings.

FIG. 43

4400

Back to ticket settings

Support Ticket Pipeline

Configure

Automation

Enable automation for this pipeline

Trigger the actions below when a ticket moves into the associated ticket stage.

When ticket reaches...	When ticket reaches...	When ticket reaches...	When ticket reaches...
NEW	WAITING ON CONTACT	WAITING ON US	CLOSED
☒ Send ticket notification Your ticket was received +	+ Create workflow	+ Create workflow	+ Create workflow

FIG. 44

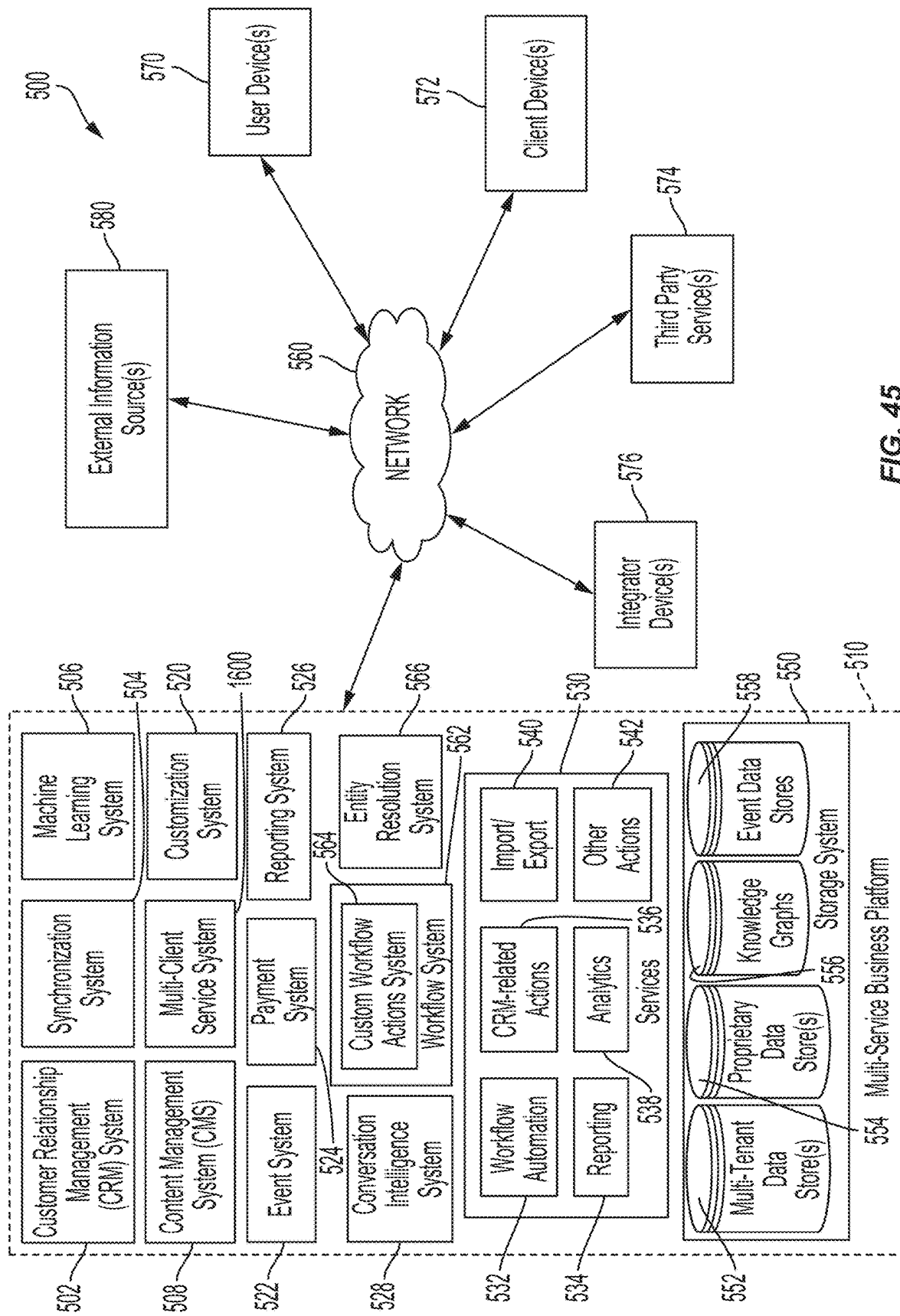


FIG. 45

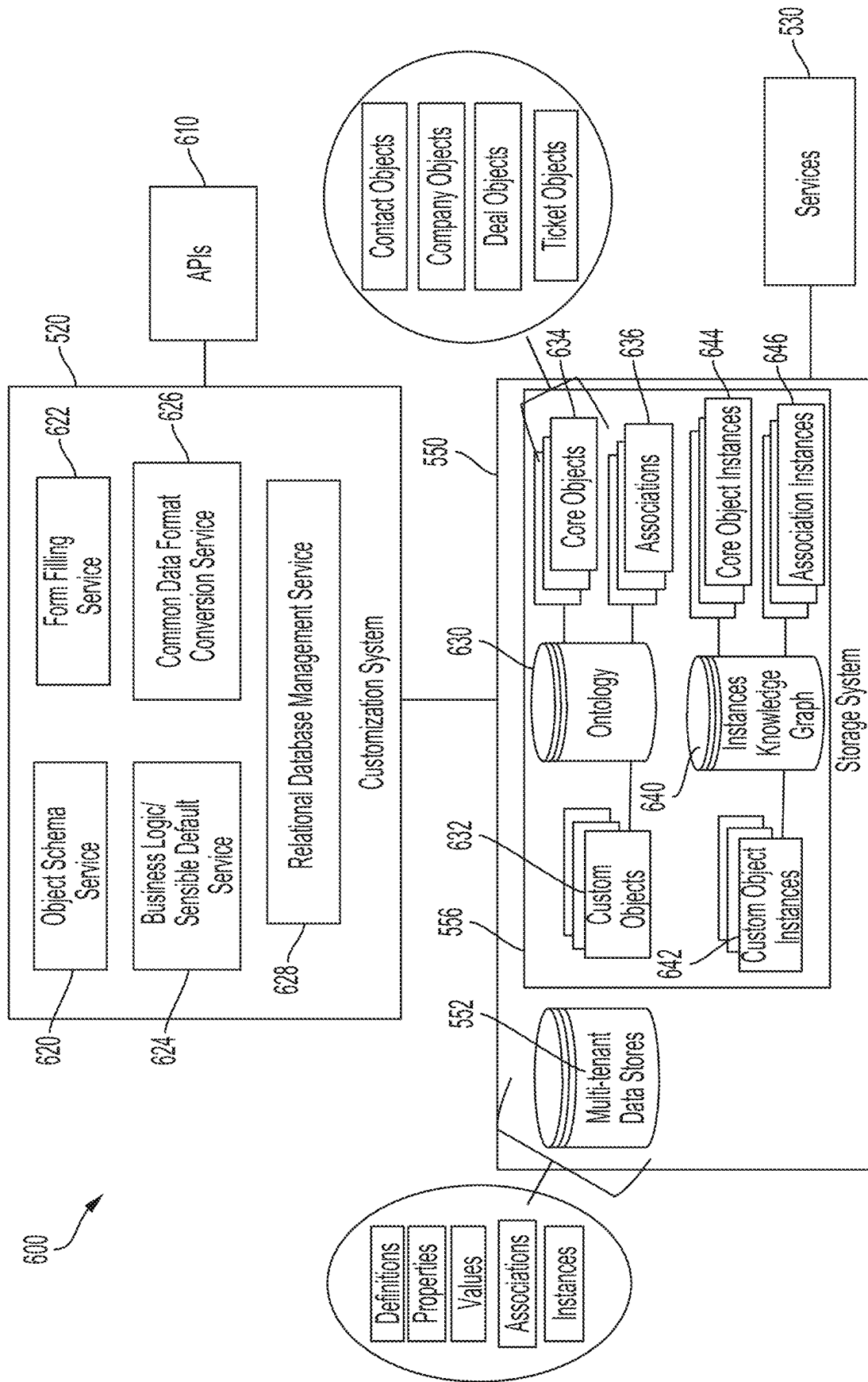


FIG. 46

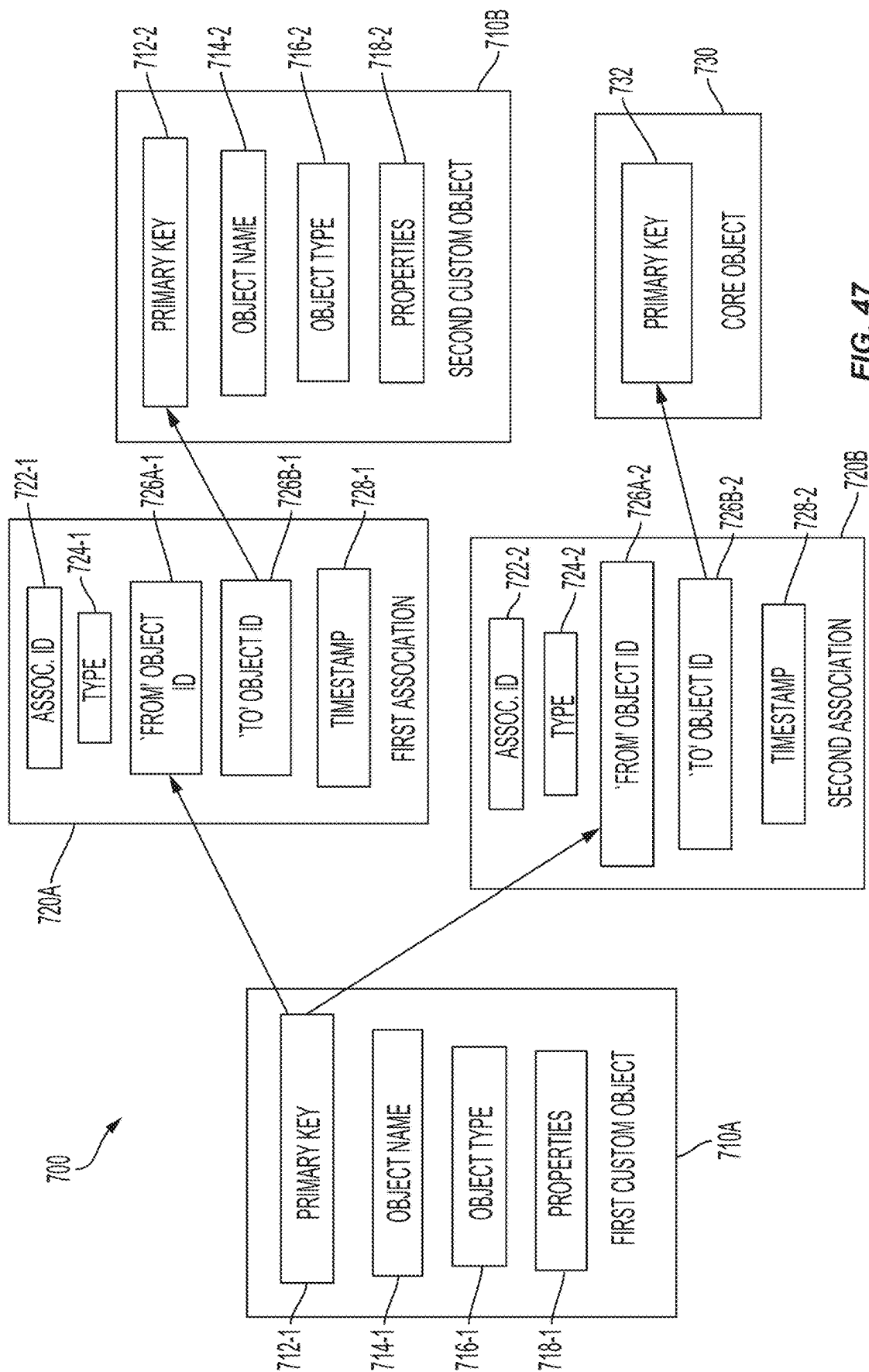


FIG. 47

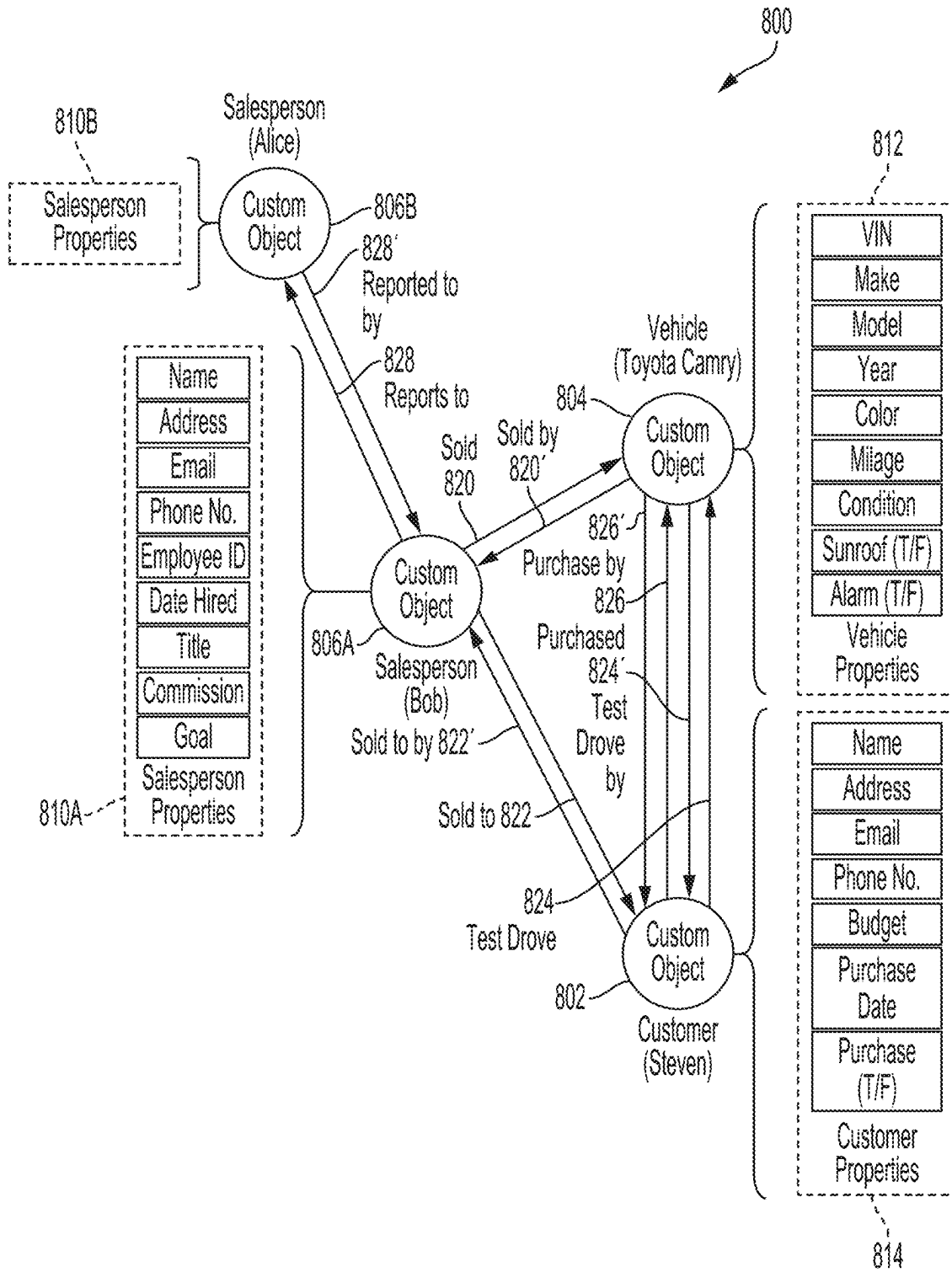


FIG. 48

1100

Create Subscription

Create new

Add existing

Edit this form

Subscriber Name*

New Subscription

Start Date*

09/16/2020

Term*

Annual

Annual

Monthly

Create Subscription

Create and add another

Cancel

Contacts

Conversations

Marketing

Sales

Service

Automation

Reports

Activity

Notes

Emails

Calls

Tasks

Meetings

Filter by: Filter activity (2024)

All users

All teams

September 2020

Lifecycle change

John Smith updated the lifecycle stage for this contact to subscriber. View details

Contact created

This contact was created

John Smith

Sr. Product Manager at HubSpot, Inc.

Note

Email

Call

Log

Task

Meet

First name

John

Last name

Smith

Email

smith@hubspot.com

Phone number

Contact owner

John Smith

Last contacted

..

Lifecycle stage

Subscriber

Lead status

About this contact

View all properties

New property history

Website activity

SITE VISITS

PAGES VIEWED

Communication subscriptions

Know how John Smith wants to be contacted from you.

John Smith isn't opted in to any of your email subscriptions right now.

Add subscription

View history

FIG. 49A

1200

○	Contacts ~	Conversations ~	Marketing ~	Sales ~	Service ~	Automation ~	Reports ~	Q	🔍	🔔	👤	Customer Order Demo ~
Subscriptions ~ 71 Subscriptions 48 Subscriptions + Add view All views												
Search Subscription Name ~ Object starts with ~ Object ends with ~ Subscription Type ~ More filters Table actions ~ Save view												
☐	SUBSCRIPTION NAME	START DATE	SUBSCRIPTION TYPE	TERM	END DATE							
☐	Service Professional - 2014	Sep 3, 2014	Service Professional	Annual	Sep 3, 2015							
☐	Service Free - 2013	Aug 8, 2013	Service Free	Annual	Aug 8, 2020							
☐	Service Free - 2012	Feb 14, 2012	Service Free	Annual	Feb 14, 2022							
☐	Service Free - 2012	Feb 14, 2012	Service Free	Annual	Feb 14, 2022							
☐	Service Free - 2013	Aug 8, 2013	Service Free	Annual	Aug 8, 2020							
☐	Service Enterprise - 2016	Apr 5, 2016	Service Enterprise	Monthly	Apr 5, 2017							
☐	Service Enterprise - 2016	May 13, 2016	Service Enterprise	Monthly	May 13, 2017							
☐	Service Enterprise - 2020	Jan 1, 2020	Service Enterprise	Monthly	Jan 1, 2021							
☐	Service Enterprise - 2011	Apr 12, 2011	Service Enterprise	Annual	Apr 12, 2018							
☐	Service Enterprise - 2008	May 13, 2008	Service Enterprise	Monthly	May 13, 2007							
☐	Service Enterprise - 2016	Apr 5, 2016	Service Enterprise	Monthly	Apr 5, 2017							
☐	Service Enterprise - 2020	Jan 1, 2020	Service Enterprise	Monthly	Jan 1, 2021							
☐	Service Enterprise - 2011	Apr 12, 2011	Service Enterprise	Annual	Apr 12, 2018							
☐	Sale Starter - 2014	Sep 3, 2014	Sale Starter	Annual	Sep 4, 2015							
☐	Sale Professional - 2019	Oct 10, 2019	Sale Professional	Monthly	Oct 10, 2021							
☐	Sale Professional - 2017	Dec 12, 2017	Sale Professional	Monthly	Dec 12, 2019							
☐	Sale Professional - 2015	Nov 27, 2015	Sale Professional	Monthly	Nov 27, 2018							
☐	Sale Professional - 2017	Nov 21, 2017	Sale Professional	Annual	Nov 21, 2020							
☐	Sale Professional - 2015	Oct 10, 2019	Sale Professional	Annual	Oct 10, 2020							
☐	Sale Professional - 2019	Oct 10, 2019	Sale Professional	Monthly	Oct 10, 2021							
☐	Sale Professional - 2017	Dec 12, 2017	Sale Professional	Monthly	Dec 12, 2019							
☐	Sale Professional - 2015	Nov 27, 2015	Sale Professional	Monthly	Nov 27, 2018							
☐	Sale Professional - 2017	Nov 21, 2017	Sale Professional	Annual	Nov 21, 2020							

FIG. 49B

Filters

Subscription properties
Comment from sales manager
Cost
Created by user ID
Description
End Date
Health Score
Number of companies interested in
Object ID
Object create datetime
Object last modified datetime
Recent feedback
Referenced by
Start Data
State
Subscription Name
Subscription Owner
Subscription Owner Notes
Subscription Start Time
Subscription Type
Term
Updated by user ID

Search

SUBSCRIPTION NAME :	START DATE	SUBSCRIPTION TYPE	TERM
Service Professional - 2014	Sep 3, 2014	Service Professional	Annual
Service Pro - 2013	Aug 8, 2013	Service Free	Annual
Service Pro - 2012	Feb 14, 2012	Service Free	Annual
Service Pro - 2012	Feb 14, 2012	Service Free	Annual
Service Pro - 2011	Aug 5, 2011	Service Free	Annual
Service Enterprise - 2018	Apr 5, 2018	Service Enterprise	Monthly
Service Enterprise - 2016	May 13, 2016	Service Enterprise	Monthly
Service Enterprise - 2020	Jan 1, 2020	Service Enterprise	Monthly
Service Enterprise - 2011	Apr 12, 2011	Service Enterprise	Annual
Service Enterprise - 2008	May 13, 2008	Service Enterprise	Monthly
Service Enterprise - 2016	Apr 5, 2018	Service Enterprise	Monthly
Service Enterprise - 2039	Jun 1, 2020	Service Enterprise	Monthly
Service Enterprise - 2011	Apr 12, 2011	Service Enterprise	Annual
Sales Starter - 2014	Sep 3, 2014	Sale Starter	Annual
Sale Professional - 2019	Oct 10, 2019	Sale Professional	Monthly
Sale Professional - 2017	Dec 12, 2017	Sale Professional	Monthly
Sale Professional - 2015	Nov 27, 2015	Sale Professional	Monthly
Sale Professional - 2017	Apr 31, 2017	Sale Professional	Annual
Sale Professional - 2019	Oct 10, 2019	Sale Professional	Annual
Sale Professional - 2019	Oct 10, 2019	Sale Professional	Monthly
Sale Professional - 2017	Dec 12, 2017	Sale Professional	Monthly
Sale Professional - 2015	Nov 27, 2015	Sale Professional	Monthly
Sale Professional - 2017	Mar 31, 2017	Sale Professional	Annual

FIG. 49C

1300

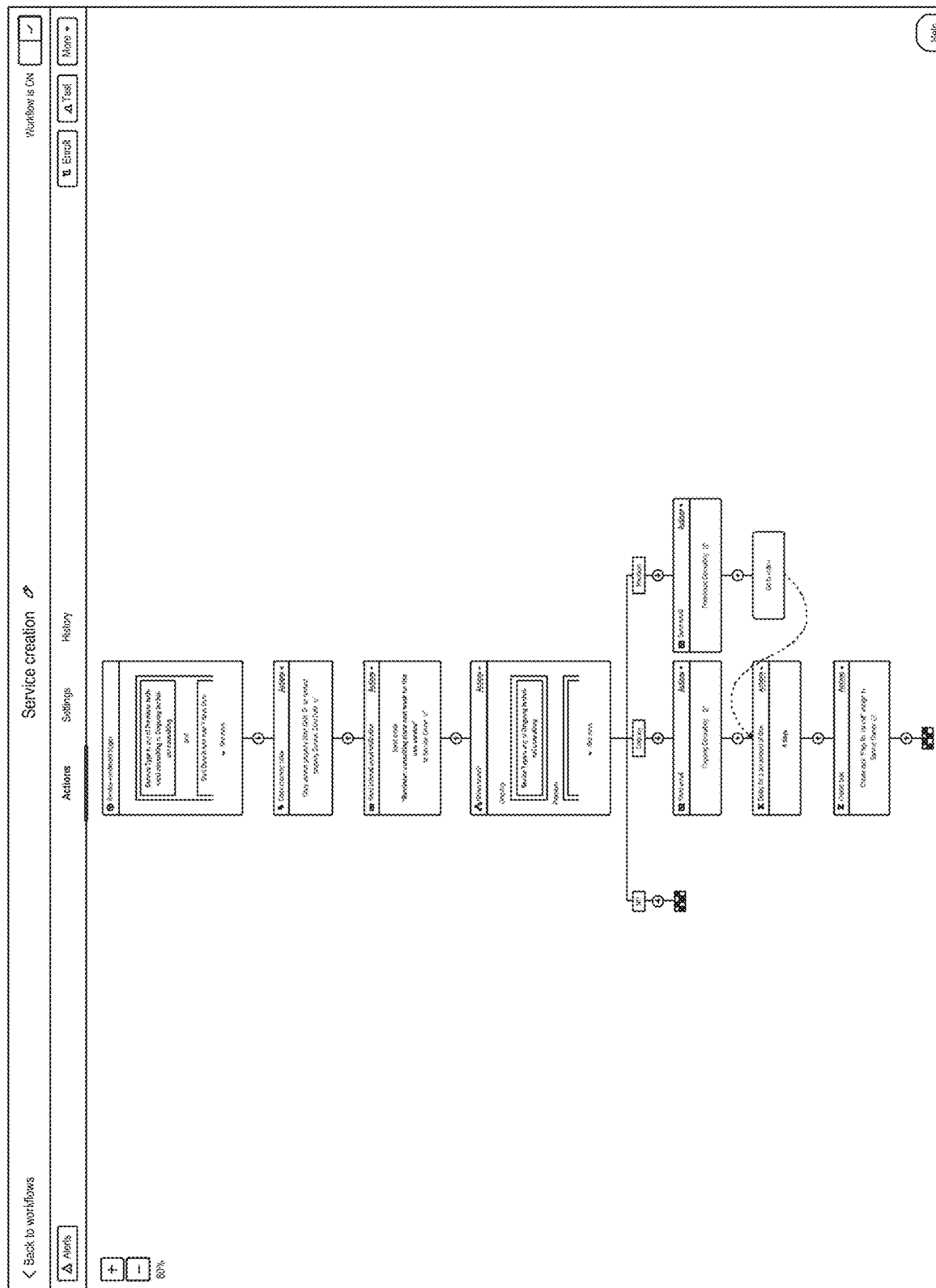


FIG. 49D

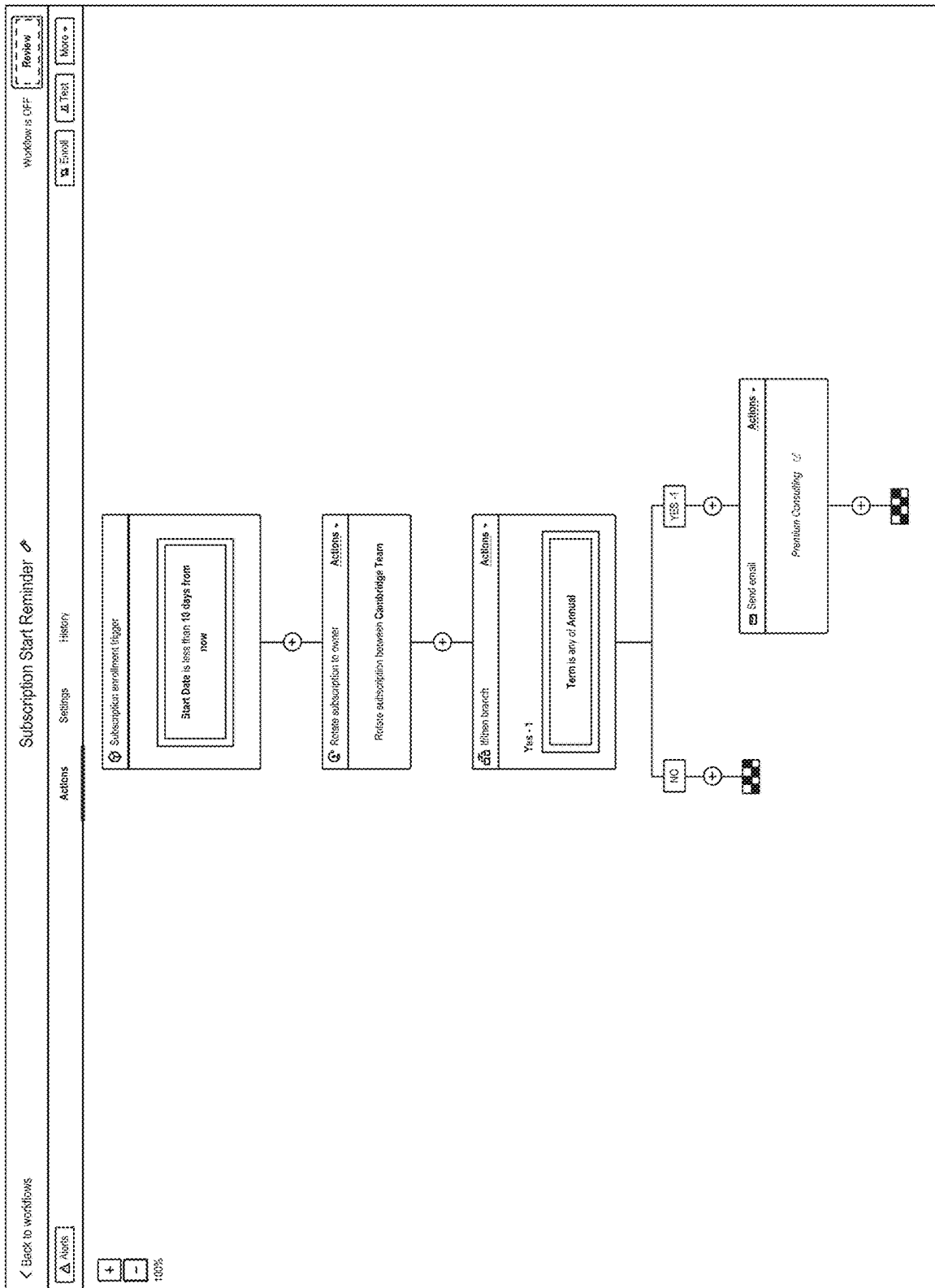


FIG. 49E

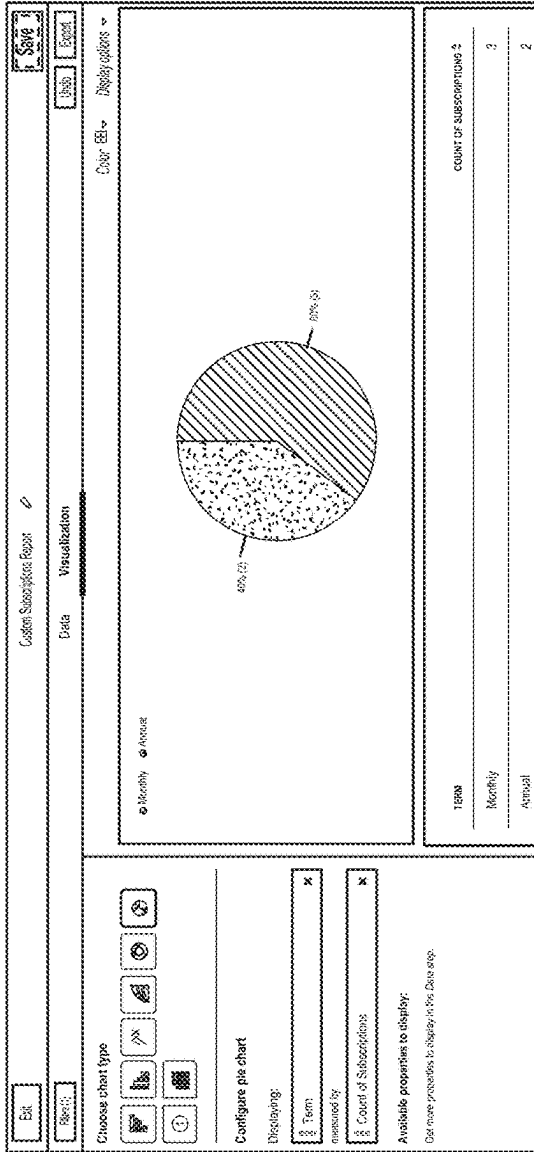


FIG. 49F

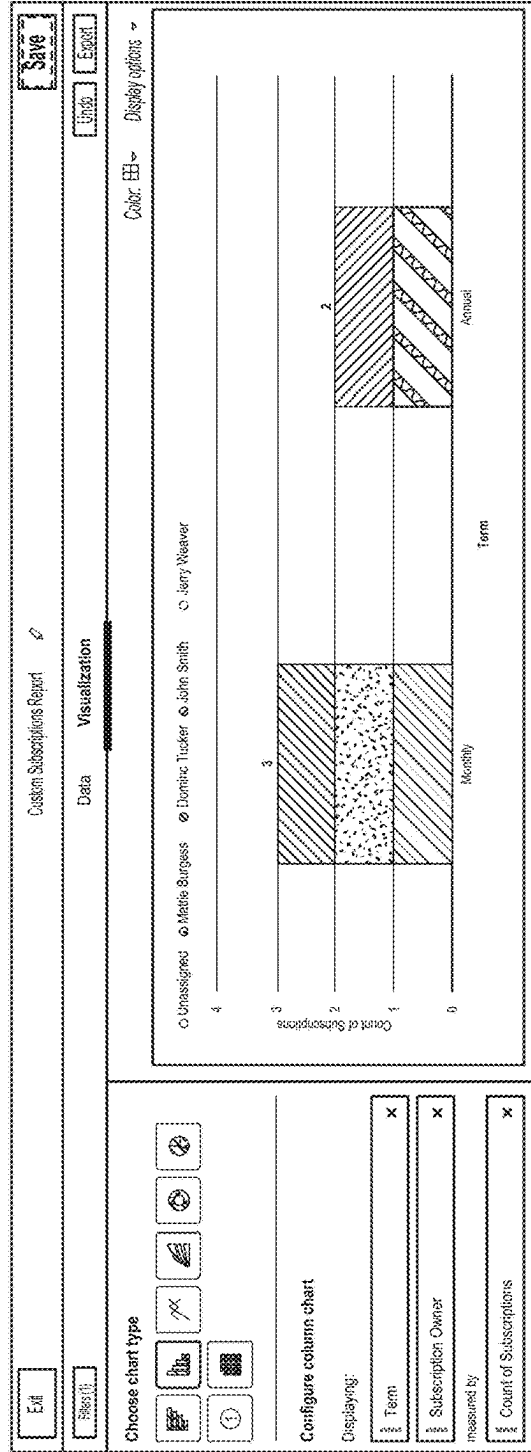
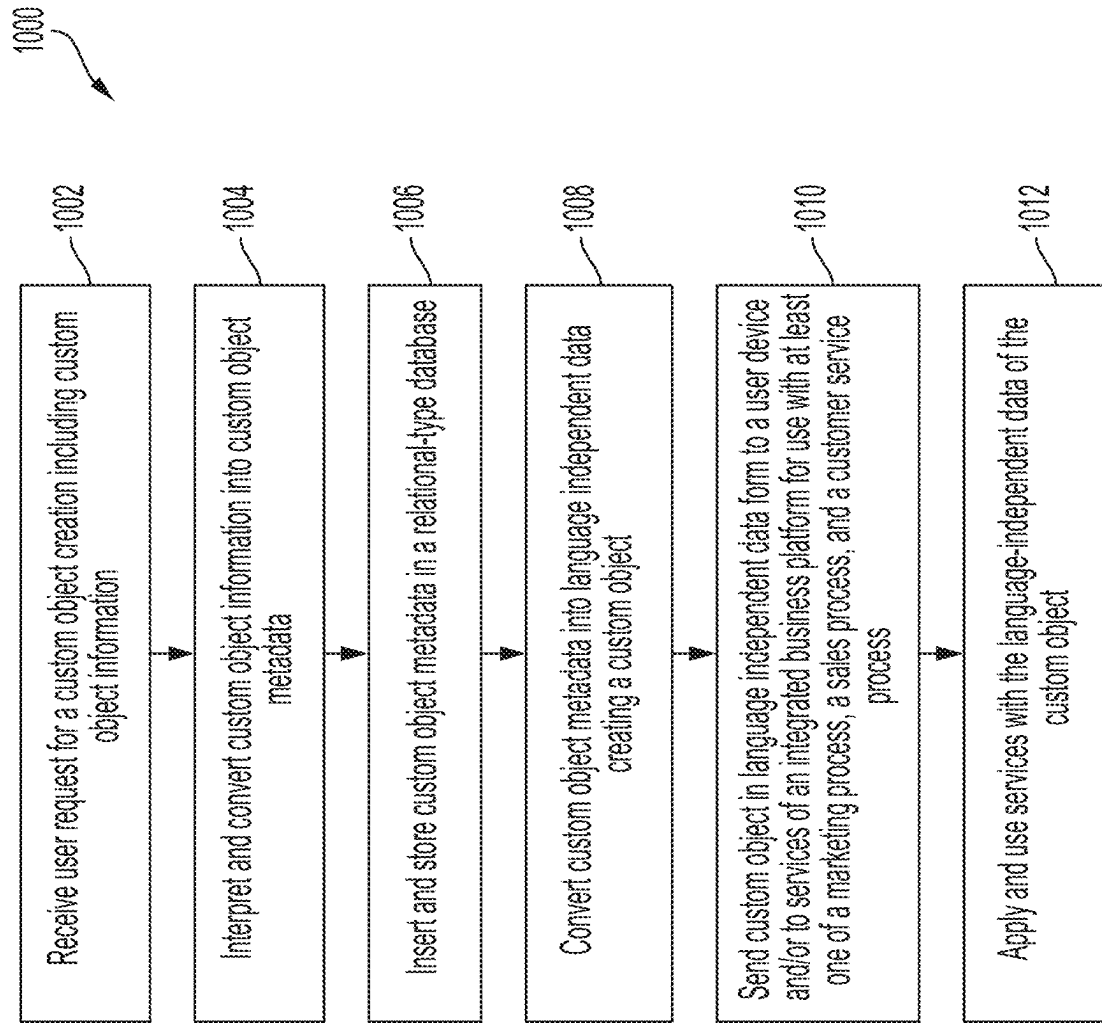


FIG. 49G

**FIG. 50**

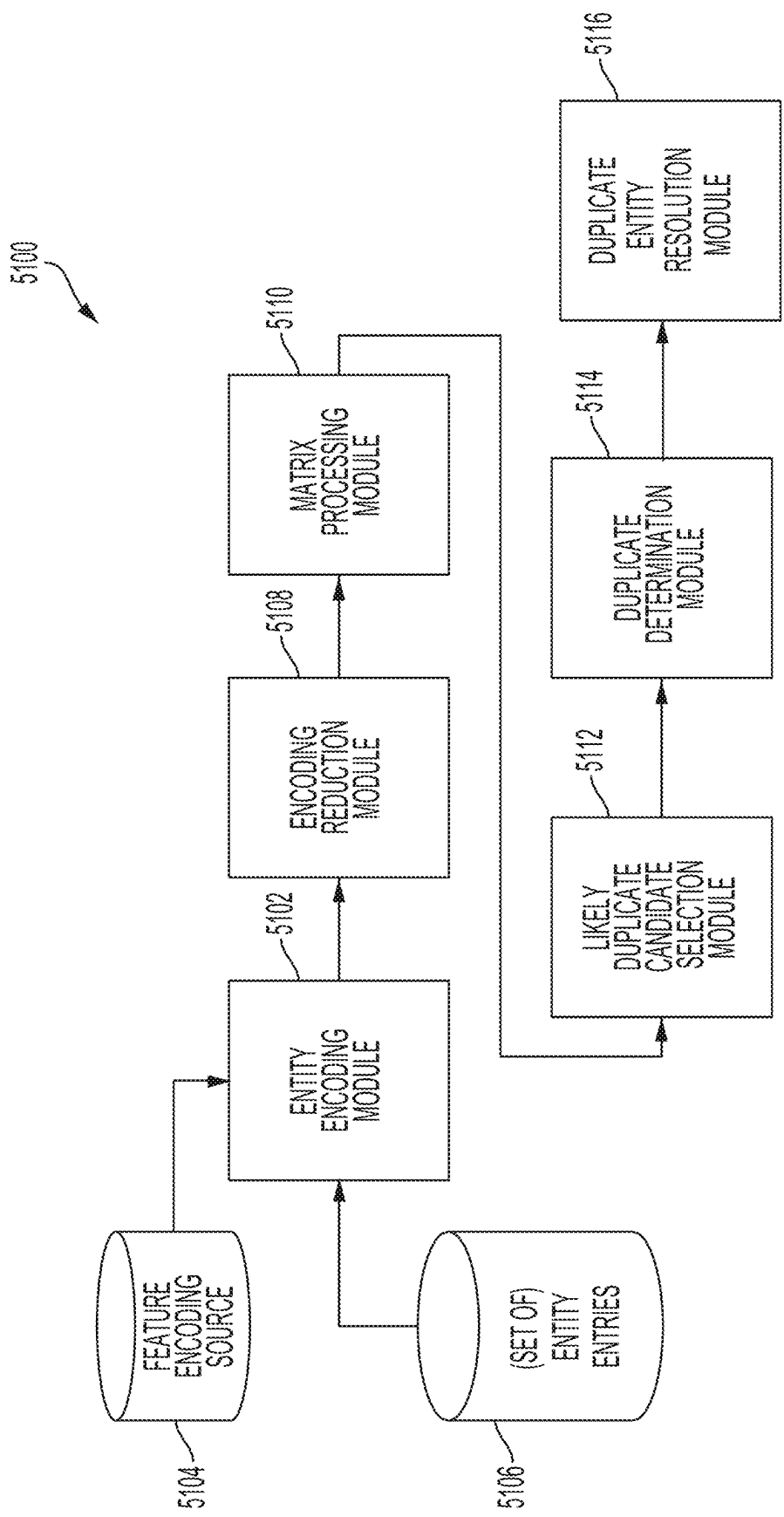


FIG. 51

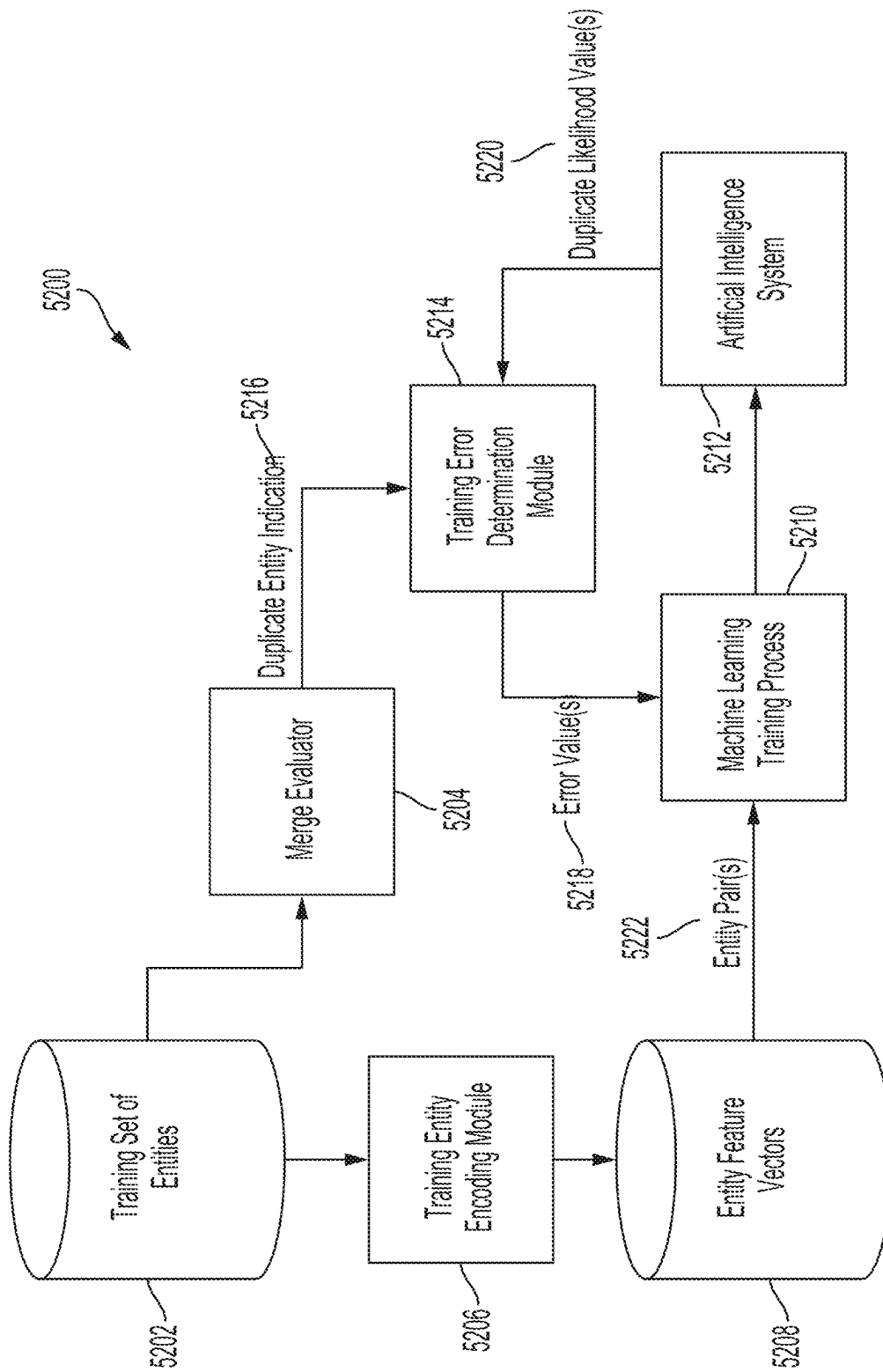


FIG. 52

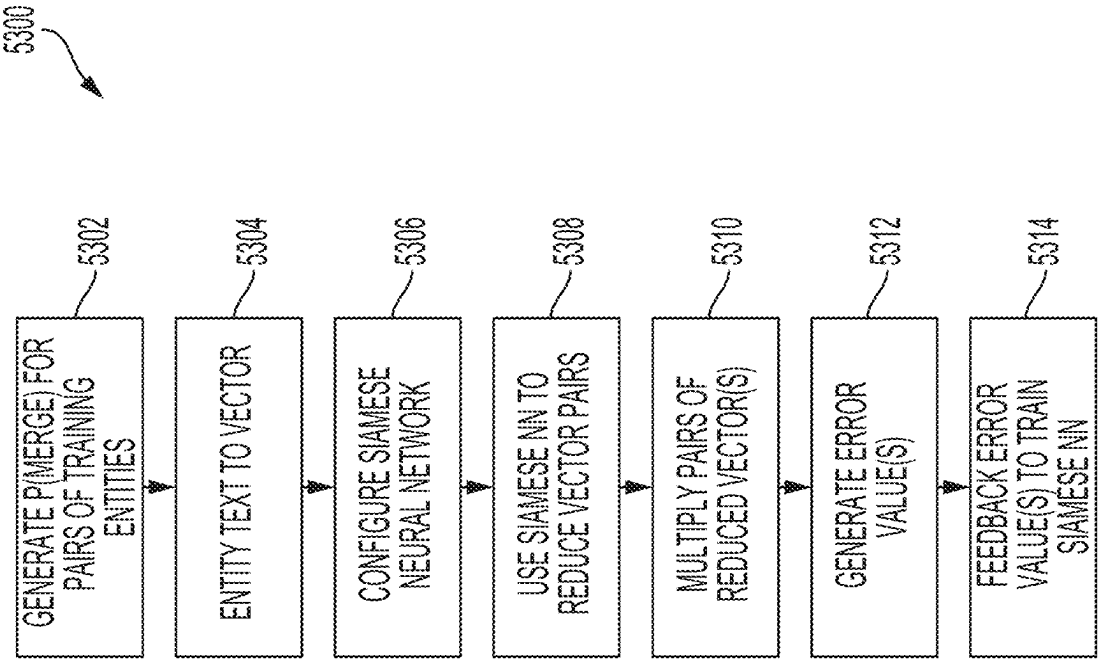


FIG. 53

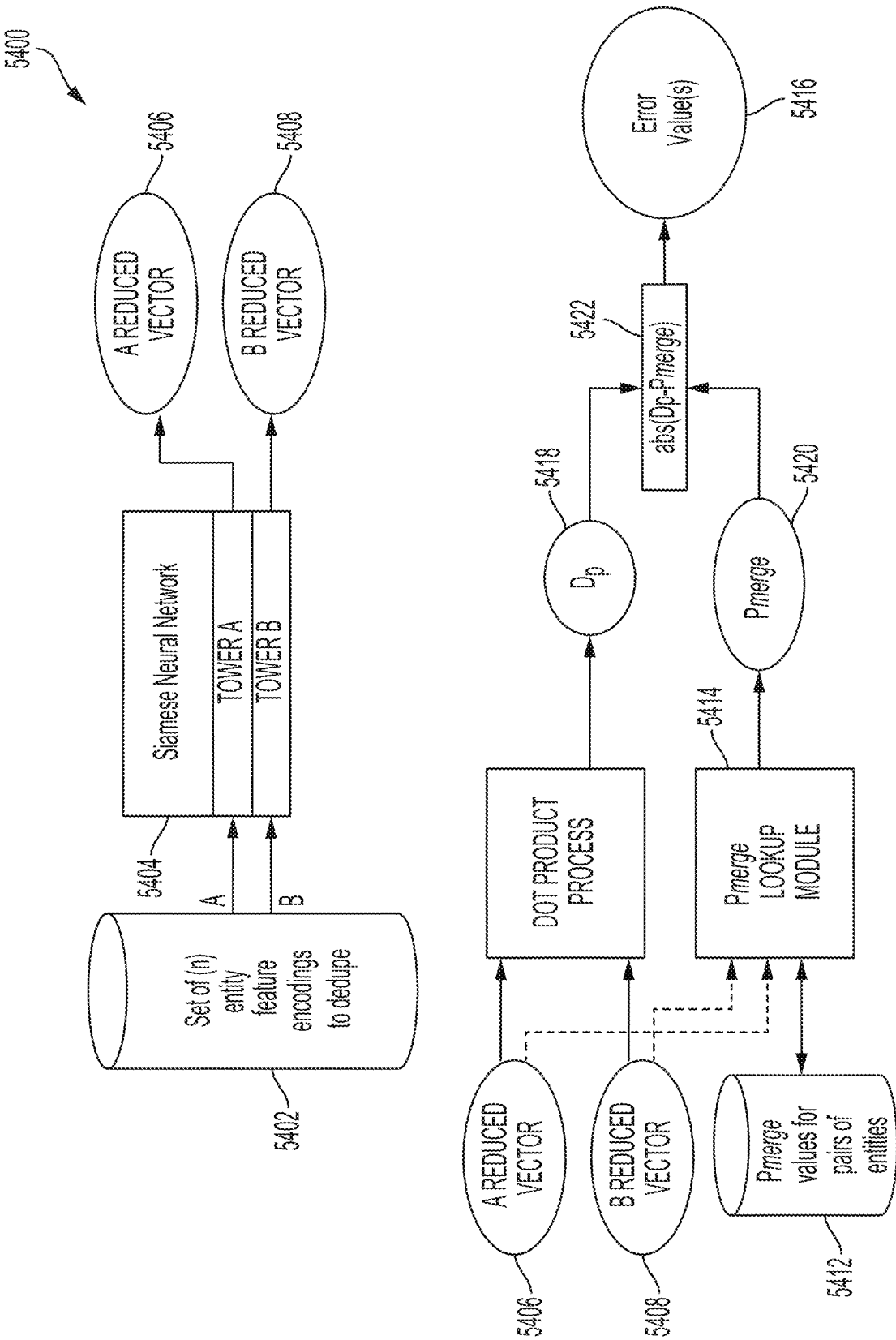


FIG. 54

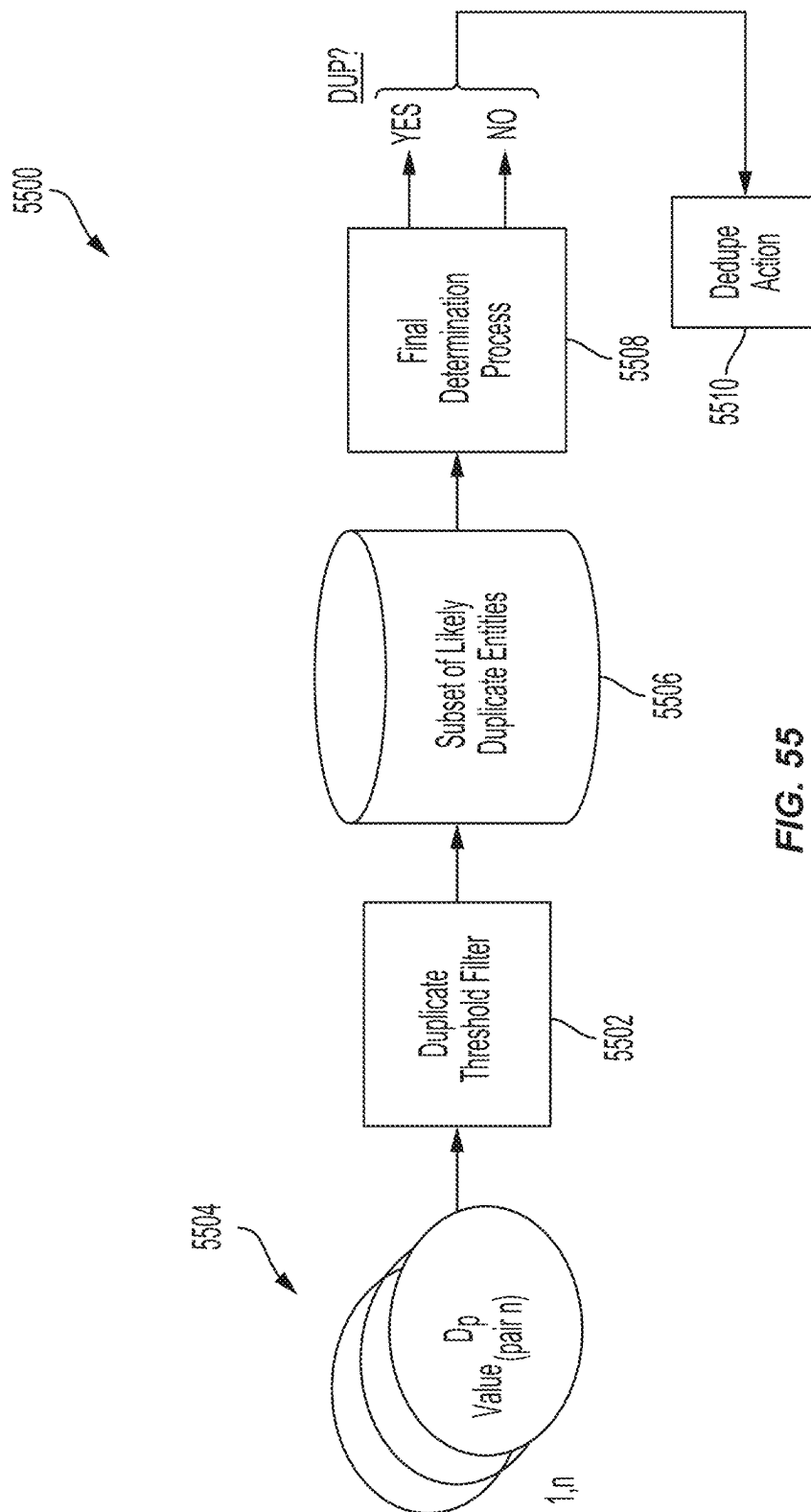
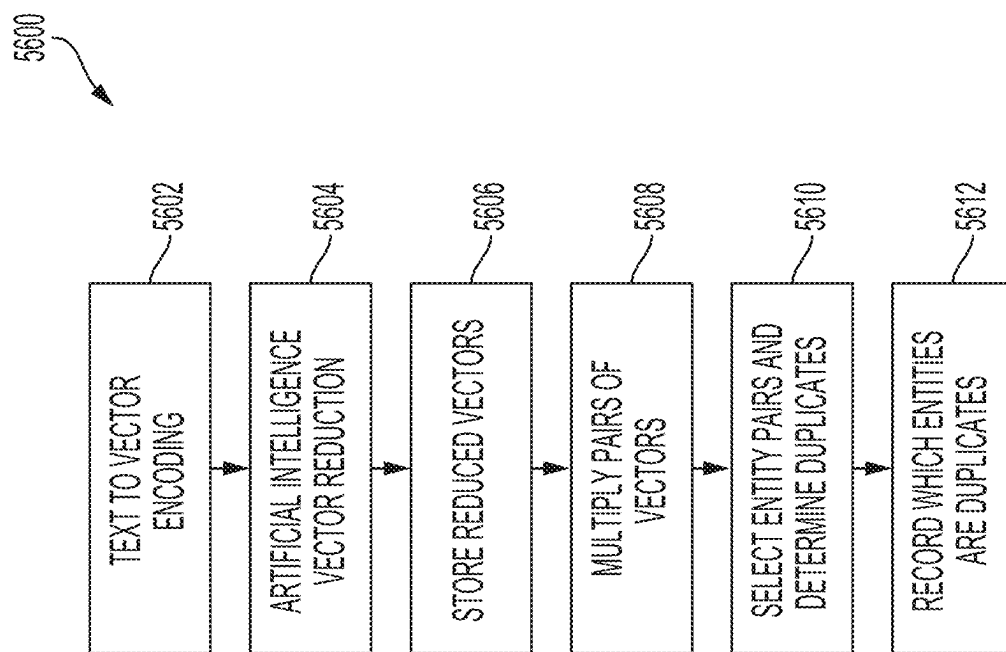


FIG. 55

**FIG. 56**

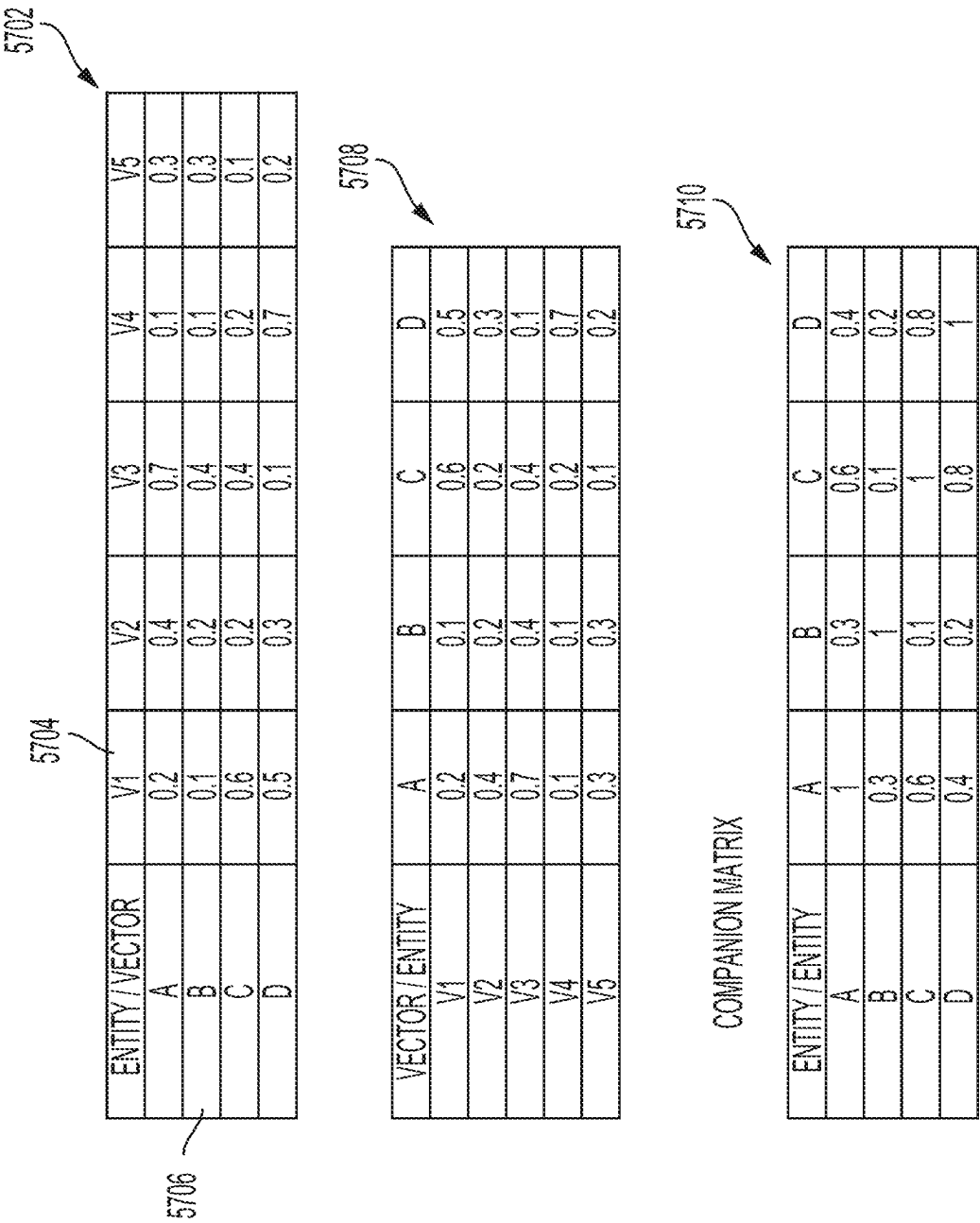


FIG. 57

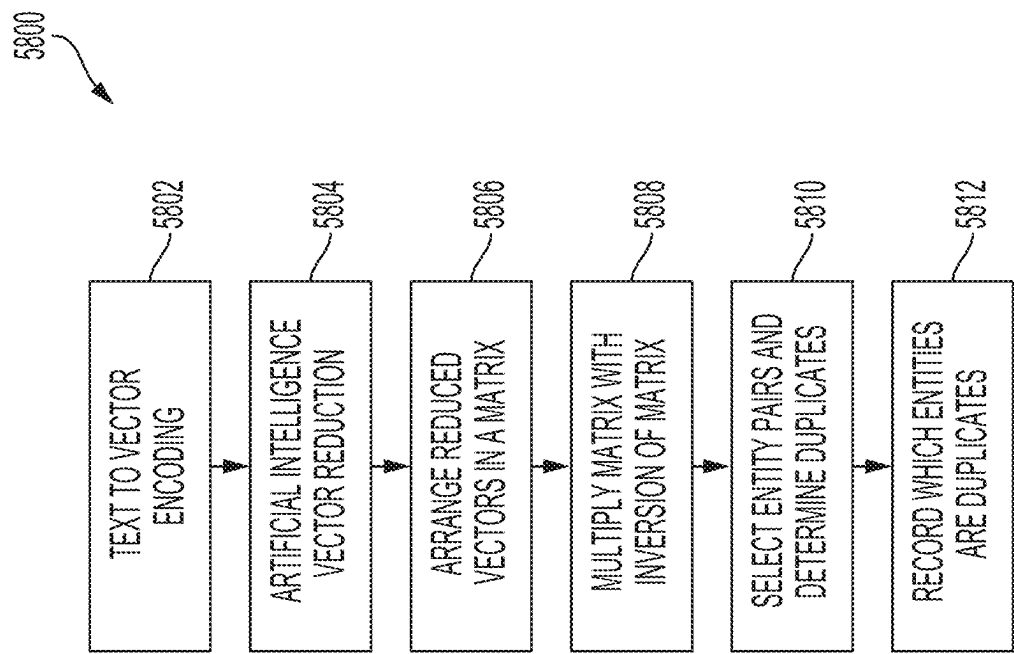


FIG. 58

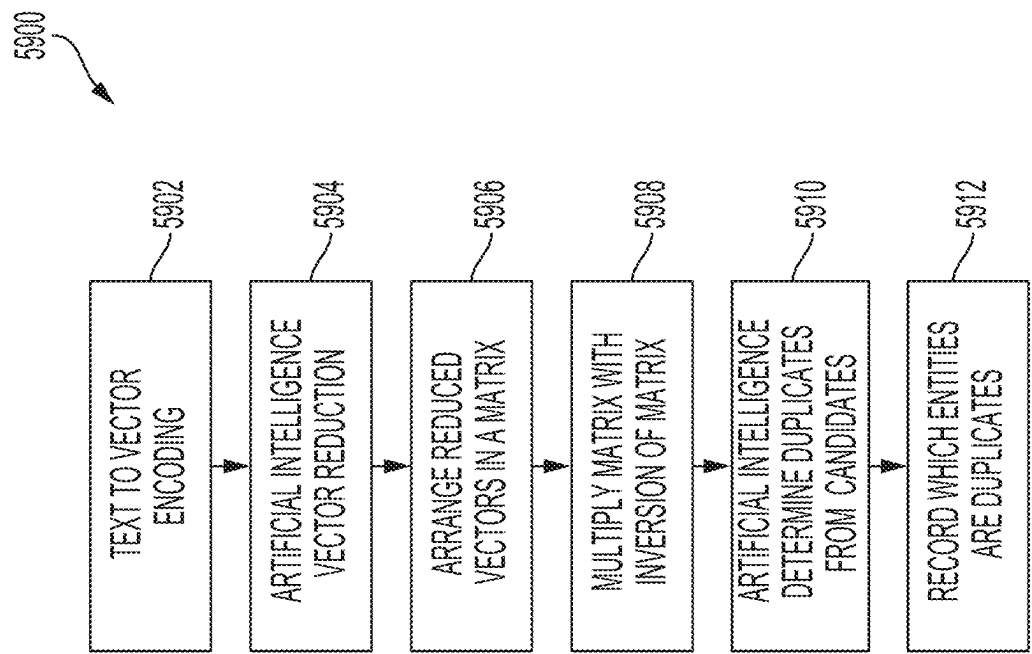


FIG. 59

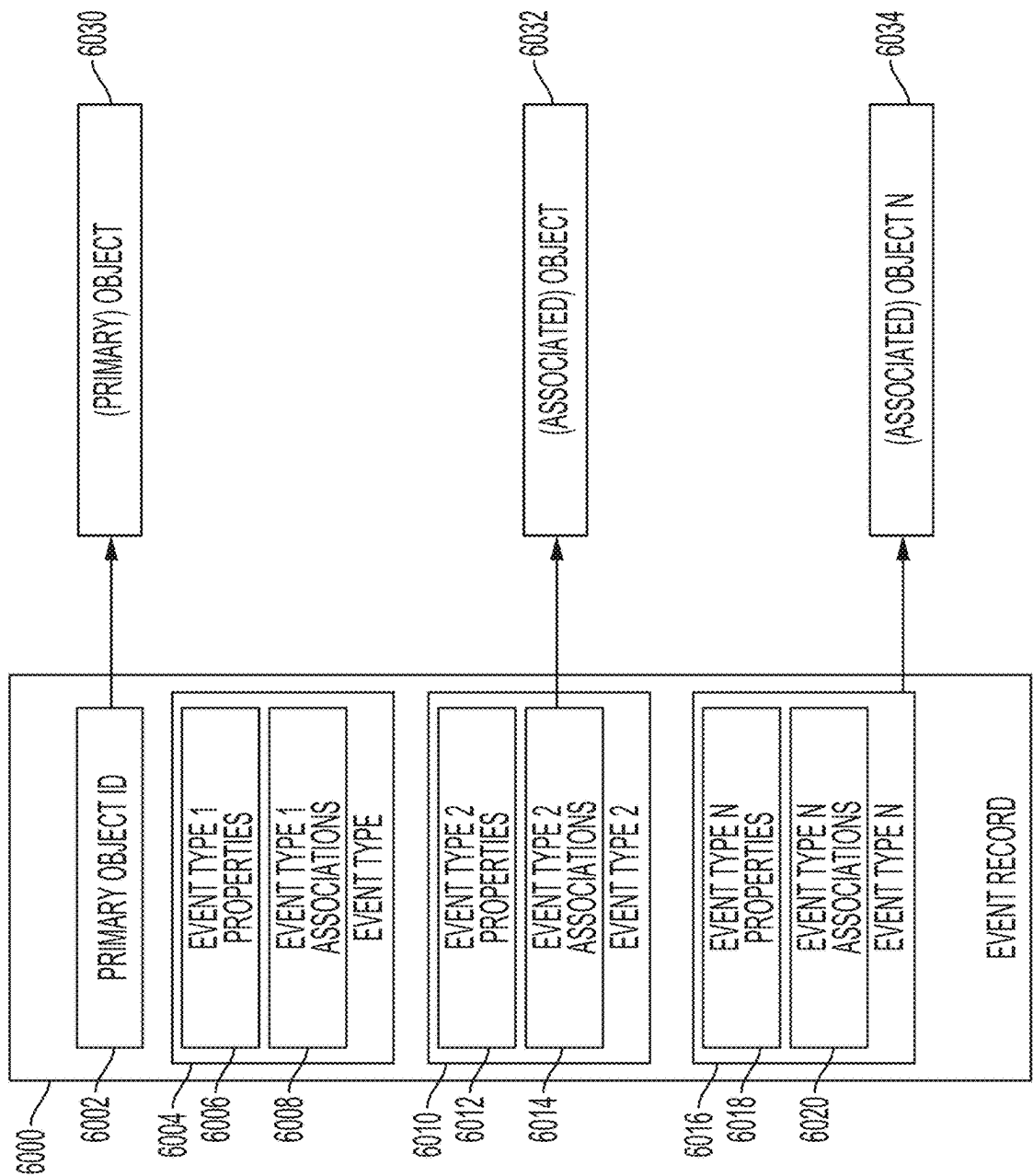


FIG. 60

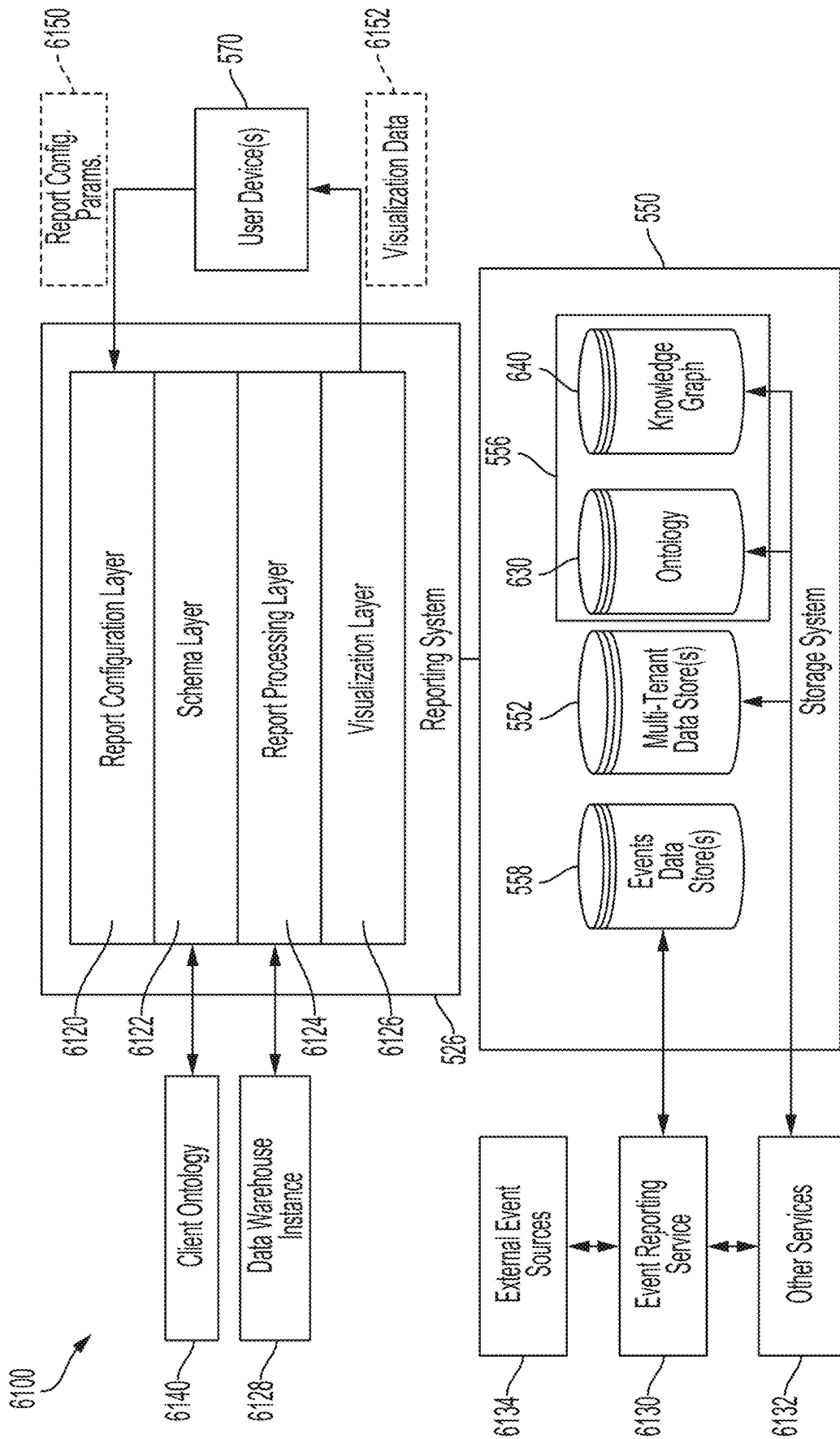


FIG. 61

6200

What do you want to report on?

Single object
Report on a single object such as your contacts.

Cross-object
Report across two different objects such as your contacts and deals, or deals and line items.

Funnels
Show progress through contact or deal stages that you define.

Attribution
See what interactions or activities drove conversions and revenue.

Custom Report Builder
Report on all of your HubSpot data within a streamlined build flow.

6202

Marketing

☒ **Contacts**

☐ **Companies**

☐ **Deals**

☐ **Tickets**

☐ **Form submissions**

☐ **Landing pages**

☐ **Forms**

☐ **Marketing email**

☐ **Ad accounts**

☐ **Ad campaigns**

☐ **Ad groups**

☐ **Ads**

☐ **Keywords**

☐ **Website pages**

☐ **Blog**

☐ **CTAs**

☐ **Video**

☐ **Web activities**

Sales

☐ **Sales activities**

☐ **Products**

Service

☐ **Conversations**

☐ **Feedback**

Custom objects

☐ **Subscriptions**

☐ **Services**

☐ **Referrals**

☐ **Company Locations**

FIG. 62A

6200

Single object Report on a single object such as your contacts.	CRM ☒ Contacts ☒ Companies ☐ Deals ☐ Tickets	6204
Cross-object Report across two different objects such as your contacts and deals, or deals and line items.		
Funnels Show progress through contact or deal stages that you define.		
Attribution See what interactions or activities drove conversions and revenue.		
Custom Report Builder Report on all of your HubSpot data within a streamlined build flow.		
Marketing		
☐ Form submissions	☐ Landing pages	☐ Forms
☐ Marketing email	☐ Ad accounts	☐ Ad campaigns
☐ Ad groups	☐ Ads	☐ Keywords
☐ Website pages	☐ Blog	☐ CTAs
☐ Video	☐ Web activities	
Sales		
☐ Sales activities	☐ Products	
Service		
☐ Conversations	☐ Feedback	
Custom objects		
☐ Subscriptions	☒ Services	☐ Company Locations

FIG. 62B

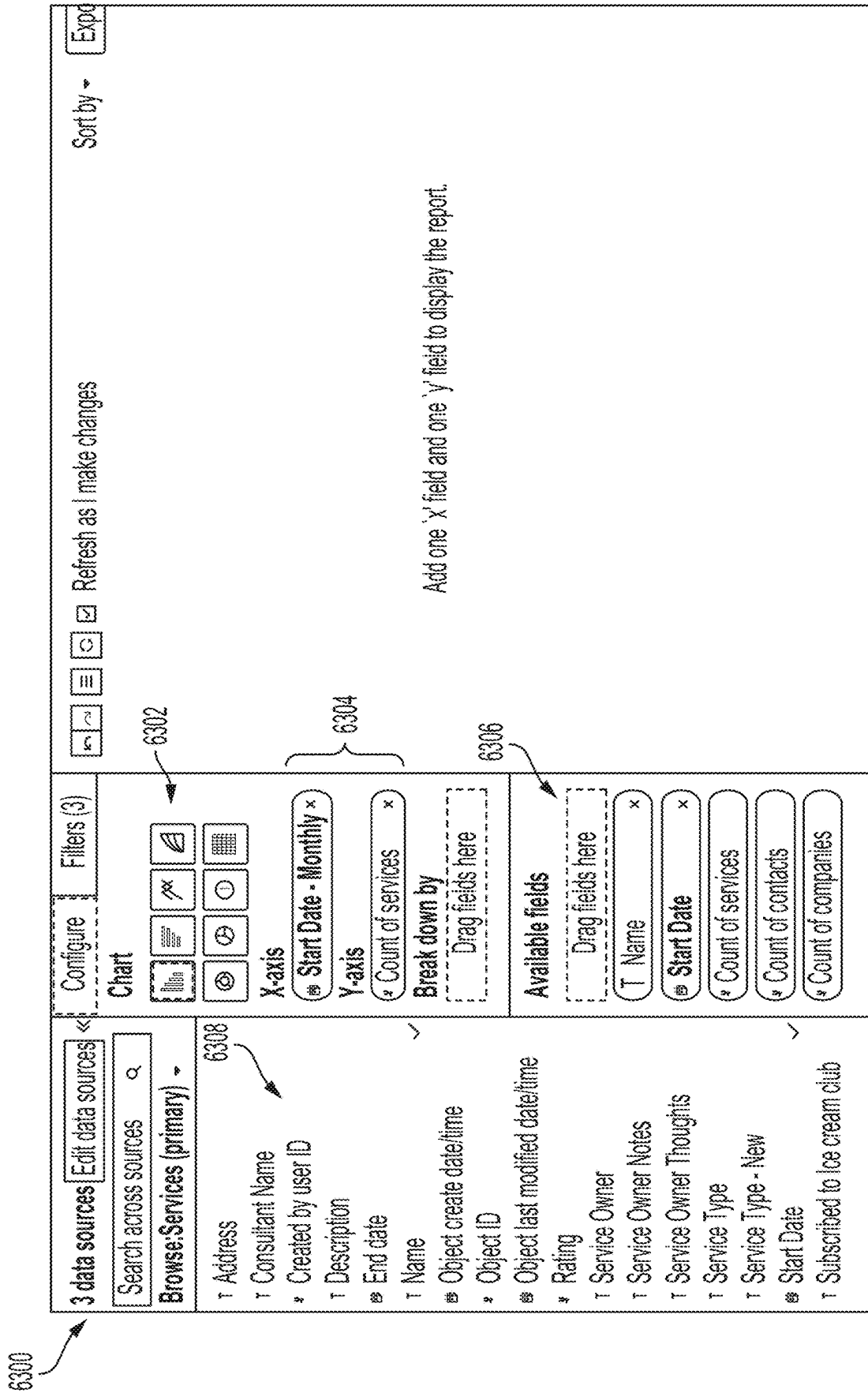


FIG. 63A

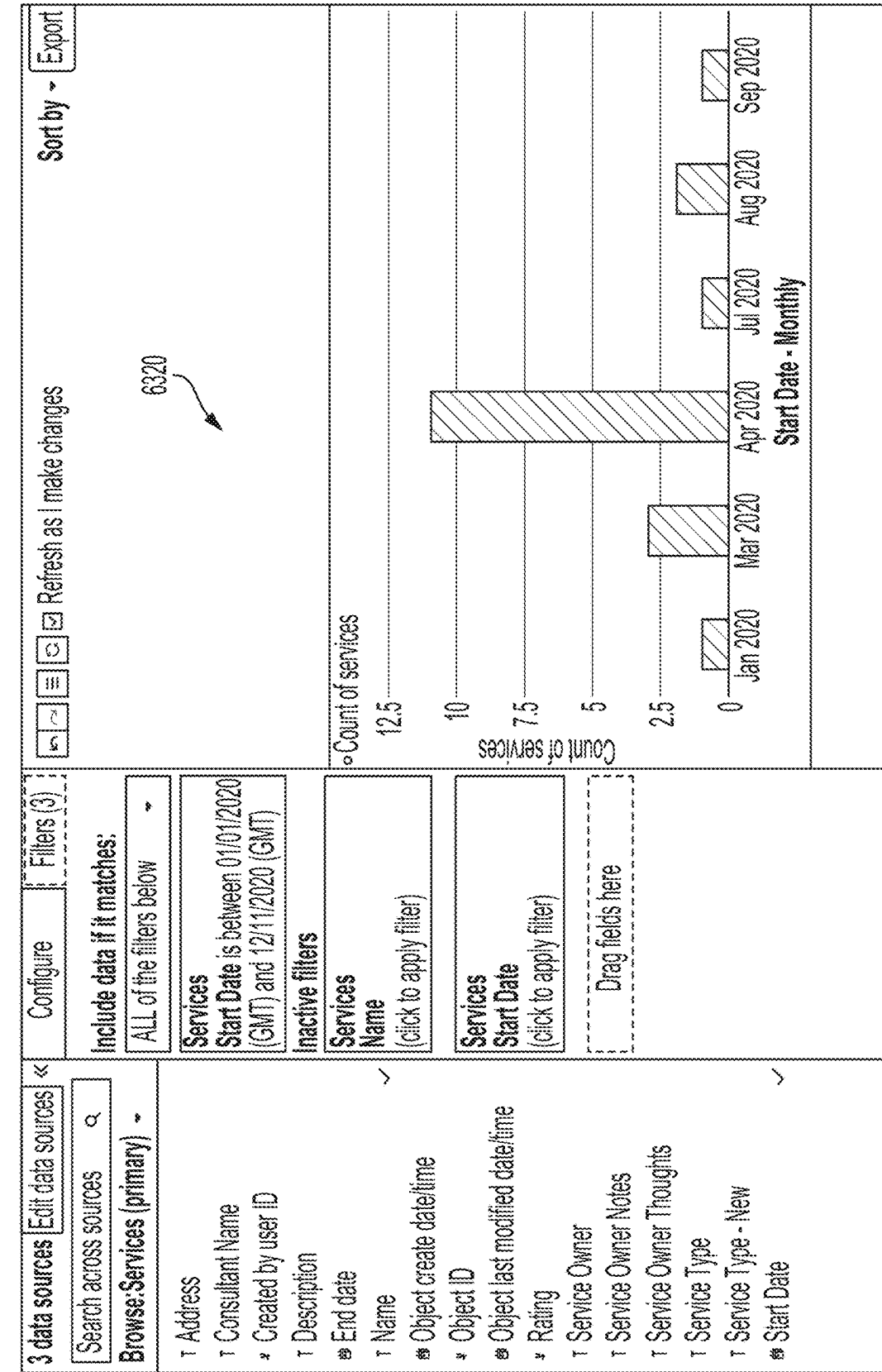


FIG. 63B

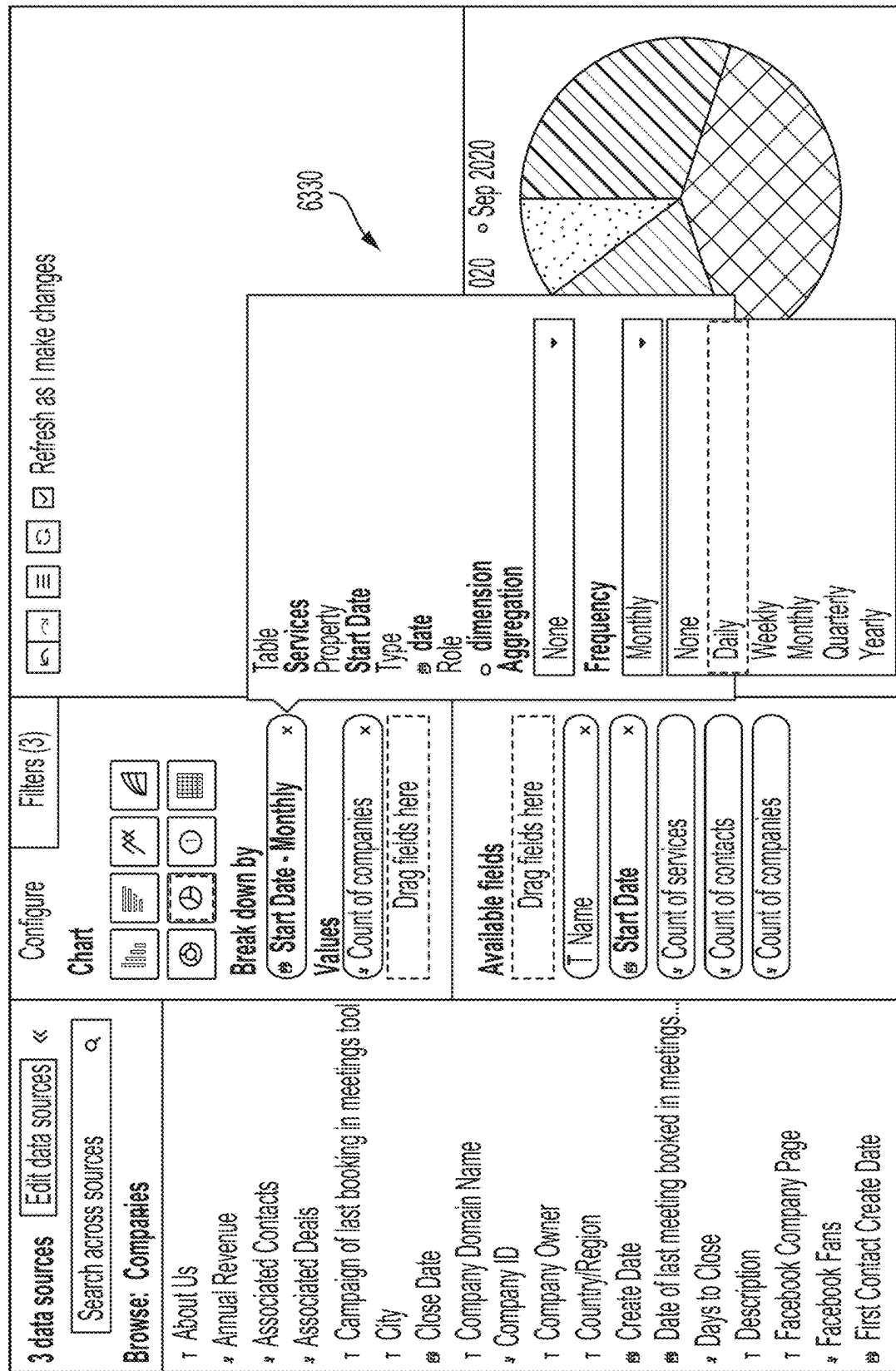


FIG. 63C

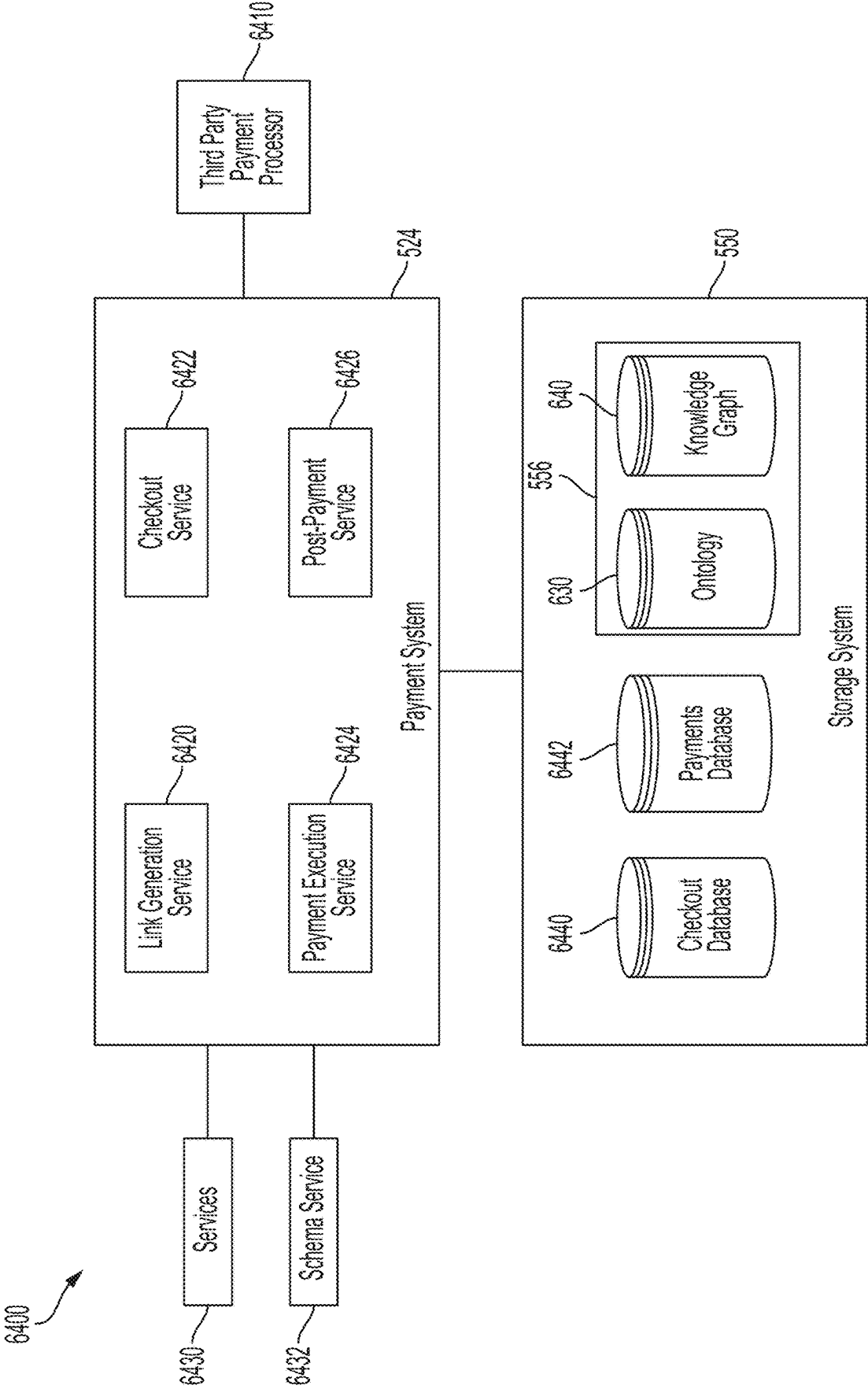


FIG. 64

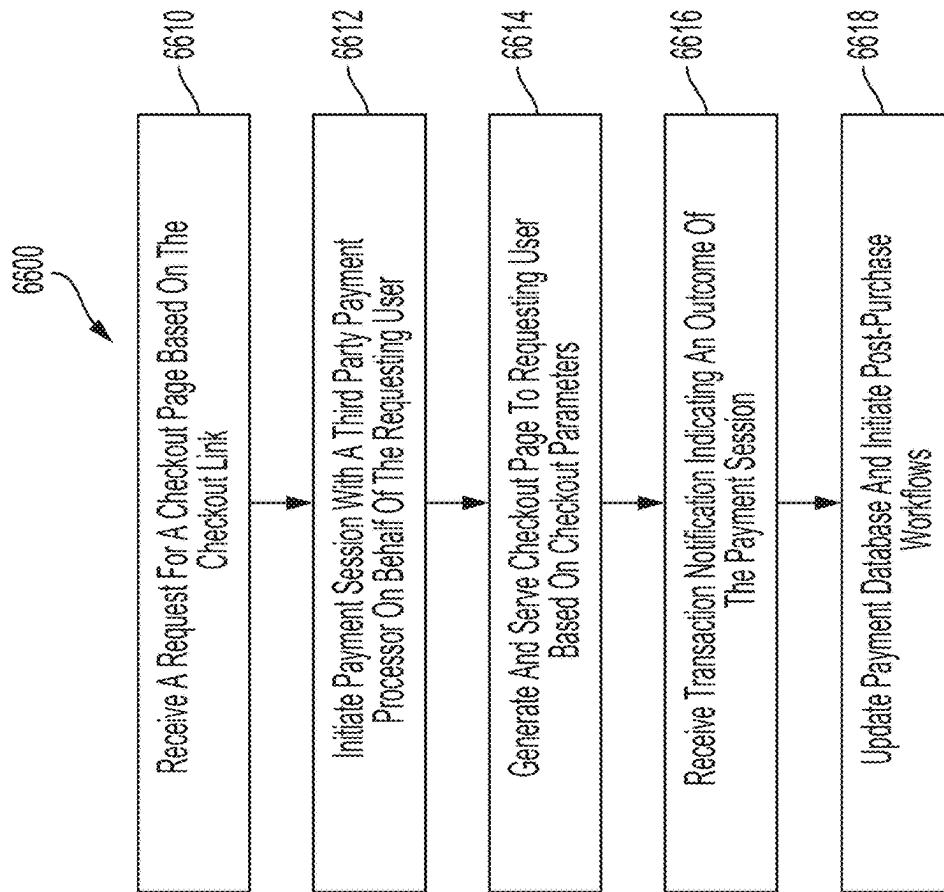


FIG. 66

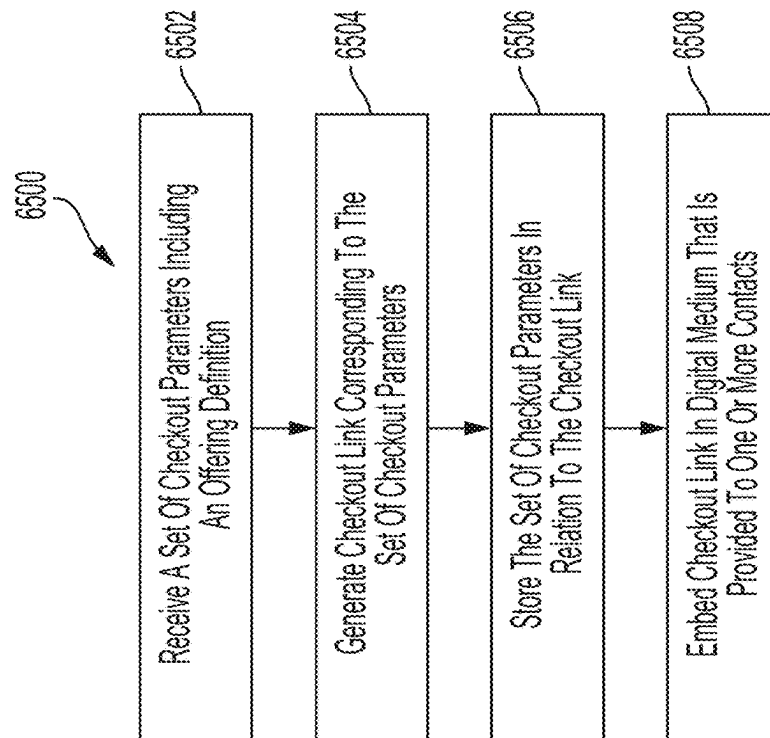


FIG. 65

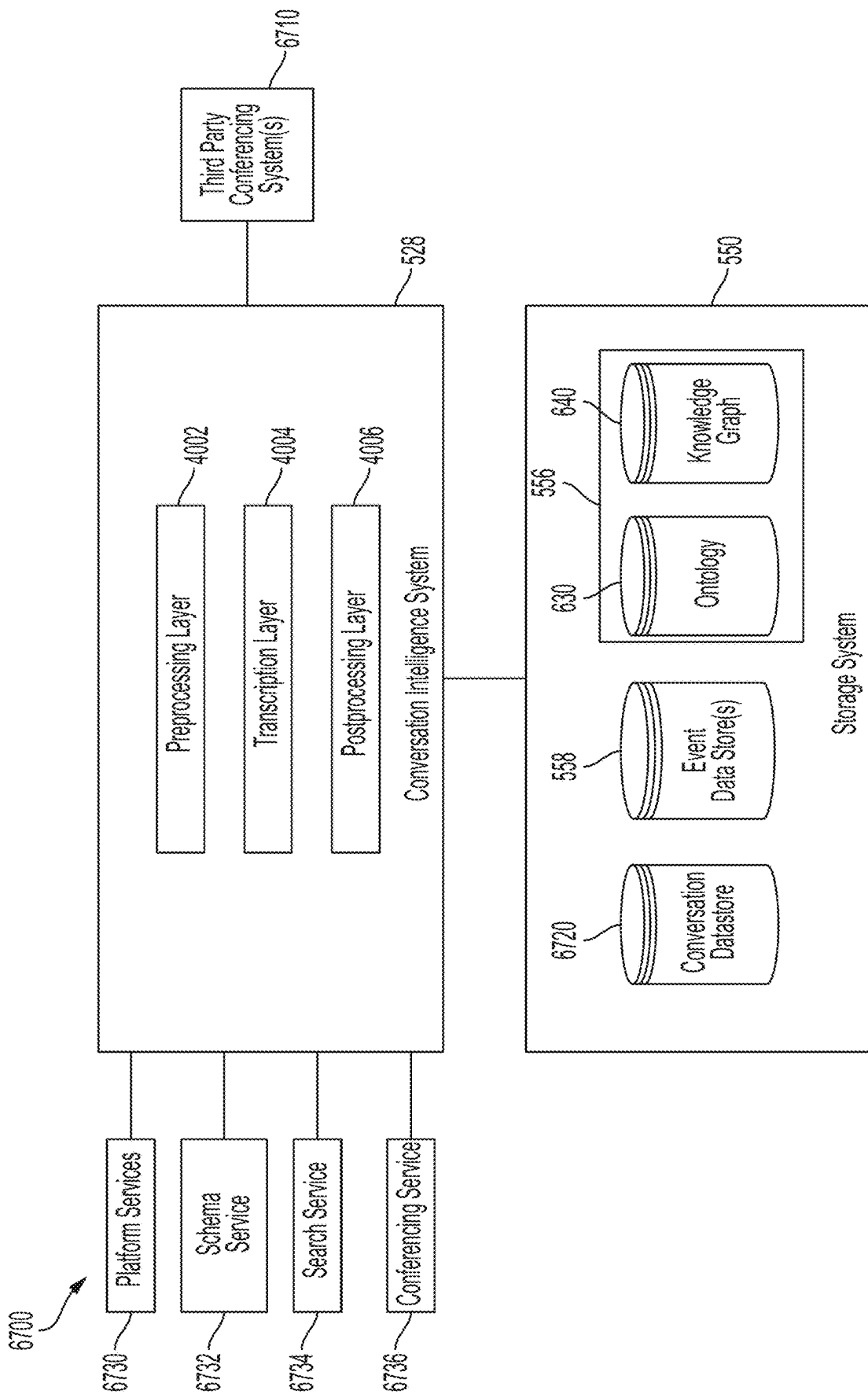


FIG. 67

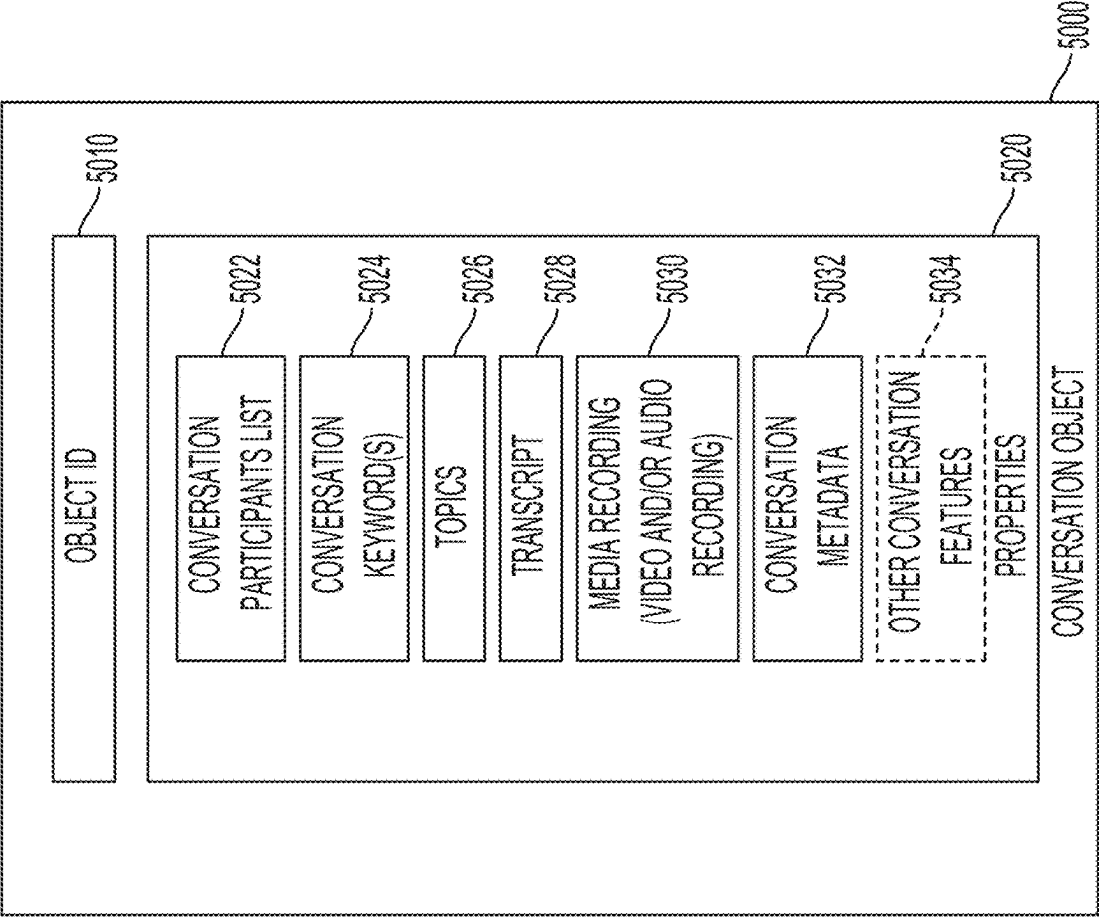


FIG. 68

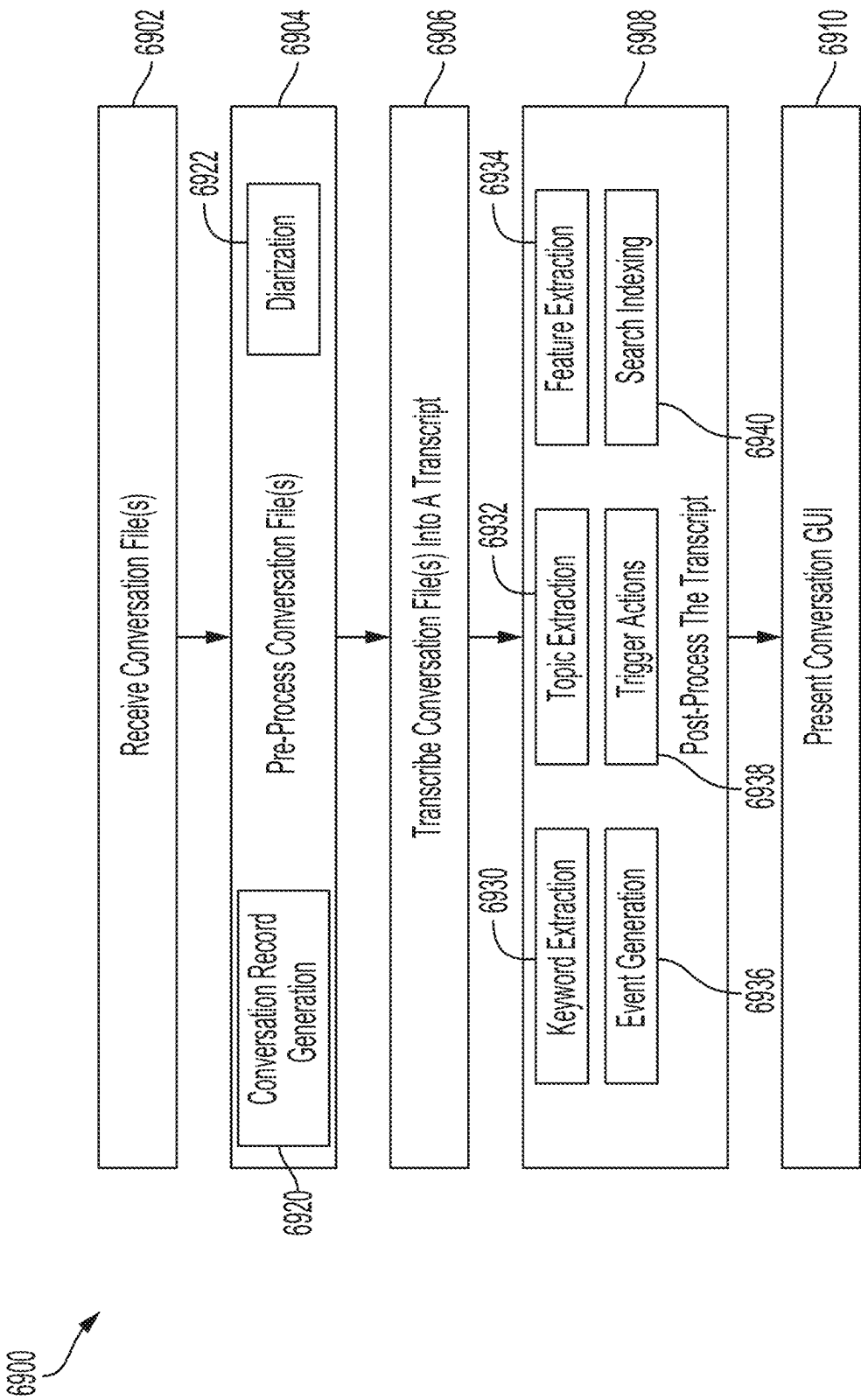


FIG. 69

7000

Contacts~ Conversations~ Marketing~ Sales~ Service~ Automation~ Reports~

Calls ▾
5 Calls

+ All records~ + Add view All views

Search

More Filters Clear All Table actions Save views

☐	CALL TITLE	CALL DURATION	CALL OUTCOME	ACTIVITY DATE (MST)	CALL NOTES
☐	Call with Elliot	a minute	--	Dec 6, 2020	--
☐	Call with Debbie	3 minutes	Connected	Nov 23, 2020	--
☐	Call with Howard	a minute	--	Nov 13, 2020	--
☐	Call with Ian	a few seconds	--	Oct 15, 2020	--
☐	Call with Justin	2 minutes	Connected	May 4, 2020	<p> Talked to just

< Prev 1 Next > 100 per page

Help

FIG. 70

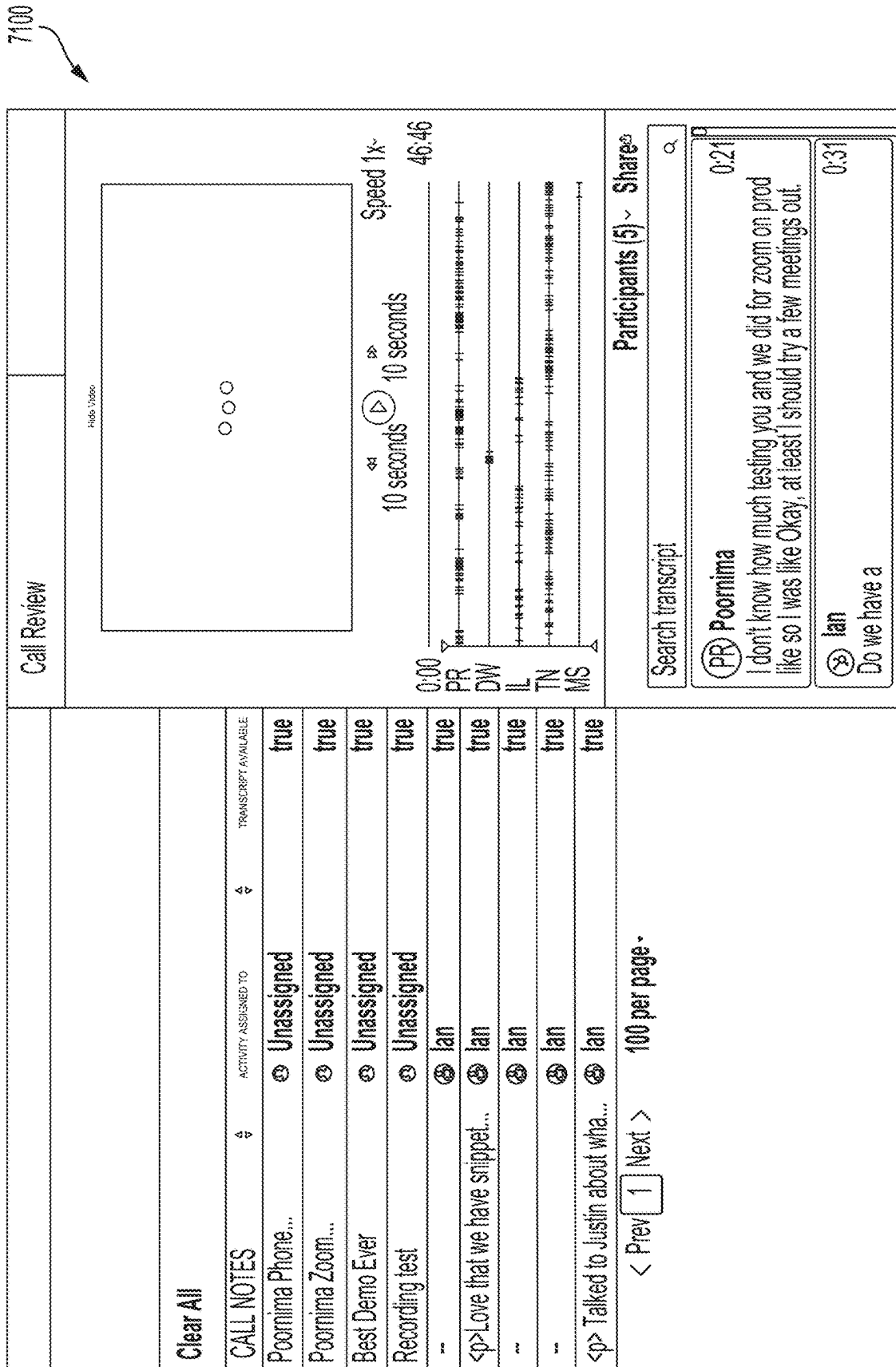


FIG. 71

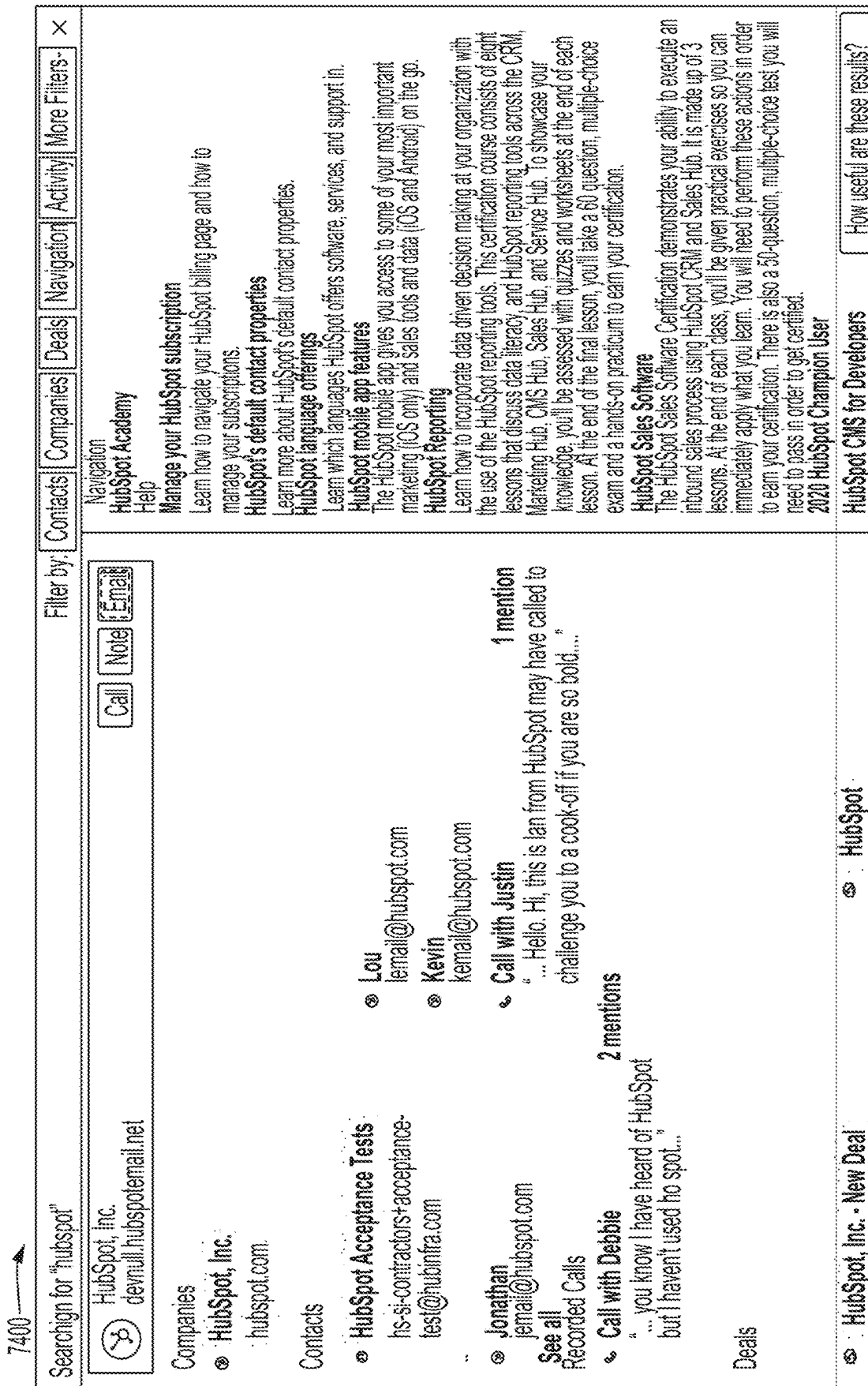
[illegible]

FIG. 72

7300

Type something to start searching		
Recently Searched		Recently Modified
 Call with Debbie "... wow that sounds kind of confusing have you ever heard of HubSpot..."	2 mentions	 Matthew memail@hubspot.com
 Justin j94@gmail.com		 Lou lemail@hubspot.com
 Call with Justin "... Hello. Hi, this is Ian from HubSpot may have called to challenge you to a cook-off if you are so bold...."	1 mention	 Ryan reemail@hubspot.com
 Call Found on Justin Talked to Justin about what he was cooking. Yum!		 aemail@hubspot.com aemail@hubspot.com
 Debbie demail@gmail.com		 Nicholas nemail@hubspot.com

FIG. 73



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MULTI-SERVICE BUSINESS PLATFORM SYSTEM HAVING EVENT SYSTEMS AND METHODS

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application claims priority to and is a continuation of U.S. patent application Ser. No. 17/654,519, filed Mar. 11, 2022, which claims priority to U.S. Provisional Application Ser. No. 63/160,446, filed Mar. 12, 2021, which are hereby incorporated by reference in its entirety.

TECHNICAL FIELD

The present application relates to a multi-client service system platform that may be part of a multi-service business platform.

BACKGROUND

Conventional systems for enabling marketing and sales activities for a business user do not also respectively enable support and service interactions with customers, notwithstanding that the same individuals are typically involved in all of those activities for a business, transitioning in status from prospect, to customer, to user. While marketing activities, sales activities, and service activities strongly influence the success of each other, businesses are required to undertake complex and time-consuming tasks to obtain relevant information for one activity from the others, such as forming queries, using complicated APIs, or otherwise extracting data from separate databases, networks, or other information technology systems (some on premises and others in the cloud), transforming data from one native format to another suitable form for use in a different environment, synchronizing different data sources when changes are made in different databases, normalizing data, cleansing data, and configuring it for use.

Some systems are customer relationship management (CRM) systems that may generally provide ability to manage and analyze interactions with customers for businesses. For example, these CRM systems may compile data from various communication channels (e.g., email, phone, chat, content materials, social media, etc.). For example, some CRM systems can be used to monitor and track CRM standard objects. These CRM standard objects can include typical business objects such as accounts (e.g., accounts of customers), contacts (e.g., persons associated with accounts), leads (e.g., prospective customers), and opportunities (e.g., sales or pending deals).

SUMMARY

According to some example embodiments of the disclosure, a multi-service business platform is disclosed. The multi-service business platform may include a storage system. The storage system may store an ontology datastore that stores an ontology that defines a set of objects and respective associations between different objects of the set of objects. The storage system may include an event datastore that stores event record instances of one or more different types of event records. Each type of event record may include an association to a primary object of the set of objects defined in the ontology and may define a plurality of event types that are tracked with respect to a respective object instance of the primary object. Each event type may

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include one or more event properties and zero or more associations to zero or more associated objects of the set of objects defined in the set of objects. The multi-service business platform may include a set of processors that may execute a set of computer-executable instructions that cause the set of processors to execute one or more listening threads. Each listening thread may be configured to: monitor a respective data stream for one or more specific types of data that may be indicative of a specific type of event of the plurality of event types; analyze the respective data stream to identify an occurrence of an event instance of the specific event type and respective event properties of the event instance; and in response to identifying the occurrence of the event instance, record the event instance and the respective properties of the event instance. In response to a new event instance being reported, the new event instance and the respective properties of the new event instance in the event datastore may be recorded. One or more types of event records may be identified that track the specific event type based on the specific event type of the new event instance. For each identified type of event record: identify a primary object instance pertaining to the new event instance based on the respective properties of the new event instance; identify an event record instance associated with the identified primary object instance; and associate the new event instance with a respective event log of the identified event record instance. The respective event log may provide a time-sequenced log of respective event instances that were recorded with respect to the primary object instance, and the new event instance may be associated with zero or more other event record instances.

In some example embodiments, the ontology may include a set of default objects that are supported across a set of services of the multi-service business platform. The set of default objects may include two or more of a contact object, a service ticket object, a deal object, or a ticket object. In example embodiments, the ontology may be a client-specific ontology of a client of the multi-service business platform that may include the set of default objects, one or more custom objects that may be specific to the client, and one or more custom associations that respectively associate one of the custom objects to other objects in the client-specific ontology. In example embodiments, the computer-executable instructions may further cause the set of processors to define a custom event record on behalf of a client, and the custom event record corresponding to a custom object of the one or more custom objects that are specific to the client and tracking one or more event types with respect to the custom object. Defining the custom event record may include: receiving a user selection from a user device indicating the custom object of the one or more custom objects; and receiving one or more event type definitions from the user device to associate with the custom event record. Each event type definition may indicate a set of event types that are tracked with respect to instances of the custom object and a set of event properties that are maintained for the event type. In other example embodiments, defining the custom event record may further include updating the client-specific ontology to associate the set of event types with the custom object. In example embodiments, receiving one or more event type definitions may include receiving a user selection of one or more previously defined event types. In other example embodiments, receiving one or more event type definitions may include receiving user input indicating a new event type, a set of new event properties of the new event type, and, for each new event property of the new event properties, a data type of the event property. In some

example embodiments, defining the custom event record may further include receiving a data source selection from the user device indicating a third-party data source to monitor for the new event type. In some example embodiments, the computer-executable instructions may further cause the set of processors to configure a custom listening thread that monitors the third-party data source for event instances of the new event type.

In some example embodiments, the event record instances may track different event instances across multiple services of the multi-service business platform. The multiple services of the multi-service business platform may include two or more of a customer relationship management service, a multi-client service system, a payment system, or a content management system.

According to some example embodiments of the disclosure, a computer-implemented method for maintaining unified events is disclosed. The method may include maintaining, by one or more processors, an ontology data store that stores an ontology that defines a set of objects and respective associations between different objects of the set of objects. The method may include maintaining, by the one or more processors, an event datastore that stores event record instances of one or more different types of event records. Each type of event record may include an association to a primary object of the set of objects defined in the ontology and may define a plurality of event types that are tracked with respect to a respective object instance of the primary object. Each event type may include one or more event properties and zero or more associations to zero or more associated objects of the set of objects defined in the set of objects. The method may include executing, by one or more processors, one or more listening threads. Each listening thread may be configured to: monitor a respective data stream for one or more specific types of data that are indicative of a specific type of event of the plurality of event types. The respective data stream may be analyzed to identify an occurrence of an event instance of the specific event type and respective event properties of the event instance. In response to identifying the occurrence of the event instance, the event instance and the respective properties of the event instance may be recorded. In response to a new event instance being reported, the new event instance and the respective properties of the new event instance in the event datastore may be recorded by one or more processors. One or more types of event records that track the specific event type based on the specific event type of the new event instance may be identified by one or more processors. For each identified type of event record, a primary object instance pertaining to the new event instance based on the respective properties of the new event instance may be identified by one or more processors. For each identified type of event record, an event record instance associated with the identified primary object instance may be identified by one or more processors. For each identified type of event record, the new event instance with a respective event log of the identified event record instance may be associated by one or more processors. The respective event log may provide a time-sequenced log of respective event instances that were recorded with respect to the object instance, and the new event instance may be associated with zero or more other event record instances.

In some example embodiments, the techniques described herein relate to a method, wherein the ontology includes a set of default objects that are supported across a set of services of a multi-service business platform.

In example embodiments, the set of default objects may include two or more of a contact object, a service ticket object, a deal object, or a ticket object. The ontology may be a client-specific ontology of a client of the multi-service business platform that may include the set of default objects, one or more custom objects that are specific to the client, and one or more custom associations that respectively associate one of the custom objects to other objects in the client-specific ontology. The method may further include defining a custom event record on behalf of a client. The custom event record may correspond to a custom object of the one or more custom objects that are specific to the client. One or more event types may be tracked with respect to the custom object. Defining the custom event record may include: receiving a user selection from a user device indicating the custom object of the one or more custom objects; and receiving one or more event type definitions from the user device to associate with the custom event record. Each event type definition may indicate a set of event types that are tracked with respect to instances of the custom object and a set of event properties that are maintained for the event type. In some example embodiments, defining the custom event record may further include updating the client-specific ontology to associate the set of event types with the custom object. In some example embodiments, receiving one or more event type definitions may include receiving a user selection of one or more previously defined event types. In other embodiments, receiving one or more event type definitions may include receiving user input indicating a new event type, a set of new event properties of the new event type, and, for each new event property of the new event properties, a data type of the event property. In some embodiments, defining the custom event record may further include receiving a data source selection from the user device indicating a third-party data source to monitor for the new event type. In some embodiments, the method may further include configuring a custom listening thread that monitors the third-party data source for event instances of the new event type.

In example embodiments, the event record instances may track different event instances across multiple services of a multi-service business platform. The multiple services of the multi-service business platform may include two or more of a customer relationship management service, a multi-client service system, a payment system, or a content management system.

According to some example embodiments of the disclosure, the techniques described herein relate to a non-transitory computer readable storage medium having a plurality of instructions stored thereon which, when executed across one or more processors, causes at least a portion of the one or more processors to perform operations. The operations may include: maintaining an ontology data store that stores an ontology that defines a set of objects and respective associations between different objects of the set of objects. An event datastore that stores event record instances of one or more different types of event records may be maintained. Each type of event record may include an association to a primary object of the set of objects defined in the ontology and may define a plurality of event types that are tracked with respect to a respective object instance of the primary object. Each event type may include one or more event properties and zero or more associations to zero or more associated objects of the set of objects defined in the set of objects. One or more listening threads may be executed. Each listening thread may be configured to: monitor a respective data stream for one or more specific types of data

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that may be indicative of a specific type of event of the plurality of event types; analyze the respective data stream to identify an occurrence of an event instance of the specific event type and respective event properties of the event instance; and in response to identifying the occurrence of the event instance, record the event instance and the respective properties of the event instance. In response to a new event instance being reported, the new event instance and the respective properties of the new event instance in the event datastore may be recorded. One or more types of event records that track the specific event type based on the specific event type of the new event instance may be identified. For each identified type of event record, a primary object instance pertaining to the new event instance based on the respective properties of the new event instance may be identified. For each identified type of event record, an event record instance associated with the identified primary object instance may be identified. For each identified type of event record, the new event instance may be associated with a respective event log of the identified event record instance. The respective event log may provide a time-sequenced log of respective event instances that were recorded with respect to the object instance, and the new event instance may be associated with zero or more other event record instances. In some example embodiments, the ontology may include a set of default objects that may be supported across a set of services of a multi-service business platform.

A more complete understanding of the disclosure will be appreciated from the description and accompanying drawings and the claims, which follow.

These and other systems, methods, objects, features, and advantages of the disclosure will be apparent to those skilled in the art from the following detailed description of the preferred embodiment and the drawings.

All documents mentioned herein are hereby incorporated in their entirety by reference. References to items in the singular should be understood to include items in the plural, and vice versa, unless explicitly stated otherwise or clear from the text. Grammatical conjunctions are intended to express any and all disjunctive and conjunctive combinations of conjoined clauses, sentences, words, and the like, unless otherwise stated or clear from the context.

BRIEF DESCRIPTION OF THE FIGURES

The disclosure and the following detailed description of certain embodiments thereof may be understood by reference to the following figures:

FIG. 1 depicts a high-level flow in which a content platform is used to process online content, identify a cluster of semantically relevant topics and produce generated online presence content involving the semantically relevant topics according to one or more embodiments of the disclosure.

FIG. 2 provides a functional block diagram of certain components and elements of a content development platform, including elements for extracting key phrases from a primary online content object, a content cluster data store for storing clusters of topics and a content development and management application having a user interface for developing content according to one or more embodiments of the disclosure.

FIGS. 3, 4, and 5 show examples of user interface elements for presenting suggested topics and related information according to one or more embodiments of the disclosure.

FIG. 6 provides a functional block diagram of certain components and elements of a content development plat-

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form, including integration of a customer relationship management system with other elements of the platform according to one or more embodiments of the disclosure.

FIG. 7 provides a detailed functional block diagram of components and elements of a content development platform according to one or more embodiments of the disclosure.

FIG. 8 illustrates a user interface for reporting information relating to online content generated using the content development and management platform according to one or more embodiments of the disclosure.

FIG. 9 depicts a user interface in which activity resulting from the use of the platform is reported to a marketer or other user according to one or more embodiments of the disclosure.

FIG. 10 illustrates an example environment of a directed content system according to one or more embodiments of the disclosure.

FIG. 11 depicts an example of the crawling system and the information extraction system maintaining a knowledge graph according to one or more embodiments of the disclosure.

FIG. 12 depicts a visual representation of a portion of an example knowledge graph representation according to one or more embodiments of the disclosure.

FIG. 13 illustrates an example configuration of the lead scoring system and the content generation system for identifying intended recipients of messages and generating personalized messages for the one or more intended recipients.

FIG. 14 illustrates an example configuration of the directed content system according to one or more embodiments of the disclosure.

FIG. 15 illustrates a method for generating personalized messages on behalf of a user according to one or more embodiments of the disclosure.

FIG. 16 illustrates an example environment of a multi-client service system platform according to one or more embodiments of the disclosure.

FIG. 17A illustrates an example of a contact database object according to one or more embodiments of the disclosure.

FIG. 17B illustrates an example of a client database object according to one or more embodiments of the disclosure.

FIG. 17C illustrates an example of a ticket database object according to one or more embodiments of the disclosure.

FIG. 18 depicts a visual representation of a portion of an example knowledge graph representation according to one or more embodiments of the disclosure.

FIG. 19 illustrates an example of a multi-client service system platform providing service systems on behalf of two independent clients according to one or more embodiments of the disclosure.

FIG. 20 is a flow chart illustrating a set of operations of a method for deploying a client-specific service system.

FIG. 21 is a screenshot showing an example GUI of a service system for showing a user a status of a ticket according to one or more embodiments of the disclosure.

FIG. 22 is a screenshot showing an example GUI of a service system for showing a user data surrounding an issued ticket, including a conversation with a contact, according to one or more embodiments of the disclosure.

FIG. 23 is a screenshot showing an example GUI of a service system for showing a user a ticket overview of multiple tickets of various contacts according to one or more embodiments of the disclosure.

FIG. 24 is a screenshot showing an example GUI of a service system for showing a user a feedback overview of

multiple contacts of a client according to one or more embodiments of the disclosure.

FIG. 25 is a screenshot showing an example GUI of a service system for showing a user feedback received from a user, including a feedback timeline, according to one or more embodiments of the disclosure.

FIG. 26 is a screenshot showing an example GUI of a service system for showing a user feedback received from a user, including a feedback timeline, according to one or more embodiments of the disclosure.

FIGS. 27-35 are screenshots of example GUIs that allow a user corresponding to a client to customize different aspects of the client's respective feedback system, according to one or more embodiments of the disclosure.

FIG. 36 is a screenshot of an example of a GUI that displays a breakdown of the net promoter scores of the contacts of a particular client according to one or more embodiments of the disclosure.

FIGS. 37 and 38 are screenshots of an example GUI for uploading content to a service platform for inclusion in a knowledge graph according to one or more embodiments of the disclosure.

FIG. 39 is a screenshot of an example GUI for viewing analytics data related to the uploaded content according to one or more embodiments of the disclosure.

FIGS. 40-44 are screenshots of an example GUI that allows a user to customize the service workflow of a client according to one or more embodiments of the disclosure.

FIG. 45 is an example environment view of a multi-service business platform communicating with various systems, devices, and data sources according to one or more embodiments of the disclosure.

FIG. 46 is an example detailed view of a customization system of the multi-service business platform according to one or more embodiments of the disclosure.

FIG. 47 is an example detailed view of a custom object and associations between the custom object and other objects according to one or more embodiments of the disclosure.

FIG. 48 depicts a visual representation of a portion of an example instance knowledge graph representation according to one or more embodiments of the disclosure.

FIGS. 49A-49G are screenshots of an example graphical user interface (GUI) that allows a user to create and use custom objects with the multi-service business platform according to one or more embodiments of the disclosure.

FIG. 50 is a flow chart illustrating a set of operations of a method for using the customization system of the multi-service business platform according to one or more embodiments of the disclosure.

FIG. 51 is a block diagram of an example entity resolution system embodiment of entity deduplication methods and systems according to one or more embodiments of the disclosure.

FIG. 52 is a block diagram of an example entity deduplication training process according to one or more embodiments of the disclosure.

FIG. 53 is a flow chart of an example entity deduplication training process according to one or more embodiments of the disclosure.

FIG. 54 is a block and data flow diagram of a training embodiment for entity deduplication according to one or more embodiments of the disclosure.

FIG. 55 is a portion of a system for entity deduplication showing backend functions that facilitate refining a neural

network generated probability of entity pairs being duplicates according to one or more embodiments of the disclosure.

FIG. 56 is a flow chart of a first embodiment of an artificial intelligence-based deduplication process where pairs of entities are processed singly according to one or more embodiments of the disclosure.

FIG. 57 is a diagram of entity feature-vector and companion matrices according to one or more embodiments of the disclosure.

FIG. 58 is a flow chart of an artificial intelligence-based deduplication process where an entire set of entities are processed concurrently according to one or more embodiments of the disclosure.

FIG. 59 is a flow chart of an artificial intelligence-based deduplication where artificial intelligence is used in entity duplication determination refinement actions according to one or more embodiments of the disclosure.

FIG. 60 is a schematic that depicts an example schema of an event record according to one or more embodiments of the disclosure.

FIG. 61 is a schematic that depicts an example configuration of a reporting system in relation to other components and services of a multi-service business platform according to one or more embodiments of the disclosure.

FIGS. 62A and 62B are schematics that depict example report request graphical user interfaces (GUIs) according to one or more embodiments of the disclosure.

FIGS. 63A, 63B, and 63C are schematics that depict example report configuration GUIs according to one or more embodiments of the disclosure.

FIG. 64 is a schematic that depicts an example configuration of a payment system according to some example implementations of the disclosure according to one or more embodiments of the disclosure.

FIG. 65 is a flow chart that depicts a set of operations of a method for generating a checkout link on behalf of a client according to one or more embodiments of the disclosure.

FIG. 66 is a flow chart that depicts a set of operations of a method for processing a payment on behalf of a client according to one or more embodiments of the disclosure.

FIG. 67 is an example detailed view of the conversation intelligence system of the multi-service business platform according to one or more embodiments of the disclosure.

FIG. 68 is an example detailed view of a conversation object according to one or more embodiments of the disclosure.

FIG. 69 is a flow chart illustrating a set of operations of a method for using the conversation intelligence system of the multi-service business platform according to one or more embodiments of the disclosure.

FIGS. 70, 71, 72, 73, and 74 are screenshots of example GUIs that allow a user to use the conversation intelligence system with the multi-service business platform according to one or more embodiments of the disclosure.

DETAILED DESCRIPTION

Multi-Service Business Platform

The complex, difficult, and time-consuming tasks described in the disclosure may tend to deter use of information from one activity when conducting the other, except in a somewhat ad hoc fashion. For example, a person providing service to a customer may not know what product the customer has purchased, leading to delay, confusion, and frustration for the service person and the customer. A need exists for the improved methods and systems provided

herein that enable, in a single database and system, the development and maintenance of a set of universal contact objects that relate to the contacts of a business and that have attributes that enable use for a wide range of activities, including sales activities, marketing activities, service activities, content development activities, and others, as well as for improved methods and systems for sales, marketing, and services that make use of such universal contact objects.

Further, a need exists for added and improved customizability with CRM systems and other-related systems for marketing and sales activities. While the CRM systems may use standard objects (e.g., accounts, contacts, leads, and opportunities), there is a need for the creation and use of custom objects. Specifically, there is a need for these systems to provide an ability for users to create custom objects relevant to the users' businesses. Also, there is a need for these systems to apply various types of features (e.g., apply processes such as analysis, reporting, workflows) to these custom objects.

In example embodiments, a method and system for creating custom objects may be offered for addressing need for customizability with CRM systems and other-related systems for marketing and sales activities. For example, a multi-service business platform (e.g., framework) may include a customization system that may be used to create custom objects. The multi-service business platform may be configured to provide processes related to marketing, sales, and/or customer service. The multi-service business platform may include a database structure that already has preset or fixed core objects (e.g., contact objects, company objects, deals objects, ticket objects as described in more detail below). However, the ability to create custom objects (e.g., using the customization system) allows for users to have the flexibility of creating any type of custom object (e.g., arbitrary objects) relevant to their business without being restricted to the fixed core objects. This allows for users to customize usage of the multi-service business platform more closely to their business with regard to marketing, sales, and/or customer service. This also may allow for improved and faster development of new custom object types by users and/or developers of the multi-service business platform. Various services of the multi-service business platform may then be applied and/or used with the custom objects. For example, some services that may be applied include workflow automation (e.g., automate based on changes to core objects and based on added custom objects or changes to custom objects and/or core objects), reporting (e.g., report on any custom objects along with core objects), CRM-related actions, analytics (e.g., get analytics for custom objects), import/export, and/or other actions. For example, other actions may include filtering used to search, filter, and list contact objects that may be used with custom objects and/or create lists for custom objects. In example embodiments, other actions may include, but are not limited to, reporting, permissioning, auditing, user-defined calculations, and/or aggregations. Machine learning that may have been used with core objects may also be applied to the custom objects. The multi-service business platform may include a synchronization system that may synchronize some arbitrary custom objects outside the platform to objects in the platform. In summary, in example embodiments, the multi-service business platform may act as an arbitrary platform that may act on arbitrary custom objects that may be used with various services (e.g., used with arbitrary actions and synced to arbitrary systems of the platform) thereby benefiting from these various capabilities.

In general, users may identify specific object types or custom object types that may have been created. The multi-service business platform (e.g., particularly services of the platform) may allow for use of created custom objects from users. Users may choose to create any custom object type that they prefer (e.g., a customer may create definition types and values that may be stored with custom objects and/or instances of custom objects). The multi-service business platform may allow users to dynamically add these unique custom object types with minimal development effort needed from users and the platform itself.

Embodiments of the disclosure are directed to computers, computer systems, networks and data storage arrangements comprising digitally encoded information and machine-readable instructions. The systems are configured and arranged so as to accomplish the present methods, including by transforming given inputs according to instructions to yield new and useful outputs determining behaviors and physical outcomes. Users of the present system and method will gain new and commercially significant abilities to convey ideas and to promote, create, sell, and control articles of manufacture, goods, and other products. The machinery in which the present system and method are implemented will therefore comprise novel and useful devices and architectures of computing and processing equipment for achieving the present objectives.

With reference to FIG. 1, in embodiments of the disclosure, a platform is provided having a variety of methods, systems, components, services, interfaces, processes, components, data structures, and other elements (collectively referred to as the "content development platform 100" except where context indicates otherwise), which enable automated development, deployment, and management of content, typically for an enterprise, that is adapted to support a variety of enterprise functions, including marketing strategy and communications, website development, search engine optimization, sales force management, electronic commerce, social networking, and others. Among other benefits, the content development platform 100 uses a range of automated processes to extract and analyze existing online content of an enterprise, parse and analyze the content, and develop a cluster of additional content that is highly relevant to the enterprise, without reliance on conventional keyword-based techniques. Referring to FIG. 1, the content development platform 100 may generally facilitate processing of a primary online content object 102, such as a main web page of an enterprise, to establish a topic cluster 168 of topics that are relevant to one or more core topics 106 that are found in or closely related to the content of the primary online content object 102, such as based on semantic similarity of the topics in the topic cluster 168, including core topics 106, to content within the primary online content object 102. The platform 100 may further enable generation of generated online presence content 160, such as reflecting various topics in the topic cluster 168, for use by marketers, salespeople, and other writers, or content creators on behalf of the enterprise.

In embodiments, the content development platform 100 includes methods and systems for generating a cluster of correlated content from the primary online content object 102. In embodiments, the primary online content object 102 is a web page of an enterprise. In embodiments, the primary online content object 102 is a social media page of an enterprise. In the embodiments described throughout this disclosure, the main web page of an enterprise, or of a business unit of an enterprise, is provided as an example of a primary online content object 102 and in some cases herein

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is described as a “pillar” of content, reflecting that the web page is an important driver of business for the enterprise, such as for delivering marketing messages, managing public relations, attracting talent, and routing or orienting customers to relevant products and other information. References to a web page or the like herein should be understood to apply to other types of primary online content objects **102**, except where context indicates otherwise. An objective of the content development platform **100** may be to drive traffic to a targeted web page, in particular by increasing the likelihood that the web page may be found in search engines, or by users following links to the web page that may be contained in other content, such as content developed using the content development platform **100**.

In an aspect, the present systems, data configuration architectures and methods allow an improvement over conventional online content generation schemes. As stated before, traditional online promotional content relied on key word placement and on sympathetic authorship of a main subject (e.g., a website) and corresponding secondary publications (e.g., blogs and sub-topical content related to the website), which methods rely on known objective and absolute ranking criteria to successfully promote and rank the website and sub-topical content. In an increasingly subjective, personalized and context-sensitive search environment, the present systems and methods develop canonical value around a primary online content object such as a website. In an aspect, a cluster of supportive and correlated content is intelligently generated or indicated so as to optimize and promote the online work product of a promoter (e.g., in support of an agenda or marketing effort). In an example, large numbers of online pages are taken as inputs to the present system and method (e.g., using a crawling, parallel or sequential page processing machine and software).

As shown in simplified FIG. 1, a “core topic” **106** or main subject for a promotional or marketing effort, related to one or more topics, phrases, or the like extracted based on the methods and systems described herein from a primary online content object **102**, may be linked to a plurality of supporting and related other topics, such as sub-topics. The core topic **106** may comprise, for example, a canonical source of information on that general subject matter, and preferably be a subject supporting or justifying links with other information on the general topic of a primary online content object **102**. In embodiments, visitors to a site where a generated online content **160** is located can start at a hyperlinked sub-topic of content and be directed to the core topic **106** within a page, such as a page linked to a primary online content object **102** or to the primary online content object **102** itself. In an example, the core topic **106** can be linked to several (e.g., three to eight, or more) sub-topics. A recommendation or suggestion tool, to be described further below, can recommend or suggest sub-topics, or conversely, it can dissuade or suggest avoidance of sub-topics based on automated logic, which can be enabled by a machine learned process. As will be discussed herein, a content strategy may be employed in developing the overall family of linked content, and the content strategy may supersede conventional key word based strategies according to some or all embodiments hereof.

In embodiments, the system and method analyze, store and process information available from a crawling step, including for a given promoter’s website (e.g., one having a plurality of online pages), so as to determine a salient subject matter and potential sub-topics related to the subject matter of the site. Associations derived from this processing and

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analysis are stored and further used in subsequent machine learning based analyses of other sites. Data derived from the analysis and storage of the above pages, content and extracted analytics may be organized in an electronic data store, which is preferably a large aggregated database and which may be organized, for example, using MYSQL™ or a similar format.

FIG. 2 provides a detailed functional block diagram of certain components and elements of a content development platform, including elements for extracting key phrases from a primary online content object, a content cluster data store **132** that stores clusters of topics, and a content development and management application **150** having a user interface that allows users to develop content. Within the platform **100**, key phrases **112** are extracted from the primary online content object **102** and are processed, such as using a variety of models **118**, resulting in one or more content clusters **130** that are stored in a content cluster data store **132**. The clusters may comprise the topic clusters **168** that are semantically relevant to core topics reflected in the primary online content object **102**, as indicated by the key phrases. The models **118**, which may access a corpus of content extracted by crawling a relevant set of pages on the Internet, are applied to the key phrases **112** to establish the clusters, which arrange topics around a core topic based on semantic similarity. From the content clusters **130** a suggestion generator **134** may generate one or more suggested topics **138**, which may be presented in a user interface **152** of a content development management application **150** within which an agent of an enterprise, such as a marketer, a salesperson, or the like may view the suggested topic **138** and relevant information about it (such as indicators of its similarity or relevancy as described elsewhere herein) and create content, such as web pages, emails, customer chats, and other generated online presence content **160** on behalf of the enterprise. Within the interface **152**, the resulting generated online presence content **160** may be linked to the primary online content object **102**, such that the primary online content object **102** and one or more generated online presence objects **160** form a cluster of semantically related content, such that visitors to any one of the objects **102**, **160** may be driven, including by the links, to the other objects **102**, **160**. In particular, the platform **100** enables the driving of viewers who are interested in the topics that differentiate the enterprise to the online presence content, such as the main web pages, of the enterprise. Performance of the topics may be tracked, such as in a reporting and analytics system **180**, such that performance-based suggestions may be provided by the suggestion generator **134**, such as by suggesting more suggested topics **138** that are similar to ones that have driven increases in traffic to the primary online content object **102**.

The system and method are then capable of projection of the crawled, stored and processed information, using the present processing hardware, networking and computing infrastructure so as to generate specially formatted vectors, e.g., a single vector or multiple vectors. The vector or vectors may be generated according to a Word2vec model used to produce word embeddings in a multi-layer neural network or similar arrangement. Those skilled in the art may appreciate that further reconstruction of linguistic contexts of words are possible by taking a body of content (e.g., language words) to generate such vector(s) in a suitable vector space. Said vectors may further indicate useful associations of words and topical information based on their proximity to one another in said vector space. Vectors based on other content information (e.g., phrases or documents,

which can be referred to as Phrase2vec or Document2vec herein) may also be employed in some embodiments. Documents or pages having similar semantic meaning would be conceptually proximal to one another according to the present model. In this way, new terms or phrases or documents may be compared against known data in the data store of the system and generate a similarity, relevance, or nearness quantitative metric. Cosine similarity or other methods can be employed as part of this nearness determination. The similarity may be translated into a corresponding score in some embodiments. In other aspects, said score may be used as an input to another process or another optional part of the present system. In yet other aspects, the output may be presented in a user interface presented to a human or machine. The score can further be presented as a “relevance” metric. Human-readable suggestions may be automatically generated by the system and method and provided as outputs, output data, or output signals in a processor-driven environment such as a modern computing architecture. The suggestions may in some aspects provide a content context model for guiding promoters (e.g., marketers) towards a best choice of topical content to prepare and put up on their websites, including suitable and relevant recommendations for work products such as articles and blog posts and social media materials that would promote the promoters’ main topics or subjects of interest or sell the products and services of the marketers using the system and method.

In an aspect, the present system and method allow for effective recommendations to promoters that improve the link structure between existing content materials such as online pages, articles and posts. In another aspect, this allows for better targeting of efforts of a promoter based on the desired audience of the efforts, including large groups, small groups or even individuals.

Implementations of the present system and method can vary as would be appreciated by those skilled in the art. For example, the system and method can be used to create a content strategy tool using processing hardware and special machine-readable instructions executing thereon. Consider as a simple illustrative example that a promoter desires to best market a fitness product, service or informational topic. This can be considered as a primary or “core topic” about which other secondary topics can be generated, which are in turn coupled to or related to the core topic. For example, weightlifting, dieting, exercise or other secondary topics may be determined to have a favorable context-based relevance to the core topic. Specific secondary sub-topics about weightlifting routines, entitled, e.g., ‘Best weightlifting routines for men’ or ‘How to improve your training form’ (and so on) may be each turned into a blog post that links back to the core topic web page.

In some embodiments, when a user uses the content strategy tool of the present system and method the user may be prompted to select or enter a core (primary) topic based on the user’s own knowledge or the user’s field of business. The tool may use this, along with a large amount of crawled online content that was analyzed, or along with extracted information resulting from such crawling of online content and prior stored search criteria and results, which is now context-based, to validate a topic against various criteria.

In an example, topics are suggested (or entered topics are rated) based on the topics’ competitiveness, popularity, and/or relevance. Those skilled in the art may appreciate other similar criteria which can be used as metrics in the suggestion or evaluation of a topic.

Competitiveness can comprise a measure of how likely a domain (Web domain) would be ranked on “Page 1” for a

particular term or phrase. The lower the percentile ranking, the more difficult it is to rank for that term or phrase (e.g., as determined by a MozRank™ provided by Moz™ indicating a site’s authority).

Popularity as a metric is a general measure of a term or phrase’s periodic (e.g., monthly) search volume from various major search engines. The greater this percentage, the more popular the term or phrase is.

Relevance as a metric generally indicates how close a term or phrase is to other content put up on the user’s site or domain. The lower the relevance, the further away the term or phrase is from what the core topic of the site or domain is. This can be automatically determined by a crawler that crawls the site or domain to determine its main or core topic of interest to consumers. If relevance is offered as a service by embodiments of the present system and method a score can be presented through a user or machine interface indicating how relevant the new input text is to an existing content pool.

Timeliness of the content is another aspect that could be used to drive content suggestions or ratings with respect to a core topic. For example, a recent-ness (recency) metric may be used in addition to those given above for the sake of illustration of embodiments of the system and method.

Therefore, analysis and presentation of information indicating cross relationships between topics becomes effective under the present scheme. In embodiments, these principles may be additionally or alternatively applied to email marketing or promotional campaigns to aid in decision making as to the content of emails sent to respective recipients so as to maximally engage the recipients in the given promotion.

Other possible features include question classification; document retrieval; passage retrieval; answer processing; and factoid question answering.

Note that the present concepts can be carried across languages insofar as an aspect hereof provides for manual or automated translation from a first language to a second language, and that inputs, results and outputs of the system can be processed in one or another language, or in a plurality of languages as desired.

FIG. 3, FIG. 4, and FIG. 5 are illustrative depictions of exemplary simplified aspects of the present system, method and tools. These depictions are not meant to be exhaustive or limiting, but are merely examples of how some features could be provided to a user of the system and method.

Some embodiments hereof employ a latent semantic analysis (LSA) model, encoded using data in a data store and programmed instructions and/or processing circuitry to generate an output comprising an association between various content by the promoter user of the system and method. LSA is being applied here to analyze relationships between a (large) set of documents and the data contained therein. In one embodiment machine learning may be used to develop said association output or outputs.

FIG. 6 provides a functional block diagram of certain additional optional components and elements of the content development platform 100, including integration of a customer relationship management system 158 with other elements of the platform according to one or more embodiments of the disclosure. In embodiments, the generated online content object 160 may comprise messaging content for a customer interaction that is managed via a customer relationship management system 158. In embodiments, the customer relationship management system 158 may include one or more customer data records 164, such as reflecting data on groups of customers or individual customers, including demographic data, geographic data, psychographic data,

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data relating to one or more transactions, data indicating topics of interest to the customers, data relating to conversations between agents of the enterprise and the customers, data indicating past purchases, interest in particular products, brands, or categories, and other customer relationship data. The customer data records **164** may be used by the platform **100** to provide additional suggested topics **138**, to select among suggested topics **138**, to modify suggested topics **138**, or the like. In embodiments, the CRM system **158** may support interactions with a customer, such as through a customer chat **184**, which in embodiments may be edited in the user interface **152** of the content development and management application **150**, such as to allow a writer, such as an inside salesperson or marketer who is engaging in a customer chat **184** with the customer to see suggested topics **138** that may be of interest to the customer, such as based on the customer data records **164** and based on relevancy of the topics to the main differentiators of the enterprise. In embodiments, a conversational agent **182** may be provided within or integrated with the platform **100**, such as for automating one or more conversations between the enterprise and a customer. The conversational agent **182** may take suggested topics from the suggestion generator **134** to facilitate initiation of conversations with customers around topics that differentiate the enterprise, such as topics that are semantically relevant to key phrases found in the primary online content object **102**. In embodiments, the conversational agent **182** may populate the customer chat **184** in the user interface **152**, such as providing seed or draft content that a writer for the enterprise can edit.

FIG. 7 provides a detailed functional block diagram of components and elements of a content development platform according to one or more embodiments of the disclosure. In embodiments, the methods and systems may include an automated crawler **104** that crawls the primary online content object **102** and storing a set of results from the crawling in a data storage facility **108**. In embodiments, the data storage facility is a cloud-based storage facility, such as a simple storage facility (e.g., an S3™ bucket provided by Amazon™), and/or on a web service platform (e.g., the Amazon Web Services™ (AWS) platform). In embodiments, the data storage facility is a distributed data storage facility. In embodiments, the automated crawler **104** crawls one or more domains associated with an enterprise customers' content (e.g., the customer's portal, main web page, or the like) as the primary online content object **102** to identify topics already in use on those sites and stores the pages in a data storage facility (e.g., S3™ storage), with metadata in a database (e.g., a MySQL™ database). The content development platform **100** may include a parser **110** that parses the stored content from the crawling activity of the automated crawler **104** and generates a plurality of the key phrases **112** and a content corpus **114** from the primary online content object **102**. The content development platform **100** may include, use, or integrate with one or more of a plurality of models **118** for processing at least one of the key phrases **112** and the corpus **114**.

In embodiments, the models **118** may include one or more of a word2vec model **120**, a doc2vec model **122**, a latent semantic analysis (LSA) extraction model, an LSA model **124**, and a key phrase logistic regression model **128**, wherein the processing results in a plurality of the content clusters **130** representing topics within the primary online content object **102**. In embodiments, the platform **100** may take content for the primary online content object **102**, such as a website, and extract a number of phrases, such as a number of co-located phrases, based on processing the

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n-grams present in the content (e.g., unigrams, bi-grams, tri-grams, tetra-grams, and so on), which may in the LSA model **124**, be ranked based on the extent of presence in the content and based on a vocabulary that is more broadly used across a more general body of content, such as a broad set of Internet content. This provides a vector representation of a website within the LSA model **124**. Based on crawling with the automatic crawler **104** of over 619 million pages on the public internet (seeking to ignore ignoring those pages that are light on content), an LSA model **124** has been trained using machine learning, using a training set of more than 250 million pages, such that the LSA model **124** is trained to understand associations between elements of content.

In embodiments, the one or more models **118** include the word2vec model **120** or other models (e.g., doc2vec **122** or phrase2vec) that projects crawled-domain primary online object content **102**, such as from customers' domains, into a single vector. In embodiments, the vector space is such that documents that contain similar semantic meaning are close together. The application of the word2vec model **120** and the doc2vec model **122** to the vector representation of primary online content object **102** (e.g., website) to draw vectors may result in a content-context model based on co-located phrases. This allows new terms to be compared against that content context database to determine how near it is to the enterprise's existing primary online content objects **102** (e.g., webpages), such as using cosine similarity. That similarity may then be converted into a score and displayed through the UI, such as displaying it as a "Relevancy" score. Ultimately, the content context model may be used to give recommendations and guidance for how individuals can choose good topics to write about, improve the link structure of existing content, and target marketing and other efforts based on their audiences' individual topic groups of interest. In embodiments, the plurality of models **118** used by the platform may comprise other forms of model for clustering documents and other content based on similarity, such as a latent semantic indexing model, a principal component analysis model, or the like. In embodiments, other similar models may be used, such as a phrase2vec model, or the like.

An objective of the various models **118** is to enable clustering of content, or "topic clusters **168**" around relevant key phrases, where the topic clusters **168** include semantically similar words and phrases (rather than simply linking content elements that share exactly matching keywords). Semantic similarity can be determined by calculating vector similarity around key phrases appearing in two elements of content. In embodiments, topic clusters may be automatically clustered, such as by an auto-clustering engine **172** that manages a set of software jobs that take web pages from the primary online content object **102**, use a model **118**, such as the LSA model **124** to turn the primary online content object **102** into a vector representation, project the vector representation on to a space (e.g., a two-dimensional space), perform an affinity propagation that seeks to find natural groupings among the vectors (representing clusters of ideas within the content), and show the groupings as clusters of content. Once groups are created, a reviewer, such as a marketer or other content developer, can select one or more "centers" within the clusters, such as recognizing a core topic within the marketer's "pillar" content (such as a main web page), which may correspond to the primary online content object **102**. Nodes in the cluster that are in close proximity to the identified centers may represent good additional topics about which to develop content or to which to establish links; for example, topic clusters can suggest an

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appropriate link structure among content objects managed by an enterprise and with external content objects, such as third-party objects, where the link structure is based on building an understanding of a semantic organization of a cluster of topics and mirroring the other content and architecture of links surrounding a primary online content object **102** based on the semantic organization.

The content development platform **100** may include a content cluster data store **132** for storing the content clusters **130**. The content cluster data store **132** may comprise a MySQL™ database or other type of database. The content cluster data store **132** may store mathematical relationships, based on the various models **118**, between content objects, such as the primary online content object **102** and various other content objects or topics, which, among other things, may be used to determine what pages should be in the same cluster of pages (and accordingly should be linked to each other). In embodiments, clusters are based on matching semantics between phrases, not just matching exact phrases. Thus, new topics can be discovered by observing topics or subtopics within semantically similar content objects in a cluster that are not already covered in a primary online content object **102**. In embodiments, an auto-discovery engine **170** may process a set of topics in a cluster to automatically discover additional topics that may be of relevance to parties interested in the content of the primary online content object **102**.

In embodiments, topics within a cluster in the content cluster data store **132** may be associated with a relevancy score **174** (built from the models **118**), which in embodiments may be normalized to a single number that represents the calculated extent of semantic similarity of a different topic to the core topic (e.g., the center of a cluster, such as reflecting the core topic of the primary online content object **102**, such as a main web page of an enterprise). The relevancy score **174** may be used to facilitate recommendations or suggestions about additional topics within a cluster that may be relevant for content development.

The content development platform may include a suggestion generator **134** for generating, using output from at least one of the models, a suggested topic **138** that is similar to at least one topic among the content clusters and for storing the suggested topic **138** and information regarding the similarity of the suggested topic **138** to at least one content of the clusters **130** in the content cluster data store **132**. Suggested topics **138** may include sub-topic suggestions, suggestions for additional core topics and the like, each based on semantic similarity (such as using a relevancy score **174** or similar calculation) to content in the primary online content object **102**, such as content identified as being at the center of a cluster of topics. Suggestions may be generated by using the keyphrase logistic regression model **128** on the primary online content object **102**, which, among other things, determines, for a given phrase that is similar to the content in a cluster, how relatively unique the phrase is relative to a wider body of content, such as all of the websites that have been crawled across the broader Internet. Thus, through a combination of identifying semantically similar topics in a cluster (e.g., using the word2vec model **120**, doc2vec model **122**, and LSA model **124**) and identifying which of those are relatively differentiated (using the keyphrase logistic regression model **128**), a set of highly relevant, well differentiated topics may be generated, which the suggestion generator **134** may process for production of one or more suggested topics **138**.

In embodiments, the parser **110** uses a parsing machine learning system **140** to parse the crawled content. In embodi-

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ments, the parsing machine learning system **140** iteratively applies a set of weights to input data, wherein the weights are adjusted based on a parameter of success, wherein the parameter of success is based on the success of suggested topics **138** in the online presence of an enterprise. In embodiments, the machine learning system is provided with a parser training data set **142** that is created based on human analysis of the crawled content.

In embodiments, the platform **100** uses a clustering machine learning system **144** to cluster content into the content clusters **130**. In embodiments, the clustering machine learning system **144** iteratively applies a set of weights to input data, wherein the weights are adjusted based on a parameter of success and the parameter of success is based on the success of suggested topics in the online presence of an enterprise. In embodiments, the clustering machine learning system **144** is provided with a training data set that is created based on human clustering of a set of content topics.

In embodiments, the suggestion generator **134** uses a suggestion machine learning system **148** to suggest topics. In embodiments, the suggestion machine learning system **148** iteratively applies a set of weights to input data, wherein the weights are adjusted based on a parameter of success, and the parameter of success is based on the success of suggested topics in the online presence of an enterprise. In embodiments, the suggestion machine learning system **148** is provided with a training data set that is created based on human creation of a set of suggested topics.

In embodiments, the methods and systems disclosed herein may further include a content development and management application **150** for developing a strategy for development of online presence content, the application **150** accessing the content cluster data store **132** and having a set of tools for exploring and selecting suggested topics **138** for online presence content generation. In embodiments, the application **150** provides a list of suggested topics **138** that are of highest semantic relevance for an enterprise based on the parsing of the primary online content object. In embodiments, the methods and systems may further include a user interface **152** of the application **150** that presents a suggestion, wherein the generated suggestion is presented with an indicator of the similarity **154** of the suggested topic **138** to a topic in the content clusters **130** as calculated by at least one of the models **118**.

In embodiments, the content development and management application **150** may include a cluster user interface **178** portion of the user interface **152** in which, after the primary online content object **102** has been brought on board to the content development platform **100**, a cluster of linked topics can be observed, including core topics in the primary online content object **102** and various related topics. The cluster user interface **178** may allow a user, such as a sales or marketing professional, to explore a set of topics (e.g., topics that are highly relevant to a brand of the enterprise and related topics) which, in embodiments, may be presented with a relevancy score **174** or other measures of similarity, as well as with other information, such as search volume information and the like. In embodiments, the cluster user interface **178** or other portion of the user interface **152** may allow a user to select and attach one or more topics or content objects, such as indicating which topics should be considered at the core for the enterprise, for a brand, or for a particular project. Thus, the cluster framework embodied in the cluster user interface **178** allows a party to frame the context of what topics the enterprise

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wishes to be known for online (such as for the enterprise as a whole or for a brand of the enterprise).

The content development and management application 150 may comprise a content strategy tool that encourages users to structure content in clusters based on the notion that topics are increasingly more relevant than keywords, so that enterprises should focus on owning a content topic, rather than going after individual keywords. Each topic cluster 168 may have a “core topic,” such as implemented as a web page on that core topic. For example, on a personal trainer’s website, the core topic might be “weightlifting.” Around those core topics 106 should be subtopics (in this example, this might include things like “best weightlifting routines” or “how to improve your weightlifting form”), each of which should be made into a blog post that links back to the core topic page.

When users use the content development and management application 150, or content strategy tool, the user may be prompted to enter a topic based on the user’s own knowledge of the enterprise. The content development and management application 150 or tool may also use information gleaned by crawling domains of the enterprise with the automated crawler 104, such as to identify existing topic clusters on their site (i.e., the primary online content object 102). For each identified core topic, the topic may be validated based on one or more metrics or criteria, (e.g., competitiveness, popularity, relevancy, or the like). In embodiments, a relevancy metric may be determined based on cosine similarity between a topic and the core topic, and/or based on various other sources of website analytics data. Competitiveness may comprise a measure of how likely a domain or the primary online content object 102 is to rank highly, such as on a first page of search engine results, for a particular word, phrase, or term. The lower the percentage on this metric, the harder it will be to achieve a high rank for that term. This may be determined by a source like MozRank™ (provided by Moz™), a PageRank™ (provided by Google™), or other ranking metrics, reflecting the primary online content object’s 102 domain authority, absent other factors. Popularity may comprise a general measure of a topic’s monthly search volume or similar activity level, such as from various search engines. The higher the percentage, the more popular the term. This may be obtained from a source like SEMRush™, such as with data in broad ranges of 1-1000, 1000-10000, etc. Relevancy may comprise a metric of how close a topic, phrase, term or the like to other content, such as topic already covered in other domains of a user, or the like. The lower the relevancy, the further away a given term is from what an enterprise is known for, such as based on comparison to a crawl by the automated crawler 104 of the enterprise’s website and other domains. Relevancy may be provided or supported by the content context models 118 as noted throughout this disclosure.

As the models 118 analyze more topics, the models learn and improve, such that increasingly accurate measures may be provided as relevancy and the like. Once the user has selected a topic, the user may be prompted to identify subtopics related to that topic. Also, the platform 100 may recommend or auto-fill subtopics that have been validated based on their similarity to the core topic and based on other scoring metrics. When the user has filled out a cluster of topics, the platform 100 may alert the user to suggested links connecting each subtopic page to a topic page, including recommending adding links where they are currently absent. The content development and management application 150 may also allow customers to track the performance of each

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cluster, including reporting on various metrics used by customers to analyze individual page performance. The content development and management application 150 or tool may thus provide several major improvements over our current tools, including a better “information architecture” to understand the relationship between pieces of content, built-in keyword validation, and holistic analysis of how each cluster of topics performs.

In embodiments, the user interface 152 facilitates generation of generated online presence content 160 related to the suggested topic 138. In embodiments, the user interface 152 includes at least one of key words and key phrases that represent the suggested topic 138, which may be used to prompt the user with content for generation of online presence content. In embodiments, the generated online presence content is at least one of website content, mobile application content, a social media post, a customer chat, a frequently asked question item, a product description, a service description and a marketing message. In embodiments, the generated online presence content may be linked to the primary online content object 102, such as to facilitate traffic between the generated online presence content and the primary online content object 102 and to facilitate discovery of the primary online content object 102 and the generated online presence content 160 by search engines 162. The user interface 152 for generating content may include a function for exploring phrases for potential inclusion in the generated online presence content 160; for example, a user may input a phrase, and the platform 100 may use a relevancy score 174 or other calculation to indicate a degree of similarity. For example, if a topic is only 58% similar to a core topic, then a user might wish to find something more similar. User interface elements, such as colors, icons, animated elements and the like may help orient a user to favorable topics and help avoid unfavorable topics.

In embodiments, the application 150 may facilitate creation and editing of content, such as blog posts, chats, conversations, messages, website content, and the like, and the platform may parse the phrases written in the content to provide a relevancy score 174 as the content is written. For example, as a blog is being written, the marketer may see whether phrases that are being written are more or less relevant to a primary online content object 102 that has been selected and attached to an enterprise, a project, or a brand within the platform 100. Thus, the content development and management application 150 may steer the content creator toward more relevant topics, and phrases that represent those topics. This may include prompts and suggestions from the suggestion generator 134. The user interface 152 may include elements for assisting the user to optimize content, such as optimizing for a given reading level and the like. The user interface 152 may provide feedback, such as confirming that the right key phrases are contained in a post, so that it is ready to be posted.

In embodiments, the application 150 for developing a strategy for development of the generated online presence content 160 may access content cluster data store 132 and may include various tools for exploring and selecting suggested topics 138 for generating the generated online presence content 160. In embodiments, the application 150 may further access the content of the customer relationship management (CRM) system 158. In embodiments, the application 150 includes a user interface 152 for developing content regarding a suggested topic 138 for presentation in a communication to a customer, wherein selection of a suggested topic 138 for presentation to a customer is based at least in part on a semantic relationship between the

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suggested topic as determined by at least one of the models **118** and at least one customer data record **164** relating to the customer stored in the customer relationship management system **158**.

The platform **100** may include, be integrated with, or feed the reporting and an analytics system **180** that may provide, such as in a dashboard or other user interface, such as, in a non-limiting example, in the user interface **152** of the content development and management application **150**, various reports and analytics **188**, such as various measures of performance of the platform **100** and of the generated online content object **160** produced using the platform **100**, such as prompted by suggestions of topics. As search engines have increasingly obscured information about how sites and other content objects are ranked (such as by declining to provide keywords), it has become very important to develop alternative measures of engagement. In embodiments, the platform **100** may track interactions across the life cycle of engagement of an enterprise with a customer, such as during an initial phase of attracting interest, such as through marketing or advertising that may lead to a visit to a website or other primary online content objects **102**, during a process of lead generation, during conversations or engagement with the customer (such as by chat functions, conversational agents, or the like), during the process of identifying relevant needs and products that may meet those needs, during the delivery or fulfillment of orders and the provision of related services, and during any post-sale follow-up, including to initiate further interactions. By integration with the CRM system **158** of an enterprise, the platform **100** may provide measures that indicate what other activities of or relating to customers, such as generation of leads, visits to web pages, traffic and clickstream data relating to activity on a web page, links to content, e-commerce and other revenue generated from a page, and the like, were related to a topic, such as a topic for which the generated online content object **160** was created based on a suggestion generated in the platform **100**. Thus, by integration of a content development and management application **150** and a CRM system **158**, revenue can be linked to the generated online presence content **160** and presented in the reporting and analytics system **180**.

FIG. **8** shows an example of a user interface of the reporting and analytics system **180**.

In general, a wide range of analytics may be aggregated by topic cluster (such as a core topic and related topics linked to the core topic in the cluster), rather than by web page, so that activities involved in generating the content in the cluster can be attributed with the revenue and other benefits that are generated as a result. Among these are elements tracked in a CRM system **158**, such as contact events, customers (such as prospective customers, leads, actual customers, and the like), deals, revenue, profit, and tasks.

In embodiments, the platform **100** may proactively recommend core topics, such as based on crawling and scraping existing site content of an enterprise. Thus, also provided herein is the auto-discovery engine **170**, including various methods, systems, components, modules, services, processes, applications, interfaces and other elements for automated discovery of topics for interactions with customers of an enterprise, including methods and systems that assist various functions and roles within an enterprise in finding appropriate topics to draw customers into relevant conversations and to extend the conversations in a way that is relevant to the enterprise and to each customer. Automated discovery of relevant content topics may support processes

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and workflows that require insight into what topics should be written about, such as during conversations with customers. Such processes and workflows may include development of content by human workers, as well as automated generation of content, such as within automated conversational agents, bots, and the like. Automated discovery may include identifying concepts that are related by using a combination of analysis of a relevant item of text (such as core content of a website, or the content of an ongoing conversation) with an analysis of linking (such as linking of related content). In embodiments, this may be performed with awareness at a broad scale of the nature of content on the Internet, such that new, related topics can be automatically discovered that further differentiate an enterprise, while remaining relevant to its primary content. The new topics can be used within a wide range of enterprise functions, such as marketing, sales, services, public relations, investor relations and other functions, including functions that involve the entire lifecycle of the engagement of a customer with an enterprise.

As noted above, customers increasingly expect more personalized interactions with enterprises, such as via context-relevant chats that properly reflect the history of a customer's relationship with the enterprise. Chats, whether undertaken by human workers, or increasingly by intelligent conversational agents, are involved across all of the customer-facing activities of an enterprise, including marketing, sales, public relations, services, and others. Content development and strategy is relevant to all of those activities, and effective conversational content, such as managed in a chat or by the conversational agent **182**, needs to relate to relevant topics while also reflecting information about the customer, such as demographic, psychographic and geographic information, as well as information about past interactions with the enterprise. Thus, integration of the content development and management platform **100** with the CRM system **158** may produce appropriate topics within the historical context of the customer and the customer's engagement with the enterprise. For example, in embodiments, tickets or tasks may be opened in a CRM system **158**, such as prompting creation of content, such as based on customer-relevant suggestions, via the content development and management application **150**, such as content for a conversation or chat with a customer (including one that may be managed by the conversational agent **182** or bot), content for a marketing message or offer to the customer, content to drive customer interest in a web page, or the like. In embodiments, a customer conversation or the customer chat **184** may be managed through the content development and management application **150**, such as by having the chat occur within the user interface **152**, such that an agent of the enterprise, like an inside salesperson, can engage in the chat by writing content, while seeing suggested topics **138**, indicators of relevance or similarity **154** and the like. In this context, relevance indicators can be based on scores noted above (such as reflecting the extent of relevance to core topics that differentiate the enterprise), as well as topics that are of interest to the customer, such as determined by processing information, such as on historical conversations, transactions, or the like, stored in the CRM system **158**. In embodiments, to facilitate increased, the customer chat **184** may be populated with seed or draft content created by an automated conversational agent **182**, so that a human agent can edit the content into a final version for the customer interaction.

In embodiments, the models **118** (collectively referred to as one or more content context models), and the platform

100 more generally, may enable a number of capabilities and benefits, including helping users come up with ideas of new topics to write about based on a combination of the content cluster data store 132, a graph of topics for the site or other content of the enterprise, and one or more analytics. This may help writers find gaps in content that should be effective, but that are not currently written about. The models 118, and platform 100 may also enable users to come up with ideas about new articles, white papers and other content based on effective topics. The models 118, and platform 100 may also enable users to understand effectiveness of content at the topic level, so that a user can understand which topics are engaging people and which aren't. This may be analyzed for trends over time, so a user can see if a topic is getting more or less engagement. The models 118, and platform 100 may also enable users to apply information about topics to at the level of the individual contact record, such as in the customer relationship management system 158, to help users understand with what content a specific person engages. For example, for a user "Joc," the platform 100, by combining content development and management with customer relationship management, may understand whether Joe is engaging more in "cardio exercise" or "weightlifting." Rather than only looking at the aggregate level, user may at the individual level for relevant topics. Development of content targeted to an individual's topics of interest may be time-based, such as understanding what content has recently been engaged with and whether preferences are changing over time.

The models 118, and the platform 100 may also enable looking at cross relationships between topics. For example, analytics within the platform 100 and on engagement of content generated using the platform 100 may indicate that people who engage frequently with a "cardio" topic also engage frequently with a "running" topic. If so, the platform 100 may offer suggested topics that are interesting to a specific person based on identifying interest in one topic and inferring interest in others.

The models 118, and platform 100 may also enable development of email content, such as based on understanding the topic of the content of an email, an email campaign, or the like. This may include understanding which users are engaging with which content, and using that information to determine which emails, or which elements of content within emails, are most likely to be engaging to specific users.

FIG. 8 illustrates a user interface for reporting information relating to online content generated using the content development and management platform. Various indicators of success, as noted throughout this disclosure, may be presented, such as generated by the reporting and analytics systems 180.

FIG. 9 depicts an embodiment of a user interface in which activity resulting from the use of the platform is reported to a marketer or other user. Among other metrics that are described herein, the user interface can report on what customers, such as ones to be entered into or already tracked in the CRM system, have had a first session of engagement with content, such as a web page, as a result of the content strategy, such as where the customers arrive via a link contained in a sub-topic or other topic linked to a core topic as described herein.

The present concepts can be applied to modern sophisticated searching methods and systems with improved success. For example, in a context-sensitive or personalized search request, the results may be influenced by one or more

of the following: location, time of day, format of query, device type from which the request is made, and contextual cues.

In an embodiment, a topical cluster comprising a core topic and several sub-topics can be defined and refined using the following generalized process: 1. Mapping out of several (e.g., five to ten) of the topics that a target person (e.g., customer) is interested in; 2. Group the topics into one or more generalized (core) topics into which the sub-topics could be fit; 3. Build out each of the core topics with corresponding sub-topics using keywords or other methods; 4. Map out content ideas that align with each of the core topics and corresponding sub-topics; 5. Validate each idea with industry and competitive research; and 6. Create, measure and refine the data and models and content discovered from the above process. Any disclosed steps are not intended to be limiting or exhaustive, as those skilled in the art might appreciate alternate or additional steps suiting a given application. One or more steps may also be omitted or combined into one step, again, to suit a given application at hand.

In some embodiments, a system and method are provided that can be used to provide relevancy scores (or quantitative metrics) as a service. Content generation suggestions can also be offered as a service using the present system and method, including synonyms, long tail key words and enrichment by visitor analytics in some instances.

FIG. 10 illustrates an example environment of a directed content system 200. In embodiments, the directed content system 200 may be configured to generate "directed content." As used herein, directed content may refer to any textual, audio, and/or visual content that is at least partially personalized for an intended recipient based on information derived by the directed content system 200. In embodiments, the directed content system 200 may be configured to identify recipients that are relevant to a client. As used herein, a client may refer to an enterprise (e.g., a company, a non-profit organization, a governmental entity, and the like) or an individual. A user affiliated with a client may access the directed content system 200 using a client device 260. The directed content system 200 may identify one or more intended recipients to which directed content may be sent to and/or may generate directed content to one or more of the intended recipients. The directed content system 200 may then transmit the directed content to the intended recipients.

In embodiments, the directed content system 200 may include, but is not limited to: a crawling system 202 that implements a set of crawlers that find and retrieve information from an information network (e.g., the Internet); an information extraction system 204 that identifies entities, events, and/or relationships between entities or events from the information retrieved by the crawling system 202; one or more proprietary databases 208 that store information relating to organizations or individuals that use the directed content system 200; one or more knowledge graphs 210 representing specific types of entities (e.g., businesses, people, places, products), relationships between entities, the types of those relationships, relevant events, and/or relationships between events and entities; a machine learning system 212 that learns/trains classification models that are used to extract events, entities, and/or relationships, scoring models that are used to identify intended recipients of directed content, and/or models that are used to generate directed models; a lead scoring system 214 that scores one or more organizations and/or individuals with respect to a content generation task, the lead scoring system referencing infor-

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mation in the knowledge graph; and a content generation system **216** that generates content of a communication to a recipient in response to a request from a client to generate directed content pertaining to a particular objective, wherein the recipient is an individual for which the leading scoring system has determined a threshold level of relevance to the objective of a client. The directed content system **200** uses the understanding from the machine learning system **212** to generate the directed content.

In embodiments, the methods and systems disclosed herein include methods and systems for pulling information at scale from one or more information sources. In embodiments, the crawling system **202** may obtain information from external information sources **230** accessible via a communication network **280** (e.g., the Internet), a private network, the proprietary database **208** (such as a content management system, a customer relationship management database, a sales database, a marketing database, a service management database, or the like), or other suitable information sources. Such methods and systems may include one or more crawlers, spiders, clustering systems, proxies, services, brokers, extractors and the like, using various information systems and protocols, such as Representational State Transfer (REST), Simple Object Access Protocol (SOAP), Remote Procedure Call (RPC), Real Time Operating System (RTOS) protocol, and the like. Such methods, systems, components, services and protocols are collectively referred to herein as “crawlers,” or a “set of crawlers,” except where context indicates otherwise.

In embodiments, the information extraction system **204** may pull relevant information from each of a variety of data sources (referred to in some cases herein as streams), parse each stream to figure out the type of stream, and optionally apply one or more explicit parsers to further parse specific streams based on the type or structure of the stream. Some streams with defined, static structures allow for direct extraction of information of known types, which can be inserted into the knowledge graph or other data resource without much further processing. In other cases, such as understanding what event happened given a headline of a news article, more processing is required to develop an understanding of the event that can be stored for further use. Certain events are of particular interest to sales and marketing professionals because they tend to be associated with changes in an organization that may increase or decrease the likelihood that an entity or individual will be interested in a particular offering. These include changes in management, especially in “C-level” executives like the CEO, CTO, CFO, CMO, CIO, CSO or the like and officer-level executives like the President or the VP of engineering, marketing or finance, which may indicate directional changes that tend to lead to increased or decreased purchasing. In embodiments, a machine learning system **212** may be configured to learn to parse unstructured data sources to understand that a specific type of event has occurred, such as an event that is relevant to the likelihood that an organization or individual will be interested in a particular offering. Relevant types of events for which particular machine learning systems **212** may be so configured may include, in addition to changes in management, attendance at trade shows or other industry events, participation in industry organizations like consortia, working groups and standards-making bodies, financial and other events that indicate growth or shrinking of a business, as well as other events described throughout this disclosure.

FIG. 11 illustrates an example of the crawling system **202** and the information extraction system **204** maintaining the knowledge graph **210**, whereby the crawling system **202** and

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the information extraction system **204** may operate to obtain information from one or more information sources **230** and to update the knowledge graph **210** based on the obtained information. In FIG. 11, a set of crawlers **220** obtain (e.g., crawling and/or downloading) information relating to entities and events from various information sources **230**. In embodiments, the set of crawlers **220** are controlled/directed by the crawling system **202** to identify documents that may contain information regarding entities, events, and relationships. Types of entities include, but are not limited to, organizations (e.g., companies), people, governments, cities, states, countries, dates, populations, markets, sectors, industries, roles, and products. Information about organizations may be obtained from, for example, company news articles, job postings, SEC filings, organizational charts, press releases, and any other digital documents relating to organizations such as patents, trademarks, blog posts, mentions in forums, market reports, internal documents, company websites, and social media. Information about people may be obtained from, for example, resumes (e.g., education, prior experience), scientific publications, patents, and any other digital documents such as contact information, news mentions, social media pages, blog posts, and personal websites.

In embodiments, the crawlers **220** crawl Internet websites to obtain documents. Internet websites may include, but are not limited to, sites that include company news articles, job postings, SEC filings, patents, trademark registrations, copyrights, blog posts, mentions in forums, market reports, internal documents, company websites, and social media, as well as sites that include information about people, which may include, but are not limited to, education (such as schools attended, degrees obtained, programs completed, locations of schooling and the like), prior experience (including job experience, projects, participation on boards and committees, participating in industry groups, and the like), scientific publications, patents, contact information, news mentions, social media, blog posts, and personal websites. Information may be extracted from websites via a variety of techniques, including but not limited to explicit selection of specific data fields (e.g., from .JSON), automatic extraction of data from raw website code (e.g., from HTML or XML code), or automatic extraction of data from images (e.g., using OCR or image classifiers) or video of websites (e.g., using video classifiers). The crawlers **220** may employ various methods to simulate the behavior of human users, including but not limited to simulating computer keyboard keystrokes, simulating computer mouse movement, and simulating interaction with a browser environment. The crawlers **220** may learn to simulate interaction with websites by learning and replicating patterns in the behavior of real website users.

In operation, the crawling system **202** may seed one or more crawlers **220** with a set of resource identifiers (e.g., uniform resource identifiers (URIs), uniform resource locators (URLs), application resource identifiers, and the like). The crawler **220** may iteratively obtain documents corresponding to the seeded resource identifier (e.g., an HTML document). In particular, the crawler **220** may parse the obtained documents for links contained in the documents that include additional resource identifiers. The crawler **220** may then obtain additional documents corresponding to the newly seeded resource identifiers that were identified from the parsed documents. The crawler **220** may continue in this manner until the crawler does not find any new links. The crawler **220** may provide any obtained documents to the

crawling system **202**, which can in turn dedupe any cumulative documents. The crawling system **202** may output the obtained documents to the information extraction system **204**.

In embodiments, the information extraction system **204** may receive documents obtained from the crawlers **220** via the crawling system **202**. The information extraction system **204** may extract text and any other relevant data that represents information about a company, an individual, an event, or the like. Of particular interest to users of the directed content platform **200** disclosed herein, such as marketers and salespeople, are documents that contain information about events that indicate the direction or intent of a company and/or direction or intent of an individual. These documents may include, among many others, blog posts from or about a company; articles from business news sources like Reuters™, CB Insights™, CNBC™, Bloomberg™, and others; securities filings (e.g., 10K filings in the US); patent, trademark and/or copyright filings; job postings; and the like. The information extraction system **204** may maintain and update the knowledge graph **210** based on the extracted information, as well as any additional information that may be interpolated, predicted, inferred, or otherwise derived from the extracted information and/or a proprietary database **208**.

In embodiments, the information extraction system **204** (e.g., the entity extraction system **224** and the event extraction system **224**) may discover and make use of patterns in language to extract and/or derive information relating to entities, events, and relationships. These patterns may be defined in advance and/or may be learned by the system **200** (e.g., the information extraction system **204** and/or the machine learning system **212**). The information extraction system **204** may identify a list of words (and sequences of words) and/or values contained in each received document. In embodiments, the information extraction system **204** may use the language patterns to extract various forms of information, including but not limited to entities, entity attributes, relationships between respective entities and/or respective events, events, event types, attributes of events from the obtained documents. The information extraction system **204** may utilize natural language processing and/or machine learned classification models (e.g., neural networks) to identify entities, events, and relationships from one or more parsed documents. For example, a news article headline may include the text “Company A pleased to announce a deal to acquire Company B.” In this example, an entity classification models may be able to extract “Company A” and “Company B” as business organization entities because the classification model may have learned that the term “deal to acquire” is a language pattern that is typically associated with two companies. Furthermore, an event classification model may identify a “company acquisition event” based on the text because the event classification model may have learned that the term “deal to acquire” is closely associated with “company acquisition events.” Furthermore, the event classification model may rely on the terms “Company A” and “Company B” to strengthen the classification, especially if both Company A and Company B are both entities that are known to the system **200**. In embodiments, the information extraction system **204** may derive new relationships between two or more entities from a newly identified event. For example, in response to the news article indicating Company A has acquired Company B, the information extraction system **204** may infer a new relationship that Company A owns Company B. In this example, a natural language processor may process the text to determine that Company

A is the acquirer and Company B is the acquisition. Using the event classification of “company acquisition event” and the results of the natural language processing, the information extraction module **204** may infer that Company B now owns Company A. The information extraction system **204** may extract additional information from the news article, such as a date of the acquisition.

In some embodiments, the information extraction system **204** may rely on a combination of documents to extract entities, events, and relationships. For example, in response to the combination of i) a publicly available resume of Person X indicating that Person X works at Company C (e.g., from a LinkedIn® page of Person X); and ii) a press release issued by Company B that states in the body text “Company B has recently added Person X as its new CTO,” the information extraction system **204** may extract the entities Person X, Company B, Company C, and CTO. The name of Person X and Company C may be extracted from the resume based on language patterns associated with resumes, while the names of Company B and Person A, and the role CTO may be extracted from the press release based on a parsing and natural language processing of the sentence quoted above. Furthermore, the information extraction system **204** may classify two events from the combination of documents based on natural language processing and a rules-based inference approach. First, the information extraction module **204** may classify a “hiring event” based on the patterns of text from the quoted sentence. Secondly, the information extraction module **204** may infer a “leaving event” with respect to Company C and Person X based on an inference that if a person working for a first company is hired by another company, then the person has likely left the first company. Thus, from the combination of documents described in the disclosure, the information extraction module **204** may infer that Person X has left Company C (a first event) and that Company B has hired Person X (a second event). Furthermore, in this example, the information extraction module **204** may infer updated relationships, such as Person X no longer works for Company C and Person X now works for Company B as a CTO.

In some embodiments, the information extraction system **204** may utilize one or more proprietary databases **208** to assist in identifying entities, events, and/or relationships. In embodiments, the proprietary databases **208** may include one or more databases that power a content relationship management system (not shown), where the database stores structured data relating to leads, customers, products, and/or other data relating to an organization or individual. This structured data may indicate, for example, names of customer companies and potential customer companies, contact points at customer companies and potential customer companies, employees of a company, when a sale was made to a company (or individual) by another company, emails that were sent to a particular lead from a company and dates of the emails, and other relevant data pertaining to a CRM. In these embodiments, the information extraction system **204** may utilize the data stored in the proprietary databases **208** to identify entities, relationships, and/or events; to confirm entities, relationships, or events; and/or to refute identifications of entities, relationships, and/or events.

In embodiments, the entity extraction system **224** parses and derives entities, entity types, entity attributes, and entity relationships into a structured representation based on the received documents. The entity extraction system **224** may employ natural language processing, entity recognition, inference engines, and/or entity classification models to extract entities, entity types, entity attributes, entity relation-

ships, and relationship metadata (e.g., dates on which the relationship was created). In embodiments, the event extraction system **226** may use known parsing techniques to parse a document into individual words and sequences of words. The event extraction system **226** may employ natural language processing and entity recognition techniques to determine whether any known entities are described in a document. In embodiments, the event extraction system **226** may utilize a classification model in combination with natural language processing and/or an inference engine to discover new entities, the entities' respective types, relationships between entities, and the respective types of relationships. In embodiments, the entity extraction system **224** may extract entity attributes from the documents using natural language processing and/or an inference engine. In embodiments, the inference engine may comprise conditional logic that identifies entity attributes (e.g., If/Then statements that correspond to different entity attributes and relationships). For example, the conditional logic may define rules for inferring an employer of an individual, a position of an individual, a university of an individual, a customer of an organization, a location of an individual or organization, a product sold by an organization, and the like. The entity extraction system **224** may employ additional or alternative strategies to identify and classify entities and their respective relationships.

In embodiments, the event extraction system **226** is configured to parse and derive information relating to events and how those events relate to particular entities. For example, news articles, press releases, and/or other documents obtained from other information sources may be fed to the event extraction system **226**, which may identify entities referenced in the documents based on the information contained in the news articles and other documents. The event extraction system **226** may be further configured to classify the events according to different event types (and subtypes). The event extraction system **226** may be configured to extract event attributes, such as from the headline and the body of a news article or another document, such as a press release, resume, SEC filing, trademark registration, court filings, and the like. Event types may include, but are not limited to, mergers and acquisitions, client wins, partnerships, workforce reductions or expansions, executive announcements, bankruptcies, opening and closing of facilities, stock movements, changes in credit rating, distribution of dividends, insider selling, financial results, analyst expectations, funding events, security and data breaches, regulatory and legal actions, product releases and recalls, product price changes, project initiatives, budget and operations changes, changes in name or contact, changes in how products and services are represented or described, award wins, conference participations, sponsoring activities, new job postings, promotions, layoffs, and/or charitable donations. Examples of event attributes may include entities that are connected to the event, a time/date when the event occurred, a date when the event occurred, a geographic area where the event occurred (if applicable), and the like. The event extraction system **226** may employ event classification models, natural language processing, and/or an inference engine to identify and classify events. In embodiments, the event extraction system **226** may receive entity information, including entity values (e.g., a name of a person or company, a job title, a state name) and entity types extracted by the entity extraction system **224** to assist in the event classification. In embodiments, an event classification model may

example of Company A acquiring Company B, an event classification model may receive the entity types of Company A and Company B, the features of the document (e.g., results of parsing and/or natural language processing), as well as a source of the document (e.g., news article from a business focused publication) to determine that the sentence fragment "a deal to acquire" corresponds to a company acquisition event, as opposed a real property purchase event or a sports trade event (both of which may also correlate to "a deal to acquire"). The event extraction system **226** may employ additional or alternative strategies to identify and classify events.

In embodiments, the information extraction system **204** structures the information into structured representations of respective, entities, relationships, and events. In embodiments, the information extraction system **204** updates the knowledge graph **210** with the structured representations.

FIG. **12** is a visual representation of a portion of an example knowledge graph **210** representation. In embodiments, structured representations in the knowledge graph may be represented by nodes and edges. In the illustrated example, the knowledge graph **210** includes entity nodes **252** that represent entities (e.g., companies, people, places, roles, and the like). In the example, the edges **254** that connect entity nodes **252** represent relationships between the entities represented by the entity nodes **252** connected by a respective edge **254**.

In embodiments, each entity node **252** in the knowledge graph **210** may correspond to a specific entity that is known to the system **200** and/or identified by the entity extraction module **224**. In some embodiments, each entity node **252** is (or points to) an entity record that includes information relating to the entity represented by the entity node, including an entity identifier (e.g., a unique value corresponding to the entity), an entity type of the entity node (e.g., person, company, place, role, product, and the like), an entity value of the entity (e.g., a name of a person or company), and entity data corresponding to the entity (e.g., aliases of a person, a description of a company, a classification of a product, and the like).

In embodiments, each edge **254** in the knowledge graph **210** may connect two (or more) entity nodes **252** to define a relationship between two (or more) entities represented by the entity nodes **252** connected by the edge **254**. In embodiments, edges in the graph may be (or point to) relationship records **254** that respectively define a relationship identifier (a unique identifier that identifies the relationship/edge), a relationship type between two or more entities that defines the type of relationship between the connected entities (e.g., "works for," "works as," "owns," "released by," "incorporated in," "headquartered in," and the like), relationship data (e.g., an entity identifier of a parent entity node and entity identifier(s) of one or more child entity nodes), and relationship metadata (e.g., a date on which the relationship was created and/or identified). In this way, two entity nodes **252** that are connected by an edge **254** may describe the type of relationship between the two entity nodes. In the illustrated example, an entity node **252** of Company A is connected to an entity node **252** of Delaware/USA by an edge **254** that represents an "incorporated in" relationship type. In this particular instance, the entity node **252** of Company A is the parent node and the entity node **252** of Delaware/USA is the child node by way of the directed "incorporated in" edge **254**.

In some embodiments, the knowledge graph **210** may further include event nodes (not shown). Event nodes may represent events that are identified by the event extraction

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system 226. An event node may be related to one or more entity nodes 252 via one or more edges 254, which may represent respective relationships between the event and one or more respective entities represented by the one or more entity nodes connected to the event node. In some embodiments, an event node may be (or point to) an event record 256 that corresponds to a specific event, which may define an event identifier (e.g., a unique value that identifies the event), an event type (e.g., a merger, a new hire, a product release, and the like), and event data (e.g., a date on which the event occurred, a place that the event occurred, information sources from which the event was identified, and the like).

In the case that the information extraction system 204 identifies a new entity, event, and/or relationship, the information extraction system 204 may store the structured representations of new entity, relationship, and/or event in the knowledge graph 210. In cases of known entities, events, and/or relationships, the information extraction system 204 may reconcile any newly derived data in the respective structured representations with data already present in the knowledge graph 210. Updating entities, events, and/or relationships in the knowledge graph 210 may include deleting or replacing structured relationships and/or changing data corresponding to entities, events, and/or relationships in the knowledge graph 210. Drawing from the example of Company A acquiring Company B above, the relationship between the two entities (Company A and Company B) may be represented by a parent entity node 252 of Company A, a child node 252 of Company B, and an “owns” edge 254 connecting the two entity nodes that indicates the relationship type as parent node “owns” child node. Drawing from the example of Person X, the information extraction system 204 may update the knowledge graph 210 by deleting or replacing the edge 254 defining the relationship between the entity node 252 of Person X and the entity node 252 of Company C, such that there is no longer an edge between the entity nodes of Person X and Company C or a previously defined edge (not shown) between the two entity nodes may be replaced with an edge 254 that indicates a “used to work for” relationship between the parent node of Person X and the child entity node of Company C. Furthermore, in this example, the information extraction system 204 may update the knowledge graph 210 to include a first edge 254 indicating a “works for” relationship between a parent node 252 of Person X and a child node 252 of Company B, thereby indicating that Person X works for Company B. The information extracting system 204 may further include a second edge 254 indicating a “works as” relationship between the parent node 252 of Person X and a child node 254 of a CTO Role, which may be related back to Company B by an “employs” role.

Referring back to FIG. 10, the machine learning system 212 is configured to perform machine learning tasks. Learning tasks may include, but are not limited to, training machine learned models to perform specific tasks and reinforcing the trained models based on feedback that is received in connection with the output of a model. Examples of tasks that can be performed by machine learned models can include, but are not limited to, classifying events, classifying entities, classifying relationships, scoring potential recipients of messages, and generating text. Depending on the task, certain types of machine learning may be better suited to a particular task than others. Examples of machine learning techniques can include, but are not limited to, decision tree learning, clustering, neural networks, deep learning neural networks, support vector machines, linear

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regression, logistic regression, naïve Bayes classifiers, k-nearest neighbor, k-means clustering, random forests, gradient boosting, Hidden Markov models, and other suitable model architectures.

In embodiments, a machine learning system 212 may be configured to learn to identify and extract information about events. In some embodiments, the machine learning system 212 may be configured to train event classification models (e.g., neural networks). An event classification model may refer to a machine learned model that is configured to receive features that are extracted from one or more documents and that output one or more potential events that may be indicated by the documents (or features thereof). Examples of features that may be extracted are words or sequences of words contained in a document or set of documents, dates that are extracted from the documents, sources of the documents, and the like.

In embodiments, the machine learning system 212 trains event classification models that identify events that indicate growth of a business, such as new job postings, financial reports indicating increased sales or market share, jobs reports indicating increased employment numbers, announcements indicating opening of new locations, and the like. Such events, because (among other reasons) they tend to be good indicators of available budgets for acquisition of goods and services, may be relevant to sales and marketing professionals of organizations who provide offerings that assist with growth, such as recruiting services, information technology systems, real estate services, office fixtures and furnishings, architecture and design services, and many others.

In embodiments, the machine learning system 212 trains event classification models that identify events that indicate changing needs of a company, such as C-level hires, layoffs, acquisitions, mergers, bankruptcies, product releases, and the like. Such events tend to be good indicators of companies that may require new services, as the changing of company needs tend to correlate to implementations of new software solutions or needs for specific types of services or employees. These types of events may be of interest to businesses such as construction companies, software companies, staffing companies, and the like.

In embodiments, the machine learning system 212 trains event classification models that identify events that tend to indicate that a business is flat, shrinking or failing, such as a decrease or lack of increase in job postings, financial reports indicating flat or decreased sales or market share, reports indicating decreased employment numbers, reports of lay-offs, reports indicating closing of locations, and the like. Such events may be relevant to sales and marketing professionals of organizations who provide offerings that assist with turnaround efforts, such as sales coaching services, bankruptcy services, and the like.

In embodiments, the machine learning system 212 trains models that are configured to parse job postings of an entity to determine the nature of an organization’s hiring as an indicator of need for particular types of goods and services. Among other things, job postings tend to be fairly truthful, as inaccurate information would tend to adversely impact the process of finding the right employees. In embodiments, the machine learning system 212 may include a classifier that learns, based on a training data set and/or under human supervision, to classify job postings by type, such that the classified job postings may be associated with indicators of specific needs of an organization (which itself can be determined by a machine learning system that is trained and supervised). The needs can be stored, such as in the knowl-

edge graph and/or in a customer relationships management or other databases as described throughout this disclosure and the documents incorporated by reference herein, such as to allow a sales and marketing professional to find appropriate recipients for an offering and/or configure communications to recipients within organizations that have relevant needs.

In embodiments, the machine learning system **212** trains models that are configured to parse news articles mentioning an entity or individuals associated with an entity to find events that are indicators of need for particular types of goods and services. News articles tend to be challenging, as they are typically unstructured and may use a wide range of language variations, including jargon, shorthand, and word play to describe a given type of event. In embodiments, the machine learning system **212** may specify a frame for the kind of event that is being sought, such as a specific combination of tags that characterize the event. For example, where the machine learning system **212** needs to be configured to discover events related to a management change, a frame can be specified seeking the name of a person, a role within the management structure of a company, a company, and an action related to that role. In embodiments, parsers may be configured for each of those elements of the frame, such that on a single pass of the article the machine learning system **212** can extract names, roles (e.g., “CEO” or “VP Finance” among many others), companies (including formal names and variants like initial letters and stock symbols) and actions (e.g., “retired,” “fired,” “hired,” “departs” and the like). In embodiments, names may include proper names of individuals (including full names, partial names, and nicknames) known to serve in particular roles, such as reflected in a customer relationship management database (such as described throughout this disclosure or in the documents incorporated by reference herein) that may be accessed by the machine learning system to assist with the parsing. The machine learning system **212**, which may be trained using a training data set and/or supervised by one or more humans, may thus learn to parse unstructured news articles and yield events in the form of the frame, which may be stored in the knowledge graph, such as at nodes associated with the organization and/or the individuals mentioned in the articles. In turn, the events may be used to help sales and marketing professionals understand likely directions of an enterprise. For example, among many other examples, the hiring of a new CTO may indicate a near-term interest in purchasing new technology solutions.

The machine learning system **212** may train event classification models in any suitable manner. For example, the machine learning system **212** may implement supervised, semi-supervised, and/or unsupervised training techniques to train an event classification model. For example, the machine learning system **212** may train an event classification model using a training data set and/or supervision by humans. In some instances, the machine learning system **212** may be provided with a base event classification model (e.g., a generic or randomized classification model) and training data containing labeled and unlabeled data sets. The labeled data sets may, for example, include documents (or features thereof) that describe thousands or millions of known events having known event types. The unlabeled data sets may include documents (or features thereof) without any labels or classifications that may or may not correspond to particular events or event types. The machine learning system **212** may initially adjust the configuration of the base model using the labeled data sets. The machine learning system **212** may then try identifying events from the unlabeled data sets.

In a supervised environment, a human may confirm or deny an event classification of an unlabeled data set. Such confirmation or denial may be used to further reinforce the event classification model. Furthermore, as the system **200** (e.g., the event extraction module **226**) operates to extract events from information obtained by the crawling system **202**, the machine learning system **212** may utilize the classification of said event classifications, the information used to make the entity classifications, and any outcome data resulting from the event classifications to reinforce the event classification model(s). Such event classification models may be trained to find various indicators, such as indicators of specific industry trends, indicators of need and the like.

In embodiments, the machine learning system **212** is configured to train entity classification models. An entity classification model may be a machine learned model that receives one or more documents (or features thereof) and identifies entities indicated in the documents and/or relationships between the documents. An entity classification model may further utilize the knowledge graph as input to determine entities or relationships that may be defined in the one or more documents. The machine learning system **212** may train entity classification models in a supervised, semi-supervised, and/or unsupervised manner. For example, the machine learning system **212** may train the entity classification models using a training data set and/or supervision by humans. In some instances, the machine learning system **212** may be provided with a base entity classification model (e.g., a generic or randomized classification model) and training data containing labeled and unlabeled data sets. The labeled data sets may include one or more documents (or the features thereof), labels of entities referenced in the one or more documents, labels of relationships between the entities referenced in the one or more documents, and the types of entities and relationships. The unlabeled data sets may include one or more documents (or the features thereof), but without labels. The machine learning system **212** may initially train the base entity classification model based on the labeled data sets. The machine learning system **212** may then attempt to classify (e.g., classify entities and/or relationships) entities and/or relationships from the unlabeled data sets. In embodiments, a human may confirm or deny the classifications output by an entity classification model to reinforce the model. Furthermore, as the system **200** (e.g., the event extraction module **226**) operates to extract entities and relationships from information obtained by the crawling system **202**, the machine learning system **212** may utilize the classification of said entities and relationships, the information used to make the entity classifications, and any outcome data resulting from the entity classifications to reinforce the entity classification model(s).

In embodiments, the machine learning system **212** is configured to train generative models. A generative model may refer to a machine learned models (e.g., neural networks, deep neural networks, recurrent neural networks, and the like) that are configured to output text given an objective (e.g., a message to generate a lead, a message to follow up a call, and the like) and one or more features/attributes of an intended recipient. In some embodiments, a generative model outputs sequences of three to ten words at a time. The machine learning system **212** may train a generative model using a corpus of text. For example, the machine learning system **212** may train a generative model used to generate professional messages may be trained using a corpus of messages, professional articles, emails, text messages, and the like. For example, the machine learning system **212** may be provided messages drafted by users for intended objec-

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tive. The machine learning system **212** may receive the messages, the intended objectives of the messages, and outcome data indicating whether the message was successful (e.g., generated a lead, elicited a response, was read by the recipient, and the like). As the directed content system **200** generates and sends messages and obtains outcome data relating to those messages, the machine learning system **212** may reinforce a generative model based on the text of the messages and the outcome data.

In embodiments, the machine learning system **212** is configured to train recipient scoring models. A lead scoring model **212** may be a machine learned model (e.g., a regression model) that receives features or attributes of an intended recipient of a message and a message objective and determines a lead score corresponding to the intended recipient. The lead score indicates a likelihood of a successful outcome with respect to the message. Examples of successful outcomes may include, but are not limited to, a message is viewed by the recipient, a response message is sent from the recipient, an article attached in the message is read by the recipient, or a sale is made as a result of the message. The machine learning system **212** may train the lead scoring model **212** in a supervised, semi-supervised, or unsupervised manner. In some embodiments, the machine learning system **212** is fed a training data set that includes sets of triplets, where a triplet may include recipient attributes (e.g., employer, role, recent events, location, and the like), a message objective (e.g., start a conversation, make a sale, generate a website click), and a lead score assigned (e.g., by a human) to the triplet given the attributes and the message objective. The machine learning system **212** may initially train a model based on the training data. As the lead scoring model is deployed and personalized messages are sent to users, the machine learning system **212** may receive feedback data **272** from a recipient device **270** (e.g., message was ignored, message was read, message was read and responded to) to further train the scoring model.

The machine learning system **212** may perform additional or alternative tasks not described herein. Furthermore, while the machine learning system **212** is described as training models, in embodiments, the machine learning system **212** may be configured to perform classification, scoring, and content generation tasks on behalf of the other systems discussed herein.

FIG. **13** illustrates an example configuration of the lead scoring system **214** and the content generation system **216** for identifying intended recipients of messages and generating personalized messages **218** for the one or more intended recipients. Traditionally, marketers and salespeople rely on bulk emails that are created by populating contact information from a database into a block of free text that has fields for elements that will be retrieved from the database and inserted into the text at the position of the fields. For example, an email template may say “Hi {First name}, . . .”, and first names from the database will be inserted at the point of the {First name} field. The methods and systems disclosed herein enable substantial improvement and enhancement of such templated emails by making available, at large scale, a range of information that is not typically available during bulk email generation. This includes publicly available information about recipients that may serve to personalize or customize the email in a way that makes the message more likely to have a positive outcome (e.g., opened, responded to, closes a deal, and the like).

The lead scoring system **214** and the content generation system **216** may utilize the knowledge graph **210** and/or the proprietary databases **208**, as well as a message data **262** and

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a recipient profile **264** to identify intended recipients and to generate personalized messages **218** for the intended recipients. A personalized message **218** may refer to a message that includes content that is specific to the intended recipient or the organization of the intended recipient. It is noted for purposes of distinguishing from traditional templated bulk messages, merely including a recipient’s name, address, or organization name in one or more template fields does not constitute a personalized message **218** in-and-of-itself.

In embodiments, the message data **262** may refer to any information that relates to the messages that a user wishes to have sent. In embodiments, the message data **262** may include an objective of the email. For example, the objective an email may be to open a dialogue, make a sale, renew a subscription, have a recipient read an article, and the like. As is discussed below, the message data **262** may be received from a user via a client device **260** and/or may be generated for the user by the systems described herein. In embodiments, message data may include a message template that includes content that is to be included in the body of the message. For example, a message template may include the name of the “sender,” a description of the sender’s business, a product description, and/or pricing information. A message template may be created by the user (e.g., using the content management system disclosed elsewhere herein) or may be retrieved by the user and sent from the client device. A message template may include fields to be filled by the content generation system **216** with information, including content that is generated by the content generation system **216** based on information selected from the knowledge graph **210**. In some embodiments, the system may automatically infer or generate message templates from historical data provided by the user and/or other users of the system

200. Historical data may include, but is not limited to, historical communication data (e.g., previously sent messages) and/or historical and current customer relationship data, such as the dates, amounts, and attributes of business transactions. In some embodiments, the system **200** may further rely on the objective of a message to generate the template.

In embodiments, a recipient profile **264** may include, among many other examples, information about an ideal recipient, including but not limited to work location, job title, seniority, employer size, and employer industry. The recipient profile **264** may be provided by a user via a client device **260** and/or may be determined using machine learning based on an objective of the message or other relevant factors. In the latter scenario, the machine learning system **212** may determine which types of information are relevant for predicting whether a recipient will open and/or respond to a message. For example, the machine learning system **212** may receive, determine, and/or otherwise obtain feedback data **272** from recipient devices **270**. Feedback data **272** may indicate, for example, whether a message was opened, whether a link in the message was clicked, whether an attachment in the message was downloaded or viewed, whether a response was elicited, and the like.

In some scenarios, a user may have a good sense of the type of recipient that the user believes would be interested in the user’s offering. For example, a user may wish to sell a new software suite to CTOs of certain types of organizations (e.g., hospitals and health systems). In such cases, a user may optionally input the recipient profile **264** and the message data **262** into the system **200** via a client device **260**. In additional or alternative embodiments, the recipient profile **264** may be generated by machine learned models

based on, for example, outcomes relating to personalized messages previously generated by the system **200** and/or the objective of the message.

In embodiments, the lead scoring system **214** is configured to identify a list of one or more intended recipients of a message based on the recipient profile **264** (whether specified by a user or generated by machine learning). In embodiments, the lead scoring system **214** may filter the most relevant entities in the knowledge graph **210** using the ideal recipient profile and create a recipient list. The recipient list may indicate a list of people that fit the recipient profile. In some embodiments, the lead scoring system **214** may determine a lead score for each person in the recipient list, whereby the lead score indicates a likelihood that a personalized message **218** sent to that person will lead to a successful outcome. In embodiments, the lead scoring system **214** may use historical and current data provided by the user and/or other users of the system to assess the probability that each recipient in the recipient list will respond to the user's message, and/or will be interested in knowing more about the user's offering, and/or will purchase the user's offering. The historical and current data used to evaluate this likelihood may include the events and conditions experienced by recipients and their employers around the time they made similar decisions in the past, the timing of those events, and their co-occurrence.

In some implementations, the lead scoring system **214** retrieves information relating to each person indicated in the recipient list from the knowledge graph **210**. The information obtained from the knowledge graph **210** may include information that is specific to that person and/or to the organization of the user. For example, the lead scoring system **214** may obtain a title of the person, an education level of the person, an age of the person, a length of employment by the person, and/or any events that are associated with the person or the organization of the person. The lead scoring system **214** may input this information to the lead scoring model, which outputs a lead score based thereon. The lead scoring system **214** may score each person in the recipient list. The lead scoring system **214** may then rank and/or filter the recipient list based, at least in part, on the lead scores of each respective person in the list. For example, the lead scoring system **214** may exclude any people having lead scores falling below a threshold from the recipient list. The lead scoring system **214** may output the list of recipients to the content generation system **216**.

In embodiments, determining which recipients (referred to in some cases as "leads") should receive a communication may be based on one or more events extracted by the information extraction system **204**. Examples of these types of events may include, but are not limited to, job postings, changes in revenue, legal events, changes in management, mergers, acquisitions, corporate restructuring, and many others. For a given recipient, the lead scoring system **214** may seek to match attributes of an individual (such as a person associated with a company) to an extracted event. For example, in response to an event where Company A has acquired Company B, the attributes of an individual that may match to the event may be "works for Company B," "Is a C-level or VP level executive," and "Lives in New York City." In this example, a person having these attributes may be receptive to a personalized message **218** from an executive headhunter. In this way, the lead scoring system **214** may generate the list of intended users based on the matches of attributes of the individual to the extracted event. In some

of the individual to the extracted event given the subject matter and/or objective of the message.

In embodiments, the content generation system **216** receives a recipient list and generates a personalized message **218** for each person in the list. The recipient list may be generated by the lead scoring system **214** or may be explicitly defined by the user. The content generation system **216** may employ natural language generation techniques and/or a generative model to generate directed content that is to be included in a personalized message. In embodiments, the content generation system **216** may generate directed content for each intended recipient included in the recipient list. In generating directed content for a particular individual, the content generation system **216** may retrieve information that is relevant to individual from the knowledge graph **210** and may feed that information into a natural language generator and/or a generative model. In this way, the content generation system **216** may generate one or more instances of directed content that is personalized for the user. For example, the content generation system **216** for an individual that was recently promoted at a company that recently went public may include the following instances of directed content: "Congratulations on the recent promotion!" and "I just read about your company's IPO, that's great news!" In embodiments, the content generation system **216** may merge the directed content with any parts of the message that is to be common amongst all personalized messages. In some of these embodiments, the content generation system **216** may merge the directed content for a particular intended recipient with a message template to obtain a personalized message for the particular intended recipient. In embodiments, the content generation system **216** may obtain the proper contact information of an intended recipient and may populate a "to" field of the message with the proper contact information. Upon generating a personalized message, the content generation system **216** may then output the personalized message. Outputting a personalized message may include transmitting the personalized message to an account of the intended recipient (e.g., an email account, a phone number, a social media account, and the like), storing the personalized message in a data store, or providing the personalized message for display to the user, where the user can edit and/or approve the personalized message before transmission to the intended recipient. A respective personalized message **218** may be delivered to a recipient indicated in the recipient list using a variety of messaging systems including, but not limited to, email, short message service, instant messaging, telephone, facsimile, social media direct messages, chat clients, and the like.

In embodiments, the content generation system **216** may utilize certain topics of information when generating directed content that can be included in an email subject line, opening greeting, and/or other portions of a message to make a more personal message for a recipient. For example, the content generation system **216** may utilize topics such as references to the recipient's past (such as to events the recipient attended, educational institutions the recipient attended, or the like), to the recipient's relationships (such as friends or business relationships), to the recipient's affinity groups (such as clubs, sports teams, or the like), and many others. In a specific example, the content generation system **216** may generate directed content that references the recipient's college, note that the sender observed that the recipient attended a recent trade show, and ask how the recent trade show went. In embodiments, the content generation system **216** may retrieve information from the knowledge graph **210**

regarding an intended recipient and may provide that information to a natural language generator to obtain directed content that may reference topics that a recipient may be receptive to. In embodiments, the machine learning system 212 can learn which topics are most likely to lead to successful outcomes. For example, the machine learning system 212 may monitor feedback data 272 to determine which messages are being opened or responded to versus which messages are being ignored. The machine learning system 212 use such feedback data 272 to determine topics that are more likely to lead to successful outcomes.

Events of particular interest to salespeople and marketers may include events that indicate a direction of the recipient's business, such as the hiring of a C-level executive, the posting of job openings, a significant transaction of the business (such as a merger or acquisition), a legal event (such as the outcome of a litigation, a foreclosure, or a bankruptcy proceeding), and the like. Acquiring such information from public sources at scale is challenging, as sources for the information are often unstructured and ambiguous. Accordingly, a need exists for the improved methods and systems provided herein for retrieving information from public sources at scale, recognizing information that may be relevant or of interest to one or more recipients, developing an understanding the information, and generating content for communication to one or more recipients that uses at least a portion of the information.

Building the knowledge graph 210 may be an automated or semi-automated process, including one where machine learning occurs with supervision and where content derived from crawled documents is merged for an entity once confidence is achieved that the content in fact relates to the entity. As the knowledge graph 210 grows with additional content, the presence of more and more content enriches the context that can be used for further matching of events, thus improving the ability to ascribe future events accurately to particular entities and individuals. In embodiments, one or more knowledge graphs 210 of individuals and entities may be used to seed the knowledge graph 210 of the system 200, such as a knowledge graph 210 from a CRM database, or a publicly available database of organizations and people, such as Freebase™, Jago™, Divipedia™, Link Data™, Open Data Cloud™, or the like. In embodiments, if a user realizes that an article is not really about the company they expected, the user can flag the error; that is, the process of linking articles to entities can be human-supervised.

Using the above methods and systems, a wide range of content, such as news articles and other Internet content, can be associated in the knowledge graph 210 with each of a large number of potential recipients, and the recipients can be scored based on the matching of attributes in the knowledge graph 210 to desired attributes of the recipients reflected in the recipient profile 264. For example, all CTOs and VPs of engineering who have started their jobs in the last three months can be scored as the most desired recipients of a communication about a new software development environment, and a relevant article can be associated with each of them in the knowledge graph 210.

Once a set of recipients is known, the system 200 may assist with generation of relevant content, such as targeted emails, chats, text messages, and the like. In such cases, the system composes a personalized message 218 for each recipient in the recipient list by combining a selected message template with information about the recipient and the recipient's organization that is stored in the knowledge graph 210. The information about a recipient and/or the recipient's organization may be inserted into a message

template using natural language generation and/or a generative model to obtain the personalized message. The content generation system 216 may use statistical models of language, including but not limited to automatic summarization of textual information to generate directed content based on the information about the recipient and/or the recipient's organization. In embodiments, the content generation system 216 may merge the directed content into a message template to obtain the personalized message for a recipient.

In embodiments, a range of approaches, or hybrids thereof, can be used for natural language generation. In some cases, the knowledge graph 210 may be monitored and maintained to ensure that the data contained therein is clean data. Additionally, or alternatively, the content generation system 216 can be configured to use only very high confidence data from the knowledge graph 210 (e.g., based on stored indicators of confidence in respective instances of data). Being clean for such purposes may include the data being stored as a certain part of speech (e.g., a noun phrase, a verb of a given tense, or the like). Being clean may also include being stored with a level of confidence that the data is accurate, such as having an indicator of confidence that it is appropriate to use abbreviations of a business name (e.g., "IBM" instead of "International Business Machines"), that is appropriate to use abbreviated terms (e.g., "CTO" instead of "Chief Technology Officer"), or the like. In embodiments, where data in the knowledge graph 210 may not be of sufficient structure or confidence, a generative model may be used to generate tokens (e.g., words and phrases) from the content (e.g., information from news articles, job postings, website content, etc.) in the knowledge graph 210 associated with an organization or individual, whereby the model can be trained (e.g., using a training set of input-output pairs) to generate content, such as headlines, phrases, sentences, or longer content that can be inserted into a message.

In embodiments, each respective personalized message may be sent with a message tracking mechanism to a respective recipient of the recipient list. A message tracking mechanism may be a software mechanism that causes transmission of feedback data 272 to the system 200 in response to a certain triggering action. For example, a message tracking mechanism may transmit a packet indicating an identifier of the personalized message and a timestamp when the recipient opens the email. In another example, a message tracking mechanism may transmit a packet indicating an identifier of the personalized message and a timestamp when a recipient downloads an attachment or clicks on a link in the personalized message. In embodiments this may be accomplished by using a batch email system as described in U.S. patent application Ser. No. 15/884,247 (entitled: QUALITY-BASED ROUTING OF ELECTRONIC MESSAGES and filed: Jan. 30, 2018); U.S. patent application Ser. No. 15/884,251 (entitled: ELECTRONIC MESSAGE LIFECYCLE MANAGEMENT and Filed: Jan. 30, 2018); U.S. patent application Ser. No. 15/884,264 (entitled: MANAGING ELECTRONIC MESSAGES WITH A MESSAGE TRANSFER AGENT and filed: Jan. 30, 2018); U.S. patent application Ser. No. 15/884,268 (entitled: MITIGATING ABUSE IN AN ELECTRONIC MESSAGE DELIVERY ENVIRONMENT and filed: Jan. 30, 2018), U.S. patent application Ser. No. 15/884,273 (entitled: INTRODUCING A NEW MESSAGE SOURCE INTO AN ELECTRONIC MESSAGE DELIVERY ENVIRONMENT and filed: Jan. 30, 2018), and PCT Application Serial Number PCT/US18/16038 (entitled: PLATFORM FOR ELECTRONIC MESSAGE PROCESSING and filed: Jan. 30, 2018), the contents of which are all herein incorporated by reference. In embodi-

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ments, feedback data 272 may include additional or alternative types of data, such as message tracking information and language that was used in a response (if such response occurred).

In embodiments, the feedback data 272 corresponding to personalized messages sent by the system 200 and potential responses to the personalized messages 218 may be received by the system and analyzed by an analytical engine 290. The analytical engine 290 may utilize the feedback data 272 to identify feedback data 272 corresponding to the personalized messages 218 sent by the system 200 and potential responses to those personalized messages 218 may include, but is not limited to, whether the message 218 was opened, whether an attachment was downloaded, whether a message was responded to, tracking information, language of a potential response from the recipient, and any other information contained explicitly or implicitly in the message communications such as the time and day the message was sent, opened and replied to, and the location where the message was read and replied to. The analytical engine 290 may perform language and/or content tests on the feedback data 272. Language and content tests may include, but are not limited to, A/B testing, cohort analysis, funnel analysis, behavioral analysis, and the like. The results from the analytical engine 290 may be presented to the user. The presentation of the results can be achieved using a variety of methods including, but not limited to, a web-based dashboard, reports or summary emails. In presenting the results of the analytical engine 290 with respect to a set of personalized messages sent on behalf of a user to the user, the user may then take appropriate actions to refine his or her recipient profile 264 and/or message template 262. In embodiments, the system 200 may take appropriate actions by modifying future personalized message using the results of language and content tests. The system 200 may furthermore take appropriate actions to refine the ideal recipient profile.

In embodiments, the analytical engine 290 may pass the feedback data 272 to the machine learning system 212, which may use the feedback data 272 to reinforce one or more models trained by the machine learning system 212. In this way, the machine learning system 212 can determine what types of recipients correlate to higher rates of successful outcomes (e.g., are more likely to open and/or respond to a personalized message), what type of events correlate to higher rates of successful outcomes, what time frames after an event correlate to higher rates of successful outcomes, what topics in the directed content correlate to higher rates of successful outcomes, and the like.

As noted above, the system 200 may be used to help salespeople and marketers, such as to send automatically generated emails promoting an offering, where the emails are enriched with information that shows awareness of context and includes information of interest to recipients who match a given profile. The emails are in particular enriched with text generated from the knowledge graph 210 in which relevant information about organizations and individuals, such as event information, is stored. In embodiments the system 200 generates subject lines, blog post titles, content for emails, content for blog posts, or the like, such as phrases of a few words, e.g., about 5 words, about 6-8 words, about 10 words, about 15 words, a paragraph, or more, up to an including a whole article of content. In embodiments, where large amounts of reference data are available (such as where there are many articles about a company), it is possible to generate full articles. In other cases, shorter content may be generated, as noted above, by

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training a system to generate phrases based on training on a set of input-output pairs that include event content as inputs and summary words and phrases as outputs. As an example, company descriptions have been taken from LinkedIn™ and used to generate conversational descriptions of companies. Inputs varied by company, as the nature of the data was quite diverse. Outputs were configured to include a noun-phrase and a verb-phrase, where the verb phrase was constrained to be in a given tense. In embodiments, a platform and interface are provided herein in which one or more individuals (e.g., curators), may review input text (such as of company websites, news articles, job postings, and other content from the knowledge graph 210) and the individuals can enter output text summarizing the content of the inputs in the desired form (noun phrase, verb phrase). The platform may enable the user to review content from the knowledge graph as well as to search an information network, such as a CRM system or the Internet, to find additional relevant information. In embodiments, the CRM system may include access to data about recipients maintained in a sales database, a marketing database, a service database, or a combination thereof, such that individuals preparing output text (which in turn is used to train the natural language generation system) have access to private information, such as about conversations between salespeople and individuals in the recipient list, past communications, and the like.

In embodiments, training data from the platform may be used to train a model to produce generated text, and the curators of the platform may in turn review the text to provide supervision and feedback to the model.

In embodiments, the system 200 may include automated follow-up detection, such that variants of language generated by the model may be evaluated based on outcomes, such as responses to emails, chats and other content, deals won, and the like. Thus, provided herein is a system wherein the outcome of a transaction is provided as an input to train a machine learning system to generate content targeted to a recipient who is a candidate to undertake a similar transaction.

In embodiments, outcome tracking can facilitate improvement of various capabilities of the system 200, such as information extraction, cleansing of content, generation of recipient profiles, identification of recipients, and generation of text. In embodiments, text generation may be based on the nature of the targeted recipient, such as based on attributes of the recipient's organization (such as the size of the organization, the industry in which it participates, and the like). In embodiments, a machine learning system may provide for controlled text generation, such as using a supervised approach (as described in the disclosure with summarization) and/or with an unsupervised or deep learning approach. In embodiments, the system may train a model to learn to generate, for example, blog post titles, email subject lines, or the like, conditioned on attributes like the industry of the recipient, the industry of a sales or marketing person, or other attributes. In embodiments, different language variants may be generated, such as text of different lengths, text of different complexity, text with different word usage (such as synonyms), and the like, and a model may be trained and improved on outcomes to learn to generate text that is most likely to achieve a favorable outcome, such as an email or blog post being read, a reply being received, content being clicked upon, a deal won, or the like. This may include A/B type testing, use of genetic algorithms, and other learning techniques that include content variation and outcome tracking. In embodiments, content may be varied with respect to various factors such as verb tense, sentiment,

and/or tone. For example, verb tense can be varied by as using rule-based generation of different grammatical tenses from tokens (e.g., words and phrases) contained in content attached to a node in the knowledge graph 210. In another example, sentiment can be varied by learning positive and negative sentiment on a training set of reviews or other content that has a known positive or negative sentiment, such as numerically numbered or “starred” reviews on e-commerce sites like Amazon™ or review sites like Yelp™. In another example, tone can be varied by learning on text that has been identified by curators as angry, non-angry, happy, and the like. Thus, variations of text having different length, tense, sentiment, and tone can be provided and tracked to determine combinations that produce favorable outcomes, such that the content generation system 216 progressively improves in its ability to produce effecting content for communications.

In various embodiments using no supervision (e.g., unsupervised learning), as soon as the system 200 has anything as an output, the system 200 can begin collecting labels, which in turn can be used for learning. In embodiments, the system can train an unsupervised model to create a heuristic, then apply supervision to the heuristic.

In embodiments, outcome tracking may include tracking content to determine what events extracted by the information extraction system, when used to generate natural language content for communications, tend to predict deal closure or other favorable outcomes. Metrics on deal closure may be obtained for machine learning from a CRM system, such as one with an integrated sales, marketing and service database as disclosed herein and in the documents incorporated by reference.

In embodiments, the system 200 may include variations on timing of events that may influence deal closure; that is, the system 200 may explore how to slice up time periods with respect to particular types of events in order to determine when an event of a given type is likely to have what kind of influence on a particular type of outcome (e.g., a deal closure). This may include events extracted by the information extraction system 204 from public sources and events from, for example, a CRM system, an email interaction tracking system, a content management system, or the like. For example, the time period during which a CTO change has an influence on purchasing behavior can be evaluated by testing communications over different time intervals, such that the system 200 learns what events, over what time scales, are worth factoring in for purposes of collecting information, storing information in the knowledge graph 210 (as “stale” events can be discarded), using information to generate content, targeting recipients, and forming and sending communications to the recipients. Outcome tracking can include tracking interaction information with email systems, as described in the documents incorporated herein by reference. As examples, the machine learning system 212 may learn what communications to send if a prospect has opened up collateral several times in a six-month space, how the cadence of opening emails or a white paper corresponds to purchasing decisions, and the like. As another example, if an event indicates a change in a CTO, the event may be tracked to discover a rule or heuristic, such as that a very recent (e.g., within two months) CTO change might indicate difficulty closing a deal that was in the pipeline, while a change that is not quite as recent (e.g., between five and eight months before) might indicate a very favorable time to communicate about the same offering. Thus, methods and systems are provided for learning time-based relevance of events for purposes of configuration of communications that

include content about the events to recipients of offers. In embodiments, the system 200 may vary time windows, such as by sliding time windows of varying duration (e.g., numbers of weeks, months, or the like) across a starting and ending period (e.g., one-year, some number of months, multiple years, or the like), before or after a given date, such as a projected closing date for a deal, or the like. The system 200 can then see what happened at each point of time and adjust the sizes of time windows for information extraction, recipient targeting, message generation, and other items that are under control of the system 200, including under control of machine learning. Thus, the system can run varying time windows against a training set of outcomes (such as deal outcomes from a CRM system as disclosed herein) to tune the starting point and duration of a time windows around each type of event, in order to learn what events are useful over what time periods.

As noted above, win-loss data and other information from a CRM system may be used to help determine what data is useful, which in turn avoids unnecessary communication to recipients who are not likely to be interested. Thus, an event timing data set is provided, wherein varying time windows are learned for each event to determine the effect of the event on outcomes when the event is associated with communications about offerings. Outcomes can include wins and losses of deals (such as for many different elements of the systems), messages opened (such as for learning to generate subject lines of emails), messages replied to (such as for learning to generate suitable content), and the like. In embodiments, time windows around events may, like other elements, be learned by domain, such as where the industry involved plays a role. For example, the same type of event, like a CTO change, may be relevant for very different time periods in different industries, such as based on the duration of the typical sales cycle in the industry.

In embodiments, learning as described herein may include model-based learning based on one or more models or heuristics as to, for example: what types of events and other information are likely to be of interest or relevance for initiating or continuing conversations; what recipients should be targeted and when, based on industry domain, type of event, timing factors, and the like; what content should be included in messages; how content should be phrased; and the like. Learning may also include deep learning, where a wide range of input factors are considered simultaneously, with feedback from various outcomes described throughout this disclosure and in the documents incorporated by reference herein. In embodiments, learning may use neural networks, such as feed forward neural networks, classifiers, pattern recognizers, and others. For generative features, such as natural language generation, one or more recurrent neural networks may be used. The system may run on a distributed computing infrastructure, such as a cloud architecture, such as the Amazon Web Services™ (AWS™) architecture, among others.

In embodiments, content generated using the system, such as directed content enriched with information extracted by the information extraction system and stored in the knowledge graph, may be used for a variety of purposes, such as for email marketing campaigns, for populating emails, chats and other communications of salespeople, for chats that relate to sales, marketing, service and the like, for dialog functions (such as by conversational agents), and the like. In embodiments, content is used in conjunction with a content management system, a customer relationship management system, or the like, as described throughout this disclosure and in the documents incorporated by reference herein.

In embodiments, methods and systems disclosed herein may be used for sales coaching. Conventionally a supervisor may review a sales call or message with a salesperson, such as by playing audio of a call or reviewing text of an email. In embodiments, audio may be transcribed and collected by the information extraction system disclosed herein, including to attach the transcript to a knowledge graph. The transcript may be used as a source of content for communications, as well as to determine scripts for future conversations. As noted above with respect to variations in language, variations in the script for a sales call may be developed by the natural language generation system, such as involving different sequences of concepts, different timing (such as by providing guidance on how long to wait for a customer to speak), different language variations, and the like. Machine learning, such as trained on outcomes of sales conversations, may be used to develop models and heuristics for guiding conversations, as well as to develop content for scripts.

In embodiments, the system **200** may be used to populate a reply to an email, such as by parsing the content of the email, preparing a reply and inserting relevant content from the knowledge graph in the reply, such as to customize a smart reply to the context, identity, organization, or domain of the sender. This may include allowing a recipient to select a short response from a menu, as well as enriching a short response with directed content generated by the content generation system **216** using the knowledge graph **210**.

In embodiments, the system **200** may be used in situations where a user, such as a sales, marketing, or service person, has a message template, and the system is used to fill in an introductory sentence with data that comes from a node **252** of the knowledge graph **210** that matches one or more attributes of the targeted recipient. This may include taking structured data from the knowledge graph **210** about organizations and populating a sentence with appropriate noun phrases and verb phrases (of appropriate tense).

In embodiments, the system **200** may provide an activity feed, such as a recommended list of recipients that match timely event information in the knowledge graph **210** based on a preexisting recipient profile or based on a recipient profile generated by the system as described in the disclosure. The system may recommend one or more templates for a given recipient and populate at least some of the content of an email to the recipient. In embodiments involving email, the system **200** may learn, based on outcomes, such as deals won, emails opened, replies, and the like, to configure email content, to undertake and/or optimize a number of relevant tasks that involve language generation, such as providing an optimal subject line as a person types an email, suggesting a preferred length of an email, and the like. The system **200** may look at various attributes of generated language in optimizing generation, including the number of words used, average word length, average phrase length, average sentence length, grammatical complexity, tense, voice, word entropy, tone, sentiment, and the like. In embodiments, the system **200** may track outcomes based on differences (such as a calculated edit distance based on the number and type of changes) between a generated email (or one prepared by a worker) and a template email, such as to determine the extent of positive or negative contribution of customizing an email from a template for a recipient.

In embodiments, the system **200** may be used to generate a smart reply, such as for an automated agent or bot that supports a chat function, such as a chat function that serves as an agent for sales, marketing or service. For example, if representatives typically send out a link or reference in

response to a given type of question from a customer within a chat, the system **200** can learn to surface the link or reference to a service person during a chat, to facilitate more rapidly getting relevant information to the customer. Thus, the system **200** may learn to select from a corpus a relevant document, video, link or the like that has been used in the past to resolve a given question or issue.

In embodiments, the system **200** may automatically detect inappropriate conduct, such as where a customer is engaged in harassing behavior via a chat function, so that the system prompts (or generates) a response that is configured to draw the inappropriate conversation to a quick conclusion, protecting the representative from abuse and avoiding wasting time on conversations that are not likely to lead to productive results.

In embodiments, the system **200** may be used to support communications by service professionals. For example, chat functions are increasingly used to provide services, such as to help customers with standard activities, like resetting passwords, retrieving account information, and the like. In embodiments, the system **200** may serve a relevant resource, such as from a knowledge graph, which may be customized for the recipient with content that is relevant to the customer's history (such as from a CRM system) or that relates to events of the customer's organization (such as extracted by the information extraction system).

In embodiments, the system **200** may provide email recommendations and content for service professionals. For example, when a customer submits a support case, or has a question, the system may use events about their account (such as what have the customer has done before, product usage data, how much the customer is paying, and the like), such as based on data maintained in a service support database (which may be integrated with a sales and marketing database as described herein), in order to provide recommendations about what the service professional should write in an email (such as by suggesting templates and by generating customized language for the emails, as described herein). Over time, service outcomes, such as ratings, user feedback, measures of time to complete service, measures of whether a service ticket was opened, and others may be used to train the system **200** to select appropriate content, to generate appropriate language and the, in various ways described throughout this disclosure. Outcomes may further include one or more indicators of solving a customer's problem, such as the number of responses required (usually seeking to keep them low); presence or absence of ticket deflection (i.e., avoiding unnecessary opening of service tickets by providing the right answer up front); the time elapsed before solution or resolution of a problem; user feedback and ratings; the customer's net promoter score for the vendor before and after service was provided; one or more indices of satisfaction or dissatisfaction; and the like.

FIG. **14** illustrates an example configuration of the directed content system **200**. In the illustrated example, the directed content system **200** includes a processing system **300**, a communication system **310**, and a storage system **320**.

The processing system **300** includes one or more processors that execute computer-readable instructions and non-transitory memory that stores the computer-readable instructions. In implementations having two or more processors, the two or more processors can operate in an individual or distributed manner. In these implementations, the processors may be connected via a bus and/or a network. The processors may be located in the same physical device or may be located in different physical devices. In embodiments, the

processing system **300** may execute the crawling system **202**, the information extraction system **204**, the machine learning system(s) **212**, the lead scoring system **214**, and the content generation system **216**. The processing system **300** may execute additional systems not shown.

A communication system **310** may include one or more transceivers that are configured to effectuate wireless or wired communication via a communication network **280**. The communication system **310** may implement any suitable communication protocols. For example, the communication system **310** may implement, for example, the IEEE 801.11 wireless communication protocol and/or any suitable cellular communication protocol to effectuate wireless communication with external devices via a wireless network. Additionally, or alternatively, the communication system **310** may be configured to effectuate wired communication. For example, the communication system **310** may be configured to implement a LAN communication protocol to effectuate wired communication.

The storage system **320** includes one or more storage devices. The storage devices may be any suitable type of computer readable mediums, including but not limited to read-only memory, solid state memory devices, hard disk memory devices, Flash memory devices, one-time programmable memory devices, many time programmable memory devices, RAM, DRAM, SRAM, network attached storage devices, and the like. The storage devices may be connected via a bus and/or a network. Storage devices may be located at the same physical location (e.g., in the same device and/or the same data center) or may be distributed across multiple physical locations (e.g., across multiple data centers). In embodiments, the storage system **320** may store one or more proprietary databases **208** and one or more knowledge graphs **210**.

FIG. **15** illustrates a method for generating personalized messages on behalf of a user. In embodiments, the method is executed by the system **200** described with respect to FIGS. **10-14**. The method **400** may, however, be performed by any suitable system. In embodiments, the method **400** is executed by the processing system **300** of the system **200**.

At **410**, the system obtains a recipient profile and message data. In embodiments, the message data may include a message template and/or an objective of the message. The message data may be received from a user of the system and/or derived by the system. For example, in some embodiments, the user may provide a message objective and/or the message template. In some embodiments, the user may provide a message object and the system may generate the message template. In embodiments, a recipient profile may indicate one or more attributes of an ideal recipient of a message that is to be sent on behalf of the user. In embodiments, the system may receive the recipient profile from the user. In some embodiments, the system may determine the recipient profile (e.g., the one or more attributes) based on the message objective and/or one or more attributes of the user.

At **412**, the system determines a recipient list based on the recipient profile and a knowledge graph. In embodiments, the system may identify individuals in the knowledge graph having the one or more attributes to obtain the recipient list. In embodiments, the system may generate a lead score for individuals having the one or more attributes and may select the recipient list based on the lead scores of the individuals. In embodiments, the lead score may be determined using a machine learned scoring model. In embodiments, the system may include individuals having the one or attributes and that are associated with a particular type of event (e.g., a change

in jobs, a change in management, works for a company that recently was acquired). In some of these embodiments, the time that has lapsed between the present time and the event is taken into consideration when determining the recipient list.

At **414**, the system determines, for each individual indicated in the recipient list, entity data, event data, and/or relationship data relating to the individual or an organization of the individual. In embodiments, the system may retrieve the event data, entity data, and/or relationship data from the knowledge graph. For example, the system may begin with an entity node of the individual and may traverse the knowledge graph from the entity node via the relationships of the individual. The system may traverse the knowledge graph to identify any other entities, relationships, or events that are somehow related to the individual and/or entities that are related to the individual. For example, the system may identify information about the individual such as the individual's organization, the individual's educational background, the individual's, the individual's recent publications, news about the individual's organization, news mentions of the individual, and the like.

At **416**, the system generates, for each individual indicated in the recipient list, a personalized message that is personalized for the individual based on the entity data, event data, and/or relationship data relating to the individual or an organization of the individual. In embodiments, the system generates directed content for each individual in the recipient list. The directed content may be based on the entity data, event data, and/or relationship data relating to the individual and/or the organization of the individual. In embodiments, the system may employ natural language generators to generate the directed content using entity data, event data, and/or relationship data. In embodiments, the system may employ a generative model to generate the directed content based on the entity data, event data, and/or relationship data. In embodiments, the system may, for each individual, merge the directed content generated for the individual with the message template to obtain a personalized message.

At **418**, the system provides the personalized messages. In embodiments, the system may provide the personalized messages by transmitting each personalized message to an account of the individual for which the message was personalized. In response to transmitting a personalized message to an individual, the system may receive feedback data indicating an outcome of the personalized message (e.g., was the message opened and/or responded to, was a link in the message clicked, was an attachment in the message downloaded. The system may use the feedback data to reinforce the learning of one or more of the models described in the disclosure and/or as training data to train new machine learned models. In embodiments, the system may provide the personalized messages by transmitting each personalized message to a client device of the user, whereby the user may approve a personalized message for transmission and/or edit a personalized message before transmission to the individual.

Referring now to FIG. **16**, an environment of a multi-client service platform **1600** is shown. A multi-client service platform **1600** may refer to a computing system that provides customer service solutions for any number of different clients. As used herein, a client may refer to an organization (e.g., a business, a government agency, a non-profit, and the like) that engages in some form of commercial or service-related activity, whereby the multi-client service platform **1600** may provide a customized or semi-customized cus-

customer service solution to service the customers of the client. In embodiments, the platform **1600** is a multi-tenant platform, such that the system serves the needs of multiple clients, who in turn use the system to provide service, support and the like to their own customers. As used herein, the term “service” should be understood to encompass, except where context indicates otherwise, any of a wide range of activities involved in providing services to customers and others, such as via various workflows of a business, including providing services for value, servicing goods, updating software, upgrading software, providing customer support, answering questions, providing instructions of use, issuing refunds or returns, and many others. As used herein, except where context indicates otherwise, a customer may refer to an entity or individual that engages with the client (e.g., a purchaser of a product or service of the client, a user of the client’s software platform, and the like), and the term “customer” should be understood to encompass individuals at different stages of a relationship with a client, such as individuals/organizations who are being targeted with marketing and promotional efforts, prospects who are engaged in negotiations with sales people, customers who have purchased a product, and users of the product or service (such as individuals within a customer organization). Furthermore, as will be discussed in further detail, the term “contact” is used to describe organizational entities or individuals that do engage with, may engage with, or previously engaged with the client. For example, a “contact” may refer to a sales lead, a potential customer, a closed customer (e.g., purchaser, licensee, lessee, loan recipient, policy holder, or the like), and/or a previous customer (e.g., a purchaser that needs service or may review the organization). In this way, the platform **1600** may help track a contact through an entire contact lifecycle (e.g., from the time the contact first comes to the attention of the client as a lead (such as when the client opens a promotional email or clicks on a promotional offer) until the end of the lifecycle of the client’s offering(s)). It should be noted that the term contact may refer to a customer, but also to individuals and organizations that may not qualify as “customers” or users of a client.

In some embodiments, the platform **1600** may include, integrate with, or otherwise share data with a CRM or the like. As used in this disclosure, the term CRM may refer to 144 sophisticated customer relations management platforms (e.g., HubSpot® or Salesforce.com®), an organization’s internal databases, and/or files such as spreadsheets maintained by one or more persons affiliated with the organization). In some embodiments, the platform **1600** provides a unified database (e.g., a proprietary database **1620**) in which a contact record serves as a primary entity for a client’s sales, marketing, service, support and perhaps other activities, such that individuals at a client who interact with the same customer at different points in the lifecycle, involving varied workflows, have access to the contact record without requiring data integration, synchronization, or other activities that are necessary in conventional systems where records for a contact are dispersed over disparate systems. In embodiments, data relating to a contact may be used after a deal is closed to better service the contact’s needs during a service phase of the relationship. For example, a contact of an invoicing software platform (e.g., the CFO of the company) may have initially been entered into a CRM as a lead. The contact may have subscribed to the software platform, thereby becoming a customer. The contact (or another contact affiliated with the client) may later have an issue using a particular feature of the invoicing software and may seek assistance. The platform **1600** may provide a customer

service solution on behalf of the invoicing software, whereby the platform **1600** may, for example, provide one or more interfaces or means to request service. Upon receiving a service request from the contact, the platform **1600** may have access to data relating to the contact from the time the contact was a lead through the service period of the relationship. In this way, the platform **1600** may have data relating to the communications provided to the contact when the contact was just a lead, when the deal was closed, and after the deal closed. Furthermore, the platform **1600** may have information regarding other contacts of the client, which may be useful to addressing a customer-service related issue of the contact.

In embodiments, a client, via a client device **1640**, may customize its customer service solution, such that the customer-service solution is tailored to the needs of the client and its customers. In embodiments, a client may select one or more customer service-related service features (or “service features” or “services”) and may provide one or more customization parameters corresponding to one or more of the service features. Customization parameters can include customized ticket attributes, service-related content (or “content”) to be used in the course of customer service, root URLs for populating a knowledge graph **1622**, customer service workflows (or “workflows”), and the like. For example, a client, via the client device **1640**, may provide selections of one or more features and capabilities enabled in the platform **1600** as described throughout this disclosure (e.g., automated content generation for communications, automated chat bots, AI-generated follow up emails, communication integration, call routing to service experts, and the like) via a graphical user interface (e.g., drop-down menu, a button, text box, or the like). A client, via the client device **1640**, may also use the platform **1600** to find and provide content that may be used to help provide service or support to its contacts (e.g., articles that answer frequency asked questions, “how-to” videos, user manuals, and the like) via a graphical user interface (e.g., an upload portal). The client may further provide information relating to the updated content, such as descriptions of the content, a topic heading of the content, and the like. This can be used to classify the media content in a knowledge graph as further described throughout this disclosure, and ultimately to help identify content that is relevant to a customer service issue, support issue, or the like. A client may additionally or alternatively define one or more customer service workflows via a graphical user interface (e.g., a visualization tool that allows a user to define a customer service workflows of the organization). A service workflow may define a manner by which a particular service-related issue or set of service related issues are to be handled by a client-specific service system (e.g., the system **1900-1** of FIG. **19**). For example, a service workflow for a particular client may define when a customer should be provided a link to content (e.g., useful articles or FAQs) to help the contact solve an issue, when a client is to be routed to a human service expert, when to send a technician, when to send follow-up communications, and the like). In embodiments, a workflow may be defined with respect to a ticket pipeline. The platform **1600** may utilize the client’s customization parameters to deploy a client-specific customer service system (discussed in greater detail with respect to FIG. **19**). In this way, the customer service offerings of each respective client may be tailored to the business of the client, rather than a one-size-fits-all solution for all clients, irrespective of their business needs.

The platform **1600** provided herein enables businesses to undertake activities that encourage growth and profitability,

including sales, marketing and service activities. Among other things, as a user of the platform, a business user is enabled to put customer service at a high level of priority, including to recruit the business user's customers to help in the growth of the business. In embodiments, enabling customer service involves and enables various interactions among activities of a business user that involve service, marketing, sales, among other activities. In some embodiments, the platform **1600** is configured to manage and/or track customer service issues using customer-service tickets (also referred to as "tickets"). A ticket may be a data structure that corresponds to a specific issue that needs to be addressed for a contact by or on behalf of the client. Put another way, a ticket may correspond to a service-related process, or any other process or workflow that has a defined start and finish (e.g., resolution). For example, a particular type of ticket used to support an on-line shopping client may be issued when a contact has an issue with a package not being received. A ticket in this example would relate to an issue with the delivery of the contact's package. The ticket may remain unresolved until the contact receives the package, is refunded, or the package is replaced. Until the ticket is resolved, the ticket may remain an open ticket, despite the number of times the contact interacts with the system.

In embodiments, tickets may have attributes. The attributes may include default attributes and/or custom attributes. Default ticket attributes are attributes that are included in any ticket issued by the platform **1600**. Custom attributes are attributes that are selected or otherwise defined by a client for inclusion in tickets issued on behalf of the client. A client may define one or more different types of tickets that are used in the client's client-specific service system. Examples of default ticket attributes, according to some implementations of the platform **1600**, may include (but are not limited to) one or more of a ticket ID or ticket name attribute (e.g., a unique identifier of the ticket), a ticket priority attribute (e.g., high, low, or medium) that indicates a priority of the ticket, a ticket subject attribute (e.g., what is the ticket concerning), a ticket description (e.g., a plain-text description of the issue to which the ticket pertains) attribute, a pipeline ID attribute that indicates a ticket pipeline to which the ticket is assigned, a pipeline stage attribute that indicates a status of the ticket with respect to the ticket pipeline in which it is being processed, a creation date attribute indicating when the ticket was created, a last update attribute indicating a date and/or time when the ticket was last updated (e.g., the last time an action occurred with respect to the ticket), a ticket owner attribute that indicates the contact that initiated the ticket, and the like. Examples of custom ticket attributes are far ranging, as the client may define the custom ticket attributes, and may include a ticket type attributing indicating a type of the ticket (e.g., service request, refund request, lost items, etc.), a contact sentiment attribute indicating whether a sentiment score of a contact (e.g., whether the contact is happy, neutral, frustrated, angry, and the like), a contact frequency attribute indicating a number of times a contact has been contacted, a media asset attribute indicating media assets (e.g., articles or videos) that have been sent to the contact during the ticket's lifetime, and the like.

In embodiments, the platform **1600** includes a client configuration system **1602**, a ticket management system **1604**, a conversation system **1606**, a machine learning system **1608**, a feedback system **1610**, a proprietary database **1620**, a knowledge graph **1622**, and a knowledge base **1624**. The platform **1600** may include additional or alternative components without departing from the scope of the

disclosure. Furthermore, the platform **1600** may include, be integrated with or into, or communicate with the capabilities of a CRM system and/or other enterprise systems, such as the content development platform **100** and/or directed content system **200** described in the disclosure. In some embodiments, the platform **1600** enables service and/or support as well as at least one of sales, marketing, and content development, with a common database (or set of related databases) that has a single contact record for a contact.

In embodiments, the client configuration system **1602** allows a client (e.g., a user affiliated with a client) to customize a client-specific service system, and configures and deploys the client-specific service system based on the client's customizations. In embodiments, the client configuration system **1602** presents a graphical user interface (GUI) to a client user via a client device **1640**. In embodiments, the GUI may include one or more drop down menus that allows users to select different service features (e.g., chat bots, automated follow up messages, FAQ pages, communication integration, and the like). Alternatively, a user affiliated with a client may select one of multiple packages (e.g., "basic", "standard", "premium", or "enterprise"), where each package includes one or more service features and is priced accordingly. In these embodiments, the service features in a respective package may be cumulative, overlapping, or mutually exclusive. In scenarios where the packages are mutually exclusive, a client may select multiple packages.

Once a service feature is selected, a user affiliated with the client may begin customizing one or more aspects of the service feature (assuming that the selected service feature is customizable). It should be noted that some service features allow for heavy customization (e.g., workflow definitions, pipeline definitions, content uploads, email templates, conversation scripting, and the like), while other service features do not allow for any customization or for minimal customization (e.g., custom logos).

In embodiments, the client configuration system **1602** is configured to interface with the client devices **1640** to receive customization parameters from respective clients corresponding to selected service features. As mentioned, customization parameters may include ticket attributes for client-specific tickets, service-related content (also referred to as "content" or "media assets") to be used in the course of customer service, root URLs for populating the knowledge graph **1622**, ticket pipeline definitions, customer service workflows (or "workflows") definitions, communication templates and/or scripts, and the like.

In embodiments, the client configuration system **1602** is configured to allow a client (e.g., a user affiliated with a client) to configure one or more different types of tickets that are used to record, document, manage, and/or otherwise facilitate individual customer service-related events issued by contacts of the client. For example, a client can customize one or more different types of tickets and, for each different type of ticket, the custom ticket attributes of the ticket. In embodiments, the client configuration system **1602** presents a GUI to a user affiliated with the client via the client device **1640** that allows a user to configure ticket objects, which are used to generate instances of tickets that are configured according to the client's specifications. The user may command the client configuration system **1602** to create a new ticket object, via the GUI. In doing so, the client may define, for example, the type of ticket, the different available priority levels, a pipeline that handles the ticket, and/or other suitable information. For example, using a menu or a text input box, the user can designate the type of ticket (e.g., "refund request ticket") and the different available priority

levels (e.g., low or high). In embodiments, the client configuration system **1602** may allow a user to designate a pipeline to which the ticket is assigned. Alternatively, the pipeline may be defined in a manner, such that the ticket management system **1604** listens for new tickets and assigns the ticket to various pipelines based on the newly generated ticket and information entered in by the user.

In embodiments, the user may further configure a ticket object by adding, removing, modifying, or otherwise updating the custom attributes of the ticket object. In embodiments, the GUI presented by the client configuration system **1602** may allow the user to define new attributes. The GUI may receive an attribute name and the variable type of the attribute (e.g., an integer, a flag, a floating point, a normalized score, a text string, maximum and minimum values, etc.) from the user via the GUI (e.g., using a menu and/or a text input box). In embodiments, the GUI may allow the user to define the manner by which the attribute value is determined. For example, the user may designate that a value of an attribute may be found in a specific field or fields of a specific database record, answers from the client in a live chat or chat bot transcript (e.g., extracted by an NLP system), and the like. The ticket management system **1604** may utilize such definitions to populate the attributes of new ticket instances generated from the ticket object. The GUI may allow a user to modify or otherwise edit the attributes of the ticket. For example, a user may rename a ticket type, add or delete attribute types, modify the data types of the different ticket attributes, and the like.

In some examples, in configuring a knowledge base of a client-specific service system, the client configuration system **1602** may present a GUI that allows users to upload articles, videos, tutorials, and other content. A knowledge base may refer to a collection of articles, videos, sound records, or other types of content that may be used to assist a contact when dealing with an issue. The contents in a knowledge base may be provided to a contact in a variety of different manners, including by a customer-service specialist (e.g., during a live chat or in a follow-up email), by a chat bot, or from a help page (e.g., a webpage hosted by the platform **1600** or the client's website). In some embodiments, the GUI allows a user to designate one or more root URLs that allow the platform **1600** to crawl a website for media assets and/or retrieve media assets. In some embodiments, the client configuration system **1602** may utilize the root URL to generate the knowledge graph **1622** (or portion thereof) that references the media assets used in connection with a client's service system. In some embodiments, the GUI may further allow a user to define topic headings to which the content is relevant or directed. For example, the user may provide the topics via a text string or may select topics from a menu or series of hierarchical menus. In embodiments, the topic headings may be used to organize the uploaded content in the client's help page and/or to assist in selection of the content when assisting a contact.

In another example, the client configuration system **1602** may present a GUI that includes a pipeline definition element. In embodiments, a pipeline definition element allows a user to visually create a ticket pipeline. In some of these embodiments, the GUI may allow a user to define one or more stages of a ticket pipeline, and for each stage in the pipeline, the user may further define zero or more workflows that are triggered when the ticket reaches the particular stage in the pipeline. In embodiments, the user may define the various stages of a ticket pipeline and may define the attributes (and values thereof) of a ticket that corresponds to each stage. For example, the user may define a new ticket

stage and may define that the ticket should be in the new ticket stage when a new ticket is generated but before any further communications have been made to the contact. The user may then define a second stage that indicates the ticket is waiting for further response from the client and may further define that the ticket should be in the "waiting for contact" stage after a new ticket notification has been sent to the client, but before the client has responded to the request (which may be part of the client's customer service requirements). The user may configure the entire pipeline in this manner.

In embodiments, the client configuration system **1602** may allow a user to create new and/or update workflows by defining different service-related workflow actions (also referred to as "actions") and conditions that trigger those actions. In some embodiments, a workflow may be defined with respect to a ticket. For example, the user affiliated with a cable or internet service provider (or "ISP") can define an action that notifies a customer when a new ticket is generated (e.g., generates and sends an email to the user that initiated the ticket). In this example, the GUI may be configured to allow the user to define the type or types of information needed to trigger the condition and/or other types of information that may be requested from the customer before triggering the email (e.g., the nature of the problem or reason for calling). The GUI may also allow the user to provide data that may be used to generate the email, such as an email template that is used to generate the email. Continuing this example, the user may define another action that routes the user to a human service specialist and one or more conditions that may trigger the action, such as unresolved issues or client requests. The user can add the action and conditions to the workflow, such that when the condition or conditions are triggered, the customer is routed to a live service specialist.

The client configuration system **1602** may allow a user affiliated with a client to configure other service features without departing from the scope of the disclosure. For example, in embodiments, the client configuration system **1602** may allow a user to upload one or more scripts that a chat bot may utilize to communicate with contacts. These scripts may include opening lines (e.g., "hello, how may we assist you today") and a decision tree that defines dialog in response to a response from the contact (or lack of response). In some embodiments, the decision tree may designate various rules that trigger certain responses, whereby the rules may be triggered as a result of natural language processing of input received from the contact.

FIGS. 37-44 have been provided for examples of a manner by which a client may provide certain customization parameters to the platform **1600** via a graphical user interface (GUI). For example, FIGS. 37-38 illustrate examples of GUIs that allow users to upload content to the platform **1600** for inclusion in the knowledge graph **1622** that may be used in connection with a client specific service system. In the GUI **3700** presented in FIG. 37, a user may provide a root URL via the GUI **3700**, whereby the root URL may be used by the platform **1600** to crawl and/or extract information from a series of webpages starting with the webpage indicated by the root. The GUI **3700** may further allow the user to designate a language that the knowledge graph will support. The GUI **3800** of FIG. 38 allows a user to customize a support page that is deployed on behalf of the client. In embodiments, the GUI **3800** may allow the user to provide customizations for a desktop-based browser, a tablet-based browser, and/or a mobile-based browser. In embodiments, the GUI **3800** may allow the user to upload or

otherwise provide an icon that appears in the browser tab of a contact landing on the client's support page and/or a logo of the client that is presented in the support page or other related pages. In this example, the GUI presents a preview of the client's support page, which includes a search bar, topic headers, and articles that link to content (e.g., articles and/or videos). In embodiments, a user may define the topic headers via the GUI and may select/upload the content to include with respect to each topic header. FIG. 39 illustrates a GUI 3900 for viewing analytics data related to uploaded content (e.g., how many views of the article and the average amount of time spent viewing the article). In this example GUI, a user can view analytics related to articles or other media assets (e.g., videos) that relate to customer service. In this way, a client can determine whether its contacts are clicking on the presented content and, if so, how engaged the average contact is by a respective media asset.

FIGS. 40-44 illustrate examples of GUI 4000 that allow a user of a client to customize service workflows of the client with respect to a ticket pipeline. In the example of FIG. 40, the GUI 4000 presents a ticket pipeline that defines a set of triggering conditions that automate the processing of a ticket, such as the ticket reaching a "new ticket" status when a customer initiates a new ticket, a ticket reaching "waiting on contact" status when customer is expected to make contact with the client, a "waiting on us" status when the client is expected to take action to close the ticket, and a "closed" status when the ticket is closed. The foregoing statuses are provided for a non-limiting example of a ticket pipeline. For each status, the user is presented with the option of defining a workflow when a ticket is in that status. In FIG. 41, the user has elected to define a workflow with respect to the "new" ticket status. The user is presented with a menu that allows the user to create a new custom task or to automatically send a ticket notification (e.g., an email indicating the generation of a new ticket). The example of FIG. 42, the GUI 4000 presents a menu with additional options that are organized based on the action type, whereby the user can select an action from a set of actions presented in the menu. The example menu of actions includes creating a new task, sending a ticket notification, adding a delay, creating a task, sending an internal email, sending an internal SMS, sending an internal SMS message, and the like. In the example of FIG. 43, the user has selected the ticket notification action. In response to the selection, the GUI 4000 presents the user with the option to draft an email template. In the example, the user provides the email template, including template fields such as "Ticket ID" that may be populated with the ticket ID of the newly generated ticket. In the example of FIG. 44, the user has created a workflow action with respect to the new ticket status. The GUI 4000 displays the workflow action with respect to the corresponding new ticket status.

The example GUIs of FIGS. 37-44 are provided as non-limiting examples of GUIs that allow a user to define example customization parameters. The client configuration system 1602 may allow a user to customize additional or alternative service features, some of which are described in greater detail below.

Referring back to FIG. 16, in embodiments, in response to receiving a client's customization parameters, the client configuration system 1602 may configure and deploy a client-specific service system. In embodiments, the platform 1600 implements a microservices architecture, whereby each client-specific service system may be configured as a collection of coupled services. In embodiments, the client configuration system 1602 may configure a client-specific

service system data structure that defines the microservices that are leveraged by an instance of the client-specific service system data structure (which is a client-specific service system). A client-specific service system data structure may be a data structure and/or computer readable instructions that define the manner by which certain microservices are accessed and the data that is used in support of the client-specific service system. For example, the client-specific service system data structure may define the microservices that support the selected service features and may include the mechanisms by which those microservices are accessed (e.g., API calls that are made to the respective microservices and the customization parameters used to parameterize the API calls). The client configuration system 1602 may further define one or more database objects (e.g., contact records, ticket records, and the like) from which database records (e.g., MySQL™ database records) are instantiated. For example, the client configuration system 1602 may configure ticket objects for each type of ticket, where each ticket object defines the ticket attributes included in tickets having the type. In embodiments, the client configuration system may include any software libraries and modules needed to support the service features defined by the client in the client-specific service system data structure. The client-specific service system data structure may further include references to the proprietary database 1620 and/or the knowledge graph 1622, such that a deployed client-specific service system may have access to the proprietary database 1620 and the knowledge graph 1622.

In embodiments, the client configuration system 1602 may deploy a client-specific service system based on the client-specific service system data structure. In some of these embodiments may instantiate an instance of the client-specific service system from the client-specific service system data structure, whereby the client-specific service system may begin accessing the microservices defined in the client-specific service system data structure. In some of these embodiments, the instance of the client-specific service system is a container (e.g., a Docker® container) and the client-specific service system data structure is a container image. In these embodiments, the container is configured to access the microservices, which may be containerized themselves.

In embodiments, the ticket management system 1604 manages various aspects of a ticket. The ticket management system 1604 may generate new tickets, assign tickets to new tickets to respective ticket pipelines, manage pipelines including updating the status of tickets as the ticket moves through the various stages of its lifecycle, managing workflows, and the like. In some embodiments, the ticket management system 1604 is implemented as a set of microservices, where each microservice performs a respective function.

In embodiments, the ticket management system 1604 is configured to generate a new ticket on behalf of a client-specific service system. In some of these embodiments, the ticket management system 1604 may listen for requests to generate a new ticket (e.g., an API call requesting a new ticket). The request may include information needed to generate the new ticket, including a ticket type, a subject, a description, a contact identifier, and/or the like. The request may be received in a number of different manners. For example, a request may be received from a contact request (e.g., a contact fills out a form from the client's website or a website hosted by the platform 1600 on behalf of the client), a chat bot (e.g., when a contact raises a specific issue in a chat with the chat bot), via a customer service specialist

(e.g., the client calls a service specialist and the service specialist initiates the request), and the like. In response to receiving a ticket request, the ticket management system **1604** generates a new ticket from a ticket object corresponding to the ticket type. The ticket management system **1604** may include values in the ticket attributes of the ticket based on the request, including the ticket type attribute, the subject attribute, the description attribute, the date/time created attribute (e.g., the current date and/or time), the last update attribute (e.g., the current date and/or time), the owner attribute (e.g., the contact identifier), and the like. Furthermore, in some embodiments, the ticket management system **1604** may assign a ticket to a ticket pipeline of the client-specific service system and may update the pipeline attribute to indicate the ticket pipeline to which it was assigned and the status attribute to indicate the status of the ticket (e.g., "new ticket"). In embodiments, the ticket management system **1604** may store the new ticket in the proprietary database **1620** (e.g., a ticket database).

In embodiments, the ticket management system **1604** manages ticket pipelines and triggers workflows defined in the ticket pipelines. In some embodiments, the ticket management system **1604** is configured to manage the ticket pipelines and trigger workflows using a multi-threaded approach. In embodiments, the ticket management system **1604** deploys a set of listening threads that listen for tickets having a certain set of attribute values. In these embodiments, each time a ticket is updated in any way (e.g., any time a ticket attribute value is newly defined or altered), each listening thread determines whether the attribute values indicated in the updated ticket are the attribute values that the listening thread is listening for. If the listening thread determines that the attribute values indicated in the updated ticket are the attribute values that the listening thread is listening for, the listening thread may add the ticket to a ticket queue corresponding to the listening thread. For example, a listening thread may listen for tickets having a ticket attribute that indicates that a communication was sent to a contact requesting further information. In response to identifying a ticket having a ticket attribute that indicates that the communication was sent to the contact, the listening thread may add the ticket to the ticket queue. Once in the ticket queue, the ticket management system **1604** may update the ticket status attribute of the ticket, and may trigger one or more workflows defined with respect to the ticket status. For example, a workflow may trigger the client-specific system to schedule a follow up email if no response is received from the contact within a period of time (e.g., three days).

In some embodiments, the ticket management system **1604** manages workflows on behalf of a client-specific service system. In some embodiments, the ticket management system **1604** is configured to manage workflows using a multi-threaded approach. In embodiments, the ticket management system **1604** deploys a set of workflow listening threads that listen for tickets having a certain set of attribute values. In these embodiments, each time a ticket is updated in any way (e.g., any time a ticket attribute value is newly defined or altered), each workflow listening thread determines whether the attribute values indicated in the updated ticket are the attribute values that the workflow listening thread is listening for. If the workflow listening thread determines that the attribute values indicated in the updated ticket are the attribute values that the workflow listening thread is listening for, the listening thread may add the ticket to a ticket queue corresponding to the workflow listening thread. Once a ticket is added to a ticket queue correspond-

ing to the workflow listening thread, the ticket management system **1604** may execute the workflow with respect to the ticket. For example, during a conversation with a chat bot, a workflow listening thread may listen for a sentiment attribute value that indicates that the contact is frustrated. When the chat bot (e.g., using NLP) determines that the contact is frustrated (e.g., the sentiment score is below or above a threshold), the chat bot may update the sentiment attribute of a ticket corresponding to the contact to indicate that the contact is frustrated. In response, a workflow listening thread may identify the ticket and add it to its queue. Once in the queue, the workflow may define actions that are to be performed with respect to the ticket. For example, the workflow may define that the ticket is to be routed to a service-specialist. In this example, the contact may be routed to a service-specialist, which may include updating the status of the ticket and providing any relevant ticket data to the service-specialist via a service-specialist portal.

In embodiments, the conversation system **1606** is configured to interact with a human to provide a two-sided conversation. In embodiments, the conversation system **1606** is implemented as a set of microservices that can power a chat bot. The chat bot may be configured to leverage a script that guides a chat bot through a conversation with a contact. As mentioned, the scripts may include a decision tree that includes rules that trigger certain responses based on an understanding of input (e.g., text) received from a user. For example, in response to a contact indicating a troubleshooting step performed by the contact, the script may define a response to output to the contact defining a next step to undertake. In some embodiments, the rules in a script may further trigger workflows. In these embodiments, the chat bot may be configured to update a ticket attribute of a ticket based on a triggered rules. For example, in response to identifying a troubleshooting step performed by the contact, the chat bot may update a ticket corresponding to the contact indicating that the client had unsuccessfully performed the troubleshooting step, which may trigger a workflow to send the contact an article relating to another troubleshooting step from the client's knowledge base.

In embodiments, the conversation system **1606** may be configured to implement natural language processing to effectuate communication with a contact. The conversation system **1606** may utilize machine learned models (e.g., neural networks) that are trained on service-related conversations to process text received from a contact and extract a meaning from the text. In embodiments, the models leveraged by the conversation system **1606** can be trained on transcripts of customer service live chats, whereby the models are trained on both what the customer is typing and what the customer service specialist is typing. In this way, the models may determine a meaning of input received from a contact and the chat bots may provide meaningful interactions with a contact based on the results of the natural language processing and a script.

In embodiments, the conversation system **1606** is configured to relate the results of natural language processing with actions. Actions may refer to any process undertaken by a system. In the context of customer service, actions can include "create ticket," "transfer contact to a specialist," "cancel order," "issue refund," "send content," "schedule demo," "schedule technician," and the like. For example, in response to natural language processing speech of a contact stating: "I will not accept the package and I will just send it back," the conversation system **1606** may trigger a workflow

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that cancels an order associated with the contact and may begin the process to issue a refund.

In embodiments where the platform **1600** supports audible conversations, the conversation system **1606** may include voice-to-text and text-to-speech functionality as well. In these embodiments, the conversation system **1606** may receive audio signals containing the speech of a contact and may convert the contact's speech to a text or tokenized representation of the uttered speech. Upon formulating a response to the contact, the conversation system **1606** may convert the text of the response to an audio signal, which is transmitted to a contact user device **1680**.

In embodiments, the machine learning system **1608** may be configured to perform various machine learning tasks for of the platform **1600**. The machine learning system **1608** may be implemented as a set of machine learning related microservices.

In some embodiments, the machine learning system **1608** trains and reinforces models (e.g., neural networks, regression-based models, Hidden Markov models, decision trees, and/or the like) that are used by the platform **1600**. The machine learning system **1608** can train/reinforce models that are used in natural language processing, sentiment and/or tone analysis, workflow efficacy analysis, and the like. The machine learning system **1608** may train models using supervised, semi-supervised, and/or unsupervised training techniques. In embodiments, the machine learning system **1608** is provided with training data relating to a particular type of task, which it uses to train models that help perform the particular task. Furthermore, in embodiments, the machine learning system **1608** may interact with the feedback system **1610** to receive feedback from contacts engaging with the platform **1600** to improve the performance of the models. More detailed discussions relating to various machine learning tasks are discussed in greater detail throughout this disclosure.

In embodiments, the feedback system **1610** is configured to receive and/or extract feedback regarding interactions with a contact. The feedback system **1610** may receive feedback in any suitable manner. For example, a client-specific service system may present a contact with questions relating to a ticket (e.g., "was this issue resolved to your liking?" or "on a scale of 1-10 how helpful was this article?") in a survey, during a chat bot communication, or during a conversation with a customer service specialist. A contact can respond to the question and the feedback system **1610** can update a contact's records with the response and/or pass the feedback along to the machine learning system **1608**. In some embodiments, the feedback system **1610** may send surveys to contacts and may receive the feedback from the contacts in responsive surveys.

In embodiments, the client configuration module **1602** provides a GUI by which clients can customize aspects of their respective feedback systems. For example, the client configuration module **1602** may allow a user of a client to define events that trigger a request for feedback from a contact, how long after a particular triggering event to wait until requesting the feedback, what mediums to use to request the feedback (e.g., text, phone call, email), the actual subject matter of respective requests for feedback, and/or the look and feel of the requests for feedback. The user can define surveys, including questions in the survey and the potential answers for the questions. Examples of GUIs that allow a client to customize its feedback system are provided below.

FIGS. **27-35** illustrate example GUIs **2700**, **2900** that allow a user corresponding to a client to customize different

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aspects of the client's respective feedback system. For example, the GUI **2700** allows the user to define the properties of a contact that is to receive a request for feedback and when to send the feedback request. For example, FIGS. **27**, **28**, **34**, and **35** illustrate examples of the GUI **2700** that allows the user to define the properties of a contact that is to receive a request for feedback and when to send the feedback request. The GUI **2700** may also allow a user to customize the types of questions asked (e.g., score an aspect of the client's business, ask for questions for determining a net promoter score, follow up questions and the like), the actual content of the questions, and the look and feel of the feedback request. For example, FIGS. **29-33** illustrate examples of a GUI **2900** that allows a user to customize the types of questions asked, the actual content of the questions, and the look and feel of the feedback request.

In embodiments, the feedback system **1610** may be configured to extract feedback from an interaction with a contact. In these embodiments, the feedback system **1610** may perform tone and/or sentiment analysis on a conversation with a contact to gauge the tone or sentiment of the contact (e.g., is the contact upset, frustrated, happy, satisfied, confused, or the like). In the case of text conversations, the feedback system **1610** can extract tone and/or sentiment based on patterns recognized in text. For example, if a user is typing or uttering expletives or repeatedly asking to speak with a human, the feedback system **1610** may recognize the patterns in the text and may determine that the user is likely upset and/or frustrated. In the case of audible conversations, the feedback system **1610** can extract features of the audio signal to determine if the contact is upset, frustrated, happy, satisfied, confused, or the like. The feedback system **1610** may implement machine learning, signal processing, natural language processing and/or other suitable techniques to extract feedback from conversations with a contact.

In embodiments, the proprietary database(s) **1620** may store and index data records. In embodiments, the proprietary database(s) **1620** may store contact records, client records, and/or ticket records. The proprietary database(s) **1620** may store additional or alternative records as well, such as product records, communication records, deal records, and/or employee records. FIGS. **17A-17C** illustrate example high level schemas of contact records **1700** (FIG. **17A**), client records **1720** (FIG. **17B**), and ticket records **1740** (FIG. **17C**).

FIG. **17A** illustrates an example schema of a contact database record **1700**. In embodiments, the contact data record **1700** may include a contact ID **1702**, a client ID **1704**, a purchase data **1706**, ticket IDs **1708**, a contact timeline **1710**, and a contact data **1712**. The contact identifier ("ID") **1702** may be a unique value (e.g., string or number) that uniquely identifies the contact from other contacts. It is noted that in some embodiments the client ID **1704** is specific to a relationship between a contact and a client, whereby the contact may only relate to a single client. If a certain individual interacts with multiple clients, then the contact may have multiple contact records associated therewith (i.e., one for each client, where the contact is a customer of different clients (businesses) for the respective products or services of the businesses). The client ID **1704** indicates the client to which the contact corresponds. In embodiments, the client ID **1704** may be a reference to the client record **1720**. In this way, the client ID **1704** defines the relationship between the contact and the client. The purchase data **1706** indicates all of the purchases made by the contact with respect to the client. In embodiments, the purchase data may indicate the products and/or services purchased by the

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client and the dates on which such products or services were purchased. In some embodiments, the products and services may be presented by product IDs. The ticket IDs **1708** indicate any tickets that have been issued with respect to the client. In embodiments, the ticket IDs **1708** include any tickets, resolved or unresolved, that have been issued with respect to the contact. In other embodiments, the ticket IDs **1708** may only include the tickets IDs **1708** that are still open.

In embodiments, the contact timeline **1710** includes a timeline documenting the contact's interaction with the client. The contact timeline may include data from the point in time when the contact was solicited as a lead, when purchases were made by the contact, when tickets were generated on behalf of the contact, when communications were sent to the contact, and the like. Thus, the contact data **1712** and the timeline **1710** provide a rich history of all interactions of the contact with the client, over the lifecycle of a relationship. As a result, an individual, such as a salesperson or service professional, can understand and reference that history to provide relevant communications. More generally, the contact data **1712** may include any data that is relevant to the contact with respect to client. The contact data **1712** may include demographic data (e.g., age and sex), geographic data (e.g., address, city, state, country), conversation data (e.g., references to communications that were engaged in with the client), contact information (e.g., phone number, email address, user name), and the like. Contact data may further include information such as a date that the contact was entered into the system, lifecycle state (what is the current relationship with the contact-current customer, lead, or service only), last purchase date, recent purchase dates, on boarding dates, in-person visit dates, last login, last event, last date a feature was used, demos presented to the contact, industry vertical of the contact, a role of the contact, behavioral data, net promoter score of the contact, and/or a contact score (e.g., net value of the contact to the client or a net promoter score).

It is noted that in some embodiments, the contact record **1700** may be used across multiple platforms, such that the contact record **1700** defines data relevant to the contact through the lifecycle of the contact (e.g., from new lead and/or buyer through customer service). In this way, customer service may be integrated with the sales arm of the client's business. For example, a contact record **1700** corresponding to a warehouse manager (a contact) that purchased an industrial furnace from an HVAC business (a client) may identify or otherwise reference the dates on which the warehouse manager was first contacted, all communications sent between the manager and the HVAC business, a product ID of the furnace, and any tickets that have been initiated by the contact on behalf of the warehouse, and the like. Thus, in some embodiments, the integration of the client's sales and marketing data with the client's customer service infrastructure allows a client-specific service system to address issues with a more complete view of the contact's data and reduces the need for APIs to connect typically unconnected systems (e.g., invoicing system, CRM, contact database, and the like). The foregoing database object is provided for example. Not all the data types discussed are required and the object may include additional or alternative data types not herein discussed.

FIG. 17B illustrates an example schema of a client database record **1720**. In embodiments, a client database record **1720** may include a client ID **1722** and one or more product IDs **1724**. The client ID **1722** may be a unique value

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(e.g., string or number) that uniquely identifies the client from other clients. The product IDs **1724** identify/reference products (e.g., goods, services, software) that are offered by the client. The product ID **1724** of a product may reference a product record (not shown) that includes data relating to the product, including warranty data, serial numbers, and the like. The foregoing database object is provided for example. Not all the data types discussed are required and the object may include additional or alternative data types not herein discussed.

FIG. 17C illustrates an example schema of a ticket database record **1740**. In embodiments, the ticket database record **1740** may include a ticket ID **1742**, a client ID **1744**, a contact ID **1746**, ticket attributes **1748** including a status attribute **1750**, and other additional ticket data **1752**. The ticket database record **1740** stores the types of ticket attributes that may be used to identify, track, and/or manage a ticket issued on behalf of a client. The ticket ID **1742** is a unique value (e.g., string or number) that uniquely identifies a ticket from other tickets. The client ID **1744** is a value that indicates the client with respect to which the ticket was issued. As can be appreciated, the client ID **1744** may point to the client record **1720** of a particular client. The contact ID **1746** indicates the contact that initiated the ticket. As can be appreciated, the contact ID **1746** may point to the contact record **1720** of the contact that initiated the ticket. The ticket attributes **1748** may include or reference any data tied to the ticket. As discussed, the ticket attributes may include default ticket attributes and/or custom ticket attributes. Examples of default ticket attributes may include a ticket priority attribute (e.g., high, low, or medium), a ticket subject attribute (e.g., what is the ticket concerning), a ticket description attribute, a pipeline ID attribute, a creation date attribute, a last update attribute, and the like. The custom ticket attributes may depend on the customizations of the customer. In an example, the custom ticket attributes may include a ticket type attributing indicating a type of the ticket (e.g., service request, refund request, lost items, etc.), a contact sentiment attribute indicating whether a sentiment score of a contact (e.g., whether the contact is happy, neutral, frustrated, angry, and the like), a product ID attribute that indicates a product to which the ticket corresponds, a contact frequency attribute indicating a number of times a contact has been contacted, a media asset attribute indicating media assets (e.g., articles or videos) that have been sent to the contact during the ticket's lifetime, and the like. In embodiments, the ticket attributes may further include the ticket status attribute **1750**. The ticket status attribute **1750** can indicate a status of the ticket. The status may be defined with respect to the ticket pipeline of the client. For example, example statuses may include: ticket is opened but not acted upon, waiting for customer response, at a chat bot stage, at service specialist, at visit stage, at refund state, issue resolved, and the like. In embodiments, the ticket record **1740** may include additional ticket data **1752**. In embodiments, the additional ticket data **1752** may include or reference the specialist or specialists that have helped service the ticket (e.g., employee IDs), any notes entered by specialists, a number of notes entered by the specialists, a list of materials that have been sent to the contact during attempts to resolve the issue, and the like. In embodiments, the additional ticket data **1752** may include references to transcripts of conversations with the contact over different mediums. For example, the additional ticket data **1752** may include or reference conversations had with a bot, over email, in text message, over social media, and/or with a customer service specialist. The additional ticket data **1752** may additionally or alternatively include analytics

data. For example, the ticket status attribute **1750** may include a sentiment or tone of the contact throughout the timeline, feedback from the contact, a contact score of the client, and the like.

In embodiments, the additional ticket data **1752** may include or reference a ticket timeline. A ticket timeline may indicate all actions taken with respect to the ticket and when those actions were taken. In some embodiments, the ticket timeline may be defined with respect to a client's ticket pipeline and/or workflows. The ticket timeline can identify when the ticket was initiated, when different actions define in the workflow occurred (e.g., chat bot conversation, sent link to FAQ, sent article, transferred to customer service specialist, made house call, resolved issue, closed ticket, and the like). The ticket timeline of a ticket record **1740** can be updated each time a contact interacts with a client-specific service system with respect to a particular ticket. In this way, many different types of information can be extracted from the ticket timeline (or the ticket record in general). For example, the following information may be extracted: How long a ticket took to come through the pipeline (e.g., how much time was needed to close the ticket); how many interactions with the system did the contact have; how much time passed until the first response was provided; how long did each stage in the pipeline take; how many responses were sent to the contact; how many communications were received from the contact; how many notes were entered into the record; and/or how many documents were shared with the contact.

In embodiments, the knowledge graph **1622** structures a client's knowledge base **1624**. The knowledge base **1624** may refer to the set of media assets (e.g., articles, tutorials, videos, sound recordings) that can be used to aid a contact of the client. The knowledge base **1624** of a client may be provided by the client (e.g., uploaded by a user via an upload portal) and/or crawled on behalf of the client by the client configuration system **1602** (e.g., in response to receiving a root URL to begin crawling). In embodiments, media assets may be assigned topic headers that indicate the topics to which the media asset pertains. In this way, a link to a media asset may be displayed in relation to a topic heading and/or may be used by customer service specialists when providing links to a contact seeking assistance with an issue pertaining to the respective topic.

In embodiments where the knowledge graph **1622** structures the client knowledge base **1624**, the knowledge graph **1622** may store relationship data relating to the knowledge base **1624** of a client. In embodiments, the knowledge graph **1622** may be the knowledge graph **210** discussed above. In other embodiments, the knowledge graph **1622** may be a separate knowledge graph. In embodiments, the knowledge graph **1622** includes nodes and edges, where the nodes represent entities and the edges represent relationships between entities. Examples of types of entities that may be stored in the knowledge graph **1622** include articles, videos, locations, contacts, clients, tickets, products, service specialists, keywords, topics, and the like. The edges may represent logical relationships between different entities.

FIG. **18** illustrates an example visualization of a portion of the knowledge graph **1622**, which is one mechanism among various alternatives (including tables, key-value pairs, directed graphs, clusters, and the like) for representing objects and relationships among objects handled by the platform **1600**, including contact objects, tickets, content objects (such as for communications), client objects, timeline objects, and many others. In the example of FIG. **18**, the knowledge graph **1622** illustrates a segment of information

relating to a knowledge base of a client that includes multiple media assets (e.g., articles). In this example, the client is a company called "Content Wizard" that provides a product called "App Creator." Amongst the topics that are relevant to App Creator are a "content integration" topic and an "app hosting" topic. An article entitled "Emoji Support" is relevant to the "content integration" topic, and an article entitled "AWS Support" is relevant to the "app hosting" topic. The knowledge graph **1622** may organize the client's knowledge base to reflect these relationships. For example, the knowledge graph **1622** may include a client node **1802** that indicates the client "Content Wizard" and article nodes **1804** that correspond to the "Emoji Support" article and the "AWS Support" article. Edges **1806** between the client node **1802** and the respective article nodes **1804** may indicate that the articles are part of the client's knowledge base. Furthermore, a product node **1808** corresponding to the "App Creator" product may be connected to the client node **1802** by an edge **1810** that indicates that Content Wizard sells App Creator. Topic nodes **1812** may connect to a product node **1808** via edges **1814** that indicate a topic that pertains to a product, such that content integration and app hosting are topics that pertain to App Creator. Furthermore, the topic nodes **1812** may be related to the article nodes **1804** (or other media asset nodes) by edges **1816** that indicate an article represented by the article node **1804** is relevant to the related topic.

Furthermore, in embodiments, the knowledge graph **1622** may organize additional data relating to a client. For example, the knowledge graph **1622** may include ticket nodes **1818** and contact nodes **1820**. A ticket node **1818** may indicate a ticket that was issued on behalf of the client. A contact node **1820** may represent a contact of the client. In this example, the contact node **1820** may be related to a ticket node **1818** with an edge **1822** that indicates the ticket was initiated by the contact (e.g., ticket #**1234** was initiated by Tom Bird). The contact node **1820** may relate to product nodes **1808** with edges **1824** that indicate the contact has purchased a respective product (e.g., Tom Bird has purchased App Creator). Furthermore, the ticket node **1818** may relate to topic node **1812** by an edge **1826** that indicates that the ticket for an issue with a topic (e.g., an issue with content integration). The ticket node **1818** may further relate to the article node **1804** with an edge **1828** indicating that the article has been tried during the servicing of the ticket. It should be appreciated that in these embodiments, additional or alternative nodes may be used to represent different entities in the customer-service workflow, (e.g., status nodes, sentiment nodes, date nodes, etc.) and additional or alternative edges may be used to represent relationships between nodes (e.g., "has the current status", "most recent sentiment was", "was contacted on", "was created on", "was last updated on", etc.).

The knowledge graph **1622** is a powerful mechanism that can support many features of a client-specific service system. In a first example, in response to a service specialist taking a call from the contact for a first time regarding a particular ticket, the client-specific service system **1900** may display a ticket history to the specialist that indicates that the user has purchased App Creator, that the contact's issue is with content integration, and that the contact has been sent the Emoji Support article. In another example, an automated workflow process servicing the ticket may retrieve the ticket node to learn that the contact has already been sent the Emoji Support article, so that it may determine a next course of action. In another example, an analytics tool may analyze all the tickets issued with a particular product and the issues

relating to those tickets. The analytics tool, having knowledge of the client workflow, may drill down deeper to determine whether a particular article was helpful in resolving an issue. In another example, a chat bot may utilize the knowledge graph to guide conversations with a contact. For instance, in an interaction between Tom Bird and a chat bot, the chat bot may state: "I see that we've sent you the Emoji Support article, were you able to read it?" If Tom Bird indicates that he has read it (e.g., "Yes, I have"), the chat bot may create a relationship between the article node **1804** and the contact node **1820** indicating that Tom Bird has read the article. If Tom Bird indicates that he has not read the article, the chat bot may then send him a link to the article, which may be referenced in the knowledge graph **1622** by, for example, a "web address" entity node.

It is appreciated that the full knowledge graph **1622** may contain thousands, hundreds of thousands, or millions of nodes and edges. The example of FIG. **18** is a limited example to demonstrate the utility of the knowledge graph **1622** in a customer service setting. In embodiments, the platform **1600** can maintain separate knowledge graphs **1614** for separate clients or may have the knowledge graph **1622** that stores information relating to all clients.

The platform **1600** may add information to the knowledge graph **1622** in any suitable manner. For example, the client configuration system **1602** may employ a crawling system and an information extraction system, as was described in the disclosure with respect to FIG. **11** for example. In some embodiments, a crawling system may be seeded with one or more root URLs, from which the crawling system may begin crawling documents. In the example GUI **3700** of FIG. **37** (discussed below), a user can enter a root URL to seed the crawling system. Additionally or alternatively, the client configuration system **1602** can add information to the knowledge graph **1622** as it is provided by the client (e.g., via upload and/or API) or learned during operation (e.g., via the interactions with the contacts or clients). The client configuration system **1602** may implement any suitable ontology for structuring the knowledge graph **1622**. Furthermore, the platform **1600** may add new entity types and relationship types to the knowledge graph of the ontology as they are discovered and/or become necessary.

FIG. **19** illustrates an example of a multi-client service platform **1600** deploying at least two separate client-specific service systems **1900**. A first client-specific service system **1900-1** is deployed on behalf of a first client and a second client-specific service system **1900-2** is deployed on behalf of a second client. Each client-specific service system **1900** may be configured according to the client's selected service features and customization parameters. Thus, the first client-specific service system **1900-1** may provide a first set of service features, while the second client-specific service system **1900-2** may provide a second set of service features that may be different than the first set of service features. As discussed, the platform **1600** may be implemented according to a microservices architecture. In these embodiments, each client-specific service system **1900** may be configured to access a respective set of microservices. While some microservices will be used by all client-specific service systems **1900** (e.g., authentication microservices, database services, etc.), other microservices may be accessed by a client-specific service system **1900** only if the client has selected service features that are supported by the other microservices. In some of these embodiments, the client-specific service systems **1900** may be configured to access a set of APIs that leverage the microservices of the multi-client service platform **1600**.

In an example configuration, a client-specific service system **1900** may include and/or may leverage one or more of a communication integrator **1902**, a ticket manager **1904**, a workflow manager **1906**, chat bots **1908**, service specialist portals **1910**, a machine learning module **1912**, an analytics module **1914**, and/or a feedback module **1916**. Depending on the service features selected by a client, a client-specific service system **1900** may not include one or more of the foregoing components. In embodiments, each client-specific service system **1900** may also access one or more proprietary databases **1620**, the knowledge graph **1622**, and/or the knowledge base **1624**. In embodiments, each client-specific service system **1900** may have client specific proprietary databases **1620**, knowledge graph **1622**, and/or knowledge base **1624** that only the client-specific service system **1900** may access. Alternatively, one or more proprietary databases **1620**, the knowledge graph **1622**, and/or the knowledge base **1624** may be shared amongst client-specific service systems **1900**.

In embodiments, a client-specific service system **1900** may include one or more APIs that allow the client to integrate one or more features of the client-specific service system **1900** in the client's websites, enterprise software, and/or applications. For example, a client website may include a chat feature, whereby the chat bot **1908** interacts with a contact through a chat bot interface (e.g., a text-based chat client) via an API that services the client website.

In embodiments, a communication integrator **1902** integrates communication with a contact over different mediums (e.g., chat bots, specialists, etc.), including the migration of the contact from one medium to another medium (e.g., website to chat bot, chat bot to specialist, website to specialist, etc.). In embodiments, the communication integrator **1902** may access one or more microservices of the platform **1600**, including the microservices of the conversation system **1606**. For example, in response to a contact engaging with the client's website, the communication integrator **1902** may access a chat bot microservice of the conversation system **1606**, which then instantiates the chat bot **1908** that effectuates communication with the contact via a chat bot interface.

In embodiments, the communication integrator **1902** may be configured to determine when or be instructed (e.g., by the workflow manager **1906**) to migrate a communication with a contact to another medium, and may effectuate the transfer to the different medium. For example, after a determination that the chat bot **1908** is ineffective in communicating with the contact, the communication integrator **1902** may transfer the contact to a customer service specialist portal **1910**, where the contact can converse with a human (e.g., via a text-based chat client or by telephone). In embodiments, the communication integrator **1902** may operate in tandem with the machine learning module **1912** to determine when to migrate a contact to another communication medium. For example, if the machine learning module **1912** determines the text being typed by the contact indicates a frustration or anger on behalf of the contact, the communication integrator **1902** may instruct a chat bot to send a message stating that the case is being transferred to a specialist and may effectuate the transfer. In effectuating the transfer, the communication integrator **1902** may provide a snapshot of the contact's data and the ticket data to the specialist via, for example, the service specialist portal **1910**.

In some embodiments, the communication integrator **1902** monitors each current communication session between a contact and the client-specific service system **1900**. For example, the communication integrator **1902** may monitor

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open chat bot sessions, live chats with specialists, phone calls with specialists, and the like. The communication integrator **1902** may then determine or be instructed to migrate the communication session from a first medium to a second medium. In embodiments, the communication integrator **1902** or another suitable component may monitor the content of a communication session (e.g., using speech recognition and/or NLP) to determine that a communication session is to be transferred to a different medium. In the latter scenario, the other component may issue an instruction to the communication integrator **1902** to transfer the communication to another medium. In response, the communication integrator **1902** may retrieve or otherwise obtain information that is relevant to the current communication session, including a ticket ID, contact information (e.g., username, location, etc.), the current issue (e.g., the reason for the ticket), and/or other suitable information. This information may be obtained from the proprietary databases **1620** and/or the knowledge graph **1622** accessible to the client-specific service system **1900**. The communication integrator **1902** may then transfer the communication session to a different medium. In some embodiments, the sequence by which a communication session is transferred (e.g., escalating from a chat bot to a specialist or escalating from a text-based chat to a phone call) is defined in a custom workflow provided by the client. The communication integrator **1902** may feed the obtained data to the medium. For example, if being transferred to a specialist, the communication integrator **1902** may populate a GUI of the specialist with the ticket information (e.g., ticket ID and current issue), contact information, the ticket status, transcripts of recent conversations with the contact, and/or the like. The communication session may then commence on the new medium without the contact having to provide any additional information to the system **1900**.

In embodiments, the ticket manager **1904** manages tickets with respect to a ticket pipeline on behalf of the client. In embodiments, the ticket manager **1904** of a client-specific service system **1900** leverages the microservices of the ticket management system **1604** of the multi-client service platform **1600** to create, modify, track, and otherwise manage tickets issued on behalf of the client.

In embodiments, the ticket management system **1604** may create tickets on behalf of a respective client. The ticket management system **1604** may create a ticket in response to a number of different scenarios. For example, the ticket management system **1604** may create a ticket when a contact accesses the client's website and reports an issue or makes a customer service request. In this example, the contact may provide identifying information (e.g., name, account number, purchase number, email, phone number, or the like), a subject corresponding to the issue (e.g., a high level reason for initiating the ticket), and a description of the issue. In another example, the ticket management system **1604** may create a ticket in response to contact calling or messaging a customer service specialist with an issue, whereby the customer service specialist requests the new ticket. In this example, the customer service specialist may engage in a conversation (via a text-based chat, a video chat, or a phone call) with the contact and based on the conversation may fill out a ticket request containing identifying information (e.g., name of the contact, account number, purchase number, email of the contact, phone number of the contact, or the like), a subject corresponding to the issue (e.g., a high level reason for initiating the ticket), and a description of the issue. In another example, the ticket management system **1604** may receive a request to create a ticket from a chat bot.

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In this example, the chat bot may engage in a conversation (via a text-based chat or a phone call) with the contact in accordance with a script, whereby the script prompts the contact to provide identifying information (e.g., name of the contact, account number, purchase number, email of the contact, phone number of the contact, or the like), a subject corresponding to the issue (e.g., a high level reason for initiating the ticket), and a description of the issue. In response to the contact providing this information to the chat bot, the chat bot may issue a request to create a new ticket containing the provided information (e.g., using a ticket request template).

In response to a ticket request, the ticket manager **1904** may generate a new ticket based on the information contained in the request. In embodiments, the ticket manager **1904** may select a ticket object from a set of ticket objects that are customized by the client and may request a new ticket from the ticket management system **1604** using a microservice of the ticket management system **1604**. For example, the ticket manager **1904** may select a ticket object based on the subject corresponding to the issue. The ticket manager **1904** may provide the request for the new ticket to the ticket management system **1604**, whereby the ticket manager **1904** includes the ticket type, the identifying information, the subject, the description, and any other relevant data with the request. The ticket management system **1604** may then generate the new ticket based on the information provided with the request and may include additional attributes in the new ticket, such as a ticket status, a date/time created attribute, a last updated attribute, and the like. In creating the new ticket and setting the status of the ticket to a new ticket, the ticket management module **1604** may add the new ticket to a ticket pipeline of the client.

In embodiments, the ticket manager **1904** may manage one or more ticket pipelines of the client. As discussed, the ticket management system **1604** may run a set of pipeline listening threads that listen for changes to specific attributes of a ticket, whereby when a ticket pipeline listening thread identifies a ticket having attribute values that it is listening for, the ticket pipeline listening thread adds the ticket to a queue corresponding to the ticket pipeline listening thread. Once in the queue, the ticket is moved to a new stage of the pipeline, and the status attribute of the ticket may be updated to reflect the new stage. For example, a ticket pipeline of an ISP client may include the following stages: new ticket; in communication with contact; troubleshooting issue; obtaining feedback; and ticket closed. Once the ticket is created, it is moved into the new ticket stage of the pipeline. The new ticket stage of the pipeline may include one or more workflows that may be performed for tickets in the new ticket stage (e.g., send notification email to contact, assign to a customer service specialist, instruct customer service specialist to contact the contact, etc.). As the workflows are executed, the ticket attributes of a ticket may change, such that a pipeline listening thread may determine that a customer service specialist has reached out to the contact. In this example, the pipeline listening thread may move the ticket into a queue corresponding to the "in communication with contact" stage of the ticket pipeline.

In embodiments, the workflow manager **1906** performs tasks relating to the execution of workflows. As discussed, a workflow defines a set of actions to be undertaken when performing a service-related task in response to one or more conditions being met. In some scenarios, a workflow may be defined with respect to a pipeline stage. In these scenarios, a workflow may be triggered with respect to a ticket only when the ticket is in the respective stage. Furthermore, a

workflow includes a set of conditions that trigger a workflow (whether the workflow is defined with respect to a ticket pipeline or independent of a ticket pipeline). In embodiments, the determination as to whether a workflow is triggered is based on the attributes of a ticket. As discussed, the ticket management system 1604 may deploy workflow listening threads that listen for tickets that meet the conditions of a particular workflow. Upon determining that a ticket meets the conditions of a workflow (or put another way, a ticket triggers a workflow), the workflow listening thread adds the ticket to a workflow queue corresponding to the workflow listening thread.

In embodiments, the workflow manager 1906 may execute workflows and/or facilitate the execution of workflows of a client. In some embodiments, the workflow manager 1906 may be implemented in a multi-threaded manner where the different threads serve respective workflows. For each workflow, the workflow manager 1906 (e.g., a workflow manager thread) may dequeue a ticket from the workflow queue of the workflow. The workflow manager 1906 may then perform actions defined in the workflow and/or may request the execution of actions defined in the workflow from another component (e.g., a microservice). For example, a client may configure the ticket pipeline to include a workflow where a notification email is sent to the contact initiating the contact. Upon a new ticket being generated the workflow manager 1906 may retrieve an email template corresponding to new ticket notifications, may generate the notification email based on the email template and data from the ticket and/or a contact record of the intended recipient, and may send the email to the contact. In another example, a workflow may define a series of actions to be performed after a ticket is closed, including sending a survey and following up with the recipient of the survey who does not respond within a period of time. In this example, the workflow manager 1906 may instruct the feedback module 1916 to send a survey to the contact indicated in the ticket. If the feedback is not received within the prescribed time, the workflow manager 1906 may instruct the feedback module 1916 to resend the survey in a follow up email. The workflow manager 1906 may perform additional or alternative functions in addition to the functions described in the disclosure without departing from the scope of the disclosure.

In embodiments, the chat bot 1908 may be configured to engage in conversation with a human. The chat bot 1908 may utilize scripts, natural language processing, rules-based logic, natural language generation, and/or generative models to engage in a conversation. In embodiments, an instance of a chat bot 1908 may be instantiated to facilitate a conversation with a contact. Upon a contact being directed to the chat bot 1908 (e.g., by the communication integrator 1902), the system 1900 may instantiate the new chat bot 1908. The system 1900 may initialize the chat bot 1908 with data relating to the contact. In embodiments, the system 1900 (e.g., the communication integrator 1902) may initialize the chat bot 1908 with a script that is directed to handle a particular type of conversation, a contact ID of a contact, and/or a ticket ID referencing a ticket initiated by the contact. For example, when a contact visits the client's webpage, an instance of a chat bot may be instantiated (e.g., by the communication integrator 1902), whereby the instance of the chat bot 1908 uses a script written for interacting with contacts coming to the client's page with service-related issues and a contact ID of the contact if available. In embodiments, the chat bot 1908 may obtain information from the proprietary database 1620 and/or the

knowledge graph 1622 to engage in the conversation. Initially, the chat bot 1908 may use a script to begin a conversation and may populate fields in the script using information obtained from the proprietary database 1620, the knowledge graph 1622, a ticket ID, a contact ID, previous text in the chat, and the like. The chat bot 1908 may receive communication from the contact (e.g., via text or audio) and may process the communication. For example, the chat bot 1908 may perform natural language processing to understand the response of the user. In embodiments, the chat bot 1908 may utilize a rules-based approach and/or a machine learning approach to determine the appropriate response. For example, if the chat bot 1908, based on the contact's ticket history asks the contact if he is having an issue with content integration and the contact responds by typing "Yes, I can't get emoji to show up in my app," the chat bot 1908 may rely on a rule that states: if no content has been sent to the contact, then send relevant content. In this example, the chat bot 1908 may retrieve an article describing how to integrate emoji into an application and may send a link to the article to the contact (e.g., via a messaging interface or via email). In another example, the chat bot 1908 may provide a ticket timeline to the machine learning module 1912, which in turn may leverage a neural network to determine that the best action at a given point is to send a particular article to the contact. In this example, the chat bot 1908 may retrieve the article recommended by the machine learning module 1912 and may send a link to the article to the contact. In embodiments, the chat bot 1908 may utilize data from the knowledge graph 1622 to provide content and/or to generate a response. Continuing the previous example above, the chat bot 1908, in response to determining that the next communication is to include a link to an article, may retrieve (or may request that another component retrieve) a relevant article or video based on the knowledge graph 1622. Having the topic/type of issue, the chat bot 1908 can identify articles or content that are related to the product to which the ticket corresponds that are relevant to the topic/type of issue. The chat bot 1908 can then provide the content to the contact (e.g., email a link or provide the link in a chat interface). In some embodiments, the chat bot 1908 can also use the knowledge graph 1622 to formulate responses to the contact. For example, if the user asks about a particular product, the chat bot 1908 can retrieve relevant information relating to the product from the knowledge graph 1622 (e.g., articles or FAQs relating to the product). The chat bot 1908 may be configured to understand the ontology of the knowledge graph 1622, whereby the chat bot 1908 can query the knowledge graph to retrieve relevant data. For example, in response to a question about a particular product, the chat bot 1908 can retrieve data relating to the product using the product ID of the product, and may use its knowledge of the different types of relationships to find the answer to the contacts questions.

In embodiments, the chat bots 1908 may be configured to escalate the ticket to a specialist (e.g., via the communication integrator 1902) when the chat bot 1908 determines that it is unable to answer a contacts question (e.g., the results of NLP are inconclusive) or when the chat bot 1908 and/or based on tone and/or sentiment analysis (e.g., the chat bot determines that the contact is becoming upset, angry, or frustrated). In embodiments, tone or sentiment analysis can be performed as a part of the natural language processing that is performed on the contact's communications, such that a tone or sentiment score is included in the output of the natural language processing. In these embodiments, the chat bot 1908 may help conserve resources of a client, by serving

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as a triage of sorts when handling a ticket. When the ticket is unable to be resolved by the chat bot **1908**, a workflow may require that the next step is to migrate the conversation to a service-specialist. In such a situation, the communication integrator **1902** may migrate the contact in accordance with the workflow manager's determination. For example, the communication integrator **1902** may transfer the contact to a service specialist, whereby the service specialist communicates with the contact via a live chat and may view relevant contact information and/or ticket information via the service specialist portal **1910**.

The service specialist portal **1910** may include various graphical user interfaces that assist a service specialist when interacting with a contact or otherwise servicing a ticket of a contact. In embodiments, the service specialist portal may include chat interfaces, visualization tools that display a specialist's open tickets and/or various communication threads, analytics tools, and the like. Upon a contact and/or ticket being routed to a service specialist, the communication integrator **1902** may provide the specialist with all relevant data pertaining to the contact and/or the ticket. The communication integrator **1902** may retrieve this information from the proprietary database(s) **1620** and/or the knowledge graph **1622**. In an example, the communication integrator **1902** may display the ticket timeline of the ticket (e.g., when events along the ticket manager **1904** were undertaken) in the service specialist portal **1910**, the purchase history of the contact, any communications with the contact, and/or any content sent to the contact into a graphical user interface that displays relevant information to the specialist.

FIGS. **21-23** illustrates an example service specialist portal GUIs that may be presented by the service specialist portal according to one or more embodiments of the disclosure. In the example of FIG. **21**, the GUI **2100** displays relevant ticket information to a specialist (or other user), as well as information relating to the contact that initiated the ticket. The GUI **2100** includes a graphical representation of the ticket's timeline (e.g., email sent), and detailed notes about different contact points with the contact. The GUI **2100** may show the name and information of a contact, a date on which the ticket was issued, what articles were opened by a contact with respect to the ticket, communications that were undertaken with the contact, and the like.

FIG. **22** illustrates a GUI **2200** that may be used to impart relevant data to a service specialist. In the illustrated state, the GUI **2200** is a portal that provides a list of a specialist's assigned tickets. In the example GUI **2200**, the specialist can view recent messages sent or received by the specialist, and may drill down into a particular conversation. Upon drilling down into a conversation, the GUI **2200** displays relevant information of the contact in relation to a text based communication session with the contact. On the right column of the GUI **2200**, relevant ticket and contact data are displayed to the specialist, including a name of the contact, a phone number and email address of the contact, a date on which the contact became a contact, a lifecycle stage of the contact, and links to any tickets that the contact may have open.

FIG. **23** illustrates a GUI **2300** that may be presented to a specialist or another service-related employee (e.g., a supervisor). The GUI **2300** of FIG. **23** is a ticket overview GUI **2300**. The GUI **2300** displays a set of open tickets and where the tickets are with respect to the client's ticket pipeline. In this example arrangement, the specialist or supervisor can view tickets that are new, tickets that are awaiting communication from the contact, tickets that have progressed to the email stage, tickets that have been

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resolved, and tickets that have been closed. Each ticket assigned to the specialist may be displayed in a respective card, whereby the card provides a synopsis of the ticket (e.g., date created, contact name, and general issue). In this view, a specialist can click on a ticket card to drill down to view the details of a particular ticket. In response to a user selection of a particular card, the communication integrator **1902** may retrieve a ticket record corresponding to the ticket represented by the selected card and may output information relating to the ticket in a GUI (e.g., a GUI **2100** of FIG. **21**)

In embodiments, the machine learning module **1912** may operate to perform various machine learning tasks related to the multi-client service system **1900**. In some embodiments, the machine learning module **1912** may be configured to leverage the microservices of the machine learning system **1608** of the platform, whereby the machine learning system **1608** may provide various machine learning related services, including training models for particular clients based on training data or feedback data associated with the client. In this way, the machine learning module **1912** may be said to train and/or leverage machine learned models (e.g., neural networks, deep neural networks, convolutional neural networks, regression-based models, Hidden Markov models, decision trees, and/or other suitable model types) to perform various tasks for the client-specific service system **1900**.

In embodiments, the machine learning module **1912** may train and deploy models (e.g., sentiment models) that are trained to gauge the sentiment and/or tone of the contact during interactions with the system **1900**. The models may receive features relating to text and/or audio and may determine a likely sentiment or tone of the contact based on those features. For example, a first contact may send a message stating "Hey guys, I really love my new product, but this is broken;" and a second contact may send a message stating "Hey, I hate this product." Based on features such as keywords (e.g., "love," "broken," and "hate"), message structure, and/or patterns of text, a model may classify the first message as being from a likely pleased contact and in a polite tone, while it may classify the second message as being from a likely angry customer and in a direct tone. This information may be stored as an attribute in a ticket record and/or provided to a chat bot or a service specialist. Tone and sentiment scores may also be fed to the analytics system **1614** and/or feedback system **1610**. For example, the analytics module **1914** may utilize tone and sentiment when determining contact scores, which may indicate an overall value of the contact to the client.

In embodiments, the machine learning module **1912** can train a sentiment model using training data that is derived from transcripts of conversations. The transcripts may be labeled (e.g., by a human) to indicate the sentiment of the contact during the conversation. For example, each transcript may include a label that indicates whether a contact was satisfied, upset, happy, confused, or the like. The label may be provided by an expert or provided by the contact (e.g., using a survey). In embodiments, the machine learning module **1912** may parse a transcript to extract a set of features from each transcript. The features may be structured in a feature vector, which is combined with the label to form a training data pair. The machine learning module **1912** may train and reinforce a sentiment model based on the training data pairs. As the client-specific service system **1900** records new transcripts, the machine learning module **1912** may reinforce the sentiment model based on the new transcripts and respective labels that have been assigned thereto.

The machine learning module **1912** can train and/or deploy additional or alternative models as well. In embodi-

ments, the machine learning module **1912** can train models used in natural language processing. In these embodiments, the models may be trained on conversation data of previously recorded/transcribed conversations with customers.

In embodiments, the analytics module **1914** may analyze one or more aspects of the data collected by the system **1900**. In embodiments, the analytics module **1914** calculates a contact score for a contact that is indicative of a value of the contact to the client. The contact score may be based on a number of different variables. For example, the contact score may be based on a number of tickets that the user has initiated, an average amount of time between tickets, the sentiment of contact when interacting with the system **1900**, an amount of revenue resulting from the relationship with the contact (or the entity with which the contact is affiliated), a number of purchases made by the contact (or an affiliated entity), the most recent purchase made by the contact, the date of the most recent purchase, a net promoter score (e.g., feedback given by the contact indicating how likely he or she is to recommend the client's product or products to someone else) and the like. In embodiments, the contact score may be based on feedback received by the feedback module **1916**. The contact score may be stored in the contacts data record, the knowledge graph **1622**, and/or provided to another component of the system (e.g., a chat bot or the service specialist portal **1910**).

In embodiments, the analytics module **1914** may generate a contact score of a contact using a contact scoring model. The contact scoring model may be any suitable scoring model (e.g., a regression-based model or a neural network). In embodiments, the analytics module **1914** may generate a feature vector (or any other suitable data structure) corresponding to the contact and may input the feature vector to the scoring model. The analytics module **1914** may obtain contact-related data from the contact record of the contact, the knowledge graph, or other suitable sources. The types of contact-related data may include, but are not limited to, a total amount of revenue derived from the contact, a number of purchases made by the contact, an amount of loyalty points (e.g., frequent flyer miles) held by the contact, a status (e.g., "gold status" or "platinum status") of the contact, an amount of time since the contact's most recent purchase, a number of tickets that the contact has initiated, an average amount of time between tickets from the contact, and/or the average sentiment of contact when interacting with the system (e.g., a normalized value between 0 and 10 where 0 is the worst sentiment, such as angry or rude). In embodiments, the analytics module **1914** may normalize or otherwise process one or more of the contact-related data items. For example, the analytics module **1914** may determine the average sentiment of the contact and may normalize the sentiment on a scale between 0 and 10. The analytics module **1914** may then feed the feature vector to the contact scoring model, which determines and outputs the contact score of the contact based on the feature vector. In embodiments, the contact scoring model is a machine learned model that is trained by the machine learning module **1912**. The contact scoring model may be trained in a supervised, unsupervised, or semi-supervised manner. For example, the contact scoring model may be given training data pairs, where each pair includes a feature vector corresponding to a contact and a contact score of the contact. In embodiments, the contact score in a training data pair may be assigned by an expert affiliated with the client and/or the multi-client service platform **1600**.

In embodiments, the analytics module **1914** may also collect and analyze data regarding the efficacy of certain

actions. For example, the analytics module **1914** may gauge the effectiveness of certain articles or videos, scripts used by chat bots, models used by chat bots, calls handled by customer service specialists, and the like. In embodiments, the analytics module **1914** may rely on a ticket's timeline and/or feedback received from the contact (e.g., surveys or the like), and/or feedback inferred (e.g., sentiment or tone) from the contact to determine the effectiveness of certain actions in the workflow of a client. For example, the analytics module **1914** may determine that certain workflow actions almost always (e.g., >90%) result in a contact escalating the ticket to another communication medium when dealing with a particular type of problem. In a more specific example, a client that is an ISP may first provide a contact with an article describing how to troubleshoot a problem, regardless of the problem. The analytics module **1914** may determine that when the ticket relates to a detected but weaker signal, contacts almost always escalate the ticket to a specialist. The analytics module **1914** may also determine that when the ticket relates to no signal being detected, the troubleshooting article typically resolves the ticket.

In embodiments, the analytics module **1914** can be configured to output various analytics related statistics and information to a user associated with the client. For example, the analytics module **1914** can present a GUI that indicates statistics relating to feedback received from contacts. For example, FIG. **36** illustrates an example of a GUI **3600** that displays a breakdown of the net promoter scores of the contacts of a particular client. FIG. **39** illustrates an example of a GUI **3900** that displays statistics relating to articles in a knowledge graph, which may indicate the respective usefulness of the individual articles.

In embodiments, the feedback module **1916** is configured to obtain or otherwise determine feedback from contacts. Feedback may be related to a purchase of a product (e.g., a good or service) and/or the customer. In embodiments, feedback may be obtained directly from a contact using, for example, surveys, questionnaires, and/or chat bots. The feedback collected by the feedback module **1916** may be stored in a contact record of the contact providing feedback, provided to the analytics module **1914**, used as training data for reinforcing the machine learned models utilized by the client-specific service system **1900**, and the like.

In embodiments, the feedback module **1916** may be configured to execute feedback related workflows, such that certain triggers cause the feedback module **1916** to request feedback from a contact. Examples of triggers may include, but are not limited to, purchases, repurchases, client visits to the contact, service technician visits, product delivery, the ticket initiation, ticket closure, and the like. In another example, a lack of feedback could be a trigger to request feedback. Furthermore, different triggers may trigger different feedback workflows (e.g., a first survey is sent to a contact when an issue is resolved over the phone and a second survey is sent to a contact after a technician visits the contact). A feedback workflow may define when to send a feedback request to a contact, what medium to use to request the feedback, and/or the questions to ask to the contact. Customer attributes of the contacts can also be used to determine a feedback workflow for a customer. Examples of customer attributes may include, but are not limited to, date the contact became a customer, lifecycle state, last purchase date, recent purchase date, on-boarding date, last login, last event, last date a feature was used, demos, industry vertical, role, demographic, behavioral attributes, a net promoter score of the contact, and a lifetime value.

In some embodiments, the feedback module **1916** may be trained (e.g., by the machine learning module **1912**) to determine the appropriate time to transmit a request for feedback. In embodiments, the feedback module **1916** is trained to determine the appropriate communication channel to request feedback (e.g., email, text message, push notification to a native application, phone call, and the like). In embodiments, the feedback module **1916** is trained to determine the appropriate questions to ask in a feedback request.

In embodiments, the feedback module **1916** is configured to extract feedback from customer communications. For example, the feedback module **1916** may analyze interactions with contacts to determine a contact's implicit and/or explicit feedback (e.g., whether the contact was satisfied, unsatisfied, or neutral). In an example, the feedback module **1916** system may analyze text containing the phrase "this product is horrible." In this example, the feedback module **1916** may determine that the contact's feedback towards the product is bad.

In embodiments, the feedback module **1916** may be configured to display feedback to a user affiliated with the client. The feedback module **1916** may present feedback of contacts individually. For example, the feedback module **1916** may display a GUI that allows a user to view the various contact providing feedback and a synopsis of the contact (e.g., a contact score of the contact, a name of the contact, and the like). The user can click on a particular contact to drill down on their feedback, or a contact profile page. In the example of FIG. **24**, an example GUI **2400** allows a user to drill down on the feedback of individual contacts. In the example, the user can click on a particular contact and the GUI **2400** may display the feedback provided by the contact. FIG. **25** is a screen shot of the GUI **2500**, whereby the contact's feedback is arranged on a timeline. FIG. **26** is a screenshot of another feedback related GUI **2600**. In the example of FIG. **26**, the feedback data of a contact is displayed in a timeline. In this example, the GUI **2600** displays individual cards that are related to various feedback events from the contact. The cards may also display at least a portion of the feedback (e.g., scores, text, and the like). The GUI **2600** further allows the user to view the tickets of the contact, the lifecycle history of the contact, a contact history of the contact, a last time the contact contacted the client, and the like.

Referring now to FIG. **20**, an example set of operations for deploying a client-specific service system, according to one or more embodiments of the disclosure. The method **2000** may be executed by one or more processors of the multi-client service platform **1600** of FIG. **16**, and is described with respect thereto. The method **2000** may be performed by other suitable systems as well without departing from the scope of the disclosure.

At **2010**, the multi-client service platform **1600** receives a request to create a new client-specific service system **1900**. The request may be initiated via a graphical user interface presented to a user affiliated with the client or by a salesperson affiliated with the multi-client service system. The request may include one or more service features which the client would like to incorporate into its client-specific service system. For example, the client may opt from one or more of a ticket support, ticket workflow management, multiple ticket workflows, email/chat and ticket integration, customized email templates, knowledge graph support, conversation routing, customer service website that includes recommended content (e.g., articles or videos on solving common problems), a chat bot (text-based and/or audio-based), automated routing to service specialists, live chat,

customer service analytics, customized reporting, and the like. A user affiliated with a client may select the service features to be included in the client-specific service system from, for example, a menu or may subscribe to one or more bundled packages that include respective sets of service features.

At **2012**, the system may receive one or more customization parameters. In embodiments, a client user may provide one or more customization parameters via one or more GUIs. The types of customization parameters that a client may provide depends on the services that the client has enlisted. The types of customization parameters may include custom ticket attributes, client branding (e.g., logos or photographs), root URLs to generate a knowledge graph on behalf of the client, topic headings for organizing a client's customer service page, media assets to be included under each respective topic heading, ticket pipeline definitions, workflow definitions, communication templates for automated generation communications, scripts to initiate conversations with a contact using a chat bot, telephone numbers of the client's service specialist system, survey questions and other feedback mechanisms, different types of analytics that may be run, and the like.

In embodiments, a client may customize tickets used in its client-specific service system. In these embodiments, the client may define one or more new ticket objects, where each ticket object may correspond to a different type of ticket. For example, a first ticket object may correspond to tickets used in connection with refund requests and a second ticket object may correspond to tickets that are used in connection with service requests. Thus, if defining more than one ticket object, a client may assign a ticket type to a new ticket object. In embodiments, a ticket object includes ticket attributes. The ticket attributes may include default ticket attributes and custom ticket attributes. The default ticket attributes may be a set of ticket attributes that must remain in the ticket. Examples of default ticket attributes, according to some implementations of the platform **1600**, may include (but are not limited to) one or more of a ticket ID or ticket name attribute (e.g., a unique identifier of the ticket), a ticket priority attribute (e.g., high, low, or medium) that indicates a priority of the ticket, a ticket subject attribute (e.g., what is the ticket concerning), a ticket description (e.g., a plain-text description of the issue to which the ticket pertains) attribute, a pipeline ID attribute that indicates a ticket pipeline to which the ticket is assigned, a pipeline stage attribute that indicates a status of the ticket with respect to the ticket pipeline in which it is being processed, a creation date attribute indicating when the ticket was created, a last update attribute indicating a date and/or time when the ticket was last updated (e.g., the last time an action occurred with respect to the ticket), a ticket owner attribute that indicates the contact that initiated the ticket, and the like. Custom ticket attributes are attributes that a user may define, for example, using a GUI. Examples of custom ticket attributes are far ranging, as the client may define the custom ticket attributes, and may include a ticket type attributing indicating a type of the ticket (e.g., service request, refund request, lost items, etc.), a contact sentiment attribute indicating whether a sentiment score of a contact (e.g., whether the contact is happy, neutral, frustrated, angry, and the like), a contact frequency attribute indicating a number of times a contact has been contacted, a media asset attribute indicating media assets (e.g., articles or videos) that have been sent to the contact during the ticket's lifetime, and the like.

In some scenarios, a client may elect to customize one or more ticket pipelines for handling tickets by the client-

specific service system **1900**, whereby a user affiliated with the client may define one or more ticket pipelines. In some embodiments, the multi-client service platform **1600** may present a GUI that allows the user to define various workflow stages (e.g., “ticket created”, “waiting for contact”, “routed to chat bot”, “routed to service specialist”, “ticket closed”, and the like). For each stage, the user may define one or more conditions (e.g., ticket attribute values) that correspond to the respective stage. In this way, a ticket meeting the one or more conditions of a respective stage may be moved to that stage. For each stage of the pipeline, the user may define one or more workflows or actions that are performed.

In some scenarios, a client may elect to define one or more workflows. The workflows may be defined with respect to a stage of a pipeline or independent of a pipeline. A workflow may include one or more actions. Thus, a user affiliated with a client may select one or more actions of a workflow. For example, the user may select actions such as “create ticket”, “send message”, “send email”, “route to chat bot”, “route to specialist”, “define custom action”, and the like. In the instance where a user elects to define a custom action, the user may provide further details on how the client-specific service is to respond. For example, the user may select that an article is to be sent to a contact upon a specific type of problem indicated in a newly created ticket. The user may further define the conditions that trigger the workflow. In embodiments, the user may define these conditions using ticket attribute values that trigger the workflow.

In some scenarios, a client may elect to have the client-specific service system **1900** generate automated messages on behalf of the client to contacts in connection with an issued ticket. In these scenarios, a user affiliated with the client may define communication templates that are used to generate automated messages (e.g., SMS messages, emails, direct messages, and the like) to contact. For example, the client may elect to have automated messages be sent to contacts at various points during the workflow. The multi-client service platform **1600** may present a graphical user interface that allows a user to upload or enter the message template. In response to receiving the communication template, the multi-client service platform **1600** may store the message template and may associate the template with the workflow item that uses the template.

In some scenarios, the user may provide a root URL to initiate the generation of a knowledge base. In response to the root URL (or multiple root URLs), the system may crawl a set of documents (e.g., webpages) starting with the root URL. In embodiments, the system may analyze (e.g., via NLP and/or document classifiers) each document and may populate a knowledge graph based on the analysis.

In some instances, the user may define one or more topic headings, and for each topic heading may upload or provide links to a set of media assets (e.g., articles and/or videos) that relate to the heading. In some embodiments, the media assets may be used to recommend additional media assets from a series of crawled websites and/or a knowledge graph. For example, the system may identify media assets having a high degree of similarity (e.g., cosine similarity) to the media assets provided by the user. In these embodiments, the system may output the recommended media contents to the user, such that the user may select one or more of the media contents for inclusion with respect to a topic heading.

In some scenarios, a client may elect to have a client-specific service system provide a chat bot to handle some communications with contacts. For example, a client may elect to have a chat bot handle initial communications with

service-seeking contacts. In some embodiments, the multi-client service platform **1600** may present a GUI to a user affiliated with the client that allows the user to upload chat bot scripts that guide the beginning of a conversation. The user may upload additional or alternative data that assists the chat bot, such as transcripts of conversations with human service specialists, such that the chat bot may be trained on conversations that have been deemed effective (e.g., helped resolve an issue). In response to receiving the script and/or any other data, the multi-client service platform **1600** may train the chat bot based on the script and/or any other data.

In some scenarios, a client may elect to have a client-specific service system **1900** request feedback from contacts during one or more stages of a workflow. In some embodiments, the multi-client service platform **1600** may present a GUI to a user affiliated with the client that allows the user to design a survey or questionnaire, including any questions and choices that may be presented to the responder. The GUI may also allow the user to customize other aspects of the survey or questionnaire. For example, the user may provide branding elements that are presented in or in relation to the survey or questionnaire. In embodiments, the user may feedback workflows that define when a survey or questionnaire is to be sent to a user and/or the communication medium used to send the survey or questionnaire to a user.

In some scenarios, a client may elect to have a client-specific service system **1900** route contacts to a telephone call with a service specialist. In some of these embodiments, a user affiliated with the client may provide routing data that can route a contact to a service specialist. For example, the user may provide a phone number associated with a call center or a roster of service specialists and their direct phone numbers and/or extensions.

At **2014**, the multi-client service platform **1600** may configure a client-specific service system data structure based upon the selected service features and the one or more customization parameters. In embodiments, the platform **1600** implements a microservices architecture, whereby each client-specific service system may be configured as a collection of connected services. In embodiments, the platform **1600** may configure a client-specific service system data structure that defines the microservices that are leveraged by an instance of the client-specific service system data structure (which is a client-specific service system). A client-specific service system data structure may be a data structure and/or computer readable instructions that define the manner by which certain microservices are accessed and the data that is used in support of the client-specific service system. For example, the client-specific service system data structure may define the microservices that support the selected service features and may include the mechanisms by which those microservices are accessed (e.g., API calls that are made to the respective microservices and the customization parameters used to parameterize the API calls). The platform **1600** may further define one or more database objects (e.g., contact records, ticket records, and the like) from which database records (e.g., MySQL™ database records) are instantiated. For example, the client configuration system **1602** may configure ticket objects for each type of ticket, where each ticket object defines the ticket attributes included in tickets having the type. In embodiments, the platform **1600** may include any software libraries and modules needed to support the service features defined by the client in the client-specific service system data structure. The client-specific service system data structure may further include references to the proprietary database(s) **1620**, the knowledge graph **1622**, and/or knowledge base **1624**, such

that a deployed client-specific service system may have access to the proprietary database 1620, the knowledge graph 1622, and/or the knowledge base 1624.

At 2016, the multi-client service platform 1600 may deploy the client-specific service system 1900. In embodiments, the multi-client service platform 1600 may deploy an instance (or multiple instances) of the platform 1600 based on the client-specific service system data structure. In some embodiments, the platform 1600 may instantiate an instance of the client-specific service system from the client-specific service system data structure, whereby the client-specific service system 1900 may begin accessing the microservices defined in the client-specific service system data structure. In some of these embodiments, the instance of the client-specific service system is a container (e.g., a Docker® container) and the client-specific service system data structure is a container image. For example, in embodiments where the client-specific service system data structure is a container, the multi-client service platform 1600 may install and build the instance of the client-specific service system 1900 on one or more servers. In these embodiments, the container is configured to access the microservices, which may be containerized themselves. Once deployed, the client-specific service system 1900 may begin creating tickets and performing other customer-service related tasks, as described in the disclosure.

Custom Object Systems and Methods

A multi-service business platform (e.g., may also be referred to as a framework) may be configured to provide processes related to marketing, sales, and/or customer service for users. The multi-service business platform may include a database structure that may have preset or fixed core objects (e.g., platform may support core objects). For example, the core objects may include contact objects, company objects, deal objects, and ticket objects. These core objects may be described and defined above in the disclosure and may be further described and defined in the disclosure below with respect to the multi-service business platform example.

Contact objects may be defined as people who may communicate with an organization (e.g., anyone who may interact with business) such as customers or prospective customers of the business (e.g., people who may convert on a form, people who contact chat team of business, and/or people who met business team at an event). Each contact object may be defined with properties (e.g., a name of the contact, a phone number of the contact, an email address of the contact, a physical address of the contact, a title of the contact, and the like). Contacts may work at companies such that company objects may also be important to represent in data. Company objects may be defined as organizations or businesses that may communicate with a user's organization (e.g., organization of user of the multi-service business platform). Each company object may include properties such as a company name, an address of the company (e.g., main location, headquarters, or the like), and other suitable properties.

A deal object may be defined as opportunities that may be available from interactions with contacts (e.g., contact objects) and/or companies (e.g., company objects). Deal objects may be defined as and represent transactions that may be typically between two businesses. Each deal object may include properties such as a sale made by a customer to a company via a contact. Some examples of deal objects may include the amount of a deal (e.g., deal_amount), an estimated close date for a deal (e.g., estimated_close_date), and a likelihood to close a deal (e.g., likelihood_to_close).

Likelihood to close may be determined from machine learning. For example, machine learning may be used to take previously closed deals and may create a model around what types of properties (e.g., attributes) and objects may create a highly likely result to close and then may output values based on this predictive machine learning.

Ticket objects may be defined as customer requests for support or help (e.g., service ticket that may relate to a service request that may be issued by a company to a user via a contact). Some examples of properties for ticket objects may include date ticket was opened (e.g., date_opened), priority of ticket, last date custom replied to ticket (e.g., last_date_customer_replied), last date rep replied to ticket (e.g., last_date_rep_replied), and the like.

The multi-service business platform may include associations between core objects. In some examples, the associations may be a set of core associations. For example, each association may be a directed association, such that a respective association may define a type of relationship from a first object to a second object. For example, an association between a contact object and a company object may be "works for" such that the contact object "works for" (association) the company object. When an instance of the contact object (e.g., the contact object instance identifies Bob as a contact) may be associated with an instance of the company object (e.g., the company object instance may identify "Acme Corp." as a the company) with a "works for" association, then the individual indicated by the company object instance may be defined in the customer databases as working for the company indicated in the company object instance (e.g., Bob works for Acme Corp.).

In example embodiments, two objects may be associated using one or more different types of associations and the associations may be directed in both directions (e.g., association and inverse association). An inverse association of the association may be created automatically for every association. For example, the association in one direction may be "works for" and the inverse association may be "employs" which may be created for the same association automatically. This same association may be viewed from the contact object and viewed from the company object such that the association may be defined as the contact object "works for" (association) the company object or the company object "employs" (inverse association) the contact object. The associations may be between the same types of objects and/or between different types of objects. For example, in continuing the example of the company object and the contact object, a contact object instance may indicate that the company defined by the company object instance may employ the individual defined by the contact object instance vis-à-vis the "employs" association (e.g., Acme Corp employs Bob) and that the individual works for the company vis-à-vis the "works for" association (e.g., Bob works for Acme Corp.).

In some example embodiments, two objects may have multiple associations in the same direction. Continuing the example of the contact object and the company object, a contact object may be associated with a company object by a "works for" association, a "previously worked for" association, a "sells to" association, and/or other suitable types of associations. Similarly, the company object may be associated with a contact object with an "employs" association, a "previously employed" association, a "buys from" association, and/or other suitable types of associations. In this way, different types of relationships between instances of objects may be defined within the customer's databases.

In example embodiments, objects may also have the same object type directed associations. For example, a contact object may be associated to itself with one or more directed associations, such as a “is supervised by” association, a “supervises” association, or the like. For example, if Bob and Alice may work for the same company and Alice may supervise Bob, then an instance of a contact object that may define Bob may be associated with a contact object instance that may identify Alice with a “is supervised by” association (e.g., Bob “is supervised by” Alice) and/or Alice’s contact object instance may be associated with Bob’s contact object instance with a “supervises” association (e.g., Alice “supervises” Bob).

In example embodiments, the multi-service business platform may include a system for creating custom objects providing customizability and may execute methods in support thereof. For example, the multi-service business platform may include a customization system that may be used by users to create custom objects. These custom objects may be created to be specific to each user’s (e.g., client’s) business and the custom objects may be used on the multi-service business platform. The ability to create custom objects within the multi-service business platform may speed up development of new types of custom objects for the platform. In some examples, the customization system may be a separate system from the multi-service business platform and may communicate with the multi-service business platform (e.g., via external application programming interfaces (APIs)).

Custom objects (e.g., may also be referred to as custom object definitions) may be defined as purposely non-prescriptive objects (e.g., flexible/customizable in contrast from fixed core objects). A user may create custom objects relevant to their business and the user’s business needs (e.g., relevant to a user’s business model). The custom objects may provide an alternative where objects of interest to businesses may not fit smoothly within core objects (e.g., not necessarily fit as contacts, companies, deals, and/or tickets). The user may create custom objects that may be particularly useful to one or more services (e.g., workflows, reporting) of the multi-service business platform as described in the disclosure. Each custom object or custom object definition may include an object type, properties (e.g., some properties may be set on an instance), and possible associations. In example embodiments, custom objects and/or types of custom objects may include products, goods such as devices/machines (e.g., cars, drones, boats, mobile phones, etc. such that these devices/machines custom objects may be used to track details about ownership, service, cost of devices/machines), business services, shipments (e.g., may be used to store data about fulfillment of orders that may be intended to be sent out), applications (e.g., may be used to store data that tracks progress of an application), projects (e.g., may be used to store data about work or deliverables), locations/stores (e.g., may be used to store granular data about companies and their many physical locations such as store locations and/or company headquarters location), customer locations (e.g., may be locations of customers that buy products and/or services from user’s business), events (e.g., may be used to store and track physical or online events a company holds), listings (e.g., may be used to store data about real estate listings for a real estate company), referrals (e.g., may be used to link two things together to notate a referral or referrer), and the like. Some businesses may have unique relationships from operating in an agency type model that the businesses may want to identify, monitor, and/or track using custom objects.

Custom objects may provide users with the ability to model their business. For example, custom objects may allow users to model their own version of contacts, companies, deals, and/or tickets or any other type of object for their businesses that may allow the users to customize what they want for objects and/or object types. The custom objects may be used with the multi-service business platform such that the upstart of the multi-service business platform may provide various functionality for usage of these custom objects. When a user may build a custom object, the user may utilize all services (e.g., features) of the multi-service business platform such that the user (e.g., user’s business) may use these services throughout the multi-service business platform towards relevant custom objects that may match user’s business needs.

In some examples, users may create custom objects with respect to usefulness with services of the multi-service business platform. For example, an auto manufacturer user may create car or vehicle custom objects that may fit into the auto manufacturer’s business workflows (e.g., workflow automation) that may be used on the multi-service business platform. In another example, a user may choose to add custom objects that may be particularly useful with reporting service for user’s business needs.

In general, this ability to create custom objects provides increased and improved customizability across the multi-service business platform. This provides several advantages to the multi-service business platform as described in the disclosure. For example, some advantages may include customization for users with respect to their business industry or field, specific customization towards each user’s business itself such that one user in a business industry (e.g., car industry) may have different custom object needs with respect to another user in the same business industry, increased speed of development of various new types of objects by users and by developers of the multi-service business platform, etc.

Using a yoga business as an example (e.g., where the user may be a yoga business owner), the user may create custom objects towards their yoga business (e.g., where the yoga business may include multiple studios that may be staffed with multiple instructors that may teach different classes that may be taught to students in accordance with the instructors’ respective schedules). In this example, a user (e.g., a user affiliated with the yoga studio or a third-party consultant) may create (e.g., via a GUI) a set of custom objects that relate to the yoga business, including defining the properties of each custom object. For example, the custom objects created may be studio objects, class objects, instructor objects, student objects, and schedule objects. Each studio custom object may include properties such as address of studio, rent of studio, and date when studio opened (e.g., date_opened). Each class custom object may include properties such as name of class, price of class, and schedule of class. Each instructor custom object may include properties such as date when instructor was hired (e.g., date_hired), latest certification date of instructor (e.g., latest_certification_date), certification expiration date of instructor (e.g., certification_expiration_date), and number of classes taught by instructor (e.g., number_of_classes_taught). Each student custom object may include properties such as date joined by student (e.g., date_joined), number of classes attended by student (e.g., number_of_classes_attended), date last attended a class by student (e.g., date_last_attended_a_class), total lifetime value of student (e.g., total_lifetime_value), credit of student, address of student, and phone number of student.

In example embodiments, the user may also define a set of associations between objects (e.g., custom objects and/or core objects). This yoga example may include several examples of associations. For example, one association may be “class_taught_by” which may be between the custom object instructor and/or a contact object (e.g., where contact may be instructor) and the class custom objects that the instructor teaches. Another example association may be “taught_at_location” which may be an association between the class custom object and the studio custom objects based on where a particular class may be held (e.g., may be determined from address or location information properties of studio custom objects). In another example, the “taught_at_location” association may be an association between the instructor custom object and the studio custom objects based on at which yoga studios a particular instructor teaches. There may be other associations created between instructor custom objects and core objects (e.g., contact objects) as well as student custom objects and core objects (e.g., contact objects). This may allow for actions to be taken based on these associations such as emails to be sent to instructors and students based on the associations of instructor custom objects and student custom objects with contact objects.

In example embodiments, machine learning may be used with custom objects to determine a likelihood to attend based on custom objects and properties that may be created. For example, the multi-service business platform may provide prompts for a user to define inputs into a machine learning model, e.g., the user may submit via prompts several properties (e.g., how often does student attend, how many classes is student signed up for, subscription plan, etc.) that may impact whether a student may be likely or unlikely to attend a class and the machine learning model may be used to perform calculations based on these inputs. In other example embodiments, the machine learning model may determine insights (e.g., properties relating to attendance may be determined) as data may be received from instances of the occurrence of actions relating to instances of the custom objects. For example, instances of associations between the object instances (e.g., between custom object instances and/or core object instances) may be used to determine these properties (e.g., based on properties of the association instances).

Referring now to example implementations of FIG. 45, there is shown an example environment 500 including a multi-service business platform 510 (e.g., may be also referred to as a multi-tenant distributed system that may serve the needs of multiple users who in turn use the system to provide services, support, and the like for their customers). The multi-service business platform 510 may communicate with various systems, devices, and data sources according to example embodiments of the disclosure. The multi-service business platform 510 may be referred to as a framework system or a multifunction business platform. The multi-service business platform 510 may include various systems 502-508, 1600, 520, 522-528, 562, 566, services 530, and a storage system 550. Specifically, the multi-service business platform 510 may include a customization system 520 (e.g., may also be referred to as a custom object creation system or custom object definition system). The customization system 520 may be used in a process to create custom objects and create associations for the custom objects.

These created custom objects may be used with various services 530 of the multi-service business platform 510 (e.g., may also be referred to as features of the multi-service

business platform). In examples, services 530 may include workflow automation 532 (e.g., workflows), reporting 534, customer relationship management (CRM)-related actions 536, analytics 538, import/export actions or services 540, other actions 542, and the like. Other actions 542 may include, for example, filtering used to search, filter, and list objects (e.g., contact objects) that may be used with other objects and/or create lists for other types of objects. In some examples, other actions 542 may include reporting, permissioning, auditing, user-defined calculations, and aggregations. The multi-service business platform 510 may include a non-exhaustive list of services 530 (e.g., set of features) that may be changed and/or added to the multi-service business platform 510 over time such that these services 530 may be automatically used with old and new core objects and/or custom objects.

The multi-service business platform 510 may be used to provide all of the objects (specifically custom objects) with various capabilities from these services 530. These various types of services 530 may be applied and/or used with the objects. For example, the workflow automation 532 (e.g., workflow system) may be used to add verbs (automation actions) with respect to nouns (e.g., custom objects). The core objects and custom objects may take advantage of all these services 530 (e.g., features) such that there may be a single source of truth (e.g., objects) that the services 530 and/or other systems of the platform may reason about that may be built onto the multi-service business platform 510.

The storage system 550 may include multi-tenant data store(s) 552, knowledge graph(s) 556, and proprietary data store(s) 554 (e.g., proprietary databases that may be similar to proprietary databases 208, 1620 described in the disclosure). Custom objects and/or core objects may include information that may be stored in the multi-tenant data stores 552 of the storage system 550. The custom objects and/or core objects as well as possible relationships (e.g., associations) between objects may be stored in an ontology of the knowledge graph(s) 556 at least implicitly and one or more instance knowledge graphs may be included in the knowledge graph(s) 556.

The multi-service business platform 510 may include other systems that may be used with the created custom objects such as a customer relationship management (CRM) system 502, a synchronization system 504, a machine learning system 506, a content management system (CMS) 508, and a multi-client service system 1600 (as described in the disclosure above). These systems may function and/or be used similarly to the same or similar systems described in the disclosure. For example, the machine learning system 506 that may already be used with core objects may also be applied similarly to the custom objects. The synchronization system 504 of the multi-service business platform 510 may synchronize some arbitrary custom objects outside the platform 510 to objects in the platform 510. In summary, in examples, the multi-service business platform 510 may act as an arbitrary platform that may act on arbitrary custom objects using various systems 502-508, 1600, 520, 522-528, 562, 566 and the services 530 (e.g., used with arbitrary actions and synced to arbitrary systems of the platform) thereby benefiting from these various capabilities.

The multi-service business platform 510 may communicate with external systems and data sources via a communication network 560 (e.g., Internet, public network, private network, etc.). Specifically, the multi-service business platform 510 may communicate with user device(s) 570 (e.g., user may be using the customization system 520 from the user device 570 to create custom objects via network 560),

client device(s) 572 (e.g., tracking various activities of the client device 572 of a customer for purposes of sales and marketing with respect to custom objects), and various external information sources 580. The external information sources 580 may include company information or data on customers, products, sales, third party data, resource description framework (RDF) site summary (RSS) feeds or really simple syndication (RSS) feeds, telemetrics (e.g., from email, websites, app usage), and the like with respect to custom objects. The multi-service business platform 510 may also communicate with third party service(s) 574 (e.g., third party applications, websites, Snowflake, etc.) via network 560.

The multi-service business platform 510 may also communicate with integrator device(s) 576. Integrator devices 576 may refer to user devices used by third-party integrator users that may create and may define a series of custom objects that may be integrated with other objects in the multi-service business platform 510 and may be offered to users (e.g., clients) of the multi-service business platform 510. The multi-service business platform 510 may include APIs (as described in the disclosure) that a user may use to define custom objects and integrate those custom objects into the CRM (e.g., CRM system 502) and thereby into the multi-service business platform 510. These same APIs may be available to integrator users to do the same thing. The integrator users may define a series of custom objects, then the integrator users may define object definitions. When a client installs that integration, the multi-service business platform 510 may enable the client (e.g., users of the client) to then start creating instances of custom objects defined by the integrator user(s).

Using yoga studio example again, an integrator user may have a company that builds CRM integration for yoga studios. This company may not be a yoga studio itself but may provide the CRM integration. For example, the integrator user may define a set of custom objects (including properties) that may be used by yoga studios or other fitness class-based businesses. The custom objects may include a studio custom object, an instructor custom object, a student custom object, a class custom object, and a schedule custom object. In this example, any client of the multi-service business platform 510 that operates a yoga studio (or other fitness, class-based business) may use the custom objects defined by the integrator (e.g., for a fee to the integrator) when on-boarding their business to the multi-service business platform 510. For example, the yoga studio users (e.g., from yoga studio businesses) may install integration (e.g., CRM integration from the integrator user) to be used on the multi-service business platform 510. After integration, the yoga studio users may be able to take advantage of the custom objects (e.g., custom definitions of the custom objects) created by the integrator user such as the “yoga class” custom object, the “yoga instructor” custom object, and the “yoga student” custom object. The yoga studio users may also have access to the services 530 of the multi-service business platform 510 such as reporting 534 (e.g., user reports), workflow automation 532 (e.g., user workflows), etc. that may be used with these custom objects. It may be as if each yoga studio user may rely on the integration from the integrator user such that the integration may be packaged with the custom objects and definitions for users of the multi-service business platform 510.

Referring now to example implementations 600 of FIG. 46, there is shown a portion of the multi-service business platform 510 with specific emphasis on details of the customization system 520 and the storage system 550 used to

create custom objects. In some examples, the customization system 520 may be a development tool such as a “generic data representation” system. In examples, the multi-service business platform 510, as described in the disclosure, may be a collection of processes that work over or on top of the customization system 520 (e.g., specifically APIs of the customization system 520). This may mean that a custom object created in and/or by the customization system 520 (e.g., including properties related to the custom object) may be immediately used by the services 530 and/or systems 502-508, 1600, 520, 522-528, 562, 566 of the multi-service business platform 510 to execute various tasks.

In example embodiments, as shown, the customization system 520 may use application programming interfaces (APIs) 610 as a computing interface to communicate and interact with users via the user devices 570 and/or integrator users via integrator devices 576. The customization system may include and use an object schema service 620 for providing a data application programming interface (API) such as an object definition API for receiving custom object information from the user devices 570 and/or integrator devices 576. The object definition API may be a CRM definition API, an object schema API, CustomObject data API, or a new schema API (e.g., user may create new schema API which may be defined as a form when filling out this API). The object definition API may be used for communicating the custom object information with the customization system 520 in creating custom objects. These data APIs (e.g., object definition APIs and/or APIs 610) may be “generic data representation” APIs that may be used by users (e.g., via user devices 570), integrator users (via integrator devices 576), and/or developer engineers (via multi-service business platform 510) to express a data model that may exist within the multi-service business platform 510 (e.g., framework).

The customization system 520 may include other services, components, and/or modules that may be used in the process of creating custom objects. For example, the customization system 520 may receive a user request for a custom object creation including custom object information (e.g., custom object name, an object type, at least one property of the custom object, and an association of the custom object with another object) from a user device 570 via the APIs 610. For example, the customization system 520 may include a form filling service 622 for receiving the custom object information for the custom object. For example, the form filling service 622 may provide a form (e.g., via a GUI) that may include prompts (e.g., spaces in a form) for the user to submit or input custom object information that may include a name (e.g., fill out name), a label, and basic information such as properties (e.g., description information about properties which may be similar to core value or core structure of metadata for the custom object being defined). Development documents may be used, or the user may use their own client for the form. In summation, the user may use the APIs 610, a custom object may be created, and then the user may login to the multi-service business platform 510 to monitor how custom objects may be integrated with the rest of the multi-service business platform 510.

The customization system 520 may include and may execute a business logic/sensible default service 624 (e.g., may use business logic and/or sensible defaults) to interpret custom object information in order to convert the custom object information into custom object metadata. The customization system 520 may include and use a relational database management service 628 (e.g., structured query

language database service such as open source MySQL™ database service) to insert and store the custom object metadata into a relational-type database (e.g., relational database management system). The customization system 520 may convert the custom object metadata into language-independent data creating a custom object. The custom object may be sent in language-independent data form to the user device 570 and/or services of the multi-service business platform 510, for use with marketing processes, sales processes, and/or customer service processes. For example, each custom object may be viewed by the user as a record on the user device 570 from the multi-service business platform 510. The customization system 520 may also include a common data format conversion service 626 that may assist with synchronization and integration of the custom object within the multi-service business platform 510 (e.g., integration of the custom object with the services 530 of the platform 510).

The customization system 520 may also communicate and direct changes to data on the storage system 550 when creating custom objects. Specifically, the multi-tenant data stores 552 of the storage system 550 may include definitions, properties, values, instances, and associations for all objects (e.g., including custom objects and core objects). These multi-tenant data stores 552 may be changed by the customization system 520 when creating custom objects. The storage system 550 may include knowledge graph(s) 556 such as an instance knowledge graph 640. The knowledge graph(s) 556 may also include, at least implicitly, an ontology 630. The ontology 630 may include the custom object with other custom objects 632 and/or core objects 634 (e.g., contact objects, company objects, deal objects, and/or ticket objects) along with one or more associations 636 (e.g., as added or selected associations 636 by the user) between the objects. Similarly, the instance knowledge graph 640 may include an instance of the custom object with other custom object instances 642 and/or core object instances 644 along with one or more association instances 646 between the object instances based on monitoring of activities of actual entities corresponding to these objects. Instances of objects (e.g., instances of custom objects 642 and/or instances of core objects 644) may be referred to as records.

The multi-tenant data stores 552 (e.g., which may include one or more databases) may be updated when adding custom objects. As described in the disclosure, the multi-tenant data stores 552 of the storage system 550 may include definitions, properties, values, instances, and associations. In some examples, the multi-tenant data stores 552 may include a set of data stores that collectively support custom objects and that may be updated by users of the multi-tenant data stores 552. For example, one data store may be a definitions data store that may be a system of records for storing objects and respective object definitions (e.g., list of core objects and custom objects). This definitions data store may be a definition of what objects (e.g., custom objects and/or core objects) exist. This definitions data store may include a list of objects, e.g., contacts, companies, deals, tickets, and custom objects (e.g., line items, products, etc.). This list of objects (e.g., custom objects) may also include and relate to any integrations that the user may have installed that define custom objects and any other custom objects that the user may find in their data (e.g., list of the tabs in a spreadsheet for user). Another data store may be a properties data store that may be a system of records for storing properties of custom objects such as tracking properties or attributes of custom objects as well as properties of core objects. Another data store may be a values data store that may be a system

of records for tracking values of properties. The larger multi-tenant data stores 552 may not discriminate based upon a user ID or a custom object itself. In some examples, each data store may include one or more databases.

In example embodiments, for the definitions data store, the system of record for what custom object types exists may be a “used car”. The properties data store may include properties or attributes that may include color, make, model, year, etc. for the used car custom object. The values data store may refer to the user, particular car, object type (e.g., which may be a car), related ID, property (e.g., car is red), etc. that may be laid out in such a way that the user may be able to dynamically create, edit, and remove values data of custom objects. Also, the user may be able to dynamically create, edit, and remove object properties and the user may dynamically create, edit, and remove properties (e.g., property values) of custom objects. This may provide flexibility immediately in terms of the user creating, editing, and/or removing custom objects, definitions of custom objects, and/or properties of custom objects.

In some example embodiments, the definitions and properties data of the multi-tenant data stores 552 may be located in a relational-type database such as relational database management system (e.g., structured query language database such as open source MySQL™ database) such that most of the data may be stored using a JavaScript™ Object Notation (JSON) (e.g., web-based tool JSON blob) to assist in creating, editing, viewing, formatting, and sharing JSON. The various metadata may be stored as columns for efficient indexing and queries. JSON may be used as data format such that JSON may be an open standard file format and data interchange format that may use text to store and transmit data objects. Other data formats may be used to accomplish the same or similar functionality described in the disclosure. In some examples, the values data store may be run by a non-structured query language (SQL) (NoSQL) or non-relational key value database which may be a similar database to Google™ Bigtable™ database.

The multi-tenant data stores 552 may include database storing metadata about object types, e.g., once metadata may be established and/or instances of custom objects may be created. Another set of APIs may be used for processing instance requests relating to specific instances of custom object. Importing may occur over a representational state transfer (REST) endpoint (e.g., REST API) over Internet as described in more detail below in the disclosure. Data may be written into a database (e.g., vastly horizontally distributed database) such that straight bytes may be written into a distributed file system. In some examples, the bytes may be interpreted using metadata in the relational database (e.g., MySQL™ systems). The multi-service business platform 510 may convert the interpreted data to a JSON representation of data (e.g., human readable or machine readable data) to be sent to a user device (e.g., user device(s) 570) or may be available on the platform 510 via a user interface of the user device. The horizontally distributed database may be used primarily as a system of record for storing object values as well as association values. In some examples, the relational database (e.g., mySQL™) may be used for storing property definitions, object definitions, and association definitions. In another example, the horizontally distributed database (e.g., may also be referred to as object instance databases) may include object property values and association instances. The relational databases (e.g., mySQL™ and/or other metadata databases) may include object types, property definitions, and association definitions. The above-described examples of storage for multi-tenant data stores

552 may be some examples of how data may be stored such that other similar and/or different examples of data storage may be utilized while maintaining core functionality of the multi-service business platform 510 and without departing from scope of this disclosure.

In some examples, the multi-service business platform 510 may include security functionality, for example, to avoid exposing entirety of multi-tenant data stores 552 (e.g., platform's object type definition data) to users. Further, in some examples, there may be assumptions about what users may want to do and these assumptions may be internal details. For example, administrators of the multi-service business platform 510 may not want a certain object type exposed to the APIs (e.g., search APIs). In another example, as described in the disclosure, business logic/sensible default service 624 such as sensible defaults may be used by the multi-service business platform 510 in accepting a new custom object (e.g., new custom object type definition) and when creating new associations.

The multi-service business platform may use a process to configure/update data stores (in some examples updating one or more databases in the data stores) based on custom objects. For example, users may use APIs (e.g., the APIs 610) that may include representational state transfer (REST) APIs that may be exposed via a network (e.g., network 560 such as the Internet). These APIs (e.g., REST APIs) may be used by users (e.g., via the Internet) to specify different operations that may be invoked to establish data needed that may define a new custom object type and/or may define instances of that new custom object type. The REST APIs may include data APIs (e.g., object definition APIs described in the disclosure) that may be used to receive custom object information from user devices 570 and/or integrator devices 576. This process may utilize a wrapper interface such as the object schema service 620 as described in the disclosure. The user may provide information using the object schema service 602 that may include a name of a custom object, properties of a custom object, and associations of the provided custom object type with other custom object types and/or core object types. Users may submit this information via a web request to the APIs. The customization system 520 may execute the business logic/sensible default service 624 (e.g., may use condensed business logic and/or sensible defaults) to interpret the information and insert necessary data in a relational database management system (e.g., set of MySQL™ tables). These MySQL™ tables may be a type of database where metadata may be stored about object types (specifically types of custom objects). Once the metadata may be established, the users may create instances of the custom objects.

In an example where a user affiliated with a drone selling/rental business creates a drone custom object, the user may want to or prefer to import data relating to several drone products and/or instances of drone products (e.g., information related to millions of drones owned by the business and/or instances of activities related to the drones) into the multi-service business platform 510 with different drone IDs and links to different deals that the drones may have been sold or rented under. When this import may be executed, a set of APIs may process these instance requests. For importing several drones (e.g., drone information and/or activities related to drone products), the user may invoke operations over REST APIs (e.g., endpoint over the Internet). The multi-service business platform 510 may take information received and may start writing data into another style of database which may be the vastly horizontally distributed database. The multi-service business platform

510 may be used to add on more virtual machines and continue to store all user data without impacting performance of the overall multi-service business platform 510. This data may be written as straight bytes into what may be essentially a distributed file system. Then, the multi-service business platform 510 may interpret the bytes accurately by using the metadata that may be available in MySQL™ systems (e.g., MySQL™ tables). When the user may want to fetch this data, the multi-service business platform 510 may read all the bytes from the distributed database system. The multi-service business platform 510 may interpret what it means to use the data from the MySQL™ systems. Then, the multi-service business platform 510 may convert this information or data into a human readable or a machine readable JSON representation of the data and may send it back to users. Alternatively, the JSON representation may be available through the existing user interface of the multi-service business platform 510.

In examples, the multi-service business platform 510 may provide a mechanism (e.g., a GUI) for a user to login to the multi-service business platform 510 to start using the created custom objects with the services 530 (e.g., framework features such as workflows, reporting). The multi-service business platform 510 may direct the usage of the integrated custom objects with various functionality. Simply, by creating the custom objects, the user may immediately be able to utilize all the functionality of the multi-service business platform 510 with the created custom objects. For example, the user may use services such as workflow automation 532 (e.g., workflows tool) and the user may see the option to include and/or use the created custom objects with workflows. The multi-service business platform 510 may direct the custom objects to be used with the services 530 providing all the automation of services described in the disclosure (e.g., automatically capable of using services with custom objects created). The custom objects may be account specific such that custom objects may only be used and viewed under one or more user accounts and/or one or more company accounts (e.g., custom object created by owner user of business may only be viewed and used with services by the same owner user). In some examples, the multi-service business platform 510 may be an external/visible entity into which users log in. In other examples, the multi-service business platform 510 may serve as a backbone of higher-level functionality that may be exposed throughout an application user interface (UI) and external APIs of the multi-service business platform 510. For example, as described in the disclosure, a manifestation of this automation integration functionality of the multi-service business platform 510 may be with services 530 (e.g., workflow automation 532 or workflows feature). As described in the disclosure, the multi-service business platform 510 may use the synchronization system 504 for providing custom object synchronization between the customization system 520 and the services 530 of the multi-service business platform 510.

In example embodiments, the multi-service business platform 510 may include the customization system 520 for providing a framework for customized programming. The multi-service business platform 510 may be configured in various ways with the customization system 520 to allow for users to be able to program custom objects. In example embodiments, the customization system 520 may be a tool whereby a user, an internal developer or team of internal developers, and/or a third-party integrator may define code that may run inside the customization system 520 of the multi-service business platform 510. An added benefit of internal developer teams being able to define new custom

object types may be the improved speed from development to shipping of these customized objects to users. For example, previously with core objects, release of these objects to users on the multi-service business platform **510** may take several months. Using this new process for creating custom objects on the multi-service business platform **510**, the custom objects and related services (e.g., features) may be released to users much sooner and faster such that a user may define a new custom object in minutes and may make use of the custom object immediately. The multi-service business platform **510** may also provide for the execution and/or use of the custom objects that may be programmed with the services **530** and/or other systems of the multi-service business platform **510**. For example, custom objects may be defined and the multi-service business platform **510** may be the execution engine that makes use of the custom objects.

The multi-service business platform **510** with the customization system **520** together may form the multi-tenant distributed system (e.g., multi-tenant data stores **552** of the multi-service business platform **510**) as described in the disclosure. In some examples, the multi-tenant distributed system and/or multi-tenant data stores as described in the disclosure may be configured generally such that all users' data may reside within a single system. For example, rather than provisioning dedicated systems for each user, the multi-service business platform **510** may be architected to allow for all customer data to co-exist within the same single system. However, the data may be segregated such that the multi-service business platform **510** may prevent mixing of the data (e.g., data from one user is never exposed to another user despite having their data stored in the same system). For example, one data store of the multi-tenant data stores **552** may include all core objects (e.g., CRM objects) and custom objects that may be defined by users, integrator users, and the developers of the multi-service business platform **510**. For example, a core object (e.g., contact object), a first custom object (e.g., drone custom object), and a second custom object (e.g., yoga class object) may all coexist within the same multi-service business platform **510** or system. The multi-service business platform **510** may use the services **530** to perform actions and operations on the defined custom objects (e.g., defining workflows, reporting with respect to custom objects, etc.) from the multi-tenant system. In some examples, data of custom objects and instances of custom objects (e.g., drone custom object data and/or instance data of the drone custom object) may be proprietary data even within the multi-tenant data stores **552**. This proprietary data within the multi-tenant data stores **552** may be segmented and separated such that the services **530** (and systems of the multi-service business platform **510**) may be executed on top of the custom objects and/or instances of objects without any need for these services **530** and systems of the multi-service business platform **510** to access the proprietary data. For example, when values data of the values data store (and possibly other data of the multi-tenant data stores **552**) may be populated from a user and knowledge graphs **556** may be created for the individual users based on this populated data, the services **530** and systems of the multi-service business platform **510** may then operate on the custom objects and instantiations of the custom objects.

In example embodiments, custom objects may be generated to be used in connections with the customer relationship management (CRM) system **502** and the content management system (CMS) **508** that may be based on custom object definitions provided by users. In some examples, as shown in FIG. **45**, the multi-service business platform **510** may

provide for custom objects to be linked/connected to and/or used with the CRM system **502** in terms of associations with core objects (e.g., contact objects, company objects, deal objects, and ticket objects) and/or other custom objects. The multi-service business platform **510** may also provide for a relationship between custom objects and the content management system (CMS) **508**. Custom objects may be shared between the CRM system **502** and the CMS **508**. For example, the CMS **508** may have a database that users may use to define data models to drive pages and content in the CMS **508**. Since the building of APIs and systems may be needed for custom objects, the CMS **508** may also migrate its database objects into the customization system **520**, storage system **550**, and/or other systems of the multi-service business platform **510**. Also, when building pages in the CMS **508**, users may leverage various tags that pull in data from other parts of the multi-service business platform **510** when a page may be rendered. For example, one such tag may be "crm_object" which may pull in the specified object into the CMS page when it may be rendered. For example, a user that has a "rental property" custom object may use the CMS **508** to define a page that may have a list of all "rental properties" that may be available and ready to rent. The user may then define subpages for when a customer clicks on a specific rental property. The content on the defined subpages may be populated from information stored on those custom objects. Thus, in example embodiments, user defined custom objects may be trackable throughout a user account lifecycle beginning in the CMS **508**, through the CRM system **502**, and potentially through the multi-client service system **1600**. In this way, users may be able to obtain insights from their data that may not have been previously available to them.

In example embodiments, the multi-service business platform **510** may use a common format for integrating custom objects with the multi-service business platform **510**. The common format may be embedded in a core of data processing systems. Various applications may be updated automatically, e.g., CRM applications and/or reporting applications may be updated automatically by syncing into third party services **574** (e.g., third party applications). The synchronization system **504** of the multi-service business platform **510** may be used to synchronize custom objects between third party services **574** and the multi-service business platform **510**. Custom objects may be configured to synchronize with external objects that exist externally from the multi-service business platform **510** (e.g., external to the CRM system **502**/CMS **508**). The synchronization system **504** of the multi-service business platform **510** may be used to sync arbitrary custom objects outside the multi-service business platform **510** to objects inside the platform **510**, which may facilitate creation of custom objects and workflows (e.g., using workflow automation **532**).

In example embodiments, the customization system **520** may provide mechanisms (e.g., GUIs) and processes for creating associations for the custom objects. For example, the customization system **520** may allow for the creation of an association definition entry (e.g., the relationship of identification (ID) representing one object type to an ID representing another object type) in a relational database management system (e.g., MySQL™ tables). The association definition entry may have an ID used to associate instances of two object types with one another. This process may use similar techniques used with graph database processes (e.g., graph database management system processes

such as Neo4j processes). Different name associations may be between different object types as well as between same object types.

For example, creating an association may first require a definition of a valid association which may also require a unique ID representing one object type and a unique ID representing another object type that may be the object types associated by this association. When users request that two object types be associated (e.g., where one object may be a custom object), then the customization system **520** of the multi-service business platform **510** may create an association definition entry in a relational database management system (e.g., MySQL™) that may link the custom object type with either another custom object type or a core object as requested. Once the association definition may be created, the association definition may be given an association type ID. The association type ID may be used by users to associate specific instances of two object types with one another. For example, associating two custom objects (e.g., associating two custom object instances may be two rows in a table) may start with a request to associate through an associations API. The customization system **520** of the multi-service business platform **510** may then write a row into the vastly horizontally distributed database (e.g., may also be referred to as an associations database) which may include a “fat row” format (e.g., may have the source object ID as the key and every linked object ID of the same object type belonging to an association type which may extend out in a wide row from that object type). This implementation, for example, may be similar to high end sophisticated graph databases such as a graph database management (e.g., Neo4j) that may use a similar strategy that may be a common proprietary open source graph database.

In some example embodiments, a qualifier may be added to an association type that may be a name of the association. The multi-service business platform **510** may have directed named associations and may expand metadata to more sophisticated metadata based on types of associations defined. In examples, company conduct associations may be used such that there may be different types of the associations (e.g., different named associations such that there may be different names of associations between the same object types).

In example embodiments, each respective association may include an inverse or opposite association that may be created automatically in response to defining the respective association. For example, when a user may create an association type (e.g., sold a car between business and customers), the multi-service business platform **510** may automatically create the inverse association type (e.g., “car was sold by”) and may give the inverse association the same name as the association. Even though an association may be created such that the user may represent a sale of a car to a customer as the customer “purchased” the car (e.g., when defining the association), the multi-service business platform **510** may also automatically create the inverse association in the opposite direction (e.g., the car was “purchased by” the customer) which may be given the same name but a different association type ID. In summary, when an association may be created for a relationship in one direction, the multi-service business platform **510** may always automatically create an association in the other direction such that the multi-service business platform **510** may support both representations for the association and inverse association which refer to same relationship between objects. For example, using the yoga class example, two custom objects may include class and student. An association between these

custom objects may be the class having a student that is Bob. An inverse association (e.g., opposite) of the association may be that student Bob may be an attendee or a member of the class (e.g., yoga class B). In summary, the inverse may mean that an association may be from the opposite view which may be from the view of the student or from the view of the class depending on the original association that was created. In this way, the time to process some search results, listing results, and/or other relevant requests may be reduced via the inverse associations.

Referring now to example implementations of FIG. 47, there is shown an example **700** of a custom object (e.g., first custom object **710A**) and associations **720A**, **720B** between the first custom object **710A** and other objects (e.g., second custom object **710B**, core object **730**) according to example embodiments of the disclosure. As shown, each custom object **710A**, **710B** may include a primary key **712-1**, **712-2** (e.g., directed associations may be defined between pairs of object identifications (IDs) within a context of a portal and object type). The primary key **712-1**, **712-2** may be used to locate specific objects. In examples, the primary key **712-1**, **712-2** may include a combination of a portal ID, an object type ID, and an object ID. These three unique identifiers may, in combination, uniquely allow for the ability to find objects. Each custom object **710A**, **710B** may also include a custom object name **714-1**, **714-2**, a custom object type **716-1**, **716-2**, and custom object properties **718-1**, **718-2**. The customization system **520** may be used to create at least one association (e.g., first association **720A** and/or second association **720B**) for the first custom object **710A** with another object **710B**, **730** based on the custom object information received (e.g., as received via the form filling service **622**). As shown in FIG. 47, in example embodiments, each association **720A**, **720B** may include an association identification (ID) **722-1**, **722-2**, an association type **724-1**, **724-2**, two IDs involved in association (e.g., a ‘From’ object identification (ID) **726A-1**, **726A-2** and a ‘To’ object identification (ID) **726B-1**, **726B-2**), and a timestamp **728-1**, **728-2** that may refer to when the association may have been created. In examples, the directionality of each association **720A**, **720B** may be indicated by the ‘From’ object ID **726A-1**, **726A-2** and the ‘To’ object ID **726B-1**, **726B-2**. For example, where a directed association between object IDs may be defined, the association in an opposite direction (e.g., inverse association) may also be defined that may result in two associations (e.g., two association records) for pairs of associated objects per association type. In some examples, a user may decide to define associations in terms of hierarchy by defining names of associations (e.g., user may name one association as “Parent” between two objects and/or may name another association as “child” between the same two objects but in the opposite direction) based on their business model. Any further information from each association may be derived from looking up and/or searching for the IDs, the object tables, and/or viewing what property values may be associated with those IDs.

The primary key **712-1** of the first custom object **710A** may be directed to either the ‘From’ object ID **726A-1**, **726A-2** or the ‘To’ object ID **726B-1**, **726B-2** based on a defined relationship between the custom object **710A** and another object (e.g., second custom object **710B** or a core object **730**). In some examples, this defined relationship may be linked both ways such that two objects may be linked via a same association, e.g., where the first custom object **710A** may be the ‘From’ (e.g., directed to ‘From’ object ID **726A-1** of the first association **720A**) and a second custom object **720B** may be the ‘To’ (e.g., direct to ‘To’ object ID **726B-1**

of the first association **720A**) and vice versa where the second custom object **710B** may be the 'From' and the first custom object **710A** may be the 'To' for the same association but from a different perspective. These inverse associations may be automatically created within each association (e.g., same association but may have different IDs distinguishing between association and inverse association) where the inverse association may be referring to the same association but from an inverse perspective of the other object.

For example, where the first custom object **710A** may be a customer and the second custom object **710B** may be a "yoga class" and the first association **720A** may be "class attendance", the same association may be described as "customer" ('From') attended "yoga class" ('To') or "yoga class" ('From') was attended by "customer" ('To'). In another example, where the first custom object **710A** may be a "salesperson" and the second custom object **710B** may be a "vehicle" and the association may be "sold", the same association may be described as "salesperson" ('From') sold (association) "vehicle" ('To') or "vehicle" ('From') was sold by (inverse association) "salesperson" ('To'). In another example, as shown, the first custom object **710A** may relate to the core object **730** via a second association **720B**. Similar to custom objects, the core object **730** may also include a name, a type, and properties but for purposes of showing relationships with custom objects, only a core object primary key **732** is shown in the core object **730**. The core object primary key **732** may be similar to the primary keys **712-1**, **712-2**. Using the previous example, the first custom object **710A** may be the "salesperson" and the core object **730** may be a company object (e.g., having an object name, company object type, and company object properties) that may be related by second association **720B**. In this example, where the first custom object **710A** may be a "salesperson" and the core object may be a "company" and the second association **720B** may be "employed by", the same association **720B** may be described as "salesperson" ('From') employed by (association) "company" ('To') or "company" ('From') employs (inverse association) "salesperson" ('To'). For the example of the vehicle and the salesperson, in some examples, absent an inverse association, a request to view who may have sold a particular vehicle may require processing each instance of the salesperson custom object before identifying which salesperson may have sold the particular vehicle. Conversely, with the inverse association, the same request may be processed by fetching the instances of the particular vehicle (e.g., particular vehicle custom object) and identifying the salesperson instances (e.g., instances of salesperson custom objects) via the inverse association ("sold by") between the vehicle record and the salesperson record (e.g., between instances of the particular vehicle custom objects and all instances of salesperson custom objects based on "sold by" inverse association).

In example embodiments, associations such as association instances may include association ID labels **722-1**, **722-2** that may define relationships but may be dynamic. For example, where a student may become a graduate or an alumnus such that the relationship with a given school may evolve from student to alumni. The association ID label **722-1**, **722-2** may exist on the details between the two objects but not necessarily on either specific object.

In creating custom objects, the customization system **520** of the multi-service business platform **510** may include safeguards. For example, one safeguard may be a guided service that may assist users in creating quality object types in the schema service. In some examples, part of the guided service may include the customization system **520** providing

requirements for users creating custom objects (e.g., users may be required to define objects). Also, part of the guided service may use the customization system **520** and the multi-service business platform **510** to provide the ability to automatically introduce some relationships (e.g., associations) to core object types. In some examples, there may be a limited number of association types, number of distinct object types, number of specific instances per given object type, etc. that may be provided in prompts to users providing various advantages. For example, the multi-service business platform **510** may limit the number of association types that may be created to prevent an overload of the association functionality of the customization system **520** of the multi-service business platform **510**. In another example, the multi-service business platform **510** may limit the number of distinct object types that may be created to avoid accidentally confusing object types with object instances. In another example, the system may limit the number of specific instances of a given type that may be created to avoid spamming with data through an import process. Other requirements may include the multi-service business platform **510** requiring that users may define a primary display label for objects so that the objects may be represented in a UI.

As described herein, the multi-service business platform **510** may be used for a business that markets towards customers interested in drone renting (e.g., renting a fleet of drones). The drone business may consider important information to track such as who may be renting drones and point of time of renting. A user from this drone business may use the customization system **520** of the multi-service business platform **510** to create a "drone" custom object that may be associated with core objects (e.g., deal objects and company objects) based on customers renting drones. After creating "drone" custom objects, the user may use the customization system **520** (e.g., import service) to import drone data having different drone IDs and links to different deals (e.g., via the APIs **610** of the customization system **520**).

For example, a user may use the multi-service business platform **510** to track drones and particularly determine who may rent drones and at what point in time. Accordingly, the user may create a drone custom object that may be linked with systems and services of the multi-service business platform **510** (e.g., particularly the CRM system **502**) as to what deal objects the drone custom object may be associated with, what customers may be renting the drones, etc. The services **530** (e.g., reporting **534** such as standard reporting and workflow automation **532**) may leverage data that the drone user defined. The drone user may use the customization system **520** (e.g., generic data representation system) to define whatever data the user may want and relationships between the data.

Then, code or automation may be run against the user's data (e.g., drone-related data from a user's business) in a multi-tenant distributed system (e.g., multi-tenant data stores **552**). Using the multi-tenant data stores **552** may mean that there may be some separation of user drone data from the rest of the data in the multi-tenant data stores **552** of the multi-service business platform **510** such that the multi-service business platform **510** may not know that the user that relates to any particular data set thereby providing privacy between data of users and user accounts. Also, the multi-service business platform **510** may not be aware of what object type may be used by accounts to uphold the privacy of users. However, the user may still use the multi-service business platform **510** to perform actions and operations on the drone custom objects and data as defined

by the users that may be exclusively accessed by the drone user accounts (e.g., from drone renting business or client).

In another example, a business may be an auto dealership. The auto dealership may wish to represent particularly important objects on the multi-service business platform **510**, e.g., vehicles, salespeople, parts suppliers, other suitable elements that may be represented in datastores (e.g., databases of datastores), and the like. In this example, a user of the auto dealership may run their business using a spreadsheet (e.g., Microsoft Excel™ spreadsheet) for recording information about the auto dealership business. Tabs of each spreadsheet may refer to inventory (e.g., new inventory and/or used inventory), salespeople, and customers. The user may create custom objects corresponding to these tabs. For example, the user may create custom objects that may include a salesperson custom object, an inventory (e.g., vehicle) custom object, and a customer custom object using the customization system **520**. Each tab may refer to a different custom object or custom object definition that may be created. These custom objects may include properties based on the tabs. For example, each custom object (e.g., custom object definition) may include a collection of properties as described in more detail in the disclosure below as property instances for custom object instances. The user may also create possible associations between these custom objects and other objects (e.g., core objects and/or other custom objects) using the customization system **520**. These custom objects (e.g., custom objects **632**) and associations **636** (e.g., association instances) may be stored in the ontology **630** of the knowledge graph(s) **556**. In general, all this data may be stored in the spreadsheet such that every tab may be a different custom object (e.g., different object definition). Each row of the spreadsheet may be a different object instance (e.g., different custom object instance).

Referring now to FIG. **48**, there is shown a visual representation of a portion of an example instance knowledge graph **800** according to example embodiments. In this example, the instance knowledge graph may relate to instances of the custom objects (e.g., from rows of a spreadsheet) that may be based on the created custom objects (e.g., from the tabs of a spreadsheet) for the auto dealership business. The custom object instances may include a salesperson custom object instance (Bob) **806A**, another salesperson custom object instance **806B** (Alice), a customer custom object instance **802** (John), and an inventory custom object instance **804** (e.g., vehicle custom object instance such as “Toyota Camry”). These custom object instances may be based on the following custom objects (e.g., custom objects **632** of the ontology **630**): salesperson custom object, customer custom object, and vehicle custom object. Corresponding defined properties are shown with each custom object instance as well as possible relationships (e.g., associations) between the custom object instances. For example, as properties may be defined when creating custom objects, the salesperson custom object instance **806A** (Bob) may include salesperson properties **810A** (e.g., a name such as “Bob”, address, email, phone number, employee ID, date hired, title, commission, and one or more goals such as per year, per quarter, per month) for a particular salesperson. The other salesperson custom object instance **806B** (Alice) may include similar salesperson properties **810B** to the salesperson properties **810A** of the salesperson custom object instance **806A** (Bob). The inventory custom object instance **804** (Toyota Camry) may include inventory properties (e.g., specifically vehicle properties **812**) as defined when creating the vehicle custom objects. The vehicle properties **812** (e.g., from car inventory tab of spreadsheet)

may include a vehicle identification number (VIN), car make such as Toyota, model such as Camry, year, color, mileage, condition, sunroof (T/F), and alarm (T/F). The customer custom object instance **802** (John) may include customer properties **814** as defined when creating the customer custom objects. The customer properties **814** may include name such as John, address, email, phone number, budget, purchase date, and purchase (T/F). In examples, some of the properties of the custom objects may be added as being set for the custom objects. Properties may be created for each custom object and may be set on an instance. Other properties (e.g., T/F properties) may be added as being flexible or optional depending on instance such that these properties may also be set on an instance such as alarm (T/F) which may refer to whether vehicle “has alarm system” property for inventory custom object but not known until instance occurs. The alarm (T/F) (e.g., “has alarm system”) may be true for some cars, false for other cars, and even unknown for some cars based on the vehicle custom object instance **804**. Accordingly, this type of flexible or optional property information may be left empty (e.g., similar to leaving a cell empty in spreadsheet) such that this property information may be filled in as instances occur. As activities occur relating to any one of these custom objects (e.g., custom objects **632**) and associations **636** (e.g., association instances) of the ontology (e.g., the ontology **630**), the instance knowledge graph (instance knowledge graph **640**) may be generated creating instances of these custom objects (e.g., custom object instances **642**) along with association instances (association instances **646**) directly corresponding to the activities that occurred (e.g., a Toyota Camry vehicle was sold by salesperson Bob to customer Steven). In this example, rows in the spreadsheet may refer to instances of the custom objects.

In an example, as shown in FIG. **48**, there are several association instances **820-828** between the different custom object instances that may be based on the possible associations defined in the ontology (e.g., associations **636** of the ontology **630**). These association instances **820-828** may be added to the instance knowledge graph **640** as activities occur relating to any one of the associations **636** between custom objects **632** of the ontology **630**. For example, an association instance between the salesperson custom object instance **806A** (Bob) and another salesperson custom object instance **806B** (Alice) may be “reports to” association instance **828** (or, for example, “is supervised by” association instance as described in the disclosure) such that the other salesperson custom object instance **806B** (Alice) relates to the salesperson custom object instance **806A** (Bob) as a manager. As described in the disclosure, there may be an inverse association (e.g., inverse association instance) created automatically for every association (e.g., association instance). The association and inverse association may be traced bidirectionally. The inverse association for “reports to” association instance **828** may be “reported to by” association instance **828'** (or, for example, “supervises” association instance as described in the disclosure) such that the other salesperson custom object instance **806B** (Alice) “reported to by” **828'** the salesperson custom object instance **806A** (Bob). The salesperson custom object instance **806A** (Bob) may relate to the inventory (e.g., vehicle) custom object instance **804** (Toyota Camry) by a “sold” association instance **820** such that the salesperson custom object instance **806A** (Bob) “sold” **820** the vehicle custom object instance **804** (Toyota Camry). The inverse association for “sold” **820** association instance may be “sold by” association instance **820'** such that the vehicle custom object

instance **804** (Toyota Camry) may be “sold by” **820** the salesperson custom object instance **806A** (Bob). The salesperson custom object instance **806A** (Bob) may relate to the customer custom object instance **802** (Steven) by a “sold to” association instance **822** such that the salesperson custom object instance **806A** (Bob) “sold to” **822** the customer custom object instance **802** (Steven). The inverse association for “sold to” association instance **822** may be “sold to by” association instance **822'** such that the customer custom object instance **802** (Steven) may be “sold to by” **822'** the salesperson custom object instance **806A** (Bob). The customer custom object instance **802** (Steven) may relate to the inventory (e.g., vehicle) custom object instance **804** (Toyota Camry) by a “purchased” association instance **826** such that the customer custom object instance **802** (Steven) “purchased” **826** the vehicle custom object instance **804** (Toyota Camry). The inverse association for “purchased” association instance **826** may be “purchased by” association instance **826'** such that the vehicle custom object instance **804** (Toyota Camry) may be “purchased by” **826'** the customer custom object instance **802** (Steven). The customer custom object instance **802** (Steven) may also relate to the vehicle custom object instance **804** (Toyota Camry) by a “test drove” association instance **824** such that the customer custom object instance **802** (Steven) “test drove” **824** the vehicle custom object instance **804** (Toyota Camry). The inverse association for “test drove” association instance **824** may be “test drove by” association instance **824'** such that the vehicle custom object instance **804** (Toyota Camry) may be “test drove by” **824'** the customer custom object instance **802** (Steven). In an example, the customer (e.g., referring to customer custom object instance **802** (Steven)) may decide to trade-in or sell their purchased vehicle (e.g., referring to the same vehicle custom object instance **804** (Toyota Camry)) back to the same salesperson (e.g., referring to the salesperson custom object instance **806A** (Bob)) such that new association instances may be added automatically with respect to and between these different custom objects.

In some examples, advanced reporting, as described in the disclosure, may be used with the auto dealership. For example, a user may want to know a home and zip code of the customers who are buying the most cars from the salesperson (Bob). The user may select all the associated sales for the salesperson (Bob). For those sales, the user may then select the associated customers and for those selected customers, the user may access or obtain their zip codes. The advanced reporting service may aggregate by zip code and count the number of instances of zip code. This may involve a standard style SQL query but may be advanced in terms of the query or by tooling which may be used in building these reports by advanced reporting.

FIGS. 49A-49G show example screenshots of user interfaces (UIs) relating to processes of creating custom objects and processes of using custom objects on the multi-service business platform **510**. These figures are screenshots of example graphical user interfaces (GUIs) allowing a user to create custom objects and then use the custom objects with services of the multi-service business platform **510** according to one or more example embodiments of the disclosure.

For example, FIG. 49A shows a screenshot of graphical user interface (GUI) **1100** for creating custom objects (e.g., creating custom object definitions including creation of properties of the custom objects). In example embodiments, other GUIs may be used such as an object definition GUI that may be presented to a user where the user may define schemas of their custom objects. In defining schemas of a custom object, the user may provide an object name (e.g.,

may describe type of object) and properties of the custom object. In this example, the subscriptions may refer to a type of custom objects such that subscription custom objects may be created in the GUI **1100** from directing APIs. The GUI **1100** and other examples of GUIs may show an ability to create custom objects and associations (e.g., creating association definitions) that may be handled via external APIs as described in the disclosure. For example, the form filling service **622** may be used with these APIs with respect to GUI **1100**. FIG. 49B shows a screenshot of GUI **1200** for listing custom objects (e.g., shown as subscription custom objects) that were created. As shown in FIG. 49B, in some examples, the user may view the custom objects along with core objects in a grid view. In this example, the user may select links associated with objects such as custom objects to view their records. This may be expanded across and throughout the multi-service business platform **510** which may be used by users for their businesses.

In examples, some of screenshots show use of the custom objects with various services (e.g., the services **530**). For example, these screenshots may be GUIs involving use of custom objects with services (e.g., using workflows and reporting). These example GUIs may be showing instance-level usage throughout a user interface (UI). For example, FIG. 49C shows a screenshot of GUI **1300** for listing custom objects with filtering capability services (e.g., search/filter capabilities). In this example, properties of custom objects (e.g., properties of subscription custom objects) may be searched using filtering capability services. In other example embodiments, FIG. 49D shows a screenshot of GUI **1400** for custom object workflows that may relate to assigning a proper service package to a subscription. In this example, actions may be shown related to service creation that may involve custom objects. Another workflow example is in FIG. 49E showing a screenshot of GUI **1500** for custom object workflows that may relate to sending an email to a customer automatically if the custom may be starting their subscription soon. In this example, actions may be shown related to a subscription start reminder that may involve custom objects. In another example, FIG. 49F shows a screenshot of GUI **1800** for reporting with custom objects that may relate to breaking down subscription custom object types. Another reporting example is in FIG. 49G showing a screenshot of GUI **3000** for reporting with custom objects that may relate to breaking down subscription terms by subscription owner. These reporting examples show options for users in configuring charts based on type of charts and/or data that may be included in charts as relating to one or more custom objects.

In example embodiments, there are other custom object API examples such as relating to customer relationship management (CRM) objects. For example, a CRM objects API (e.g., public API) may be able to represent and organize data based on business requirements. Specifically, the CRM objects API may be created to define and manage custom objects. Users and/or integrators may use the CRM objects API to define custom objects, properties, associations, and/or leverage the multi-service business platform **510** to increase business. Custom objects may be specific to a portal. The custom objects may exist with core objects (e.g., contacts, companies, deals, and/or tickets) in the multi-service business platform **510**. Custom objects and custom properties may be defined to organize and represent business data for a company or industry. Custom objects may be linked and/or associated to the core objects or other custom objects.

In example embodiments, a custom object may be defined by defining a custom object type and property definitions. This defining process may also include defining association definitions with core objects (e.g., contact objects) and/or custom objects. By default, the property type and field type may be a string and text, respectively (e.g., a property may exist under properties to be specified as various code terminology). As part of the schema, there may be association definitions automatically created between the custom object and email, meeting events, notes, tasks, calls, and conversation sessions. Custom objects may be retrieved (e.g., retrieved as meta-type custom objects, fully qualified named custom objects, and/or portal-specific custom objects). Related object schema may be modified. Once a custom object may be defined, in some examples, the name and labels (singular and plural) may not be changed. Properties of the custom object may be modified after creation. Custom objects may be associated with core objects (e.g., contact, company, deal, or ticket objects) or other custom objects. Custom objects may be deleted after all object instances of a same or similar type are deleted (e.g., custom object type may be deleted after all object instances of the same or similar type, associations, and custom object properties may be deleted). In examples, custom objects may be associated with core objects or other custom objects.

In example embodiments, when creating or updating properties for custom objects, both type and field type values may be required. The "type" value may determine the type of the property (e.g., a string or a number). The field type property may determine how the property may appear in the multi-service business platform **510** or on a form (e.g., as a plain text field, a dropdown menu, or a date picker). There are various examples of how type and field type may be configured for properties on new custom objects. Values for field type and compatible field type values may include enumeration (e.g., string representing a set of options), date (e.g., representing a specific day, month, and year), date time (e.g., representing day, month, year, and time of day), string (e.g., plain text strings), and/or number (e.g., value containing numeric digits such as one decimal). Other field type values may include Boolean checkbox (e.g., input that may allow users to select either "Yes" or "No" as used in a form displaying a single checkbox), checkbox (e.g., a list of checkboxes that may allow a user to select multiple options from a set of options allowed for the property), date (e.g., date value that may be displayed as a date picker), file (e.g., allows for file to be uploaded to a form such as stored and displayed as a URL link to the file), number (e.g., string of numbers in decimal or scientific notation), radio (e.g., input that may allow users to select one of a set of options allowed for the property which may be displayed as a set of radio buttons when used in a form), select (e.g., dropdown input that may allow users to select one of a set of options allowed for the property), text (e.g., plain text string that may be displayed as a single line text input), and/or text area (e.g., plain text string that may be displayed as a multi-line text input).

In example embodiments, a custom object process may include creation of a custom object schema (e.g., object type, properties, and associations). The custom object schema may be fetched. A custom object instance may be created. An association may be created between the custom object instance and an existing contact. A new association definition may be created between the custom object and a ticket object. A new property definition may be created. The object schema may be updated with the new property.

In example embodiments, the multi-service business platform **510** may provide marketing and sales functionality for communicating with customers, tracking interest, and/or offering possible deals based on inventory. For example, where the user is a car dealership, object schema may be created to define and represent car inventory in the multi-service business platform **510**. There may be attributes of the car that may be stored. The car may be represented as having the following properties: condition (new or used): enumeration, Date received (at a dealership): date, Year: number, Make: string, Model: string, VIN: string, Color: string, Mileage: number, Price: number, and/or Notes: string. To connect with a potential customer, a relationship between core contact objects and the custom car object may be defined. By defining an association between the contact (core object) and the car (custom object), the multi-service business platform **510** may be utilized to communicate with potential buyers and track progress on a potential deal. Once a data model may be finalized, the object schema may be created using the API. This may define the custom object and its properties as well as its association with other core objects (e.g., contact objects). A property group relating to car information may be created. Some object properties may be created when creating the schema. Additional properties may be added, and the object schema may be updated at a later time.

The custom object process for creation of the custom object schema may be applied to the car dealership example. For example, an object schema related to cars may be fetched. An object instance may be created which may allow for the use of the multi-service business platform **510** to communicate with potential customers, track potential interest, and offer possible deals for them based on inventory (e.g., creating a new custom car object instance and associating this instance with an existing contact object instance). A new object identification (ID) may be used for associations created such as the association between the custom car object and an existing contact. Objects and properties may be retrieved as needed. An association may be created to an object instance. For example, with the new object ID and a target contact ID, the multi-service business platform **510** may be used to create an association between the new car instance and the contact. As described in the disclosure, associations may be described in both directions or bidirectionally (e.g., between car and contact such that each association may relate to the car's association or relationship to the contact in one direction as well as the contact's association or relation to the car in the other direction). There may be a new association definition. For example, the multi-service business platform **510** may be used to track services performed on cars sold. The multi-service business platform **510** may further use tickets to log any maintenance performed. A new relationship between custom car objects and core ticket objects may be created by creation of a new association. There may be a new property definition. For example, maintenance packages may be offered to customers (e.g., oil change every 5 k miles) which may be tracked by adding a new property. A new field may show up in a sidebar interface of the multi-service business platform **510** as a secondary property which may provide quick identification of a level of service when a car goes into a car shop.

As shown in FIGS. 49A-49G, in an example CRM, properties may be found and may show up in several services. For example, properties may show up in reports, workflows, imports, profile page, etc. Specifically, there may be a listing page GUI (e.g., the GUI **1200** of FIG. 49B) that

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may show a listing of all the object instances for a type. In this page, the user may shift from one object type to other object types that may be defined in this portal. For example, there may be a subscriptions custom object and services object. There may be several instances of services. If a user clicks into an individual profile for each service, the user may view different fields. Each and every single one of these fields may be a property in the multi-tenant data store that may be related to the object definition for the corresponding custom object. In another example, if a user goes to settings and goes to properties, the user may go into service properties and may create a new property for the service custom object (e.g., custom object definition). Then, when the user goes back to the services, the user may then be able to add the property as a column in this table, may drill in and add the property as an item that the user may want to view, and the user may start reading and writing data from that property which may be from the CRM side. When the user goes into automation and defined workflows, the user may request that they want a workflow from scratch, e.g., a service-based workflow. For the triggers and the actions on this service-based workflow, the user may have access to all these properties that may have been previously defined. If the user went in and added a new property, the new property may show up in this list, and the user may trigger logic based on values of the “service” custom objects.

Additionally, in some examples, the user may filter on associated custom objects. For example, if a service instance may be associated with a deal. The user may filter on the associated deal information by leveraging associations with custom objects. If the associations are not defined and in the knowledge graph, then the user may not leverage them in workflows. In summary, anytime users load a part of the multi-service business platform 510, the platform 510 may pull in the knowledge graphs (e.g., knowledge graph(s) 556) and then may expose or may hide different services (e.g., features) based on the data that may be stored therein. If properties exist, then they may show up. If associations to other custom objects and other object types may exist, then those associations may be visible within the multi-service business platform 510. For example, when going into the listing tool of the multi-service business platform 510, a user may create a list. This may function similarly to how the user may view this function in workflows. It may be visible throughout the software such that the multi-service business platform 510 may pull in the knowledge graph(s) 556 throughout the platform 510 so that the user may then leverage the instance data throughout the platform 510.

Referring now to example implementations of FIG. 50, there is shown a flow chart including a set of operations of a process 1000 for creating a custom object according to example embodiments of the disclosure. In this example, a user request for a custom object creation including custom object information 1002 may be received. The custom object information may be interpreted and converted into custom object metadata 1004. The custom object metadata may be inserted and stored into a relational-type database 1006. The custom object metadata may be converted into language-independent data creating a custom object 1008. The custom object in language-independent data form may be sent to a user device and/or one or more services of a multi-service business platform for use with a marketing process, a sales process, and/or a customer service process 1010. Services may be applied and used with language-independent data of the custom object 1012.

As described in the disclosure, the multi-service business platform 510 may include the services 530 (e.g., features)

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that may be used with and customized for interacting with custom objects. As new instances of the services 530 may be added or instances of the services 530 may be changed, the multi-service business platform 510 may automatically apply the added and/or changed instances of the services 530 to all custom objects. The multi-service business platform 510 may also automatically integrate use of all custom objects with any new services that are added to the services 530.

In another example, a user may be from a yoga business. In this example, the user may create the following custom objects that may include studio custom objects, class custom objects, instructor custom objects, student custom objects, and schedule custom objects as described in the disclosure. The user may use the customization system 520 of the multi-service business platform 510 to create these custom objects. The user may also use the other systems and the services 530 of the multi-service business platform 510 for assisting with management of their yoga business (e.g., especially use of custom objects of the yoga business).

For example, the yoga user may use the workflow automation 532 (e.g., workflows) to send out a “special” to students who may have not attended a specific class more than 90 days ago. Using the workflow automation 532 (e.g., workflows tool), the user may create a “student” workflow. Enrollment criteria may be that the student may have attended at least one class and the date the student last attended was more than 90 days ago. An action for this workflow may be to email the contact record (e.g., contact information in properties such as email) associated with the student a discount for a new package of classes. This workflow may be updated and kept in sync automatically as students attend classes in real time and/or periodically as time passes. For example, the user may use reporting 534 to find out how many classes students may be attending. Using reporting 534 (e.g., reporting tool), the user may build a histogram of student counts by number of classes attended. A CRM-related action 536 may include searching capabilities that may be used with CRM records. The search may be used by the user to create a public listing of all classes and locations so students may be able to view and access what classes may be available and may sign up to attend online. A search page may be supported by a CRM search engine of this CRM-related action 536 (e.g., search) which may support defining well-formed queries against a data set. For example, a search may be a query such as “Show me all classes at a specific location that are taught by instructor X and are held on weeknights”. The user may use import/export services 540 (e.g., importing and/or exporting properties from external data source(s) such as external information source(s) to the multi-service business platform 510). In this example, the user may specifically use import service (e.g., import tool). For example, when the user (e.g., owner of yoga business) starts using the multi-service business platform 510, the user may need to import data when creating custom objects. After the user defines the custom objects, the user may leverage the import tool in the CRM to import all of their data (e.g., yoga business data from external information sources 580) in a comma-separated values (CSV) file form into the multi-service business platform 510.

Another action 542 of the services 530 may be a listing service (e.g., a listing tool or a lists tool). The user may use the listing tool to create a list of all instructors who need to be re-certified. Using the listing tool, the user may define a list of all instructors whose certification may be 90 days from expiring. A manager of the yoga business may then

connect with any instructors on that list and may ensure that the listed instructors may get re-certified. This list may be automatically updated and kept in sync in real time and/or updated periodically as time passes. Another action **542** or service that may be used by the user may be a public API. For example, as the yoga business grows and becomes more sophisticated, the yoga business may want to develop their own programs that may run in-house that may leverage the data stored on the multi-service business platform **510**. For example, the multi-service business platform **510** (e.g., specifically the customization system **520** of the platform **510**) may include custom object APIs that may allow for the user to read and write object data from a custom object datastore of the multi-tenant data stores **552**. Any services (e.g., features) of the multi-service business platform **510** may automatically take newly written data into account. Another action **542** or service that may be used by the user may be permissions (e.g., custom object permission service or system). As the yoga business or company grows, the yoga company may need to hide sensitive information about units from different agents. By leveraging the custom object permission service, the user of the yoga company may effectively partition what data specific reps may access and edit. For example, the user (e.g., yoga company user, which may be an administrator user) may use this permissions service to hide classes taught by one instructor from another instructor.

Another action **542** or service that may be used by the user may be advanced reporting. For example, the yoga business or company may want to leverage more sophisticated business intelligence tools. Data in a custom objects data store (e.g., of the multi-tenant data stores **552**) may be automatically synced into a platform instance (e.g., platform instance that may be from a third party hosted and managed database such as Snowflake™) that may then be used to drive advanced queries and joins. For example, all the same information may be mirrored and/or copied into the third party hosted and managed database (e.g., using third party service **574** such as Snowflake™) that may be leveraged to do efficient analytic queries. Most of the CRM may be meant to be real time or near real time as possible so that as users may make changes to properties of their objects, workflows may be updated, reports may be updated, lists may be updated, content may be updated, etc. For more advanced reporting, where real time may be less important, all that data may be sent off to the third party hosted and managed database (e.g., Snowflake™) where users may then perform SQL-style queries into the data set. Whenever a user may define a new custom object or install an integration that may bring in custom objects, the user may then leverage this advanced SQL style query against that data set. With this link to third party hosted and managed database, the multi-service business platform **510** may provide hosted data warehousing solutions for users. Users may then join their data sets that they already have in third party hosted and managed databases with the data of the multi-service business platform **510** (e.g., storage system **550**), which may provide an expansion of users' data.

Another action **542** or service that may be used by the user may be internal listing and profile pages. All objects (e.g., specifically custom objects) may be defined in the custom object data store (e.g., of the multi-tenant data stores **552**) that may have internal listing pages and profile pages out of the box. This may make it relatively easy for a user to view and/or have access to all of the instances of the custom object that the user may have created and interact with the related data.

In example embodiments, the multi-service business platform **510** may include a reporting system **526** that allows users to create different types of visual reports using various data sources associated with a client's business. In example embodiments, the reporting system **526** may allow users to generate visual reports that incorporate data collected with respect to core objects and/or custom objects, thereby providing for multi-dimensional custom reports. The custom event reporting may generate reports using disparate data that was collected, derived, generated, and/or otherwise obtained by the various services of the multi-service business platform **510** (e.g., CRM **502**, CMS **508**, payment system **524**, event system **522**, and/or the like) based on the common object definitions that may be applied across the platform. In some example embodiments, the custom event reporting may implement or leverage a unified analytics pipeline such that all event reporting may be based on that pipeline (e.g., utilizing common infrastructure).

The multi-service business platform **510** may include an event system (e.g., event system **522**). The event system **522** may be configured to monitor for and record the occurrence of events. In some example embodiments, the event system **522** may be configured to maintain unified events that are tracked across several systems of the multi-service business platform **510**. In some of these example embodiments, event records may track all the different types of events that may occur with respect to a particular type of object such that the event record provides a log of all instances of different types of events that occurred with respect to the object. The event system **522** may fit with several of the services in this disclosure including reporting aspects and triggering of workflows and actions as related to default and custom objects.

The multi-service business platform **510** may include instances of custom objects that may be used to perform customer-defined analytics (e.g., analytics **538**) across the CRM system **502** and the CMS **508**. The multi-service business platform **510** may include a common infrastructure such that all objects (new, old, core, and custom objects) may be tracked via a unified analytics pipeline. The custom analytics may be associated with CRM/CMS based on custom object definitions that may be provided by users. The custom analytics may be used with a unified analytics pipeline such that all event reporting may be based on that pipeline (e.g., utilizing common infrastructure).

The multi-service business platform **510** may generate custom actions that may operate on or with respect to instances of the custom objects. The custom actions may be part of another system that may reside within the multi-service business platform **510**. The multi-service business platform **510** may be built on top of the customization system **520** (e.g., custom object data system) and may not be aware of workflows or their custom actions. The custom actions may be based on objects being considered nouns and actions being considered verbs such that automation of the multi-service business platform **510** may allow for verbs as actions may be added easily along with adding nouns as custom objects. The multi-service business platform **510** may include APIs such that any user may write their own extensions (e.g., using Lambdas or serverless functions). For example, custom actions may be new types of actions that may be implemented due to creation of new custom objects (e.g., new actions tracked based on new custom objects).

The multi-service business platform **510** may provide custom object filtering. For example, custom objects may include a set of custom properties that may be used to filter instances of the custom objects based on values of the

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respective custom properties. Examples of custom object filtering may include list segmentation, filtering, and searching across custom object types and being able to automate off changes (e.g., changes in custom objects and/or core objects).

The multi-service business platform **510** may use the machine learning system **506** with custom objects. For example, instances of custom objects may be used with machine learning system **506** to train machine-learned models that may be used with the user (e.g., related to a user's business) for all objects (e.g., custom objects and core objects). In one example, the machine learning system **506** may provide custom object filtering for custom objects.

For example, the machine learning system **506** may be used with real estate custom objects such that a user may use the CRM system **502** to track homes for sale and may have built landing pages in the CMS **508** for each house that may be for sale. The machine learning system **506** (e.g., machine learning as a service (MLaaS)) may be used to predict interesting things. Two examples may be a likelihood that a home may sell and the most likely prices. In each example, the user may request for the machine learning system **506** of the multi-service business platform **510** to predict the value of these properties on the house and may use the machine learning system **506** (e.g., MLaaS) to predict other insights. The machine learning system **506** may use machine-learning models to take into account information about a specific home, data from other home custom objects in the CRM system **502**, and data from objects associated with the house (e.g., core objects and/or custom objects), such as the realtor and contacts that may have viewed landing pages for the homes.

The multi-service business platform **510** may include custom objects that may be configured to support a custom application architecture of a user that may connect with the CRM system **502**/CMS **508** of the multi-service business platform **510**. The multi-service business platform **510** may be an arbitrary platform that may act on arbitrary objects to do arbitrary actions and sync to arbitrary systems and may get the benefit of various capabilities. In an example, the CMS **508** may be made front end to the CRM system **502** (e.g., under protection of login) such that a user may view what they need in the CMS **508**. The custom application may be any type of application, e.g., a web application. For example, a yoga studio business may include custom objects such as schedule objects, class objects, "my calendar" (gigantic web application) that may be built on top of the CMS **508** and the CRM system **502** such that users may build and present CMS-driven apps integrated with the multi-service business platform **510**.

The multi-service business platform **510** may support a custom application architecture of a user that may integrate with a payment processing service (e.g., payment system **524**) and may connect with the CRM system **502**/CMS **508** such that a payment processing service may feed payment data to the CRM system **502** and the CMS **508** of the multi-service business platform **510** in real-time. In example embodiments, the payment system **524** is configured to establish payment sessions for customers with third-party payment processors on behalf of clients of the multi-service business platform **510**. The payment processing service may, for example, assist a manufacturing company in creating invoices for its customers and may initiate payment sessions with a third-party payment processor that allows the customers to pay the amounts indicated in the invoices directly via the third-party payment processor. In this way, the multi-service business platform **510** may not have to

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request, receive, or securely store sensitive payment information from the client's customers, while still being involved in the payments' workflows. Using this payment processing service, customers may easily go onto a website and make a purchase similar to other purchasing sites. The payment processing service may be used with business to business (B2B) transactions, e.g., custom objects for B2B; custom actions for B2B; and tight integration between objects in a B-commerce framework (e.g., product catalog may be in the CRM, the website may be in the CMS, payments may be immediately reflected in the CRM and deal records, custom objects may be shipping and/or tracking, etc.).

The multi-service business platform **510** may include an attribution reporting tool which may be an extremely powerful tool that may leverage much of this disclosure. Attribution reporting may be a measure of efficacy of effort. For example, a user's business may have website visitors that may be reviewing web pages, may be filling out forms, etc. and the user's business may have sales reps making calls. All these actions by visitors and by members of the business may be measured by how effective each of these individual touch points was in order to have some outcomes (e.g., closing deal). Attribution reporting may take all these different inputs which may be happening in the multi-service business platform **510** (e.g., the CRM system **502** of the multi-service business platform **510**) and may weight them using different models. The weighting may be prescriptive or customizable. When prescriptive, weighting may be based on some industry standard attribution models that may be built (e.g., W model or a U model or an "all touch" model may be used). These standard attribution models may focus on what may be the weighting percentages that may be attributed to specific touch points which may be the first interaction some customers have with the business or it may be the way that customers became a contact in CRM (e.g., these may be important touch points that users may want to add or increase weight for). In another example, weighting may use a machine learning (ML) powered model which may take in various actions that may be happening and may try to determine what may actually be the most likely property along the same customer journey (e.g., may use ML model that may be indicative of user's business process). Then, the attribution reporting may report on which may be the most valuable touch points along a customer's journey through the process till outcome (e.g., closing a deal). Custom objects may fit in with attribution reporting such that custom objects may be the output of this attribution system. The multi-service business platform **510** may include an attribution engine for providing these functionalities. The attribution engine may be able to leverage custom objects as the output, which may mean all services and/or systems described in the disclosure may also be utilized. For example, this attribution reporting may be used alongside other services such as custom object reporting. There may be workflows that may be triggered based on the attribution engine which may be a result of the fact that custom objects are the way that various data may be processed on the multi-service business platform **510**.

Dedupe-Entity Resolution Systems and Methods

Business entity databases often include entries that are duplicative or at least contain duplicative information about a common entity. Entries may be duplicative despite there being variances in entity-related details, such as spelling of a business name, missing contact middle initial, multiple business email addresses and the like. Existing techniques for entity resolution including determining entries that are

similar and that may be duplicates may prove useful for small or moderately sized entity databases. However, such techniques, such as string comparison, entity heuristics, and the like consume excessive amounts of computing resources when attempting to handle large or massively large entity databases, which are becoming more common. Determining duplicate entries in a set of entities is generally an N-squared problem (e.g., $(N*(N-1)/2)$, meaning that the number of comparisons required to determine if any two entries in an entity set are duplicates grows as a square of the count (e.g., N^2) of entries. Therefore, the number of comparisons for large entity sets of, for example, 100,000 or more entries is prohibitive (e.g., 5 billion). Resolving entities by, for example, detecting and addressing duplicate entries in business entity databases provides great benefits to a business operation. Determining duplicate or likely duplicate entries applies to databases of various sizes and may be of particular need in large databases, as the number of likely duplicates tends to increase with database size. Additionally, publicly available entity information that can readily be harvested are continuously becoming available through the expansion of use of electronic marketing, sales, advertising, social media posting and the like. Therefore, ensuring that duplicate newly harvested entities may be identified as well as ensuring that the newly harvested entities may not be mistakenly deemed to be duplicates of existing entries requires ongoing entity resolution processing (e.g., daily processing in some cases). Therefore, techniques that consume moderate computing resources for detecting candidate duplicate entity entries may be beneficial for achieving acceptable levels of entity database processing for large or very large entity databases. Such techniques are described herein and may facilitate reducing the computing resources for fully determining duplicate entries in a large or very large entity database (e.g., a large database having 100,000 entries or more) by at least five orders of magnitude or more relative to earlier techniques.

Existing approaches for determining duplicates among a set of entity entries may be useful for achieving a high degree of confidence of a likelihood of two entities being duplicates. However, as noted above, the computing resource costs of existing approaches limit existing approaches to small and moderately sized entity sets (e.g., sets with a few thousand or fewer entries). One such approach involves generating heuristics for each entry and processing those heuristics to determine likely duplicate entries. While heuristics are referenced in this disclosure as an example existing approach, any other approach that provides high-confidence duplicate detection, optionally with both low false negative and low false positive results may be readily used as a basis for determining likely duplicate entries.

In embodiments, techniques for determining a candidate set of likely duplicate entries may rely, at least initially, on a duplicate determination approach to train a set of artificial intelligence entity resolution models (e.g., including entity deduplication models). When these trained models are combined with the further techniques described herein, determining a candidate set of likely duplicate entries may reduce the computing resources consumed by existing approaches, thereby enabling, deduplication of massively large entity databases in a scalable manner.

The entity resolution methods and systems of entity deduplication described herein may include various degrees of technical complexity that, when applied over time, may achieve a fully synthesized artificial intelligence approach to entity resolution through deduplication.

Referring now to an example implementation, FIG. 45 shows the example environment 500 including, in embodiments, the multi-service business platform 510 having an entity resolution system 566. As shown, the entity resolution system 566 may communicate with various systems, devices, and data sources according to one or more embodiments of the disclosure.

Referring to FIG. 51, an entity resolution system 5100 (e.g., an artificial intelligence-based entity resolution system) for entity resolution through deduplication is shown. In example embodiments, the entity resolution system 566 may be the entity resolution system 5100. Elements of the entity resolution system 5100 presented in FIG. 51 will be described in greater detail below. Each entity may be described by and/or include, or reference one or more entity features. For example, in a Customer Relationship Management (CRM) system, there may be a contact entity. The contact entity may have or be associated with one or more of the following example entity features: first name, last name, address, email address, age, company name, location, and the like. In example embodiments, entity entries (e.g., entity entries 5106) may be received. The entry entities 5106 may be stored in the multi-service business platform 510 (e.g., stored in an entity database of the platform 510). In example embodiments, the entry entities 5106 may be objects obtained from the storage system 550. The objects may be Customer Relationship Management (CRM) objects. In some examples, the objects may be core objects (e.g., core objects 634) or custom objects (e.g., custom objects 632 that may be defined in the multi-service business platform 510). Some examples of these objects, as described herein, may include but may not be limited to a contact object, a prospect object, a marketing object, a services object, a company, a ticket (e.g., customer service ticket), a product, a deal, any other object or entity associated with activities or relationships of an organization with its current and prospective customers, and the like.

In example embodiments, the entity resolution system 5100 may include an entity encoding module 5102 that may receive an entity entry, may extract one or more entity features from the received entity entry, and may encode each of at least a portion of the one or more entity features as a vector (e.g., a multidimensional entity feature vector) suitable for use by an artificial intelligence system (e.g., a neural network system). In example embodiments, the entity features may be object properties that may be associated with the core object or custom object. The entity encoding module 5102 may encode the entity features into a vectorized representation, for example, text strings, identifiers, numbers, Boolean connectors, and the like of each entity feature (e.g., each element of the entity feature) of each entity of the entities (e.g., entity entries 5106). In example embodiments, a name feature of a first entity may be encoded into a first multidimensional feature vector of the first entity and an address feature of the first entity may be encoded into a second multidimensional feature vector of the first entity. The entity encoding module 5102 may reference a feature encoding source 5104, which may include one or more feature encoding schemes. In some examples, the encoding scheme (e.g., entity feature encoding scheme) may be a text encoding scheme (e.g., Universal Sentence Encoder (USE) type of encoding scheme, FastText word-centric encoding scheme, and the like). The entity encoding module 5102 may select an encoding scheme from the encoding source 5104 and apply the selected encoding scheme (e.g., the USE scheme) to the feature(s) of the entity entries 5106. The type of entity encoding scheme used may

include encoding schemes that may be based on at least one of text, sentence(s), phrase(s), and/or word(s)). Independent of the type of entity encoding scheme used, a result of the encoding performed by the entity encoding module **5102** may be a vector with a value that may be specific to each entity feature of an entity in the entity entries **5106**. While the USE encoding scheme is an exemplarily referenced type of encoding scheme, other vector-encoding schemes or approaches (e.g., other text string vector-encoding schemes) may be used. In example embodiments, the text encoding scheme (e.g., feature encoding scheme) may not be limited to a commercially available scheme. As an example, the text encoding scheme may be produced or generated through use of one or more artificial intelligence approaches. In example embodiments, use of the USE encoding scheme may be instructive for further teaching the methods and systems of artificial intelligence-based entity resolution through entity data set entry deduplication described herein. As an example of use of a particular configuration of the USE encoding scheme, each entity feature processed with the particular configuration of the USE encoding scheme may result in or produce a 512-element feature vector. Other configurations of the USE encoding scheme may produce feature vectors with fewer or greater quantity of elements. Other encoding schemes applied to a given entity feature may result in a different size feature vector. In some examples, the feature vector may be referred to as a name vector (e.g., for a name entity feature), an address vector (e.g., for an address entity feature), etc. (e.g., where each unique feature may relate to a different vector).

In example embodiments, the entity encoding module **5102** may provide its output entity feature vector(s) to an encoding reduction module **5108** (e.g., that may use a trained neural network and/or trained entity deduplication model). In example embodiments, the encoding reduction module **5108** may be implemented to leverage a neural network (e.g., a Siamese neural network) or a suitable model that may be trained to produce a reduced entity-specific vector by processing the feature vector(s) associated with a specific entity. In example embodiments, the reduced entity-specific vector produced from the encoding reduction module **5108** may be suitable for further processing to generate a numeric value indicative of a likelihood that the entity that the reduced entity-specific vector represents may be a duplicate of another entity that is similarly represented by a corresponding reduced entity-specific vector. In other words, the reduced entity-specific vector facilitates determining, for any pair of entities, if the pair of entities may likely be duplicates. In example embodiments, the further processing may include a matrix processing module **5110** may receive the reduced entity-specific vector(s) output by the encoding reduction module **5108**. The matrix processing module **5110** may organize the received reduced entity-specific vectors as an entity feature matrix (e.g., two-dimensional entity feature matrix, two-dimensional matrix, entity matrix, entity-specific vector matrix, 2D matrix). In some example embodiments, this two-dimensional (2D) entity feature matrix may be a structured list of the reduced entity-specific vectors indexed by entity, such that a reduced entity-specific vector representative of an entity appears on a single row of the 2D entity feature matrix. The matrix processing module **5110** may produce a transposed version of the entity feature matrix (e.g., transposed 2D entity feature matrix) such that rows and columns may be swapped. The matrix processing module **5110** may further multiply the entity feature matrix with its transposed version, such as through a dot-product (e.g., Dp) process, to

produce a companion matrix comprising numeric values indicative of a likelihood that each pair of entities represented in the entity feature matrix may be duplicates. In other words, the companion matrix may hold values indicative of a likelihood that all pairwise combinations of entities in the entity feature matrix are duplicates. Example embodiments of an entity feature matrix and a companion matrix are depicted in FIG. **57** that is described in the disclosure. In example embodiments, when all entities in the set of entities **5106** are represented in the entity feature matrix, the companion matrix may hold a value indicative of a likelihood that all pairwise combinations of entities in the set of entities **5106** may be duplicates. In example embodiments, the entity resolution system **5100** may include a likely duplicate candidate selection module **5112**. The likely duplicate candidate selection module **5112** may use the likelihood values from the companion matrix for pairs of entities to select entity pairs (e.g., candidate duplicate entities for a business entity) for further processing. In one example, the selected entity pairs may be a result of a selection of top ten pairs (e.g., top ten pairs having the highest likelihood values). In example embodiments, a duplicate determination module **5114** (e.g., may also be referred to as a duplicate refinement module or a duplicate refinement/determination module) may produce a set of the selected entity pairs for further processing. In example embodiments, a duplicate entity resolution module **5116** may process the selected entity pairs (e.g., top ten pairs) with one or more automated entity comparison algorithms, human operators, and artificial intelligence systems to determine which of the selected entity pairs represent one entity and therefore may be deemed to be duplicate entries (e.g., may also be referred to as common entities). In example embodiments, two entities may represent one entity (e.g., may represent the same business, contact, product, and the like) when the only difference is a phone number feature (e.g., where the one entity may be referred to as a common entity). Two entities may represent one entity when, for example, an entity name feature, entity address feature, and entity primary phone number feature match, independent of any lack of matching of other features. In yet another example of two entities representing one entity, an entity name feature (e.g., a business name of the entity) may match while an address feature may not match (e.g., a regional hospital may have several locations for serving patients).

FIG. **52** shows an example entity dedupe setup/training process **5200** according to example embodiments. In an initial phase of the entity dedupe setup/training process **5200**, entity deduplication-specific artificial intelligence models may be prepared using machine learning. In example embodiments, entity deduplication artificial intelligence models may be trained using a training set of entity data **5202** for which duplicate status may be known. In other words, a duplicate status of each pairwise combination of entities represented in the entity data **5202** may be known (e.g., precomputed) so that when any pair of entities from the training set of entity data **5202** is presented for training, a corresponding duplicate status may be referenced to facilitate entity deduplication artificial intelligence model training. In example embodiments, the training set of entity data **5202** may include one or more of duplicate entities, near duplicate entities, and/or non-duplicate entities. Corresponding duplicate status for each pair of entities may be included in the training set of entity data **5202** and/or may be stored external to the training set of entity data **5202**.

In example embodiments of the training process **5200**, each pairwise combination of entities (e.g., referred herein to

as a pair or a pair of entities or entity pair) in the training set of entity data **5202** may be processed through the training process **5200**. A merge evaluator **5204** may receive a pair of training entities (e.g., pair (A,B)) from the training set of entity data **5202** (e.g., entries of training entities that may refer to training entities that were entered into the platform) and may produce a corresponding duplicate entity indication **5216** (e.g., P (merge) value for (A,B)), referred to herein as Pmerge and/or p-merge, that may reflect the duplicate entity status for the pair of entities received from the training set of entity data **5202**. For example, the merge evaluator **5204** may generate a Pmerge value for a pair of training entities from the training set of entity data **5202**. As described in the disclosure, this duplicate pair status value may be referred to herein as a Pmerge value (or a "Pmerge"), which may be a probability of seamless merging of the two entities (e.g., the two entities being duplicates). In example embodiments, a duplicate detection approach, such as the use of heuristics or string matching may be used by the merge evaluator **5204** to determine the Pmerge value. The training process **5200** may be repeated for each pair of entities in the training set of entity data **5202**. Therefore, for each pair of entities in the training set of entity data **5202**, the corresponding duplicate entity indication **5216** (e.g., Pmerge value) may represent a probability that the pair may be duplicates. For simplicity, this duplicate entity indication **5216** value (e.g., Pmerge value) may be computed to be in a range from about 0 to about 1. The probability of the two entities being duplicates may correspond to the duplicate entity indication **5216** (e.g., the Pmerge value). In example embodiments, a Pmerge value of 1 may represent a 100% probability that the two entities may be duplicates, whereas a Pmerge value of 0 may represent a 0% probability that the two entities may be duplicates. This Pmerge value for a pair of entities (e.g., entry entities) may be used as a label in training an artificial intelligence entity deduplication model to facilitate determining likely duplicate entries. As an example use of a label, a Pmerge value may be input to a machine learning process as a control against which an accuracy of an entity resolution model (e.g., an entity deduplication model) may be measured.

In example embodiments, each entity in the training set of entity data **5202** (e.g., training data set) may include any of multiple values in one or more features (e.g., first name, last name, address, email address, age, company name, location, and many others). In example embodiments, the entity dedupe setup/training process **5200** may include a training entity encoding module **5206** that may be configured and operate comparably or similarly to the entity encoding module **5102** of FIG. **51**. The training entity encoding module **5206** may encode the training set of entities **5202** (e.g., training entity data) into one or more entity feature vectors **5208** per entity (e.g., where each entity may be encoded into entity feature vectors). These encoded entity feature vectors **5208** may be stored for each training entity (e.g., training entity entry in the training set of entity data **5202**) in a machine learning training encoded entity feature vector data set (e.g., entity feature vector data set **5208**). In example embodiments, a neural network configured for entity resolution may reduce the one or more of entity feature vectors for each respective entity to a single N-dimensional entity-specific vector. The methods and systems of entity resolution through entity deduplication described herein may reduce the processing required to produce at least a manageable entity duplicate candidate set for large or massive entity databases. In example embodiments, entity deduplication may produce an accurate indi-

cation of a likelihood of any two entities being duplicates. Therefore, the neural network may be configured to reduce the number of entity feature-specific vectors for each entity down to a single entity-specific vector (e.g., single 256 element entity-specific vector). This reduction in dimensions may facilitate reducing computation requirements since a substantively smaller number of dimensions to be processed may achieve a lower computation load per entity pair. Generally, models with more nodes may potentially improve model performance (e.g., ability to reproduce a function that the model emulates), although that performance may likely plateau at some point such that further increases in nodes may not be economical (e.g., the additional computing costs for a model with a count of nodes that has plateaued may provide insignificant improvement in performance). In example embodiments, processing increases as a number of nodes in the model increases. In a non-limiting example, doubling a count of nodes in a model may correspond to roughly doubling the computing effort (e.g., time for a given processor). Also, time to train the model may increase as a count of nodes in the model increases. Additionally, a required amount of training data may be greater for models with higher node counts. In example embodiments, a count of nodes of an artificial intelligence model (e.g., artificial intelligence entity deduplication model) may be set to be a power of 2 (in examples $2^8=256$) to make computation more efficient. In a non-limiting example of a sample technique for reducing dimensions for entity deduplication, a tower neural network (e.g., Siamese tower neural network) may be configured with multiple input nodes (e.g., 3,072 input nodes for reducing six (6) 512 feature-specific vectors). In this example, the tower neural network may reduce these 3,072 inputs to a single entity-specific vector of about 256 output nodes (e.g., a 256 element entity-specific vector). The number of entity vectors, size of each entity vector, and the resulting entity-specific vector size may vary from these examples. Therefore, other combinations of inputs and output nodes may be contemplated and included herein.

In example embodiments, an overview of a feedback portion of the training process **5200** may include retrieving feature vectors for a pair of entities (mathematically represented as pair (A,B)) from the feature vector storage **5208**, such as by a machine learning process **5210**. The feature vectors for the pair (A,B) may be processed through an artificial intelligence system **5212**. Optionally, the machine learning processing **5210** may provide the retrieved feature vectors to the artificial intelligence system **5212**. The artificial intelligence system **5212** may generate an output duplicate likelihood value **5220** for the pair (A,B). In example embodiments, the artificial intelligence system **5212** may employ a dot-product function (e.g., Dp) when producing the duplicate likelihood value **5220**. A training error determination module **5214** may determine an error value **5218** for the two entities (A,B) by processing the duplicate likelihood value **5220** with the duplicate entity indication **5216** (e.g., a precomputed indication of a likelihood that the pair (A,B) are duplicates). This error value may be fed back to the machine learning process **5210** where it may be matched with the corresponding entity feature vectors for pair (A,B). The machine learning process **5210** may use the feedback **5218** to train the artificial intelligence system **5212** to produce the duplicate likelihood value **5220** that approximates the corresponding duplicate entity indication value **5216** (e.g., minimizes the error value **5218**). In example embodiments, all pair-wise combinations of enti-

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ties represented in the feature vector storage **5208** may be processed at least one time for training of an artificial intelligence system **5212**.

As described in the disclosure, the methods and systems of entity resolution (e.g., of the entity resolution system **5100**), such as artificial intelligence-based deduplication may benefit from being trained by feedback. In example embodiments, the entity dedupe setup/training process **5200** may include a training error determination module **5214** that may generate error feedback as an error value **5218** (e.g., a duplicate entry-pair error value). For example, the training error determination module **5214** may determine the error as the absolute value of the difference between a result of the dot product function of the artificial intelligence system **5212** and the Pmerge value for the pair (e.g., |Dp-Pmerge|). A machine learning training process **5210** may receive the error value **5218** (e.g., from the training error determination module **5214**) as feedback. In example embodiments, the machine learning training process **5210** may provide machine learning for training an artificial intelligence system **5212**, optionally comprising a neural network to generate vectors to facilitate entity deduplication. The training process **5200** may include the machine learning training process **5210** retrieving the one or more feature vectors for a pair of entities **5222** (e.g., mathematically represented as entity pair (A,B) of the entity pairs) from the entity feature vectors **5208**. In example embodiments, as shown in FIG. **52**, the machine learning training process **5210** may train an artificial intelligence system **5212** responsive to the entity pair **5222** and the corresponding error value **5218** (e.g., the machine learning training process **5210** may adjust the artificial intelligence system **5212** such as by adjusting weights of a neural network of the artificial intelligence system **5212**, optionally to minimize the corresponding error value **5218**). Training of the artificial intelligence system **5212** may include adjusting weights and the like of an entity deduplication model of the artificial intelligence system, such as an entity deduplication model within a neural network. In examples, the artificial intelligence system **5212** may, at least in part through use of the entity duplication model, facilitate entity resolution through deduplication of entities (e.g., output the duplicate likelihood value **5220**). The training error determination module **5214** may mathematically compare the duplicate entity indication **5216** (e.g., Pmerge value) for each entity pair in the training set of entities with the generated duplicate likelihood value(s) **5220** for each entity in each training set pair from the artificial intelligence system **5212** (e.g., generates the duplicate likelihood value(s) **5220**). In examples, the generated duplicate likelihood value **5220** may be mathematically expressed as Dp for (A,B). In example embodiments, the artificial intelligence system **5212** may employ a vector dimension reducing neural network to produce a reduced dimension entity-specific vector for each entity in the pair (A,B) that represents the feature vectors for the corresponding entity in the pair (A,B). The artificial intelligence system **5212** may further include a vector processor that may employ a dot-product function (e.g., Dp) to produce a dot-product value as a duplicate likelihood value (e.g., the duplicate likelihood value **5220**) for each pair of training reduced dimension entity-specific vectors generated from the training set of entity data **5202**. An objective of the training may be to produce dot-product pair values (e.g., the duplicate likelihood value **5220**) that approximate the Pmerge value for each entity pair in the training set of entity data **5202**. Therefore, as described for the example embodiments of FIG. **51**, the entity feature vectors may be gener-

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ated in such a way that their dot product may be generally a value in the range of about zero (0) to about one (1). In an example, the training error determination module **5214** may generate the training error value **5218** by taking an absolute value of a difference of the duplicate likelihood value **5220** and the corresponding duplicate entity indication **5216** (e.g., corresponding Pmerge value). The corresponding Pmerge value may be for the same pair of training set entries (e.g., training data set) that were used by the artificial intelligence system **5212** to produce the duplicate likelihood value **5220**. The machine learning process **5210** may receive the training error value **5218** as feedback to be used for adjusting aspects of the artificial intelligence system **5212**, (e.g., adjusting weights and the like of a neural network). An objective of the feedback process may be to produce entity feature vectors that, when processed through the dot-product function, may produce a value that may approximate a corresponding duplicate entity indication **5216** (e.g., Pmerge value) with improved quality. In example embodiments, an entity deduplication model may be updated based on a machine learning system applying the training error (e.g., training error value **5218**). In embodiments, the entity deduplication model may be updated to reduce and/or minimize the training error value **5218**.

In example embodiments, training of the neural network may proceed by a training system that may feed or input pairs of training entity vectors through the neural network and may use a portion of the output produced by the neural network, or another value derived from the neural network output as feedback to affect the learning or training of the neural network. Further, while a Siamese neural network may be used as an example of a type of neural network suitable for the technology expressed herein, a single tower neural network may be used in another example, with each entity being processed sequentially through the single tower neural network. Using the single tower neural network may involve adjustments to the overall system in that each entity-specific vector produced may need to be stored for subsequent duplicate detection processing and/or feedback during training. Similarly, a multi-tower approach or multi-neural network approach (e.g., more than two towers or more than two neural networks) may be applied with any number of identically configured neural networks being used to process the training set of entities (e.g., the training set of entity data **5202**). In some example embodiments, for a training entity set of N entries (where N may refer to any number entries or any range of numbers of entries), as many as N neural networks may be used, with each of the neural networks processing a corresponding entry. The quantity and type of neural network may be determined by factors other than those relevant for determining duplicate entity entries, such as available computing resources, time available for the training, and the like.

FIG. **53** shows an example set of operations of a method for training an entity deduplication model **5300** according to some example embodiments of the disclosure. For example, at **5302**, pairs of training entities in a training set may be processed through an entity duplicate detection process, such as one based on heuristics and the like that may produce and record a probability of an entity duplicate value for each pair (e.g., a Pmerge value). For example, at **5302**, the Pmerge value may be generated for pairs of training entities. At **5304**, a text-to-vector encoding module may generate vector representations of one or more features for the entities in the training set (e.g., entity feature text to vector of one or more features for the entities). A neural network (e.g., a Siamese tower neural network) may be

configured, at **5306**, to produce reduced complexity entity-specific vectors that may be suitable for generating, via dot-product vector processing, a value that may be comparable to the Pmerge value generated at **5302** (e.g., the Siamese neural network may be configured at **5306**). At **5308**, the neural network (e.g., Siamese neural network) may be used to reduce vector pairs. For example, at **5308**, pairs of entity feature-specific vectors (e.g., one set of feature-specific vectors for each entity in the pair of entities) may be processed through the Siamese neural network configured at **5306** to produce a pair of reduced complexity entity-specific vectors (e.g., one entity-specific vector for each entity in the pair of entities). At **5310**, pairs of reduced vectors may be multiplied. For example, at **5310**, the pair of reduced complexity entity-specific vectors produced at **5308** may be processed with a dot product function, thereby generating an entity duplicate likelihood value for the pair of entities. At **5312**, an error value may be generated. For example, at **5312**, the duplicate likelihood value may be processed (e.g., compared) with the corresponding Pmerge value for the entity pair produced at **5302** to produce an error value. In example embodiments, each error value may be produced from generating an absolute value of a difference of the duplicate likelihood value and the corresponding Pmerge value. In example embodiments, the error value may be one or more of a compound value, a mean value and a range, a standard deviation, a ranking, a percentage of difference, normalized values, an absolute difference and a count of occurrences, a difference from a prior error value, a multi-dimensional vector, and the like. At **5314**, error value(s) may be used as feedback to train the neural network (e.g., Siamese neural network). For example, at **5314**, the error value(s) may be returned to a machine learning training system that may facilitate training an artificial intelligence system for entity resolution, including at least the neural network being configured at **5306**. This process may be repeated for all pairs of entities in the training set of entities and with varying neural network configurations.

In example embodiments, artificial intelligence methods and systems may be used to replace, with substantially similar accuracy, an example high computing load entity deduplicating scheme. An example flow that uses artificial intelligence as a proxy for a high computation demand process, such as text string matching and/or heuristics, may include a front-end text encoder (e.g., Universal Sentence Encoder), a middle stage trained neural network (e.g., a trained Siamese neural network), and a back-end merge indicator function (e.g., dot-product). This approach may process pairs of entities efficiently, one pair at a time and may further be scaled to handle any quantity of pairs concurrently. Scaling may be accomplished by, for example, replicating portions of the system, such as the middle stage trained neural network.

An example system configuration and data flow of this approach are shown in FIG. **54** and FIG. **55**. FIG. **54** shows a training system and process for entity deduplication **5400** according to example embodiments. This training system and process may include entity deduplication operations that may have a set of entity feature encodings **5402** to be processed for determining duplicate entities. The set of entity feature encodings **5402** may be grouped into entity feature encoding groups, where an entity feature encoding group (e.g., feature group) may represent features of the entity. In example embodiments, a Siamese twin tower neural network **5404** may receive a feature group of entity feature encodings from the set of entity feature encodings **5402** (e.g., feature encodings) for a pair of the entities, where

one group of feature encodings may correspond to one entity per tower (e.g., tower A or tower B in the Siamese neural network **5404**). The Siamese neural network **5404** may produce a first reduced vector **5406** (e.g., vector A) from tower A (e.g., of the Siamese neural network **5404**) and a second reduced vector **5408** (e.g., vector B) from tower B (e.g., of the Siamese neural network **5404**), representing a reduced complexity vector per entity. In example embodiments, a dot product process **5410** (e.g., using a dot product module) may process the reduced vectors **5406** and **5408** to obtain a dot product **5418** (e.g., Dp) of the reduced complexity vectors. This dot product may represent, at least in part, a likelihood that the pair of entities processed by the Siamese neural network **5404** are duplicates. A Pmerge lookup module **5414** (e.g., Pmerge lookup module) may retrieve one or more Pmerge values **5420** (e.g., Pmerge values may be a control for the training) from a set of Pmerge values **5412** that may correspond to the pair of entities for which the current reduced vector pair was produced by the Siamese neural network **5404** (e.g., produced by Siamese neural network towers A and B). The dot product(s) **5418** and Pmerge value(s) **5420** may then be processed as described herein (e.g., processed by the training error module **5214** in FIG. **52** as described in the disclosure) to generate an error value **5416** for each pair of training entities (e.g., dot product error value for use by machine learning). As shown in FIG. **54** and described elsewhere herein, the dot product error value may be computed as an absolute value of a difference between value Dp **5418** and value Pmerge **5420** (e.g., resulting in |Dp-Pmerge| or an absolute of Dp-Pmerge). This error value **5416** may be returned to a machine learning training system as feedback to improve learning.

FIG. **55** shows a flow that may correspond to the backend merge indicator process **5500** (e.g., entity dedupe for the backend) referenced in the disclosure according to some example embodiments. A duplicate threshold filter **5502** (e.g., likely duplicate threshold filter or duplicate probability threshold filter) may reference a likelihood of duplication threshold value to filter an artificial intelligence system-generated dot-product duplicate probability value **5504** (e.g., AI-derived pair duplicate probability) for each pair of entities. In example embodiments, a dot-product duplicate probability value **5504** (e.g., Dp (pair n)) may be comparable to, for example, Dp **5418** of FIG. **54** or as otherwise described herein. The filter **5502** may limit entity pairs for further processing to those pairs that may exceed a duplicate probability/likelihood threshold value (e.g., more likely to be duplicates) and may store the filtered pairs in a subset of likely duplicate entities **5506** (e.g., a subset of likely duplicate entities from the set of entities to dedupe). Optionally rather than using a numeric probability threshold, a fixed number of the pairs with the highest probability value may be passed on for further processing. In example embodiments, a final duplicate determination process **5508** (e.g., final determination of duplicates for each pair of entities in the subset) may organize and process this subset of likely duplicate entities **5506** by optionally combining computer automated entity comparison functions and human entity comparison operation(s) to determine for each pair of likely duplicate entities **5506** if the two entities in each pair are duplicates. In example embodiments, the final duplicate determination process **5508** (e.g., embodied as a duplicate entity determination module) may classify each entity of the pair as one of duplicate of the other entity of the pair or a non-duplicate of the other entity of the pair (e.g., each entity in the pair is classified as either duplicate or non-duplicate

of the other). The backend merge indicator process **5500** may include a dedupe action at **5510** (e.g., embodied as a duplicate entity resolution module) in which an action is taken in response to final duplicate determination process **5508**. In example embodiments, a dedupe action taken at dedupe action at **5510** (e.g., and optionally by a duplicate entity resolution module) may include deleting one or more of the pair of duplicate entries, merging features of the duplicate entries, and the like. The dedupe action taken at **5510** may be performed automatically by a computer processor that has access privileges to a database that includes the duplicate entries. A dedupe action taken at **5510** may include saving the classification into a duplicate entity classification log for later processing, such as by an operator and the like.

FIG. **56** is a flow chart that shows a set of operations of an example process for performing artificial intelligence-based entity deduplication **5600**. At **5602**, text to vector encoding may occur. For example, at **5602**, entities may be processed with a text-to-vector encoding module to produce a set of vectors representing text features (e.g., name, address, and the like) of an entity. At **5604**, artificial intelligence vector reduction may occur. For example, at **5604**, the resulting entity feature-specific vectors for at least one of the entities may be processed through a trained artificial intelligence system to produce a reduced complexity entity-specific vector that may be suitable for the methods of dot-product-based entity deduplicating described herein. In example embodiments, the trained artificial intelligence system may include a Siamese neural network that may process sets of vectors for pairs of entities. At **5606**, reduced vectors may be stored. For example, at **5606**, the reduced complexity entity-specific vector(s) may be stored for later optional use. At **5608**, pairs of vectors may be multiplied. For example, at **5608**, pairs of reduced complexity entity-specific vectors produced from the neural network may be processed with a dot product function, thereby producing an entity duplicate likelihood value for the pair of entities. At **5610**, entity pairs may be selected, and duplicates may be determined. For example, at **5610**, based on the value of the entity duplicate likelihood value, the entity pair may be processed with an entity comparison module that may use one or more of heuristics, string comparison, and the like to determine whether the pair of entities may be duplicates. At **5612**, the process **5600** records which entities may be duplicates. For example, at **5612**, a result (e.g., duplicate or non-duplicate) may be recorded for each entity pair. Duplicate entities, as determined by the process **5600** of FIG. **56** may be addressed by one or more human operators through use of one or more user devices, such as by deleting duplicates, merging duplicates, and the like (e.g., received as instructed commands from users via user devices).

In example embodiments, the computing required for checking combinations of entity pairs in larger entity data sets may be further simplified. One example approach may include processing, substantially in parallel, all entities in a large entity database to identify a candidate set of likely duplicate entries. This simplified computing technique or approach may be enabled by use of the artificial intelligence entity feature encoding processes and systems described in the disclosure. As described in the disclosure, a trained neural network may be used to generate entity-specific vectors that, when processed as pairs through a dot-product process, may generate a value indicative of a probability of the two entities being duplicates.

FIG. **57** shows examples of one or more entity-specific vector matrices (e.g., entity feature matrix) and a companion

matrix. An example of a further simplification of computation requirements may be to arrange the entity-specific vectors generated from the trained neural network for a set of N entities (e.g., where "N" may refer to any number of entities) to be deduplicated into a first two-dimensional matrix (A) at **5702**, wherein each column at **5704** in entity-specific vector matrix A at **5702** may represent one element value in the entity-specific vector and each row **5706** in entity-specific vector matrix A may represent a corresponding entity. A number of columns in matrix A may correspond to a number of elements/values in the entity-specific vector. In example embodiments, for a 256-value entity-specific vector, entity-specific vector matrix A may be constructed with 256 columns; for a 64-value entity-specific vector, entity-specific vector matrix A may be constructed with 64 columns. In example embodiments, the entity-specific vector matrix comprises an array of M columns wherein M corresponds to the number of values in the entity-specific vector and N rows where N corresponds to a count of the entities to be deduplicated. In the example entity-specific vector matrix A at **5702**, an entity-specific vector matrix includes five columns (e.g., five elements in each entity-specific vector) and four rows (e.g., four entities). The generated entity-specific vector value for each entity may be entered in the cells of a row for the entity in corresponding columns. The entity-specific vector matrix A may be copied and transposed (e.g., matrix B at **5708**), and the two matrices (e.g., matrix A and the matrix transposed copy) may be multiplied to produce a duplicate-likelihood matrix D at **5710** (e.g., companion matrix) with pair-wise dot-product values for each entity pair appearing in the cells. Therefore, cell D (A,B), for example, may hold a value that may be indicative of the likelihood that entities A and B are duplicates. Likewise, cell D (A,C), for example, may hold a value that may be indicative of entities A and C being duplicates.

The resulting cell values in each of the rows of the duplicate likelihood matrix D (e.g., companion matrix) may be sorted from highest to lowest value while maintaining reference to the two entities for which the duplicate indication value(s) may apply. Independent of sorting matrix D at **5710**, a subset of the values in each row representing likely duplicates of the entity for which the row may be labeled, such as the top n values (e.g., top ten) or, for example, only values above a duplicate likelihood threshold and the like may be selected for further processing. In the example companion matrix D at **5710**, a most likely duplicate of entity A may be entity C due to the cell value at D (A,C) being greater than other entries in row A. Also, in the example companion matrix D at **5710**, the pair of entities that are most likely to be duplicates are entities C and D due to the cell value at D (C,D) being higher than any other cell value in the matrix D. at **5710**. While the companion matrix D at **5710** shows values in all cells, in example embodiments, values along the diagonal and in cells below the diagonal may be unfilled. Values along the diagonal may represent only one entity (e.g., D (A,A) only represents entity A). Values below the diagonal may be duplicates of corresponding cells above the diagonal (e.g., D (B,A) may be a duplicate of D (A,B)).

The further processing may include processing the corresponding pairs of entities through another type of entity duplication detection function, such as the heuristic or string-matching functions generally described in the disclosure. In this example further processing, only the selected subset of entity pairs may be processed through this other potentially more accurate duplicate detection process. As a result, duplicate entries may be automatically found within

a relatively large set of entities with much less computing load than applying this other duplicate detection process to all pairwise combinations of entities. For example, rather than requiring N-squared computations (where N represents a numerical count of entities) for determining which entities may be duplicate, only those entity pairs that may exhibit a likelihood of being duplicates, based on for example a value in companion matrix D at 5710, may be processed with the relatively larger computation demanding functions. In example embodiments, the further processing may include presenting the selected subset of entity pairs to a user (e.g., via a user device) who may use various digital and/or visual comparison tools and/or judgment to determine which, if any, of the selected set of entity pairs may be duplicates.

FIG. 58 shows an example set of operations of a process for performing artificial intelligence-based entity deduplication 5800. At 5802, text to vector encoding may occur. For example, entities may be processed with a text-to-vector encoding module at 5802. In example embodiments, the text-to-vector encoding at 5802 may be comparable to entity encoding described in the disclosure, such as for the entity encoding module 5102, training entity encoding module 5206, entity text to vector at 5304, and the like that may produce entity feature-specific vectors. At 5804, there may be artificial intelligence vector reduction. For example, at 5804, the resulting entity feature-specific vectors may be processed through a trained artificial intelligence system to produce a reduced complexity entity-specific vector suitable for dot-product-based entity deduplicating. At 5806, the reduced complexity entity-specific vectors may be arranged in a matrix. For example, at 5806, the reduced complexity entity-specific vectors may be arranged in an entity interim matrix. In example embodiments, the entity interim matrix may be transposed to obtain a transposed matrix. At 5808, the artificial intelligence-based entity deduplication process 5800 may multiply the interim matrix with the transposed matrix. For example, at 5808, the transposed matrix and the interim matrix may be multiplied, thereby producing a matrix comprising an entity duplicate likelihood value for each pair of entities (e.g., a companion matrix). At 5810, entity pairs may be selected, and duplicates may be determined. For example, at 5810, the top N duplicate candidates for each entity may be selected and processed, as pairs, through an entity comparison module that may use heuristics, string comparison, and the like to determine whether the pairs of entities may be duplicates. At 5812, the artificial intelligence-based entity deduplication process 5800 may record which entities are duplicates. For example, at 5812, a result (duplicate or non-duplicate) may be recorded for each entity pair. Duplicate entities, as determined by the artificial intelligence-based entity deduplication process 5800, may be addressed by deleting duplicates, merging duplicates, and the like.

In example embodiments, an entity resolution system may optionally use fully synthesized artificial intelligence models to reduce a massive entity database to a manageable candidate set of likely duplicates. Such an entity resolution system may also perform duplicate entity detection with accuracy comparable to existing high computing resource demand techniques, such as string comparison, heuristics, and the like. Such an entity resolution system may be constructed by using a machine learning-trained artificial intelligence process for determining which pairs of a selected subset of entity pairs in a companion matrix are to be classified as duplicates (e.g., an artificial intelligence-based backend deduplication process). This entity resolution system may be constructed by replacing, at least for pro-

duction embodiments, the high computation backend (e.g., heuristic, string comparison, and the like) process applied, for example in 5812 of the artificial intelligence-based entity deduplication process 5800, with a machine learning-trained artificial intelligence backend deduplication process. In example embodiments, predetermining which companion matrix entities may be duplicates and which companion matrix entities may be non-duplicates may be used to train a duplicate detection artificial intelligence system that may be used to automatically determine which entities are to be classified as duplicates.

FIG. 59 shows an example set of operations of a process for performing fully automated entity deduplication process 5900 according to some example embodiments of the disclosure. At 5902, text to vector encoding that produces entity feature-specific vectors for entity features may occur. For example, at 5902, entities may be processed with a text-to-vector encoding module. At 5904, artificial intelligence vector reduction may occur. For example, at 5904, the resulting entity feature-specific vectors may be processed through a trained artificial intelligence system to produce a reduced complexity entity-specific vector suitable for dot-product-based entity deduplication. The process 5900 may arrange the reduced complexity entity-specific vectors in a matrix at 5906. For example, the reduced complexity entity-specific vectors may be arranged in a duplicate entity interim matrix at 5906. In example embodiments, the entity interim matrix may be transposed to obtain a transposed matrix. At 5908, the process 5900 may multiply the interim matrix with the transposed matrix. For example, at 5908, the transposed matrix and the interim matrix may be multiplied, thereby producing a matrix comprising an entity duplicate likelihood value for each pair of entities (e.g., a companion matrix, such as companion matrix D at 5710). At 5910, artificial intelligence may determine duplicates from candidates. For example, at 5910, the top N duplicate candidates (those most likely to be a duplicate) for each entity may be selected and processed, as pairs, through the duplicate detection artificial intelligence system. At 5912, the process 5900 may record which entities may be duplicates. For example, at 5912, the duplicate detection artificial intelligence system may determine, for each entity pair, if they are duplicates or not (e.g., non-duplicates) and may record the information. Duplicate entities, as determined by the process 5900, may be addressed by performing one or more actions, such as by deleting duplicates, merging duplicates, and the like.

Referring back to FIG. 45, in example embodiments, the multi-service business platform 510 may include an event system 522 that may be configured to monitor for and record the occurrence of events. In example embodiments, the multi-service business platform 510 may include a payment system 524 that processes payments on behalf of clients of the multi-service business platform 510. In example embodiments, the multi-service business platform 510 may include a reporting system 526 that may allow users to create different types of reports using various data sources associated with a client's business (e.g., including data sources corresponding to custom objects defined with respect to the client's business and/or any core objects that are maintained with respect to the client's business). In example embodiments, the multi-service business platform 510 may include a conversation intelligence (CI) system 528 that may be configured to process recorded conversations (e.g., video calls, audio calls, chat transcripts, and/or the like). In example embodiments, the multi-service business platform 510 may include a workflow system 562 that may relate to controlling, configuring, and/or executing of workflows in

the platform **510**. In example embodiments, the workflow system **562** may include a custom workflow actions system **564** that may communicate with various systems, devices, and data sources according to one or more embodiments of the disclosure. The custom workflow actions system **564** may provide users with the ability to create custom workflow actions (e.g., custom code actions).

Custom Events

Referring back to FIG. **45**, in example embodiments, the multi-service business platform **510** may include an events system **522** that is configured to monitor for and record an occurrence of events. Events may include various types of events that relate to different facets of a client's business. In example embodiments, events may include a standard set of events that are tied to the CRM system **502**, the CMS **508**, the multi-client service system **1600**, a payment system **524**, and/or the like. For example, with respect to the CRM system **502**, example events may include instances of potential customers (which may be also referred to as "contacts") visiting a client's website, a potential customer being contacted by a salesperson, a potential customer contacting the client, an offer being made to the potential customer, a deal being closed with the potential customer, an offer being rejected by the potential customer, a form being submitted by a contact, and the like. With respect to the CMS **508**, examples of events may include an electronic message being sent to a contact, a contact opening an electronic message, a contact clicking on a link in an electronic message, a contact responding to an electronic message, a contact sharing a link contained in an electronic message, and the like. Examples of service-related events may include an initiation of a new ticket by a customer, a customer service representative contacting the customer, an action being taken with respect to the ticket, an issue corresponding to the ticket being resolved, a ticket being closed, feedback being provided by a customer, and the like. With respect to the payment system **524**, example types of events that may be related to payments may include a credit card payment being processed, an Automated Clearing House (ACH) payment being processed, a one-time payment being processed, a payment being made for a subscription, a deposit being processed, a refund being processed, a payment being declined, and the like.

In example embodiments, events may be recorded in event records that may be stored in an event data store(s) **558** (e.g., may include an events database) within a storage system **550**. In example embodiments, an event record may provide a log of one or more types of events that may be recorded in relation to respective instances of a particular type of object. For example, a "contact" event record may be one type of event record. The "contact" event record may track events with respect to a contact such that the contact event record may track instances of the contact receiving an email, the contact responding to an email, the contact viewing the client's website, the contact submitting a form to the client, the contact purchasing a product, the contact initiating a ticket with respect to a purchase made by the contact, the contact requesting a refund, the contact being issued a refund, the contact leaving a review, the contact closing a ticket, and/or the like. In example embodiments, the event record may store or reference each instance of each type of event. Thus, if a contact is sent multiple emails, a corresponding instance of a contact event record may identify each time the contact was sent an email from the client (e.g., from a salesperson or an automated message). Furthermore, in some example embodiments, an event record may store properties for each type of event that may be

tracked. In example embodiments, the properties of an event record may include a timestamp (e.g., date and/or time), an event identifier, one or more associations to different objects, and any other suitable properties (e.g., default properties or customer defined properties that may be tracked throughout the platform). Continuing the "contact" event record example, the properties of each instance of an email sent to the client may include a timestamp indicating when the email was sent and an association that indicates from whom the email was sent. Similarly, the contact event record may track each of the contact's purchase events, whereby each purchase made by the contact may be recorded in the contact event record with a time stamp and an association to an instance of the product and/or service that was purchased by the contact.

FIG. **60** illustrates an example schema of an event record **6000**. In the illustrated example, the event record **6000** may include a reference **6002** to a primary object **6030**. The reference **6002** to the primary object **6030** may indicate the object (e.g., core object or custom object) to which the events being tracked in the event record **6000** correspond. For instance, the primary object **6030** may be a contact object, a company object, a ticket object, a deal object, a product object, a service object, a custom object defined by a client, and/or the like. Thus, when a new instance of an event record **6000** is created in relation to an instance of an object (e.g., contact, ticket, deal, company, or the like), the created event record instance may be associated with a respective object instance corresponding to the instance of the primary object **6030** (e.g., contact, ticket, deal, company, or the like). In this way, events relating to the instance of the core object, the custom object, or the like may be tracked in the newly created event record instance.

In example embodiments, an event record **6000** may log N different types of events (e.g., first event type **6004**, second event type **6010** . . . , Nth event type **6016**), where each type of event may represent different respective actions that may occur with respect to the subject of the primary object of the event record. Put another way, an event record **6000** relating to a particular thing (e.g., a contact, a company, a salesperson, a ticket, a deal, a product, a service, a yoga studio, a car dealership, a subscription, or the like) may define trackable actions that may occur with respect to the thing. For example, with respect to a contact, a non-limiting example of event types (e.g., event type 1, event type 2, event type N) that may be logged in a "contact" event record **6000** may include, but may not be limited to: visits to the client's website by a particular contact, the contact being contacted by a sales person, the contact contacting the client, an offer being made to the contact, a deal being closed with the contact, an offer being rejected by the contact, a form being submitted by the contact, the electronic message being sent to a contact, the contact opening the electronic message (e.g., email), the contact clicking on a link in an electronic message, the contact responding to an electronic message, the contact sharing a link contained in an electronic message, the contact engaging on a social media post, the contact interacting with an online advertisement, the contact opening a ticket, the contact leaving feedback, the contact closing a ticket, the contact requesting a refund, the contact being issued a refund, and the like.

As may be appreciated, different types of event records **6000** correspond to different types of objects and will therefore include different, but potentially overlapping sets of event types. For each event type, the event record **6000** instance may log (e.g., store or index) instances of each time a respective type of event occurred with respect to an

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instance of the primary object. Thus, in some example embodiments, the event records **6000** may track the lifecycle of an object (via instances thereof), such as a customer lifecycle (or “contact lifecycle”), a product lifecycle, a deal lifecycle, a ticket lifecycle, and/or the like. In example

embodiments, the instances of respective event types logged in an instance of an event record may be stored in a sequential manner (e.g., according to timestamps of the instances of a particular event type). For example, in some example embodiments, each event type in a respective event record may include a data structure such as an array or linked list, where each respective instance of the event type may be represented as an entry in the data structure.

In example embodiments, each event type (e.g., event types **6004**, **6010**, **6016**) logged in an event record **6000** may include a set of event properties (e.g., event properties **6006**, **6012**, **6018**) and a set of zero or more associations (e.g., associations **6008**, **6014**, **6020**). In example embodiments, the event properties may include a timestamp that indicates a date and/or time at which an instance of the event occurred. Additionally, the event properties of a particular event type may further include additional metadata, such as a geolocation corresponding to the event and/or other contextual data. In example embodiments, the properties of each event instance may also include a unique event ID that identifies the event instance from other event instances of the same type, such that a single event may be referenced by multiple event record instances. In example embodiments, the associations of a particular event type may indicate how a type of event may relate to two or more different objects. For example, a contact may purchase a vehicle from a client business (e.g., a car dealership). Thus, an event instance (e.g., tracked in an event record instance) corresponding to the purchase of the vehicle may be associated both with a contact object and a vehicle object, whereby both a contact event record and a vehicle event record may both log the event instance (e.g., based on a timestamp and/or a unique event ID). In this example embodiment, the contact event record instance associated with the customer that purchases the vehicle may reference the contact object instance of the customer as the primary object **6030** and may reference the vehicle object instance of the purchased vehicle as an associated object for at least one event type **6004** (e.g., a “purchased” event type). Similarly, the vehicle event record instance corresponding to the purchased vehicle may reference the vehicle object instance of the vehicle as the primary object **6030** and may reference the contact object instance of the customer as an associated object **6032** for at least one event type **6010** (e.g., a “purchased by” event type). There may be any N number of other associated objects **6034** (related to Nth event type **6016**). In this way, the object’s relating to a client’s business may be tracked and reported on in a unified and/or customizable manner, such that the respective event records may represent a single source of truth throughout the various lifecycles associated with the client’s business.

In some example embodiments, the customization system **520** (e.g., FIG. **45**) may be configured to allow a user to define event records that correspond to custom objects (e.g., a yoga studio event record or a vehicle event record) and/or associate existing event records to custom objects (or certain properties thereof). In some of these example embodiments, the user may select a custom object and may define the types of events that may occur with respect to the custom object, such that the custom object may be the primary object of the new event record. In some example embodiments, the user may select from a predefined list of event types, such as

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“sale” event, “ticket opening” event, “payment” event, or the like. Additionally or alternatively, the user may define new event types and may define one or more properties of each event type. In these example embodiments, the user may define the associations that give rise to the event or other actions that occur that may trigger the event. In some example embodiments, a user may define a new type of event and may associate the event with one or more types of event records (e.g., contact event record, company event record, ticket event record, product event record, or the like). For example, each event may occur with respect to one or more object types that may be associated with each respective event. In example embodiments, the user may define a set of event property types that may be tracked and may define a data type (e.g., int, string, Boolean, decimal, date, time, or the like) for each type of event property. In example embodiments, the customization system **520** may update the ontology **630** (e.g., ontology datastore) to reflect the new type of event and/or may update schemas of one or more object event records to reflect the new event type that is being tracked with respect to one or more respective object types. In some example embodiments, the customization system **520** may receive one or more event source selections from the user, where the event sources may indicate a third-party data source that may be monitored for event instances. In example embodiments, the customization system **520** may further configure a custom listening thread that listens for a new event type, for example, from an event source identified by the configuring user. In these example embodiments, a listening thread may monitor an event source (or multiple event sources) for event instances, whereby the event system **522** (discussed in the disclosure) may update the event data store(s) **558** and/or the instances knowledge graph **640** (e.g., of the knowledge graph(s) **556**) when such events are detected. For example, the listening thread may listen for actions, such as a contract being e-signed by a contact and/or a payment being received from the contact to update the “contact” event record (and any other implicated event records).

In example embodiments, the event system **522** may be configured to monitor for and log events that may occur in relation to a set of objects (e.g., core objects and/or custom objects) defined by the user. As discussed, a user may define a custom object (or multiple custom objects) and may define the events that occur with respect to the custom object and other objects (custom objects, core objects, and/or standard objects) that may be related to the client’s business via the customization system **520**. In defining the custom objects, the user may also define the properties of the custom objects and sources for the types of data represented by the properties. In example embodiments, event instances corresponding to the custom object may be detected and recorded when data is received from a specific source. In some of these example embodiments, the event system **522** may instantiate and execute listening threads that may listen to various services **6132** of the multi-service business platform **510** (e.g., payment services, ticketing services, CRM services, CMS services, machine-learning services, geolocation services, identity resolution services, telemetry services, and/or the like) and/or external event sources for particular types of data and may update the event data store(s) **558** and/or the knowledge graph **640** based on the data obtained from the various services **6132**. In example embodiments, other services **6132** may include services **530** of the multi-service business platform **510**, other services within the multi-service business platform **510**, and/or other services that may be external from the platform **510**. For example, if

a user has defined a vehicle object (e.g., vehicle custom object **804**) and configured the event system **522** to log “test drive events” (**824**, **824'**) with respect to the vehicle custom object **804** (and a contact object or a “customer” custom object **802**), when a contact test drives a particular vehicle, the event system **522** may log the test drive event (e.g., date, time, test drive length, and/or the like) and associate the event with a vehicle event record instance corresponding to the particular vehicle. In this example, the event system **522** may implement an event listening thread that receives a data stream from a CRM service, a workflow service, and/or an enterprise resource planning (ERP) service and listens for specific types of data that may indicate that a particular person has test drove a particular make of vehicle (e.g., sales person entries or notes entered into a CRM or an automated entry generated by a workflow service). It is noted that while the foregoing example relates to a custom object, the event system **522** may track events associated with a default set of objects in a similar manner. In some example embodiments, the event system **522** may log the test drive event in the knowledge graph **640** via an association between the vehicle instance and the contact instance. It is noted that the other services **6132** may also include third party event sources that may be accessed via application programming interfaces (APIs) and/or webhooks.

In some example embodiments, the event system **522** may expose (e.g., provide access to an API of) an event reporting service (e.g., as shown in FIG. **61**) that may execute listening threads on behalf of one or more clients. For example, in the example of FIG. **61**, an event reporting service **6130** may monitor a set of data streams (e.g., other services **6132** of the multi-service business platform or external event sources **6134**) to identify events that may be recorded in the event data store(s) **558**. In example embodiments, the event reporting service **6130** may be configured to execute a set of listening threads on behalf of a client. In example embodiments, a listening thread may monitor one or more data streams for one or more specific types of data that may be generated or otherwise provided by other services **6132** of the multi-service business platform **510** and/or external event sources **6134**. In these example embodiments, the listening thread may analyze the data streams and may extract specific information that may indicate the occurrence of an instance of a specific type of event and one or more properties of an event instance. In example embodiments, the listening thread may report the event and the event properties thereof to the event reporting service **6130**, which in turn may update the event data store(s) **558** based on the recorded event instance. In some example embodiments, the event reporting service **6130** may determine the types of objects that may be associated with the particular event instance (e.g., based on the event type of the particular event) and may identify the particular object instances that may be associated with the particular event instance based on the event properties of the event instance. In these example embodiments, the event reporting service **6130** may associate the event instance with respect to event record instances that may be associated with the particular object instances. Further examples of an event reporting service **6130** are described with respect to the reporting system discussed in the disclosure. It is noted that in some example embodiments, the event reporting service **6130** may be used in connection with other services of the platform **510**. For example, the event reporting service **6130** may trigger workflows performed by a workflow service and/or other actions that may be performed by other systems of the multi-service business platform **510**.

Custom Reporting

Referring back to FIG. **45**, in example embodiments, the multi-service business platform **510** includes a reporting system **526** that allows users to create different types of reports (e.g., custom reports) using various data sources associated with a client’s business, including data sources corresponding to custom objects defined with respect to the client’s business and/or any default objects that may be maintained with respect to the client’s business. One benefit of custom reporting may be that users are provided greater leeway in framing an inquiry of a report, which may allow for a much wider range of data to be used to generate reports. For instance, a user may wish to view a report that depicts how many customers that have made multiple purchases with the client have submitted customer service tickets. Furthermore, the user may break down the results of the report by the type of ticket (e.g., delayed shipping, malfunctioning product, missing parts, or the like). In example embodiments, the reporting system **526** may leverage a schema layer that may operate on a client’s respective ontology to determine data types of object(s) and event properties and associations between the client’s objects and/or events to support the generation of custom reports in a flexible and schema-agnostic manner.

In example embodiments, the reporting system **526** may provide a graphical user interface (GUI) to a user via a user device (e.g., user device(s) **570**), where the user may select a reporting type from a set of different reporting types that may be supported by the reporting system **526**. Examples of reporting types may include, but may not be limited to, single object reporting, cross-object reporting, funnel reporting, attribution reporting, and/or custom reporting. In single object reporting, the user may select a single object/data source to generate a report on, such that the report may be generated only from properties and events that are associated with the selected object. In cross-object reporting, the user may select two data sources (e.g., objects) and the user may select a report based on the properties and/or events implicated by the intersection of the two data sources. In funnel reporting, a user may select a specific funnel report from a curated set of funnel reports. As previously mentioned, a funnel report may be generated based on a predetermined set of sequential database queries (e.g., show contacts that may be engaged with the client via a website, then purchased a product/service, and then wrote a review). In attribution reporting, a user may select an attribution report from a curated set of different types of attribution reports. As described in the disclosure, different types of attribution reports may visualize a correlation of revenue to different marketing efforts of the client. According to some example embodiments of custom reporting, a user may select a set of data sources from a set of available data sources and the reporting system **526** may generate a wide array of non-opinionated user configured reports based on the set of data sources and the intersections thereof. In example embodiments, the available data sources from which a user may select for a custom report may include data sources relating to default objects, custom objects, and/or events that may be tracked on behalf of the client.

As mentioned, a custom report may include a visualization (or “chart”) that may be generated by the reporting system **526** based on a set of data sources selected by the user, where the selected data sources may include object properties and/or event properties maintained by the multi-service business platform **510** on behalf of the client. Examples of the visualization may include, but may not be limited to, a graph, a histogram, a pie chart, and/or the like.

In some example embodiments, when a user selects a custom reporting feature, the reporting system 526 may determine and display a set of supported data sources, from which the user may select a primary data source and zero or more secondary data sources. In example embodiments, a primary data source may be an intended focus of the report and the secondary data sources may correspond to objects and/or events that may be directly related to the primary data source. In some example embodiments, the reporting system 526 may allow a user to select up to a maximum number of secondary data sources (e.g., up to five secondary sources). For example, a client may use the multi-service business platform 510 in connection with a car dealership and may have an ontology that may define a “vehicle” custom object that may be related to a “deals” object and a “contact” object, as well as one or more event records (e.g., a vehicle event object that may define a set of different events that may be recorded with respect to a vehicle instance and/or a contact event record that may include a “sale event” that is also associated with a vehicle that is purchased during the sale event). In this example, a user may wish to view a report that graphs how many vehicles were sold each month in the previous year and may wish to further view how many of each type of model were sold or the color breakdown for the vehicles sold each month. In this example, the user may select a “vehicle” data source corresponding to a “vehicle” custom object and may further select one or more sales-related secondary data sources. In this example, if the user wishes to view reports relating to sales of vehicles by color or by model, the user may select the vehicle data source as the primary data source and then may select deals as a secondary data source, which may allow the user to configure the desired report.

In example embodiments, the reporting system 526 may present a data set selection GUI to a user that allows the user to select one or more data sources. In example embodiments, the reporting system 526 may prune the available data sources from which the user may select the secondary data sources based on the selected primary data source. In example embodiments, the reporting system 526 may determine which data sources may be selected as secondary data sources based on the objects and/or events that may be related to the primary data source. As the user selects the primary data source (e.g., vehicles), the reporting system 526 may constrain the available data sources that may be selected as a secondary data source based on the associations between the primary object corresponding to the primary data source and any other objects and/or events that may be related with the primary object. For example, when the user selects a “vehicle” data source, the reporting system 526 may identify associated data sources vis-à-vis the objects and events that may be associated with the vehicle object, e.g., deals, ads, contacts, tickets, and the like. In this way, a user may request to create reports that may visualize properties and/or events that may relate to vehicles, deals (e.g., sales/leases of vehicles), and/or contacts. In response, the reporting system 526 may then constrain the available data sources that the user may select for customization of the report such that the constrained set of data sources may be data sources that correspond to the objects and/or events that may be directly related to the primary data source (e.g., as defined in the client’s ontology 6140). The user may then select a set of secondary data sources from the constrained set of data.

In response to the user selecting a primary data source and zero or more secondary data sources, the reporting system 526 may then provide a mechanism (e.g., a report configu-

ration GUI) that allows the user to configure a custom report. In configuring the custom report, the user may select a type of chart (or “visualization”) that may depict the data in the report, assign properties from the selected data sources to different components of the selected chart, aggregate/group one or more of the properties, and/or apply filters to the assigned properties. In response, the reporting system 526 may generate and execute a report plan based on configuration parameters provided by the user, where the report plan may include pre-query database operations (e.g., join operations on one or more columns of two data sources or the like), query database operations (e.g., select clauses, where clauses, and/or the like), and/or post query database operations (e.g., sum operations, average operations, count operations, and/or the like). In some example embodiments, a report plan may be a structured query language (SQL) query that may include pre-query database operations, query database operations, and/or post query database operations. In these example embodiments, the reporting system 526 may generate an SQL query and may query the data warehouse (e.g., a Snowflake® instance) using the SQL query, where the results of the SQL query may be used to generate visualized reports.

In example embodiments, the user may select a type of chart from a set of available charts. Examples of the types of charts that a user may select from include, but may not be limited to: histograms, bar charts, pie charts, doughnut charts, line graphs, sunburst diagrams, or the like. The reporting system 526 may then determine the reportable properties that may be assigned to the different components of the selected chart. In example embodiments, the reporting system 526 (or a supporting service) may leverage the data types of each type of property that may be referenced by the selected data sources to determine the properties that may be reported on and/or assigned to a respective component of a selected chart. In some example embodiments, the user may instruct the reporting system to group or aggregate certain properties, so that the results of grouping or aggregation may be assigned to certain components of the selected chart. For instance, if the user selects a bar graph, the reporting system 526 may determine the types of data that may be assigned to the x-axis and the y-axis of the graph, such that a user may only assign certain types of properties to the x-axis or the y-axis. As may be appreciated, in a bar graph the y-axis may be typically a numerical value, while the x-axis may be typically a category (e.g., day, month, individual, grouping, or the like). Thus, any property that may be assigned to the y-axis may be a numerical value (e.g., an integer, a decimal, or the like), while the property that may be assigned to the x-axis may be categorical. It may be noted, however, that some properties that are not necessarily numerical may be aggregated to obtain a numerical value that may be assigned to the y-axis. For instance, even though aggregated sales volume data may not be explicitly maintained on behalf of the client, sale events from a particular period of time may be aggregated to determine a number of sales that occurred over the period of time. Continuing the car dealership example above, the user may assign “count-vehicle sales” to the y-axis and “sale date-monthly” to the x-axis, such that the resultant chart may depict the number of vehicles that were sold each month that the sales data tracked. In some example embodiments, the reporting system 526 may determine that sale events include a date property, which may be a “date” data type. In example embodiments, the reporting system 526 may apply a rule that permits dates to be grouped by week, month, quarter, or year. In example embodiments, the reporting system 526 may apply another rule that says

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events may be aggregated. Thus, in an example report, the reporting system 526 may group sale events by the month in which the sale events took place and then, for each month, the sale events may be aggregated to determine the number of vehicle sales for the month. In example embodiments, the reporting system 526 may constrain the available data types that may be assigned to a chart component based on the data type that was assigned to another component of the chart. For instance, in selecting “sale date-monthly”, the reporting system 526 may constrain the available data types to “number of vehicles sold”, “number of vehicles leased”, “number of vehicles test driven”, and the like. Had the user selected “salesperson” for the x-axis, the reporting system 526 may have constrained the available data types to a different but potentially overlapping set of data types (e.g., “number of vehicles sold”, “number of customers engaged”, “number of complaints”, “emails sent”, or the like).

In example embodiments, the user may select a property type to break down the depicted information in a chart. For example, if the user has assigned “vehicle sales-monthly” to the x-axis and “number of vehicles sold” to the y-axis, the user may select “model type” to break down the data, such that each bar in the depicted bar graph may be broken down by vehicle model to show how many of each vehicle model was sold during that month. In some of these example embodiments, the reporting system 526 may determine the associations between different types of properties to determine whether a particular component may be broken down by a particular property type.

In example embodiments, the reporting system 526 may allow users to apply filters to the properties applied to chart components. In example embodiments, the types of filters that may be applied to a particular property may be based on the data type of the property. For example, the user may filter the “sales dates-monthly” to sales before Jan. 1, 2020 or between the dates of Jan. 1, 2019 and Dec. 31, 2019. In another example, the user may filter a “number of sales” property by model type or by vehicle color.

In example embodiments, the reporting system 526 may generate a report plan based on report configuration parameters 6150 (e.g., chart selection, property assignments, filters, aggregations, and/or groupings) provided by the user. The report plan may include a set of database operations that may be performed with respect to the selected data sources, including pre-query operations (e.g., join operations on one or more columns of two data sources or the like), query operations (e.g., select clauses, where clauses, and/or the like) and/or post query operations (e.g., sum operations, average operations, count operations, and/or the like). The reporting system 526 may then provide report data to the requesting user device (e.g., user device(s) 570) based on the results of the report plan. In example embodiments, the report data may include a template for displaying the selected chart and data for parameterizing the selected chart based on the results database operations and any post-query processing. The user device may then display the selected chart to the user. Once the report is generated, the user may continue to filter the report and/or change the properties that may be assigned to the chart components. In example embodiments, the reporting system 526 may allow a user to share and/or export the report.

FIG. 61 illustrates an example configuration 6100 of a reporting system 526 in relation to other components and services of a multi-service business platform 510. The reporting system 526 may include additional or alternative components without departing from the scope of the disclosure. In example embodiments, the reporting system 526

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may include or leverage a report configuration layer 6120, a schema layer 6122, a report processing layer 6124, and a visualization layer 6126. It is noted that the various layers of the reporting system 526 may be embodied as computer-readable instructions that may be executed by a set of one or more processors. In example embodiments, the reporting system 526 may be configured as part of a microservice architecture, such that the reporting system may be accessible by other services of the multi-service business platform 510 and may access various services of the multi-service business platform 510. For example, in some example embodiments, the schema layer 6122 of the reporting system 526 may include computer-executable instructions that may be configured to request schema services from an object schema service or schema service (e.g., of the multi-service business platform 510) or another schema service (e.g., object schema service external to the platform 510) when facilitating the generation of reports by the reporting system 526. In some example embodiments, the schema layer 6122 may be configured to perform the object schema services independently. For purposes of explanation, descriptions of operations and functions performed by the schema layer 6122 may include operations that may be performed by the object schema service (e.g., object schema service 620) and/or the schema layer.

In example embodiments, the reporting system 526 may interface with a storage system 550 of the multi-service business platform 510, which may be accessed to configure and serve reports generated by the reporting system 526. In example embodiments, the storage system 550 may store the event data store(s) 558 that stores event instances and/or event records, an ontology data store 630 that stores client ontologies 6140 of various clients of the multi-service business platform 510, a knowledge graph 640 (e.g., instances knowledge graph 640) that may store instances of the client's objects and events, and a multi-tenant data store(s) 552. As shown, in example embodiments, the ontology data store 630 and the instances knowledge graph 640 may be part of the knowledge graph(s) 556 of the storage system 550. In example embodiments, the reporting system 526 may be configured to access a data warehouse instance (e.g., a Snowflake® data warehouse instance), whereby the data warehouse instance may store a query-able knowledge graph 640 which the report processing layer 6124 may access and query in response to determining a report plan. The data warehouse instance may be stored in the storage system 550 of the multi-service business platform 510 or may be hosted by a third party (e.g., Snowflake Inc.) and accessed via a network using an API or other suitable mechanism.

In some example embodiments, an event reporting service 6130 may monitor for and record events in the event data store(s) 558. In example embodiments, the event reporting service 6130 may be configured to execute a set of listening threads on behalf of a client. In example embodiments, a listening thread may listen for one or more specific types of data that may be generated or otherwise provided by other services 6132 of the multi-business platform and/or external systems. For example, a listening thread that may be configured with respect to a yoga studio may listen for payments that are processed on behalf of the yoga studio. Each time a payment is processed on behalf of the yoga studio, a payment service (e.g., as exposed by the payment system 524) may issue a payment notification of the payment to the event reporting service 6130. In this example, the payment notification may indicate the customer that made the payment, the offering (e.g., item(s) that was/were purchased and

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a price), and any other suitable data. In response, the event reporting service **6130** may generate a payment event based on the received notification. The payment event may indicate a payment ID that uniquely identifies the payment and a date and time at which the payment was processed (e.g., a time stamp). In example embodiments, the payment event may include additional data (e.g., a payment type, a sale location, or the like). In example embodiments, the event reporting service **6130** may store the event in the event data store(s) **558** and may associate the payment event with one or more event records in the knowledge graph **640**. For example, the payment event may be associated with a contact event record corresponding to the customer who made the purchase, a product or service event record corresponding to the item that the payment was made for, and/or any other relevant event record. In this way, the event record may be associated with multiple objects within the client's data in a unified manner, thereby facilitating complex reporting features. In example embodiments, the event records that are stored in the event data store(s) **558** and the associations thereto may be copied into the data warehouse instance **6128**, such that the data warehouse instance **6128** may be queried by the reporting system **526** when processing report requests. While the foregoing example of events centers around payments, the event reporting service **6130** may be configured to report other types of events as well.

As the various services **6132** of the multi-business platform **510** write certain types of data to the knowledge graph **640** or other databases, the event reporting service **6130** may generate event records corresponding to the data being written. Examples of events may include, but may not be limited to, an electronic message being sent to a contact, a contact opening an electronic message, a contact clicking on a link in an electronic message, a contact responding to an electronic message, a contact sharing a link contained in an electronic message, a contact visiting the client's website, the initiation of a new ticket by a customer, a customer service representative contacting a contact, an action being taken with respect to the ticket, the issue corresponding to the ticket being resolved, a ticket being closed, feedback being provided by a contact, a credit card payment being processed, an ACH payment being processed, a one-time payment being processed, a payment being made for a subscription, a deposit being processed, a refund being processed, a payment being declined, and the like. Furthermore, to the extent that a client has configured custom objects, the event reporting service **6130** may monitor for and record events relating to the custom objects.

In example embodiments, the report configuration layer **6120** may provide one or more GUIs that allow a user to request and configure different types of reports. In some of these example embodiments, the report configuration layer **6120** may instruct the schema layer **6122** to identify the data sources that may be selected to generate a report, and once the user has selected the data sources and the type of report that will be generated, the types of properties may be selected by a user to parameterize the report. In example embodiments, the report configuration layer **6120** may receive a set of report configuration parameters **6150** from the user via the one or more GUIs. In example embodiments, the report configuration parameters **6150** may include a chart type selection that indicates the type of chart the user may like to create, one or more property assignments that may indicate the types of properties (e.g., "reporting properties") that may be assigned to respective components of the chart, break down selections that may indicate one or more properties (e.g., "break down properties") with which

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a chart is further broken down by, aggregation definitions that may indicate the manner by which certain properties in the chart are aggregated and/or grouped (e.g., by day, week, month, or year, by individuals or teams, by name, by letter, or the like), and/or filter definitions that may define the manner by which certain properties are filtered (e.g., before, after, or between certain dates, accounts having no less than X in revenue, or the like). In response to receiving the report configuration parameters **6150** to the report configuration layer **6120** (e.g., via the report configuration GUIs discussed in this disclosure), the report processing layer **6124** may generate and execute a report plan based on the report configuration parameters **6150**. In example embodiments, the report plan may include a set of database operations that may be performed with respect to the selected data sources, including pre-query operations (e.g., join operations on one or more columns of two data sources or the like), query operations (e.g., select clauses, where clauses, and/or the like), and/or post query operations (e.g., sum operations, average operations, count operations, and/or the like). The report processing layer **6124** may output the results of the report plan to the visualization layer **6126**. The visualization layer **6126** may output visualization data **6152** to the user device (e.g., user device(s) **570**) from which the request to generate the report was received. In example embodiments, the visualization data **6152** may include a chart template to render the selected chart and a set of visualization parameters corresponding to the results of the report plan, whereby the template may be parameterized with the visualization parameters to render the chart to reflect the results of the report plan.

FIGS. **62A**, **62B**, **63A**, **63B**, and **63C** illustrate an example set of GUIs that may be used to request, configure, and display a report according to some example embodiments of the disclosure. The operation of the reporting system **526** is further explained with respect thereto.

FIG. **62A** illustrates an example of a report request GUI **6200** according to some example embodiments of the disclosure. In example embodiments, the report configuration layer **6120** may present the report request GUI **6200** to a user in response to the user requesting to create a report. In the illustrated example, the user may select a type of report from a set of available report types **6202**. In this example, the user may select from single object reporting (e.g., "Single Object"), cross-object reporting (e.g., "Cross Object"), funnel reporting (e.g., "Funnels"), attribution reporting (e.g., "Attribution"), and/or custom reporting (e.g., "Custom Report Builder"). In the example, the user has opted to create a custom report.

FIG. **62B** illustrates an example of the report request GUI **6200** according to some example embodiments of the disclosure. In response to the user selecting the custom reporting feature, the report request GUI **6200** may present a set of available data sources **6204** from which the user may select one or more data sources **6204** from which the report may be generated. As discussed, the available data sources **6204** that a user may select from correspond to objects and/or events that correspond to the client (e.g., a company that employs the user). In example embodiments, the schema layer **6122** may determine the available data sources **6204** for a client associated with the user. In some of these example embodiments, the schema layer **6122** may access an ontology **630** associated with the client to identify the objects and event records that may be maintained on behalf of the client and may determine the available data sources **6204** based on the identified objects and event records, including any custom objects or custom events defined or

otherwise incorporated by the client. In the illustrated example, the user may select the data sources **6204** corresponding to CRM-related objects (e.g., “contacts”, “companies”, “deals”, “tickets”), CRM-related events, marketing related objects (e.g., “landing pages”, “marketing email”, “website pages”, “video”, “blog”, “ads”, or the like), marketing-related events (e.g., “form submissions” or “web activities”), sales-related objects (e.g., “products”), sale-related events (e.g., “sales-activities”), service-related objects (e.g., “conversations” or “feedback”), service-related events, custom objects (e.g., “subscriptions”, “services”, “company locations”, as shown in the illustrated example) and/or custom events (e.g., “referrals”). It may be appreciated that the events and/or objects that are shown in relation to the “custom objects” heading may vary depending on the client’s configuration.

In some example embodiments, the user initially may select a primary data source and then zero or more secondary sources. In some example embodiments, the report configuration layer **6120** may receive the primary data source selection from the user via the report request GUI **6200**. In some example embodiments, the schema layer **6122** may determine a constrained set of available data sources based on the primary data source selection provided by the user. For example, in some example embodiments, the schema layer **6122** may identify the object or event corresponding to the primary data source in the client ontology **6140** and then may determine any objects or events that may be associated with the object that corresponds to the primary data source. In some example embodiments, the schema layer **6122** may include data sources corresponding to objects and/or events that may be directly associated with the object corresponding to the primary data source in the constrained set of data sources. An object may be directly associated with another object if the object is connected in the ontology to the other object via one or more associations. In example embodiments, the schema layer **6122** may determine objects that may be directly related to a selected object vis-à-vis event records that may be related to the selected object. In these example embodiments, the schema layer **6122** may identify event records that identify the selected object as a primary object and then may determine the objects that may also be associated with the event record via an event type association.

In response to the schema layer **6122** determining the constrained set of available data sources, the report configuration layer **6120** may constrain the set of available data sources, such that the user may only select data sources from the constrained set of available data sources. In the illustrated example, the user has selected the “Services” data source as the primary data source, and the report configuration layer **6120** has grayed out any data sources that may not be combined with the “Services” data source. The user may then select any number of secondary data sources from the constrained set of available data sources. In some example embodiments, however, the reporting system **526** may limit the number of secondary data sources that may be selected to a maximum number of secondary data sources (e.g., up to five total data sources). In this example, the user has selected the “Contacts” data source and the “Companies” data source as secondary data sources as shown in FIG. **62B**. Once the user has selected the secondary data sources, the user may configure one or more reports from the selected data sources.

FIG. **63A** illustrates an example report configuration GUI **6300** that may be presented to a user in response to a user selecting a set of available data sources (e.g., via the report

request GUI **6200**). In example embodiments, the report configuration GUI **6300** provides an interface by which the user may provide a set of report configuration parameters that may be used to generate a custom report. In example embodiments, a user may configure a report by selecting a type of chart to visualize the report and by assigning different properties to the various fields of the selected type of chart. For example, the user may select from a bar graph (vertical or horizontal), a line graph, a pie chart, a donut chart, a sunburst diagram, or the like. In the illustrated example, the user has selected a vertical bar graph as a chart type from a set of available chart types **6302**.

In response to the user selection of the chart type, the report configuration layer **6120** may display a set of parameterizable fields to the user that corresponds to different components **6304** of the chart. In this example, the user has selected a vertical bar graph, which includes an x-axis and a y-axis, where the x-axis depicts groupings or individual items and the y-axis depicts a numerical value. The user may then assign certain properties to different fields **6306** of the chart, whereby the user’s assignments to the chart components represent the inquiry being made by the user. Additionally, in some example embodiments, the report configuration layer **6120** may allow a user to group or aggregate certain types of properties and/or apply different filters to certain types of properties based on the data types thereof and/or the manner by which the properties are related to one another. For example, strings may be tallied into values and/or may represent different groupings. Similarly, dates may be grouped into days, weeks, months, years, days of the week, or the like. In another example, a one-to-many association between a first object to a second object may indicate that the second object may be grouped under the first object. For example, if a contact object is related to a salesperson object, contacts may be grouped under the salesperson that serviced the contacts.

In example embodiments, the schema layer **6122** may determine which properties of an object may be used to filter the object based on the data type of the properties. For example, objects having properties that may be represented in dates may be filtered based on the date of each instance of the object, such that object instances may be filtered to include only objects having a date that is before a certain date, after a certain date, after a first date and before a second date, or the like. Thus, a user may request to generate a report showing all instances of sales occurring after Jan. 1, 2020, and before Jan. 1, 2021.

In some example embodiments, the report configuration GUI **6300** may be configured to depict a reportable set of properties **6308** (e.g., the left-hand column of the report configuration GUI **6300**) given the selected data sources, whereby the user can browse for different property types to report on. In some example embodiments, the user may search by a name of the property or may select one of the various data sources selected by the user and browse through the property types corresponding to the data source. For instance, in this example, the user is browsing the property types of the “Services” data source and may select from property types such as address, consultant name, description, end date, name, object ID, service owner, service type, start date, and the like. In this example, the user has selected “Name” and “[Service] Start Date”.

In some example embodiments, the user may drag-and-drop a property type to the “Available Fields” section of the fields **6306** of the report configuration GUI **6300**. When a user adds a property type to the available fields section, the schema layer **6122** may determine the different ways the

selected property type may be reported on given the respective data type of the property and the type(s) of data that are assignable to each column component of the selected chart type. For example, the user here has selected “Start Date” and “Name”, which are explicitly defined property types, where “Start Date” may be a date data type that indicates a date on which a service was started and “Name” may be a string data type that indicates the name of a service. In example embodiments, the schema layer 6122 may be configured to identify a set of implicit properties that may be assigned to the chart components based on the data types of the selected properties and/or the relationships between the selected properties and events and/or other objects that may be related to the selected property. For instance, because the user has selected the “Name” service, the services may be tallied based on the name of each service appearing in the client’s data, and therefore a “Count of Services” property may be added to the available fields. Furthermore, as the “Name” property is a property of the “Service” object, and because the contact object and service object may, for example, be associated to the service object via a service event record (e.g., to indicate a service that may be purchased by a respective company on behalf of a company), the schema layer 6122 may determine that a user may also assign a “Count of Companies” and/or a “Count of Companies” to one or more of the chart fields. In this example, the user may assign the “Start Date” or “Name” to the X-axis and one of the “Count of Services”, “Count of Companies”, and/or “Count of Contacts” to the Y-axis, as the former properties may be categorical properties and the latter implicit properties may be numerical values. In the illustrated example, the user has assigned “Start Date” to the x-axis and has further grouped the dates by month and has selected “Count of Services” to the y-axis. In this way, the user has requested to see the number of services that were started each month.

Furthermore, in some example embodiments, the report configuration layer 6120 may allow a user to break down a selected chart based on an additional property type, thereby allowing the user to add dimensionality to a depicted chart. For example, a bar graph may be broken down according to a specified property type, such that each “bar” of a bar graph may be segmented on counts of different instances of the specified property type. As may be appreciated, a vertical bar graph may depict respective values (e.g., counts, scores, or the like) associated with respective “buckets” (e.g., date ranges, groups, persons, or the like). When a user selects to break down the chart by an additional field, each group may be broken down by tallies corresponding to the additional field. For example, the user may break down each bar in a graph by “Name”, such that each bar may be further broken down into smaller components showing the number of each type of service started in a particular month by name. It may be appreciated that different types of charts may have different fields to which properties may be assigned. In example embodiments, the schema layer 6122 may determine which property types may be assigned to the “Break down by” component of the chart based on the property type(s) that may be assigned to the main component(s) of the chart and the relationships that may be shared between the assigned property types. For instance, the “Start Date” property may be a service event property and the “Count of Services” may be a tally that is indicative of the number of services that were started by the client’s business for a given month. As both these property types may be related to a service object and “Name” is a property type of the service object that is represented by a string, the schema layer 6122

may determine that each column in the bar graph may be broken down by the “Name” property type. Conversely, because the “Count of Services” and “Count of Companies” are numerical values, the schema layer 6122 may prevent the user from being able to break down the columns by those properties.

In example embodiments, the schema layer 6122 may determine which data types may be assigned to different fields based on the type of chart and the data types of the properties of the selected data sources. A data type may refer to the manner by which a particular property is depicted. For instance, a property may be represented by a string, an integer, a decimal, a date, a time, or the like. The schema layer 6122 may determine a property is a value if the data type of the property is an integer or a decimal. Additionally, the schema layer 6122 may determine that a particular property may be aggregated into a numerical value (e.g., a count) when the data type of the property is a string. For example, if the properties of a Service object include “Name”, the schema layer 6122 may determine that “Name” can be aggregated by grouping the instances of the Service object by “[Service] Name” and tallying the number of instances in each group to obtain a count of services for each type of service offered by the client, and ultimately, a total number of services. In this way, the custom report may allow a user to assign “Count of Services” to the Y axis field, despite the count of services not being an explicit property type that is defined with respect to the services object. Similarly, the schema layer 6122 may determine that a particular property may be grouped into months, weeks, years, or the like if the data type of the property is a date.

In example embodiments, the report configuration GUI 6300 allows a user to select a filter to apply to one or more of the chart components and to define one or more filter parameters, such that a selected filter may cause the report processing layer 6124 to filter the query results based on the filter parameters. In example embodiments, the schema layer 6122 may determine the types of filters that may be applied to a property based on the data type of the property. For instance, dates may be filtered to include instances that may happen before a particular date, after a particular date, between two particular dates, excluding certain dates, or the like. In another example, values may be filtered to include values that are less than a certain threshold, greater than a certain threshold, that fall within a defined range, that are equal to a number, or the like. In example embodiments, each type of filter may include a database operation template that defines one or more database operations and a set of parameterizable fields, such that parameterizable fields may be parameterized with the filter parameters defined by the user.

Once a user has defined the report configuration parameters (e.g., chart type, field assignments, groupings, aggregations, and/or filters), the report processing layer 6124 may generate a report plan, including pre-query operations (e.g., join operations on one or more columns of two data sources or the like), query operations (e.g., select clauses, where clauses, and/or the like), and/or post query operations (e.g., sum operations, average operations, count operations, and/or the like). In example embodiments, the report processing layer 6124 may generate the report plan based on the configuration parameters, one or more database operation templates corresponding to the configuration parameters, and the relationships between the properties assigned to the chart parameters (e.g., as defined in the configuration parameters). For example, the report processing layer 6124 may determine pre-query operations, such as whether two or

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more tables need to be joined based on the property types that were assigned to the selected chart components, and if so, which tables are to be joined. In response to determining that two or more tables need to be joined, the report processing layer **6124** may parameterize a JOIN clause with the names of the tables (e.g., properties) that are to be joined. The report processing layer **6124** may further generate one or more queries that may be used to query a data set (which may be the result of a JOIN operation). For example, the report processing layer **6124** may parameterize a SELECT clause that retrieves a specific set of records (e.g., rows) from one or more tables. Furthermore, if there are filters that are to be applied to a chart component, the report processing layer **6124** may generate a WHERE clause that is parameterized with one or more filter parameters. For example, if the filter is a date-based filter that filters out dates that appear before a certain date, the corresponding template may be “WHERE [Property Name]>=[FILTER_DATE]”, wherein [FILTER_DATE] is defined by the user in the filter parameters with respect to a property corresponding to [PROPERTY_NAME]. In another example, if the user has selected a property type that is grouped by a certain grouping parameter, the report processing system **6124** may parameterize a GROUP BY operation based on the user selection. In another example, the report processing layer **6124** may parameterize one or more post-query operations. For example, the report processing layer **6124** may parameterize COUNT clause templates to generate counts of a selected set of records, an AVG clause to determine an average value of a certain property in a selected set of records, a SUM clause to sum all the values of a certain property in a selected set of records, and the like.

In example embodiments, the report processing layer **6124** may generate the report plan based on the parameterized operation templates and may execute the query plan with respect to the identified data sources. In example embodiments, the report processing layer **6124** may query a copy of the client's databases in the data warehouse instance **6128** using the generated report plan. In response, the data warehouse instance **6128** may return the database query results to the report processing layer **6124**. The report processing layer **6124** may output the results to the visualization layer **6126**, which in turn may provide report visualization data to the user device (e.g., user device(s) **570**) from which the request was received.

In example embodiments, the visualization layer **6126** may retrieve a chart template corresponding to the selected chart type and may determine report data that may be depicted in the chart template. In example embodiments, the chart template may have an associated function that receives the query results and outputs a set of visualization parameters with which the chart template is parameterized. For example, in a bar graph, the visualization parameters may include the number of bars, the heading for each column, the values along the x-axis, the height of each bar, the width of each bar, the color of each bar, and the like. In another example, for a pie chart, an associated function may determine the label for each respective “slice” of the chart, an angle for each slice of the chart, and/or a color assignment for each slice of the chart. In example embodiments, the visualization layer **6126** may generate a file (e.g., a JSON file) that contains the chart template and the visualization parameters and may transmit the file to the user device of the user. In response, the user device may display the depicted report **6320** in the report configuration GUI **6300**, as shown in FIG. **63B**.

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In example embodiments, the data warehouse instance **6128** may maintain the joined data sets in memory after a requested report has been generated, such that the user may adjust one or more of the report configuration parameters during the same reporting session without having to replicate a computationally expensive table join operation. In these example embodiments, certain adjustments to the report configuration parameters may be processed much more quickly as the relevant tables have already been joined in the data warehouse instance **6128**. In adjusting one or more of the report configuration parameters, the user may select a new chart type, assign new properties to chart components (of the new chart type or the original chart type), define new filter parameters, define new aggregation parameters, and/or the like. For example, in FIG. **63C** the user has selected a pie chart from the available chart types and has assigned “Count of Companies” to the values field and “Start Date-Monthly” to the break down field as shown in the report configuration GUI **6300**. In this example, the user may wish to see the number of companies that started a service in each month. Furthermore, in this example, the user is adjusting the aggregation parameters relating to the “Start Date” property. As shown, the user may adjust the aggregation to daily, weekly, monthly, quarterly, yearly, or the like. Assuming the user does change the aggregation parameter to “Weekly” or “Daily”, the report processing layer **6124** may adjust the report plan based on the adjusted aggregation parameter and execute the adjusted report plan. The report processing layer **6124** may receive the results of the report plan from the data warehouse instance **6128** and may output the results to the visualization layer **6126**. The report visualization layer **6126** may then update the visualization of the selected chart based on the updated result (e.g., updating a pie chart report **6330** depicted in the report configuration GUI **6300**).

The GUIs depicted in FIGS. **62A**, **62B**, **63A**, **63B**, and **63C** are provided for example and not intended to limit the scope of the disclosure. The GUIs depicted herein include many embodiments of ornamental features separate and distinct from the many functional embodiments depicted and described herein.

Payments

In example embodiments, the multi-service business platform **510** may include a payment system **524** that processes payments on behalf of clients of the multi-service business platform **510**. The payment system **524** may be configured to facilitate one-time payments, recurrent payments (e.g., subscription payments), installment payments, and/or the like on behalf of clients. For example, if a client of the multi-service business platform **510** is a yoga studio, the yoga studio may offer items (e.g., yoga mats, yoga accessories) and/or individual packages for sale. Such offerings are likely to be processed as one-time payments. In this example, the yoga studio may also offer subscription plans (e.g., unlimited yoga classes for \$100 a month), which may be processed as subscription payments, according to some example embodiments of the disclosure. In this example, the yoga studio may also offer personal training packages that automatically renew once a customer has finished their last personal training session. In this example, the personal training offering may be processed as a recurring payment that may be triggered based on the occurrence of a condition (e.g., the last session being used). The payment system **524** may be configured to facilitate these different types of payments on behalf of the clients and to integrate any payment-related data and event data into the client's data stores. In this way, rich payment data and payment event

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data may be integrated with the client's CRM services, CMS services, ticketing services (e.g., customer service), and the like, such that certain payment events may trigger different workflows associated with the client's business. Additionally, the payment data and payment event data may be reported on via the reporting system **526**.

In example embodiments, the payment system **524** may be configured to generate "checkout links" corresponding to respective checkout pages for respective offerings provided by a client. In example embodiments, a checkout page may be a digital medium that is configured using a set of checkout parameters provided by the client and allows a user to initiate a payment for the respective offering that may be defined in the checkout parameters corresponding to the checkout link. In example embodiments, a checkout link may refer to a resource identifier (e.g., a uniform resource locator (URL) that links to a webpage, a deep link to a native application, and/or the like) that causes the multi-service business platform **510** to serve a checkout page to a user device when the checkout link is selected (e.g., clicked, pressed, or otherwise activated) from the user device. An offering may refer to one or more items (e.g., services, products, warranties, leases, or any combination thereof) that may be offered by the client in exchange for a set payment price. For example, a yoga studio may offer different types of classes for students, where the different classes may be priced differently. Additionally, the yoga studio may offer discounts to students that purchase different packages of classes (e.g., three different classes for a combined price of \$75), purchase multiple classes (e.g., 20% discount if a student purchases three of the same classes together), or purchase a monthly subscription (e.g., \$100 a month for unlimited classes). Thus, in this example, different offerings may include individual classes at their respective standard price, different class packages each offered at a respective price, same class bundles at a respective discounted price, and a monthly subscription at a fixed price. In this example, a user affiliated with the yoga studio may define (e.g., via a graphical user interface) a set of checkout parameters for each offering, including item(s) (e.g., product, service, warranty, property, lease, and/or combination thereof) being offered, a price of the offering, and any other additional requirements for checkout (e.g., number of payments such as one time, set number, or continuing subscription, schedule of recurrent payments (if applicable), contract signature requirements, and the like). In response to the checkout parameters, the payment system **524** may generate a checkout link that corresponds to a checkout page, whereby the yoga studio may provide the checkout link to contacts or potential contacts (e.g., in emails, ads, text messages, or any other suitable medium).

In example embodiments, a user affiliated with a client may request that a checkout link be generated for an offering via a graphical user interface (GUI) (referred to as a "checkout configuration GUI"). In doing so, the user may provide a set of checkout parameters via the checkout configuration GUI, including defining the offering. In providing the offering, the user may select one or more items (e.g., products, services, warranties, subscriptions, or the like) and may define a price for the offering. In some example embodiments, the types of items that a client offers (e.g., sells, leases, or the like) may be represented as custom objects in the client's ontology. Thus, in some example embodiments, the payment system **524** (or a schema service) may identify the types of items a client may offer based on the types of objects that are in the client's ontology. Additionally or alternatively, the checkout configuration GUI may allow the

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user to provide offering information, such as an item name, an item description, and/or media content depicting the item. In example embodiments, the user may define the price of the offering. For example, the user may define the price if it is a special offer and/or the price is not listed in the client-specific data stored by the multi-service business platform **510**. In example embodiments, the user may also define whether the payment price is a one-time price, a set of installment payments, or a subscription-based payment. In the case that the payment is an installment payment, the user may define how many payments are to be made, an amount per payment, and a schedule for the recurring payments (e.g., weekly, monthly, yearly, quarterly, beginning of the month, monthly from the date of purchase, or the like). In the case that the payment is a subscription payment, the user may define a payment schedule (weekly, monthly, yearly, quarterly, draw payments at the beginning of the month, draw payments every month on the corresponding date of purchase, or the like) and a payment amount for each scheduled payment (e.g., \$12.99 a month, \$99.99 a year, or the like). In some example embodiments, the checkout parameters may include one or more checkout requirements. Examples of checkout requirements may include a contact signing a contract before a payment may be processed or the contact agreeing to a set of terms and conditions before a payment may be processed. In these example embodiments, a user may define the checkout requirements, such as requiring a purchase contract be electronically signed by a customer prior to purchase.

In response to the user providing the configuration parameters and requesting the checkout link, the payment system **524** may generate a checkout link for the offering and may store the checkout parameters in a checkout database. In some of these example embodiments, the checkout parameters may be indexed based on the checkout link. In this way, the checkout link may reference the checkout parameters, such that when the checkout link is selected by a user, the payment system **524** may parameterize a checkout page template with client-specific data, including the checkout parameters and any client-provided content (e.g., logos, fonts, styles, layouts, salesperson information, business contact information, and/or the like).

Once a checkout link is generated, the checkout link may be embedded in any suitable medium and/or stored with the client's data. For example, a checkout link may be embedded in electronic messages (e.g., email, SMS, direct messages, and/or the like), on the client's website, in digital advertisements, in live chats, in promotional materials, in quick response (QR)-codes, or the like. In some example embodiments, the checkout link may be provided to the CMS system and/or the CRM system, such that the checkout link may be integrated into messaging workflows and event data relating to the checkout link that may be tracked (e.g., each time the checkout link is activated, each time the checkout link is sent in a message, each time the checkout link is provided in an advertisement, and/or the like). In some examples, an effectiveness of different checkout link mediums may be reported on by the reporting system (e.g., in a multi-object custom report). It is appreciated that other payment-related reports may also be generated. For example, how many times a customer opens a checkout link before initiating a purchase, how many times a checkout page is accessed, how many times a checkout page is shared, and/or the like.

In example embodiments, the payment system **524** may store the checkout parameters (e.g., in a checkout database **6440**) and may generate a checkout link that may be

associated with the checkout parameters. Thus, in some example embodiments, the payment system 524 may generate a checkout link for each respective offering, whereby a checkout link may be clicked (or otherwise selected) by a potential purchaser (e.g., from the client's website, an email, a text message, a chat, scanning a QR code, or the like) to initiate a purchase for the offering. In response to a checkout link being clicked or otherwise selected, the payment system 524 may generate and serve a checkout page (or "checkout UI") corresponding to the offering indicated by the checkout link. In example embodiments, the payment system 524 may retrieve the set of checkout parameters corresponding to the offering indicated by the checkout link and, in some example embodiments, any client-specific content (e.g., branding, fonts, layouts, or the like) corresponding to the client selling the offering, and then may serve the checkout page based on the checkout parameters and the client-specific content. Put another way, the payment system 524 may serve a checkout page that may be configured in accordance with the checkout parameters and client-specific content corresponding to the offering indicated by the checkout link. In example embodiments, checkout parameters may include, but may not be limited to, the offering (including the price), a description of the offering, a contact for the offering (e.g., a salesperson), whether the payment is a one-time payment or a recurring payment (e.g., a multi-installment payment or subscription payment), a payment schedule (if a recurrent payment) and/or the like. In example embodiments, client-specific content may include a logo of the client, the client's branding, fonts and/or style sheets provided by the client, a photograph or other media corresponding to the offering, and/or other types of content that may be presented in a checkout page. In example embodiments, the payment system 524 may be configured to facilitate credit card payments, ACH payments, cryptocurrency payments, or other suitable payment methods (e.g., PayPal, Venmo, etc.). In these example embodiments, the customer may provide payment information, such as credit card information or bank account and routing information, etc. In example embodiments, the payment system 524 may initiate a payment session with a payment processor (e.g., Stripe®, Chase® Merchant Services, Podium®, and the like) on behalf of a customer and may provide the payment session information (e.g., session ID and/or session key) to the customer's user device. Once presented with the checkout GUI, a customer may provide payment information via the checkout UI, whereby the checkout page (as executed at the customer user device) may route the payment information directly to the third party payment processor 6410 using the payment session information. In response to the payment information, the third party payment processor 6410 may transfer funds from the customer's account (e.g., credit card account if by credit card or bank account if by ACH) to an account affiliated with the client and may provide a confirmation of a successful transaction to the payment system 526 (e.g., via an exposed webhook).

In response to confirming a successful transaction, the payment system 524 may initiate one or more bookkeeping actions. Examples of bookkeeping actions may include, but may not be limited to, generating a payment event indicating a successful purchase, notifying the CRM system 502 of the sale, updating one or more properties of an object instance (e.g., deal object instance, a product object instance, a custom object instance, or the like), triggering a custom workflow, or other suitable actions. In example embodiments, the custom workflow may include scheduling a follow up communication with the customer (e.g., for ship-

ping information, for customer service information, for training, for feedback, or the like), scheduling recurrent payments, notifying an external system of the payment (e.g., to schedule a service, such as a training, a class, a construction project, professional services, or the suitable services), and/or the like. Similarly, if a payment is unsuccessful, the payment system 524 may initiate one or more bookkeeping actions corresponding to an unsuccessful payment.

In example embodiments, the payment system 524 may be implemented as an internal microservice. In this way, payments may be initiated from various services of the multi-service business platform 510 and at various points in a customer lifecycle. For example, payments may be initiated from the CRM system 502, the CMS system 508, the service system 1600, and/or from other services of the multi-service business platform 510. Checkout links to specific offerings may be embedded in websites, native applications, electronic messages (e.g., email, text message, direct messages), chats, QR codes, social media pages, and the like. Additionally, payment-related functions (e.g., subscriptions, invoicing, refunds, and the like) may be integrated into various workflows that are supported by the multi-service business platform 510. For example, a user may configure a workflow to send automated follow-up emails to buyers of a certain offering (e.g., a product or service) that has a checkout link to purchase additional offerings (e.g., product, service, warranty, or the like) after a certain amount of time after the initial purchase.

In example embodiments, the payment system 524 may generate checkout links that may be accessible to a quoting service of the multi-service business platform 510. In these example embodiments, a user affiliated with a client may configure a quote/invoice and request that may be sent to a contact via the quoting service. In response, the quoting service may generate the quote (which may or may not include an invoice) and may request a checkout link from the payment system 524. The request may include a set of checkout parameters that may be obtained from the quote. In response to the request, the payment system 524 may generate the checkout link and may provide the checkout link to the quoting service. In response to the checkout link, the quoting service may embed the checkout link in the quote/invoice and may provide the quote/invoice to the customer, where the customer may use the link to access a checkout page and initiate a payment to the client. In some example embodiments, the client may require an electronic signature from the customer before the payment may be processed. In these example embodiments, the quoting service may further send an electronically signable contract with the quote/invoice or the electronically signable contract may be accessed via the checkout page. In this way, a client may initiate an entire sale closing workflow that may include providing a quote, executing a contract, and executing a payment from a single quote via the quote and the checkout link. It may be appreciated that the payment system 524 may interface with other services of the multi-service business platform 510 without departing from the scope of the disclosure.

FIG. 64 illustrates an example configuration 6400 of a payment system 524 according to some example implementations of the disclosure. In example embodiments, the payment system 524 may be executed by a set of processors. In example embodiments, the payment system 524 may include and/or access a link generation service 6420, a checkout service 6422 (e.g., may also be referred to as checkout system), a payment execution service 6424, and a post-payment service 6426. In example embodiments, the

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payment system **524** may read data from and write data to a checkout database **6440**, a payments database **6442**, an ontology **630**, and/or a knowledge graph **640** (e.g., instances knowledge graph **640**). As shown, in example embodiments, the ontology data store **630** and the instances knowledge graph **640** may be part of the knowledge graph(s) **556** of the storage system **550**. In example embodiments, the payment system **524** may communicate with a set of internal services (e.g., services **6430**), such as messaging services, customer relationship management services, quoting services, invoicing services, and/or the like.

In example embodiments, the payment system **524** may communicate with a third party payment processor **6410**, such as Stripe®, Chase® Merchant Services, Podium®, and the like. In these example embodiments, the payment system **524** may be configured to access the third party payment processor **6410** via an API of the third party payment processor **6410**. In this way, clients of the multi-service business platform **510** may not have to integrate their own respective payment processors with the multi-service business platform **510**. Rather, the payment system **524** may manage the back-end API integration with the third party payment processor **6410** on behalf of the clients of the platform **510**. Furthermore, in some example embodiments, the payment system **524** may initiate payment sessions on behalf of customers, but may provide the payment session information (e.g., session ID and session key) to the customer device, such that the payment system **524** does not receive any payment information from customers. Rather, payment information may be provided directly to the payment processors from the customer device.

In example embodiments, the link generation service **6420** may generate checkout links corresponding to user-defined offerings. In some of these example embodiments, a user may access the link generation service **6420** via a user interface presented by the payment system **524** or another suitable system (e.g., the CRM, the CMS, the service system, or the like) and may provide a set of checkout parameters. In response, the link generation service **6420** may generate a checkout link (e.g., a URI) corresponding to the checkout parameters and may store the checkout parameters in the checkout database **6440**, where the checkout parameters may be associated with the checkout link in the checkout database **6440**. In example embodiments, the user may identify the checkout parameters, including the offering, via a checkout configuration GUI. For instance, the user may select one or more items (e.g., products, services, warranties, or the like) to include in an offering and may provide a price for the offering. In some example embodiments, the types of items that a client may offer (e.g., sells, leases, or the like) may be represented as custom objects in the client's ontology. Thus, in some example embodiments, the payment system **524** (or the schema service **6432**) may identify the types of items a client may offer based on the types of objects that may be in the client ontology **6140** (or in the ontology **630**). In this way, the user may select the items that may be included in an offering from a menu (e.g., a drop down menu). In these example embodiments, an item may include one or more properties (price, suggested price, item descriptions, or the like), such that the link generation service **6420** may obtain some of the item's properties to include in the checkout parameters. In example embodiments, the user may provide additional checkout parameters and/or client-specific content. For example, the user may define whether the payments are recurring payments, one-time payments, or paid in accordance with another payment plan. Additionally or alternatively, the user may provide the

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checkout parameters manually and/or may provide media contents corresponding to the offering, such as images or videos of the offering, which may be stored as client-specific content. In example embodiments, client-specific content that may be used for all checkout pages that may be served on behalf of the client (e.g., fonts, logos, style sheets, trademarks, stock images, or the like) may be uploaded by a user associated with the client during on-boarding or a similar configuration phase. In this way, when checkout pages are generated on behalf of a client, the checkout pages may all have the same look-and-feel. In some example embodiments, the link generation service **6420** may receive additional configuration requirements from the user. For example, the user may add a location verification requirement, an age requirement verification, a contract that must be e-signed, terms and conditions that must be agreed to, and/or the like. In some of these example embodiments, the user may upload or otherwise link a contract or a terms and conditions (e.g., terms and conditions document) to the checkout parameters, such that the contract or terms and conditions may be provided to the customer prior to checking out.

In response to the user providing the checkout parameters, the link generation service **6420** may generate a uniform resource identifier (URI) (e.g., a uniform resource locator (URL), a deep link, or the like). In example embodiments, the URI may include a domain (e.g., second-level domain, subdomain, and/or top-level domain), a path (e.g., subdirectory portion), and a tail portion (e.g., remaining portion of the URI after the path). In example embodiments, the domain of the URI may indicate the multi-service business platform **510**; the path of the URI may indicate a subdirectory that references the client checkout directory; and the tail of the URI may be a unique identifier corresponding to the checkout parameters of the offering. The link generation system **6420** may store the additional checkout parameters and/or the client-specific content in the checkout database **6440**. In some example embodiments, a client may generate multiple checkout links for the same item, such that different checkout links may correspond to different prices for the item. For instance, if a salesperson wants to offer a special discount or otherwise differing price to a customer or class of customers, the salesperson may configure the offering with the item and the alternate price, such that the resultant checkout link may reference the item at the alternate price. In this way, when the resultant checkout link may be accessed, the payment system **524** may verify that the customer attempting to purchase the item at the alternate price has permission to purchase at the price, and if so, to serve a checkout page that depicts the alternate pricing.

The link generation system **524** may output the checkout link to the service that requested the checkout link. As discussed, the checkout link may be embedded into a medium (e.g., email, text message, website, direct message, a quote, an invoice, or the like) directly by the user from the link generation GUI by the CRM system **502**, content management system **508**, the multi-client service system **1600**, a quoting service, or the like.

In example embodiments, the checkout service **6422** may serve checkout pages to user devices. In example embodiments, the checkout service **6422** may be called when a customer selects a checkout link via a user device. The checkout service **6422** may receive the tail of the checkout link and may retrieve the checkout parameters from the checkout database **6440** based on the tail of the checkout link. Additionally, the checkout service **6422** may retrieve any client-specific content based on the tail of the checkout

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link. The checkout service **6422** may generate the checkout page based on the checkout parameters and any client-specific content retrieved from the checkout database **6440**. The checkout service **6422** may output the checkout page, which may be transmitted to the user device of the customer. In example embodiments, the checkout page may provide a GUI by which the customer may provide payment information, such as a name, an address, credit card information or ACH information, and other suitable payment information.

In example embodiments, the payment execution service **6424** may initiate payment sessions with a third-party payment processor **6410** on behalf of the client's customers. In example embodiments, the payment execution service **6424** may request a payment session from the third party payment processor **6410**. The payment execution service **6424** may request the payment session in response to the selection of the checkout link (e.g., in response to the request provided by the customer device). In response to a payment session being initiated, the payment execution service **6424** may receive payment session information from the third party payment processor **6410** (e.g., a session ID and/or session key) from the third party payment processor **6410** and may transmit the payment session information to the requesting customer device. In response, the customer device may provide the payment information entered by the customer via the GUI to the third party payment processor **6410** using the payment session information.

In example embodiments, the payment execution service **6424** may expose a webhook to the third party payment processor **6410** and may add the pending transaction to a listening thread that may listen for confirmation of the transaction via a webhook that may be exposed by the payment execution system **6424**. In these example embodiments, the third party payment processor **6410** may provide a confirmation of the transaction to the payment execution service **6424** via the exposed webhook.

In response to receiving the confirmation for a transaction, the payment execution system **6424** may update the payments database **6442** to indicate the successful transaction. In some example embodiments, the payments database **6442** may store a unique identifier that may identify the transaction (e.g., a unique string), an identifier of the customer that participated in the transaction, a date of the transaction the item(s) that was/were purchased by the customer, the price of the item(s), and/or any other relevant data. In some example embodiments, the payment execution service **6424** may redirect the customer to a merchant page of the client upon completion of the transaction. In some example embodiments, the payment execution system **6424** may generate a payment event, which may be logged in one or more event records. For example, the payment event may be logged in a company or contact event record corresponding to the entity that provided the payment, a deal event record corresponding to the deal from which the payment event was initiated, and/or a custom event record corresponding to the item that was purchased.

In example embodiments, the post-payment service **6426** may execute or initiate post-payment workflows. The post-payment workflows may include default workflows and/or customer-defined workflows. In example embodiments, post-payment workflows may include issuing notifications to the client and updating the knowledge graph **640** based on the transaction (e.g., updating the knowledge graph **640** to indicate that the customer purchased the item at a certain price on a particular date). Additionally, the client may define post-payment workflows that are specific to its business. For instance, the client may define a workflow where

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a follow up email may be automatically sent to the customer requesting feedback after a prescribed time period or to remind a salesperson to contact the customer after a prescribed time period. The client may define additional or alternative post-payment workflows without departing from the scope of the disclosure.

In example embodiments, the payment system **524** may be configured to process refunds on behalf of clients. In these example embodiments, the payment system **524** may receive a request for a refund from a customer user device. In some example embodiments, the checkout service **6422** (e.g., may also be referred to as checkout system) may provide a refund page to the customer, whereby the customer may provide transaction information, such as the unique transaction identifier of the transaction at issue. The checkout service or system **6422** may verify that the transaction may be a valid transaction based on the unique transaction identifier and that the customer is the customer associated with the transaction identifier. Upon verifying that transaction information, the payment execution service **6424** may initiate a refund session with the third party payment processor **6410**. In example embodiments, a refund session may be a payment session, where the third party payment processor transfers funds from the client back to the customer. In response, the third party payment processor **6410** may initiate the refund session and may return a session ID. The payment execution service **6424** may then transmit refund information (e.g., an amount and account information of the customer) to the payment processor **6410**. The third party payment processor **6410** may then issue the refund to the customer's account and may communicate a confirmation of the refund to the payment execution service **6424**. In response, the payment execution service **6424** may update the payment database **6442** to indicate that the refund was issued to the customer. In example embodiments, the post-payment service **6426** may execute or initiate refund workflows. For example, the post-payment service **6426** may issue a notification that the refund was issued, may update the knowledge graph **640** to indicate that the customer was issued a refund, send an email to the customer indicating that the refund was issued, or the like. Additionally, the client may define custom refund workflows that may be executed in response to a successful refund being issued.

FIG. **65** illustrates an example set of operations of a method **6500** for generating a checkout link on behalf of a client. The method **6500** is described as being performed by the payment system **524** of the multi-service business platform **510**. In example embodiments, the method **6500** may be executed by a set of processors of the multi-service business platform **510**. In example embodiments, the operations of the method **6500** may be performed by a set of services (e.g., microservices) of the multi-service business platform **510**.

At **6502**, the payment system **524** may receive a set of checkout parameters corresponding to an offering (e.g., offering definition) from a user device. In example embodiments, the link generation service **6420** may provide a checkout configuration GUI to a user device, which may be presented to a user. The user may define an offering and a set of checkout parameters corresponding to the offering via the checkout configuration GUI. As discussed, the offering may include one or more items offered by a client. In some example embodiments, the payment system **524** may allow a user to import items that may be represented in the client's ontology **630**/knowledge graph **640**. For example, the user may define custom objects that may represent items that may be offered by the client and instances of these objects may

include properties of the item defined therein. In some of these example embodiments, the schema service **6432** may be configured to identify custom objects that may represent items that the client transacts (e.g., sells, leases, or the like) based on the properties of the custom object. In some of these example embodiments, the schema service **6432** may identify these items and may display the items in a menu (e.g., a drop-down menu) or other GUI elements. In some example embodiments, the user may search for items listed in the client's knowledge graph and the schema service **6432** (or another suitable search-related service) may return items that match the search query. In the foregoing example embodiments, the link generation service **6420** may receive a user selection of a menu item or a search result and may import a set of properties of the item from a respective object instance (e.g., from the knowledge graph **640**). In some example embodiments, the user may manually enter items and their descriptions. In example embodiments, the user may further provide checkout parameters via the checkout configuration GUI, such as a payment amount, a payment type (e.g., one-time, recurring, or the like), any checkout requirements and supporting documents or links, and/or other suitable checkout related data.

At **6504**, the payment system **524** may generate a checkout link corresponding to an offering based on a set of checkout parameters that may include the offering. In example embodiments, the link generation service **6420** may generate (or otherwise obtain) one or more URIs corresponding to the checkout parameters.

At **6506**, the payment system **524** may store the set of checkout parameters in relation to the checkout link. In some of these example embodiments, the link generation service **6420** may store the checkout parameters and any other user-provided content in a checkout database **6440** and may index the checkout parameters with the one or more URIs. In some of these example embodiments, the link generation service **6420** may store the checkout parameters in a checkout record that may include or otherwise reference the checkout parameters and any additional user-provided content that may index the checkout parameters with the tail portion of the one or more URIs.

At **6508** the checkout link may be embedded in a digital medium (or multiple digital media) that may be provided to one or more contacts. Once a checkout link is generated and the checkout parameters are indexed, a client (e.g., a user associated therewith) may embed the checkout link in any suitable digital medium. For example, a client may embed the checkout link on websites, native applications, electronic messages (e.g., email, text message, direct messages), chats, QR codes, social media pages, digital advertisements, or the like. In some example embodiments, the checkout links may be output to a user via a graphical user interface, whereby the user may copy the link and/or select a specific digital medium in which the checkout link may be embedded. In some examples, a user may embed the checkout link in a chat conversation with a contact or in an email. In this way, the user (or another user) may generate a special offer for one or more contacts and may provide a direct link to checkout at the special price. In another example, a user may embed the checkout link on a social media page and/or in social media advertisements, whereby interested contacts may select the checkout link and may be directed to a checkout page that may be generated using the checkout parameters.

Additionally or alternatively, the checkout link may be imported to the CRM system **502** and/or a CMS **508**. In these example embodiments, the checkout link may be

incorporated into the client's marketing efforts and/or workflows. For example, a user of the CRM system **502** may initiate a targeted digital media (e.g., email or SMS message) campaign that may include the checkout link (e.g., a special offer to certain contacts of the client). In this example, events relating to the email and/or the checkout link may be monitored and recorded, such as how many contacts opened the email, clicked on the checkout link, purchased the item(s) indicated (e.g., by initiating a payment), and/or other suitable events. In this way, the multi-service business platform **510** may leverage the tracked events to generate reports relating to payments and/or the checkout link.

The method **6500** of FIG. **65** may include additional or alternative operations. In example embodiments, the method **6500** may be executed so that a user may generate checkout links corresponding to the same combination of one or more items, such that the client may provide different offerings (e.g., pricing, contracts, terms, or the like) for the same combination of items to different contacts in a trackable and/or reportable manner. Moreover, in some example embodiments, a client may associate checkout links with specific contacts (e.g., with user accounts thereof), such that only the specified contacts may initiate a payment with the checkout link. For instance, the yoga studio may elect to reward valued customers with a special discount price on certain packages of classes. In this example, the yoga studio may provide respective sets of checkout parameters to the multi-service business platform **510** (e.g., while accessing the payment system **524**, the CRM system **502**, the CMS system **508**, and/or the service system **1600**), whereby the payment system **524** may generate a first checkout link that may correspond to an offering of a set of three different classes to customers at a first price and may also generate a second checkout link corresponding to a second offering of the same set of three classes at the discounted price. In this example, the payment system **524** may not restrict the use of the first checkout link, but may associate an account restriction with the second link, such that only customers having designated accounts (e.g., valued customers) may be allowed to access the offering associated with the second link. Furthermore, in this example, the reporting system may allow a user to define custom reports that may be configured to depict various analytics relating to checkout links and/or payments.

FIG. **66** illustrates an example set of operations of a method **6600** for processing a payment on behalf of a client. The method **6600** is described as being performed by the payment system **524** of a multi-service business platform **510**. In example embodiments, the method is executed by a set of processors of the multi-service business platform **510**. In example embodiments, the operations of the method **6600** may be performed by a set of services (e.g., microservices) of the multi-service business platform **510**.

At **6610**, the payment system **524** may receive a request for a checkout page corresponding to or based on the checkout link. In example embodiments, the checkout service **6422** may receive a request for a checkout page from a user device (e.g., of a potential customer) in response to a user selection of the checkout link. In example embodiments, the request may include a portion of the URI (e.g., the tail portion of the URI).

At **6612**, the payment system **524** may initiate a payment session with a third party payment processor on behalf of a requesting user (e.g., a customer) from which the request was received. In example embodiments, the payment execution service **6424** may request a payment session from the

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third party payment processor. In response, the third party payment processor may initiate a payment session and may provide payment session information to the payment execution service **6424**. In example embodiments, the payment execution service **6424** may provide the session information to the customer user device, such that the payment information may be provided directly from the customer user device to the third party payment processor **6410**. In some of these example embodiments, the payment execution service **6424** may provide the payment session information to the customer user device with the checkout page (e.g., as discussed at **6614**). In some example embodiments, the payment system **524** may provide the payment session information after serving the checkout page to the customer user device. In some of these example embodiments, the payment system **524** may provide the payment session information to the customer user device after confirming that the customer has fulfilled other requirements associated with the transaction. For instance, the payment execution service **6424** may verify that the customer has applied an electronic signature to (e.g., eSigned) any required contracts (or any analogous electronic documents) prior to transmitting the payment session information to the customer device.

At **6614**, the payment system **524** may generate and serve a checkout page to the requesting user based on the checkout parameters corresponding to the checkout link. In example embodiments, the checkout service **6422** may retrieve the checkout parameters from the checkout database **6440** based on the selected URI (e.g., using the tail portion of the URI). In some of these example embodiments, the checkout database **6440** may return the checkout parameters corresponding to the selected URI as well as any additional user-provided content (e.g., styles, fonts, templates, images, branding, and/or the like). In example embodiments, the checkout service **6422** may serve the checkout page to the user device. In some of these example embodiments, the checkout service **6422** may generate the checkout page (e.g., a parameterized JSON or XML file) based on the checkout parameters and any additional user-provided content and may send the checkout page to the requesting user device.

In response to receiving the checkout page from the checkout service **6422**, the user device may display the checkout page, such that the user may initiate a payment via the checkout page. The user may then input payment information (e.g., credit card number, expiration date, card verification value (CVV), ACH routing and account number, pin number, zip code, address, and/or the like) via a GUI of the checkout page. In these example embodiments, the user checkout device may provide the payment information directly to the third party payment processor **6410** using the payment session information (e.g., session ID and/or session key) provided by the payment execution service **6424**.

At **6616**, the payment system **524** may receive a transaction notification indicating an outcome of the payment session. In example embodiments, the payment execution service **6424** may expose a webhook to the third payment processor **6410** and may add the pending transaction to a listening thread that may listen for confirmation of the transaction via the exposed webhook. In these example embodiments, the third party payment processor **6410** may provide a notification indicating an outcome of the transaction to the confirmation of the transaction to the payment execution service **6424** via the exposed webhook. The notification may indicate whether the payment was successfully processed or unsuccessfully processed.

At **6618**, the payment system **524** may update the payments database **6442** and initiate any post-purchase work-

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flows based on the transaction notification. In some example embodiments, the payment execution system **6424** may update the payments database **6442** to indicate the successful transaction in response to receiving the confirmation for a transaction. In some example embodiments, the payment execution system **6424** may store transaction information (e.g., a unique identifier that may identify the transaction, an identifier of the customer that participated in the transaction, a date of the transaction, the item(s) that was/were purchased by the customer, the price of the item(s), and/or any other relevant data) in the payments database **6442**. In some example embodiments, the payment execution system **6424** may generate a payment event, which may be logged in one or more event records. For example, the payment event may be logged in a company or contact event record corresponding to the entity that provided the payment, a deal event record corresponding to the deal from which the payment event was initiated, and/or a custom event record corresponding to the item that was purchased. In some example embodiments, the payment execution system **6424** may store transaction information indicating that the payment was unsuccessful if the third party payment processor **6410** is unable to process the payment.

In some example embodiments, the payment execution service **6424** may redirect the customer to a merchant page of the client upon successful completion of the transaction. Similarly, the payment execution service **6424** may redirect the customer to an error page and/or may provide an error message indicating that the payment was not processed.

In example embodiments, the post-payment service **6426** may execute or initiate post-payment workflows. The post-payment workflows may include default workflows and/or customer-defined workflows. In example embodiments, a default post-payment workflow may include issuing notifications to the client and updating the knowledge graph **640** based on the transaction (e.g., updating the knowledge graph **640** to indicate that the customer purchased the item at a certain price on a particular date). In these scenarios, the client may receive the notification of a successful transaction and may initiate the transfer of the item(s) indicated by the offering to the customer (e.g., either electronically or physically (e.g., by shipping)). Additionally or alternatively, the client may define custom post-payment workflows that may be specific to its business. For instance, the client may define a workflow where a follow up email may be automatically sent to the customer requesting feedback after a prescribed time period or to remind a salesperson to contact the customer after a prescribed time period. In other non-limiting examples, a custom post-payment workflow may include reinforcing a machine-learned model associated with the client, updating client-specific analytics, and/or sending automated messages (e.g., to the client's sales team, the customer, or the like). The client may define additional or alternative post-payment workflows without departing from the scope of the disclosure.

The foregoing method **6600** of FIG. **66** is provided for example and not intended to limit the scope of the disclosure. The method **6600** may include additional or alternative subprocesses without departing from the scope of the disclosure.

Conversation Intelligence

Intro/Overview Compared to Industry

In general, the disclosure provides a conversation intelligence system (e.g., conversation intelligence system **528**) that may provide services related to recording and transcribing conversations such as calls, and adding these transcribed calls into the system framework or platform (e.g., multi-

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service business platform **510**) so that users may use the conversation intelligence system **528** to coach and provide analysis over these conversations. Whereas other transcription services or solutions may tend to be relatively enterprise focused and niche-focused, the conversation intelligence system **528** may provide a mass market solution that may be easy to use and built into/integrated with the system framework or platform (e.g., multi-service business platform **510**). Platform Having a Conversation Intelligence (CI) System

Referring to the example implementation, FIG. **45** shows the example environment **500** including, in example embodiments, the multi-service business platform **510** having a conversation intelligence (CI) system **528**. In example embodiments, the CI system **528** may be configured to process recorded conversations (e.g., video calls, audio calls, chat transcripts, and/or the like).

General Layers of CI System

Referring now to an example implementation of FIG. **67**, there is shown a portion **6700** of the multi-service business platform **510** with specific emphasis on details of the conversation intelligence (CI) system **528**. The CI system **528** may include a preprocessing layer **4002** that may generally provide preprocessing types of functionality which may include but may not be limited to: identifying identities of speakers (e.g., using a diarization service), identifying/distinguishing speaker tracks (e.g., using a speaker track service), conversation record generation, and the like as described in the disclosure. The CI system **528** may include a transcription layer **4004** that may include a transcription service for providing transcription of media recordings (e.g., audio recordings, video recordings, and the like) into a transcript as described in the disclosure. In other examples, the transcription layer **4004** may communicate with an external third-party transcription service that may provide a transcription of media recordings. The CI system **528** may include a postprocessing layer **4006** that may provide general post-processing types of functionality (e.g., with respect to the transcript) that may include but may not be limited to including: providing transcription hints (e.g., use a keyword service), keyword extraction, topic extraction (e.g., call topic tagging), search indexing, event generation, trigger actions (e.g., trigger actions may be initiated from a unified event), feature extraction, other metadata extraction (e.g., metadata tagging, a presentation service for displaying various information related to conversations, a calls preview/review service with regard to the transcript, a global unified search service for searching the transcript, a commenting service that allows the user to comment on the transcript, etc. as described in the disclosure. In example embodiments, the CI system **528** may communicate and/or utilize platform services **6730** (e.g., any one of the various systems and/or services of the multi-service business platform **510**), a schema service **6732** (e.g., object schema service) which may be from the multi-service business platform **510** or an external schema service with respect to the platform **510**, a search service **6734**, and/or a conferencing service **6736** (e.g., platform conferencing service part of the multi-service business platform **510**) as shown in FIG. **67** and described in the disclosure for providing various functionality. Also, in example embodiments, the CI system **528** may use various data in the storage system **550** to provide functionality as described in the disclosure. In particular, the CI system **528** may utilize conversation data (e.g., in a conversation datastore **6720**), event data (e.g., in the event data store(s) **558**), and knowledge graph data (e.g., in knowledge graph(s) **556**). Most relevant, the CI system **528** may use the ontology **630** and knowledge graph **640** (e.g., instances knowledge graph)

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of the knowledge graph(s) **556** for providing various functionality. In some example embodiments, the CI system **528** may communicate with third party conferencing system(s) **6710** to provide various functionality as described in the disclosure.

Conversation Object

Referring now to an example implementation of FIG. **68**, there is shown an example of a conversation object **5000** (e.g., may also be referred to as a call object, a conversation record, or a call record) according to example embodiments of the disclosure. In example embodiments, the conversation object **5000** defines a set of properties of a recorded conversation (e.g., a video call, an audio call, a chat transcript, or the like). In this way, any conversations that are processed by the CI system **528** may be represented in a client's datastores (e.g., knowledge base instance of the instances knowledge graph **640** and/or events data store(s) **558**) as conversation object instances (or "conversation records"). In example embodiments, the conversation object **5000** may include an object identifier **5010** (object ID) that uniquely identifies a specific conversation from other conversations and a set of conversation properties **5020**. Examples of conversation properties **5020** may include, but may not be limited to, a list of conversation participants **5022**, a set of conversation keyword(s) **5024**, a set of detected topics **5026**, a transcript of the conversation **5028** (e.g., link to a transcript), a media recording of the transcript **5028** (e.g., link to a video and/or audio call recording), conversation metadata **5032**, and/or other features of the conversation **5034** (e.g., other features related to the transcript data). In some examples, the conversation object **5000** may be tied into the multi-business platform **510** (e.g., framework or system framework) to be used with other types of objects (e.g., core objects and/or custom objects). For example, the CI system **528** may be used with other components in the multi-business platform **510** to allow for the conversation object **5000** to be associated with core objects (e.g., standard objects) and/or custom objects.

In example embodiments, the conversation participants list **5022** may indicate a list of participants in a respective conversation. In example embodiments, the conversation participants may be listed in a conversation record by their name (e.g., a string indicating the name of the conversation participant) and/or via a reference to another record that corresponds to the respective participants. In the latter scenario, the other referenced records may be contact records, employee/salesperson records, or the like. In some example embodiments, additional information of the conversation participants may be stored in the conversation record, such as a position of each participant (or the role of the participant, such as a buyer, customer, seller, or the like). In example embodiments, the conversation participants list property **5022** may allow conversations to be searchable by one or more participants in the call (e.g., searchable transcripts of calls). The conversation list of participants **5022** may be part of the conversation object **5000** and/or may be found in the transcript. This may allow for the ability to search conversations for one or more participants.

In example embodiments, the conversation keywords property **5024** may include a list of keywords that may be extracted from the respective conversation. It is noted that keywords may be one word or multiple word phrases. In example embodiments, the keywords may be keywords that are automatically extracted from any conversation (e.g., prices, "next steps", deal-specific language, or the like) and/or keywords that may be relevant to the conversation and/or the client (e.g., competitor names, product names,

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person names, industry specific words or phrases, words having a term frequency-inverse document frequency (TF-IDF) score above a threshold, or the like). In example embodiments, the conversation record (or an associated record, such as an event record) may store, for each extracted keyword, metadata relating to the keyword, such as how many times the keyword was mentioned, timestamps/locations in the transcript when the keyword was mentioned, who mentioned the keyword, and/or the like.

In example embodiments, the conversation topics property **5026** may indicate the various topics that were discussed during the respective conversation. In example embodiments, topics may be general topics (e.g., small talk, sales discussion, customer service discussion, or the like) or client and/or industry-specific topics (e.g., discussions relating to products, product launches, competitors, partners, and/or the like). In example embodiments, the conversation record (or an associated record, such as an event record) may store, for each detected conversation topic, metadata relating to the topic, such as timestamps/locations in the transcript when the topic was discussed, a duration of the discussion of the topic, who participated in the discussion of the topic, and/or the like.

In example embodiments, the transcript property **5028** may include a reference to where the transcripts may be obtained. In example embodiments, the transcript property may contain a link, a memory address, or any other suitable reference that allows the CI system **528** (or any other suitable component of the platform **510**) to retrieve a transcript of the conversation.

In example embodiments, the media recording property **5030** may include a reference to where the media conversation (e.g., audio and/or video of the conversation) may be obtained. In example embodiments, the media recording property may include one or more links, memory addresses, or any other suitable references that allow the CI system **528** (or any other suitable component of the platform **510**) to retrieve the file(s) that contain a recording of the conversation. In some examples, the media recording property may include links to third party services (e.g., third party conferencing system(s) **6710**) or platforms that have the media conversation (e.g., video and/or audio recording). As discussed, in some example embodiments, audio and/or video may be stored as multiple files. In this way, the media recording property may include multiple references, such that video files and/or audio files are retrievable.

In example embodiments, the conversation metadata property **5032** may define metadata relating to the conversation. Examples of conversation metadata **5032** may include, but may not be limited to, a date and time of the conversation, a length of the conversation, a conversation medium on which the conversation took place (e.g., Zoom®, Skype®, Google Duo®, a conversation service offered by the multi-business platform, a phone call, a text-based chat, and/or the like), and/or other suitable types of metadata. This type of information and/or data may also be referred to as or related to conversation features (e.g., other conversation features **5034**). In some examples, the conversation metadata property **5032** may include conversation metadata such as sentiment of speakers, how fast speakers are speaking, identified good call/bad call with respect to business, and/or other coaching metrics.

In example embodiments, the conversation object **5000** may include additional or alternative conversation properties (e.g., other conversation features **5034**) without departing from the scope of the disclosure.

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Conversation Intelligence Process

Referring now to FIG. **69**, there is shown a flowchart **6900** having an example set of operations for processing a conversation. In the illustrated example of FIG. **69**, the method is described with respect to processing a video call conversation. It is appreciated that the conversation may be an audio-only recording of a call. In some example implementations, the conversation may be a text-based chat. In these example implementations, some operations may be skipped (e.g., transcription).

At **6902**, the CI system **528** may receive a set of one or more conversation recording(s) (e.g., conversation file(s)). In example embodiments, the conversation files may include one or more media files (e.g., video files and/or one or more audio files). In some example embodiments, each media file may correspond to a different speaker in the conversation. In example embodiments, the CI system **528** may receive the conversation files (e.g., call files) from a third-party call service, video conference service, conferencing service (e.g., conferencing service **6736**), and/or call provider services, such as Zoom®, Skype®, Microsoft Teams®, Google Duo®, or the like. These third-party conferencing services may be third party conferencing system(s) **6710**. Alternatively, the media file may be received from a video conferencing service (e.g., conferencing service **6736**) offered by the multi-service business platform **510** (e.g., internal tools). In some example embodiments, the multi-service business platform **510** may provide automatic call recording (e.g., using conferencing service **6736** to automatically record or capture calls and video conferencing placed via the platform **510** or from an integrated provider). These calls may be placed via the platform **510** (e.g., conferencing service **6736**) or a third-party integrated provider (e.g., third party conferencing system(s) **6710**). In some examples, an API may be used to ingest and transcribe third party calls/video.

At **6904**, the CI system **528** may pre-process the conversation file(s). In example embodiments, the CI system **528** may pre-process the conversation files to improve the transcription of the file. In example embodiments, pre-processing may include identifying identities of the speakers in the stream (e.g., of a media file and/or transcript file). In some of these example embodiments, the CI system **528** may further obtain information relating to one or more of the speakers, such as a name of the speaker, a title of the speaker, a company of the speaker, and/or the like. As described, this may occur in the preprocessing layer **4002** at **6904**. In other example embodiments, this may occur in the postprocessing layer **4006** later at **6908** depending on where this data may be pulled from. For example, the CI system **528** may identify speakers and pull contact information but may need to use the transcript to complete this process. In other examples, the CI system **528** may pull data that exists from the CRM to improve the quality of the transcripts. In example embodiments, the CI system **528** may further include additional information, such as “transcription hints”. In example embodiments, transcription hints may be a list of words that may likely appear in the transcript, such as competitor names, a company name, names of individuals on the call or that work in the company, keywords that come up during the conversation, or the like. In some examples, there may be pre-processing of transcripts. In some example embodiments, pre-processing conversation file(s) may include performing speaker diarization **6922**. In these example implementations, the CI system **528** may operate on the received audio files to assign respective audio feeds to respective speakers in the conversation, such that the audio segments may be transcribed into transcripts that assign segments of transcribed text to the correct speaker. In this way, the

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transcript may be processed into a readable format. In example embodiments, the CI system 528 may perform other pre-processing operations on the received files to improve the quality of the transcription and other actions may be performed on the received files (e.g., audio and/or video files) and the transcript. In some example embodiments, the CI system 528 may perform conversation record generation 6920 when pre-processing the conversation file(s).

In example embodiments, the CI system 528 may create a call engagement or a user may use an API to create a calling engagement, and then a recording URL may be transferred (e.g., including app ID, speakers, etc.). Then, the CI system 528 may pull that calling engagement to the conversation object in the CRM. The CI system 528 may identify associations from the call or calling engagement which may be created and/or added to the conversation object by the CI system 528. In example embodiments, the CI system 528 may use an API (e.g., partner API) that may push call recordings (and future state transcripts) to the multi-service business platform 510. In example embodiments, the call engagement API may be extended to accept call recording links and all calls may be added as objects, transcribed, and usable in the platform 510.

At 6906, the CI system 528 may transcribe the conversation file(s) into a transcript. In example embodiments, the CI system 528 may use a transcription service to transcribe the conversation files. In some example embodiments, the CI system 528 may leverage a third-party transcription service, such as the speech-to-text service offered by Google Cloud® or other suitable services. In other example embodiments, the CI system 528 may leverage a proprietary transcription service that may perform speech-to-text transcription. In example embodiments, the CI system 528 may provide each audio file individually to the transcription service. In some example embodiments, the CI system 528 may further provide information determined in the pre-processing step 6904 (e.g., in the preprocessing layer 4002 or from the preprocessing stage), such as identifies of the participants, information relating to the participants, transcription hints, and/or the like. In example embodiments, the transcription service may use this information to improve the quality of the transcription. In example embodiments, a natural language processing (NLP) model may be used to assist with producing a transcript. In some examples, third party services may be used for transcribing the conversations or calls. In other example embodiments, a multi-language transcript process may be used for transcribing conversations in various different languages.

In example embodiments, the call engagement may be created, at which point information related to the call engagement may be directed to a transcription process or service. Then, the transcription service may pull down a uniform resource locator (URL), download it to transcription service. The CI system 528 may use this data to generate the transcript.

At 6908, the CI system 528 may post-processes the transcript. In example embodiments, post-processing may include performing various feature extraction techniques (e.g., feature extraction actions), various recordation-related actions (e.g., conversation object or record generation), transcript processing, and/or other suitable processes.

In some example embodiments, post-processing the transcript 6908 may include performing keyword extraction 6930. In example embodiments, the CI system 528 may perform various techniques to extract one or more keyword(s). In some example embodiments, the CI system

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528 may have a predetermined list of keywords that may be identified. These terms may be terms that appear in the ordinary course of business, terms that are specific to the user/client (e.g., employees, customers, competitors, partners, products, services, and/or the like), or words that may be specific to certain topics (e.g., sales-related terms, customer service-related terms, or the like). In some example embodiments, the CI system 528 may use an analytics-based approach to extract keywords. In these example embodiments, the CI system 528 may use various approaches to extract keywords based on the contents of the transcript. The CI system 528 may use other suitable techniques to identify keywords.

In some example embodiments, post-processing the transcript 6908 may include performing topic extraction 6932. In example embodiments, the CI system 528 may process the transcript, and portions thereof, to classify topics that were discussed during the conversation. These may include topics that took place only during certain portions of a call or conversation. For instance, the CI system 528 may classify a “small-talk” section of a conversation or a “negotiation” section of the call. In example embodiments, the CI system 528 may leverage one or more topic extraction models that may be trained to classify topics in a conversation based on one or more features of the conversation. The CI system 528 may use additional or alternative methods to extract tone.

In some example embodiments, post-processing the transcript 6908 may include performing additional feature extraction 6934. For example, the CI system 528 may extract features such as a tone of the conversation (or a portion thereof), a pace of the conversation, and/or any other suitable features.

In some example embodiments, post-processing the transcript 6908 may include generating one or more event notifications (e.g., event generation 6936). In example embodiments, the CI system 528 may generate event notifications when certain keywords, topics, and/or other certain features are detected. In example embodiments, the CI system 528 may generate an event notification that indicates the type of event (e.g., the keyword or topic discussed), the speaker or speakers involved in the event, and/or metadata surrounding the event (e.g., a timestamp indicating when the event was recorded in relation to the audio/transcript). In this way, events may be used to trigger workflows or prompts, to update event records, and/or the like.

In some example embodiments, post-processing the transcript 6908 may include triggering one or more actions (e.g., trigger actions 6938), where the actions may be an initial action in a workflow (which may be a custom workflow defined by or on behalf of the client). In these example embodiments, the CI system 528 may be configured to trigger certain actions, tasks, and/or workflows when certain topics and/or keywords are detected. For instance, when a sale price is discussed and agreed to, the CI system 528 may trigger a workflow that automatically generates a price quote that may be sent to a customer. Other examples of workflows and/or actions that may be triggered are discussed throughout the disclosure.

In some example embodiments, post-processing the transcript 6908 may include indexing (e.g., search indexing 6940) the transcript such that the transcript may be searchable via a search service (e.g., search service 6734) of the multi-business platform 510. In example embodiments, the CI system 528 may provide the transcript to a search service (e.g., search service 6734) of the multi-business platform 510, which may index the conversation, whereby users may subsequently search across many transcripts to find conver-

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sations that discussed certain keywords and/or topics. In example embodiments, the search indexing may provide call indexing such as indexing of all calls or conversations for a user in one place (e.g., index of all team's calls viewable in the same place).

The CI system 528 may perform other suitable post-processing techniques without departing from the scope of the disclosure at 6908.

At 6910, the CI system 528 may present or display a conversation graphical user interface (GUI) based on the post-processing, the transcript, and/or the conversation files. Various examples of a conversation GUI may be discussed in greater detail in the disclosure.

Example Conversation Intelligence Process

In example embodiments, the generated transcript may be provided to a transcription database and the CI system 528 may tag each conversation object with a link to the transcript database. In between, the CI system 528 may perform various actions (e.g., identifying speakers via diarization). In some examples, the conversation (e.g., call) may be sent out as separate speaker media files for each speaker to the transcription service. As in the disclosure, the CI system 528 may provide voice tagging such that the CI system 528 may split up conversations. Then the CI system 528 may match conversations to the speaker tracks based on the information the CI system 528 has access to in relation to associated contacts and callers. When the conversation object is created, the CI system 528 may give application programming interfaces (APIs) the ability to give hints as to who is speaking and when they are speaking. Then, the CI system 528 may use this information when performing speaker diarization. For example, with Zoom® integration, the call may be used and provided to an API to add calling engagement with the associated contacts. Then, the CI system 528 may pass hints as to who is talking, and the CI system 528 may then split up these speakers to be provided to the transcription service (internal, external, or third-party service). The CI system 528 may store the call and may send the call to the transcription service (e.g., when split up) to be transcribed.

In example embodiments, once the CI system 528 has a transcript in the database, the transcript may be processed for keywords and may also be processed for a global search. The CI system 528 may send the processed transcript to a search worker that may be listening for transcripts and may monitor to make the transcripts searchable on the back end. There may be several workers or services such as keyword workers or services. In some examples, there may be automatic data entry as a service as well. Each conversation may be broken into utterances. There may be different worker threads or services that may provide keywords next extraction, keyword tagging, diarization, etc. For example, there may be a portion of media (e.g., audio) that may be split up into separate speakers. Then, this media data may be sent to the transcription service (e.g., third party transcription service) such that the data may be transcribed for each speaker track separately. The transcription service may support several languages out of the box which may utilize auto-detection. For example, the CI system 528 may auto-detect what language is being spoken using the transcription service.

In example embodiments, the CI system 528 may use a model in providing transcription improvements to transcripts (e.g., model may be used to improve transcripts received internally or received from a third party service). This may be accomplished by providing a number of manually transcribed calls to use as data to improve transcript-

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related models. This may be completed for any language such as English. In some examples, several hours of audio (e.g., 10 hours of audio) may be used to measure transcript accuracy, and then the CI system 528 may use the feedback to improve the quality of the transcripts.

In example embodiments, after processing, a call or conversation object instance may be created. Speakers may be identified which may be tagged and keywords may be tagged. These tags may be by utterance such that every one of these utterances may be labeled (e.g., one utterance belongs to one topic). In some examples, the CI system 528 may use the label data sent to automate this process. Then, the CI system 528 may aggregate from the utterance. For example, where referring to core object-type keywords (e.g., pricing), these keywords may be linked to the utterance which may be used to identify topics or other items downstream. For example, the keywords may be associated with the call in general. For example, this may be an aggregate of the types of the utterances. For example, the CI system 528 may tag the utterances and then the CI system 528 may take all the utterances and create a mega tag.

In general, there are call speakers and there are utterances. Before the transcription service (e.g., internal or third-party transcription service) is utilized, there may be speakers such that each speaker thread (e.g., speaker-related audio) may be sent to the transcription service. Then afterward, the CI system 528 may analyze the transcript and break it down into utterances. After sending each speaker's audio, the CI system 528 may piece these speaker transcriptions together into one transcript of utterances. For example, the speaker transcriptions may be stored separately. For example, on the back end, there may be a separate transcript file for each speaker that may be stitched together to form the transcript with the logical flow of the conversation. As is typical, the transcript may include these segments based on timestamps. The speaker utterances may be stitched together to form the transcript. These utterances with the different time stamps may be converted into one or more conversation elements or utterance elements that may be depicted. Each of these utterance elements may be clicked on to provide further information or position play recording at clicked utterance, comments may be added to these utterances, the utterances may be shared, etc.

Conversation Intelligence Example GUIs

FIGS. 70-74 show example screenshots 7000, 7100, 7200, 7300, 7400 of user interfaces (UIs) relating to processes of using the CI system 528 on the multi-service business platform 510. These figures are screenshots of example graphical user interfaces (GUIs) allowing a user to perform post-processing of transcripts and present this information along with the transcript in GUIs. In general, in example embodiments as shown in the GUIs, recordings and transcripts may be displayed for viewing which may take advantage of functionality by the CI system 528 and the multi-service business platform 510.

Conversation Objects Example GUI

In example embodiments, FIG. 70 shows a screenshot of graphical user interface (GUI) 7000 for displaying conversation objects (e.g., conversation objects 5000). In this example, the conversation intelligence system 528 may be used to log into the CRM. The conversation objects may be call objects that may be part of the CRM or may be associated with other objects in the CRM. As shown, the CI system 528 may allow a user to log in and view the conversation objects (e.g., call objects). The user may view information about the call such as targeted managers, who on the team is talking, and when a person speaks during a

conversation. Other information that may be viewable may include call outcome and duration. As shown, the CI system 528 may be used to label calls with a call title so that users may monitor what is happening in the conversation. These calls may be internal or through third party calling services. Call Preview/Review Example GUI

In example embodiments, FIG. 71 shows a screenshot of graphical user interface (GUI) 7100 for displaying a review preview and/or review features. The CI system 528 may provide an ability to preview. Without having to actually click into the call, users may easily preview what is going on in the call or conversation to identify speakers (e.g., who is talking). The CI system 528 may allow the user to scan and/or read through the transcript and/or edit/modify the transcript without having to leave the context of the call. The CI system 528 may allow the user to click into the page either from clicking view full screen in a “Call Review” window (or may be referred to as a “call preview” window) or clicking in a calls table. In example embodiments, the CI system 528 may access this call review or preview anywhere in the CRM such that a user may view the call from other records (e.g., from a contact) without having to leave the context GUI or other display or other section of the multi-business platform 510.

Conversation or Call Example GUI

In example embodiments, FIG. 72 shows a screenshot of graphical user interface (GUI) 7200 for displaying a conversation view (or call view) of a transcript. The CI system 528 may allow a user to view which speaker is talking, view the call speaker tracks, and add notes throughout a transcript in line with the media recording (e.g., video and/or audio recording). As described in the disclosure, the CI system 528 may provide call topic tagging in this GUI 7200 (e.g., part of the call may be small talk, questions, pricing, etc.). The CI system 528 may use one or more models for providing some of these features.

In example embodiments, the “conversation view” may also show keyword tagging (e.g., indicating that “sales” was mentioned five times in the call). As shown in FIG. 72, the GUI 7200 may include commenting on the transcript (e.g., similar to commenting on a document). The GUI 7200 also shows the ability to scroll through comments, click on comments, and add comments at particular points in the call. The comments may scroll with the user’s view as they shift through a call (e.g., shifting with mouse or pointer in conversation display). The GUI 7200 may also show call participants and ability to search through the transcript.

Global Search Example GUIs

In example embodiments, FIGS. 73-74 show screen shots of graphical user interface (GUIs) 7300, 7400 for displaying global searching capabilities (e.g., may use the search service 6734). The CI system 528 may allow for searching across every conversation as shown in GUIs 7300, 7400. The CI system 528 may also provide for searching across every call and other objects. For example, searching companies, contacts, and pulling up all the conversations (e.g., calls). For example, as shown in GUI 7400, a search of company “HubSpot” in various types of conversations (e.g., calls and/or emails). In example embodiments, this searching may be a unified search that may combine keywords with the search feature (e.g., unified search) providing interesting and related features.

Integration of Conversation Intelligence with Platform/Framework

In example embodiments, the CI system 528 may provide integration of the transcript and the GUIs (e.g., user may slide bar of the conversation to observe where it is going)

with the multi-service business platform 510 (e.g., framework or platform) which may provide context and the ability to utilize a variety of interesting technologies. Tying the CI system 528 with the framework (e.g., aspects of the underlying framework) may provide interesting results when executed in this context. For example, tying the CI system 528 into the broader framework may allow for connecting of conversations with reporting functionality (e.g., reporting on deals, reporting of a service organization related to conversations, etc.). In other examples, the CI system 528 may involve triggering of keywords when the CI system 528 may detect that specific topics, phrases, and/or words have been mentioned in conversations (e.g., “Hey, should I generate a quote for this?”). The CI system 528 may generate a check out page based on a conversation (e.g., from a call to a closed deal with minimal effort). The CI system 528 may be able to provide this link between the transcript and actions due to the integration and set up of the CI system 528 with the multi-service business platform 510 (e.g., framework) having joined services.

Events

In example embodiments, the CI system 528 may be used to generate events based on the conversation or the call. For example, this may be a generation of a keyword event such that whenever a keyword is mentioned in a transcript, a keyword event may be generated and associated with one or more event records. In some example embodiments, certain keyword events may correspond to a category of keywords (e.g., keywords grouped in a category which may relate to and/or be associated with one or more topics such as “pricing” may be a keyword category that may include groups of keywords such as “sell” or another keyword category may be “products”) that may be unified in the event framework. Keywords may be definable with respect to these categories of keywords. In some examples, there may be models that may assist users in selecting keywords as the models may provide suggested keywords or transcription hints to words of interest. Categories may be created manually or by a model that may analyze conversations (e.g., conversations between a user and customers), or may use natural language understanding. Examples of categories may include, but may not be limited to, call topics, action items (e.g., next steps), competitors, products, company initiatives, etc.

Keywords

The CI system 528 may allow for a keyword to be defined by a user such that every time the keyword shows up in a conversation (e.g., keyword detected), the keyword may be reported. For example, calls may be used to predict categories of keywords. The CI system 528 may use models to predict keywords, and these predicted keywords may then be used across the framework, either in reporting, workflows (e.g., email sends), etc. These keywords may be used with any combination of workflows such that phrases and/or quotes may be provided to/used to power the CI system 528 (e.g., when a person mentions “sales”, send automated emails). In some examples, statements may be made during a call that may be detected and may trigger actions (e.g., “add note or comment here”). In some examples, the CI system 528 may use a model that may detect pricing, next steps, packaging, etc. The architecture for this model may be different from all the other call topic tagging models (e.g., architecture of the model may be different from tagging architecture).

In example embodiments, once keywords are entered, the keywords may be used in a number of ways. For example, the entered keywords may be used to improve quality of the

transcripts (e.g., improve transcription process or post-quality transcription process). These keywords and related data may be fed to the transcription service (e.g., third party transcription service API) to improve the quality of the transcripts. In another example, after processing the transcripts, the CI system 528 may monitor for words, phrases, and/or terms that probably should be keywords and fix or adjust the transcript accordingly. These identified keywords may be used to define categories which may then be used in reporting, to run workflows on keywords, etc. The CI system 528 may be built for utilization with specific use cases on top of the workflows such as a competitor name being mentioned on a call or the like may trigger a workflow (e.g., an email being sent). For example, the CI system 528 may allow for setting up a follow-up quote or trigger based on a Slack® notification (e.g., for any competitor being mentioned in a conversation, trigger Slack® notifications). In another example, the CI system 528 may show these mentions on the reviewed call (e.g., “Hey, you mentioned workflow six times”) such that the CI system 528 may allow users to click on transcript and find all comments mentioning workflows in the transcript.

In example embodiments, keywords may be extracted using keyword extraction models (e.g., may use heuristics and/or other types of models for keyword extraction). There may be transcription hints to words of interest and keywords. Keywords may refer to, for example, pricing in the call, closing the deal, etc. Whereas hinting to keywords may refer to finding a key word (e.g., company X’s name) that may sound similar to another word such that this keyword may not be known in a transcript. When trained, the models may be improved accordingly with new keyword data and/or hint-related data. For example, there may be a need by a user to make sure that the word for a competitor and/or the word referring to the user’s own company may be used correctly. Commenting in Transcripts

The CI system 528 may provide the user with an ability to comment in the transcript. For example, the user may comment about utterances in the transcript. This layout and design of commenting may provide similar functionality to a word processing document (e.g., comments located in line with the text of transcript). The comments may be shared, and events may be triggered from comments. Comments may be added for each utterance (e.g., conversation snippet). In some examples, a comment may be added for a group of utterances. In other examples, the CI system 528 allows for snippet or utterance sharing.

Generate Speaker Tracks without Transcript

In some examples, the CI system 528 may generate call speaker tracks without a transcript. This may remove the costs of generating transcripts while still providing benefits of providing conversation intelligence (e.g., information from conversation) without needing transcripts to be generated. The process for generating this type of timeline and/or call speaker tracks without transcripts may be based on the CI system 528 analysis of conversation files (e.g., media files such as audio files and/or video files). Instead of relying on the transcripts to build these timelines, the CI system 528 may process the audio directly to build these call out speaker tracks. The CI system 528 may show the screen with keyword information without ever needing to display a transcript.

Search Techniques

In example embodiments, it may be interesting how searching in transcripts with respect to a global search may be conducted (e.g., may use the search service 6734). The CI system 528 may connect the transcript to searching and

indexing capabilities. The entire transcript may be indexed such that the CI system 528 may be used to search for anything within the transcript. In terms of searching, the CI system 528 may utilize a lexical search that may include performing exact matches of strings within queries, to documents, to search credit strips, to a truly semantic search such that when searching for “set up”, the user may also find “onboarding” and other related terms. The CI system 528 may also use machine learning and natural language processing to help sort fuzzier matches.

The CI system 528 may use transcript searching to help implement the technology platform (e.g., multi-service business platform 510) to do a variety of searches. In some examples, the CI system 528 may utilize elastic search (e.g., a distributed Lucene™ type query). In other examples, the CI system 528 may provide lexical searches. In another example, machine learning techniques may be used by the CI system 528 allowing for dynamic and relevant search results that may use transcript searches. When searching across multiple objects, there may be a model that may provide ranking specifically for each user. This may cause some of the objects to change based on relevance (e.g., a relevance score calculated for each user). For example, a sales user may have customized unique search settings such that the sales user may prioritize searching for “tickets” over other terms whereas a CEO user may prioritize “revenue” over other terms based on these settings (e.g., search setting may be based on user’s preferences, user’s position, user’s role in an organization, user’s department in an organization, etc.). In summary, there may be query processing on the front end and/or there may be machine learning models on the back end that may help score relevance to the particular user or the context of the search and the search terms that users enter. In some examples, the CI system 528 may allow for users to enter data for creating a model. This may be provided with conversations from media files (e.g., audio and/or video files) along with emails. From these conversations, the CI system 528 may determine user role (e.g., product manager in the signature) such that the CI system 528 may enter this data into the platform for each user.

In example embodiments, there may be a search infrastructure that provides indexing and latent searching. For example, the search infrastructure may provide lexical (e.g., letter by letter kind of searching) and also semantical search. In some examples, the CI system 528 may provide fuzzy matching from conversations (e.g., based on a most recent call), then the CI system 528 may also determine relevant scores and provide the relevant scores so that more important parts of the conversation may be pulled (e.g., the CI system 528 may use the search service 6734 to provide searching of conversation intelligence data). For example, more important portions of the conversation for a user’s particular role may be higher ranked than other portions based on various factors (e.g., based on calls that the user was on, calls that fall into the user’s business unit, calls that align with user’s role, more recent calls, etc.). This technique may be referred to as scoring or sort scoring.

Overall Summary of Conversation Intelligence Process

In example embodiments, the CI system 528 may receive a media recording such as an audio recording, a video recording, and/or combination audio recording/video recording from a third-party service such as the third-party conferencing system(s) 6710 (e.g., from Zoom® call or Skype® call or Microsoft® Teams call or any other third-party calling service) or an internal call or any other calling/conference service. The CI system 528 may receive a series of video feeds that have audio associated with them. Each

feed may be associated with a different speaker. These feeds may be sent to the transcription service which may be an internal service or an external service. There may be a preprocessing stage (e.g., run via preprocessing layer 4002) which may be provided to the transcription layer 4004 (e.g., transcription service). The CI system 528 may receive from the transcription service a transcript that may have time stamps on when specific utterances were made. Then, there may be a post processing stage (e.g., run via post processing layer 4006) where the utterances may be combined to form a transcript. The conversation object instance (or call object instance or transcript object instance) may record the properties of the conversation (e.g., the transcript may be attached to this call object or conversation object or the call/conversation object may have transcript capabilities). The conversation object or call object may either use the transcription service to create transcripts or a third-party service may send the system transcripts.

In examples, there may be post processing (e.g., via the postprocessing layer 4006) where the CI system 528 may apply a set of models. Those models may be used to obtain sentiment, topic extraction, and key word extraction aspects. Then, there may be indexing of the transcript so that the transcript becomes searchable. The transcript may be searchable and the transcript may be input to the multi-business platform 510 (e.g., framework). The transcript (e.g., as part of the conversation object) may be reportable. There may be an option to provide coaching after the transcript has been processed which may be a downstream feature. The CI system 528 may have the transcript and may direct the conversation or call back into the GUI. The utterances may match up which may be placed in UI elements. The utterances may be connected to the transcript such that the timestamp may associate back to the conversation or call allowing the user to move in the interface back and forth from the transcript to the call recording.

Pre-Processing Models (Metadata)

The value of conversation intelligence may come from applying metadata to conversation text. This may include transcripts, emails, and/or chats. The CI system 528 may take this text and feed it through a set of models to get metadata such as sentiment, topics, keywords (e.g., keyword prediction), and determination/prediction of other information. This may allow for the CI system 528 to provide suggestions to users regarding the most accurate keywords based on the model prediction. This metadata may then be used by the reporting service (e.g., reporting system 526 or reporting 534 of the services 530) to generate reports (e.g., cross-object custom reports), by an automation service to trigger workflows that may set off a number of actions (e.g., trigger emails to customers, send out Slack® hooks so a team may be notified of an event, etc.), or other suitable CRM-based services. In another example, the CI system 528 may provide a coaching tool as shown in the GUI 7200 of FIG. 72. This GUI 7200 may provide a coaching view that may show questions asked, keywords mentioned, topics, etc. This coaching view may allow for the user to quickly locate a correct point in a call. This may relate to searchable features as well as metadata that may be used to find calls more promptly. An example of where this becomes relevant with extracting metadata may be an email parser. The email parser may read through emails in a connected inbox or logged engagements and may extract specific properties (e.g., job title, phone number, address, etc.) which may then be entered into the contact record. This may be done for any conversation (e.g., calls along with chats) but may also provide extraction of other properties. For example, the CI

system 528 may extract sentiment across emails, calls, and chats. Then, that metadata may be reported on and searched on. This metadata may also be added to the CRM objects. In some examples, the users may be coached on emails that may be around the metadata extracted. In example embodiments, the coaching tool (e.g., call coaching) may provide various features and advantages that include but may not be limited to: listening for important aspects or parts of a call using speaker tracks or the transcript, speeding up a call, sharing and searching the call, and leaving comments/notes to give feedback. In some examples, calls on mobile may be determined for providing coaching of a list of calls on mobile.

In some examples, the CI system 528 may determine a contact based on data of speakers in conversations (e.g., conference calls). This contact may not already exist in the database but may be created as new for the speaker and associated with the speaker from the call. This extraction may be rule-based (e.g., using a large set of rules) or use a developed model. The model-based approach or rule-based approach may be built in a way such that it may be trained. For example, the rule-based approach may be built such that the rules may be trained/adjusted over time (e.g., adjusted based on trained data over time). The model may also be updated based on the changing data across the Internet as well as changing data specific to the user over time. In some examples, the model and/or rule-based approaches may utilize manual oversight or semi-supervised overview (e.g., partially automated) and in other examples, the model and/or rule-based approaches may be executed automatically (e.g., fully automated).

Metadata Extraction Examples

In example embodiments, there may be a variety of types of metadata that may be extracted. For example, metadata that may be extracted from conversations (e.g., calls) may include questions asked, numbers, call topics (broad and narrow), and the like. Call topics may refer to small talk, pricing, discovery, talking about products, etc. In other examples, call topics may refer to a discovery call, an intro call, a next steps call, and the like. Another type of metadata that may be extracted may be sentiment which may be whether the customer is happy, sad, mad, etc.

In some examples, topic metadata may be pulled and the CI system 528 may determine correlations between specific topics (e.g., including specific words and/or phrases) with results of the customer (e.g., customer purchases a product or service). For example, a type of topic pulled may be finding the most commonly used phrases and correlating these phrases to a deal (e.g., one set of types of phrases may lead to higher revenue based on correlations). For example, a customer has a homegrown business where anytime that a sales representative said, "customers like you", it would increase the chance that a deal may close by a specific percentage (e.g., 25%). Then, the user may use this data to implement a company policy for every sales representative to start saying the phrase "customers like you". In some examples, there may be feedback from previous calls that may go through analytics of the CI system 528 that may combine calls that resulted in sales versus calls that did not result in sales. Then the CI system 528 may provide feature extraction from the keyword extraction to find when phrases are used (e.g., identify phrases in an analytic manner). This correlation type of analysis may utilize one model or multiple models.

In examples, any property and/or object (e.g., core object and/or custom object) in the multi-services business platform 510 (e.g., in CRM) may be extracted as metadata from

conversations. For example, objects such as deals may have a decision maker where a contact is usually added. The CI system 528 may extract this object from conversations such as phone calls and/or emails (e.g., email extraction may be a precursor to all the objects and/or properties that may be extracted) such that the decision maker may be extracted from the conversations (e.g., calls and/or emails). This may be in the form of metadata and may be based on how questions are answered. In some examples, deference may be given to certain people in the conversations and/or if a speaker/communicator recommends speaking to a particular person. Other types of metadata that may be extracted may include money phrases, decision makers, recommended actions, and any other relevant properties and/or objects in the CRM that may be potentially extracted as call metadata. This data may be related to information in transcripts, information in emails, or other information accessible from the CRM.

In example embodiments, the CI system 528 may also pull and provide recommended actions (e.g., as metadata) that may be mentioned in the conversation. For example, the CI system 528 may identify the phrase "I'm going to call you tomorrow" in a call then engage the multi-service business platform 510 to implement an action providing an invite for tomorrow. In examples, the CI system 528 may use a tasks type of tool to provide this functionality.

In example embodiments, as part of these metadata processes, the CI system 528 may utilize natural language processing (NLP) and natural language understanding (NLU) with models in order to understand language. In some examples, deep neural networks may be used for the NLP. Feature extraction may be fed into the models which may output metadata (e.g., sentiment, decision maker is X, call topic is Y, the call topic for this particular text is Z, etc.). The NLP model may be pre trained in a variety of ways. For example, the NLP may be pretrained by Internet data. The model may also be post-trained with framework-specific data (e.g., data in the multi-services business platform 510 such as data in the storage system 550). In some examples, information from a user of the platform may be used in training the model in real time. In summary, there may be three phases of pretraining the NLP which may include training from the Internet, training from the data of the multi-services business platform 510 (e.g., framework data) and the user data, and/or customer data training (e.g., based on information from the user and/or customer and context about a particular call used to differentiate the particular user from other users).

Relevant Tagging Based on Ontology

In example embodiments, there may be an implicit ontology and the neural networks may learn the ontology automatically. This means that there may be implicit ontology in the neural network model even it cannot be directly viewed. In some examples, users may be able to adjust and/or give hints to the models. This may be an example of how keywords may be created in the multi-services business platform 510 such that by adding a keyword, there may be a hint to the model to bias towards one or more specific keywords. In example embodiments, an ontology may already exist and/or be added and updated in the ontology 630 based on the neural network process.

In some examples, when there are words that sound the same (phonetically), the CI system 528 may bias towards the word that the CI system 528 knows is likely an intended topic. In some examples, phonetic similarity may be used to improve the quality of transcription during post processing (e.g., fix errors in the transcript). For example, the NLP may

look to synonyms of words/phrases related to a specific topic (e.g., use synonym and lexical understanding to help with topics). When the CI system 528 is processing the transcript, the CI system 528 may know that specific words are related (e.g., king and prince may be related as words) even though these words may be lexically different which may be based on training from the Internet. This encapsulates that semantic information may be used by the NLP in identifying similarities among words, similarity of sentences to other sentences, and the like.

In example embodiments, with call topic tagging, there may be a model that may work in a relatively unique way compared to other similar models. For example, the CI system 528 may not have any labels for call topics such that the model may provide unsupervised learning to generate clusters of topics that may then be used to train a separate model (e.g., generate the labels for topics). The CI system 528 may then automatically add relevant labels and/or provide a capability for developers to manually review and add relevant labels. This call topic tagging of relevancy may be further verified by users.

Transcript Quality (Preprocessing and Postprocessing)

In example embodiments, transcript quality models may be used with transcripts at preprocessing and postprocessing stages. For example, the CI system 528 may relate to preprocessing of transcripts once the transcripts have been added to the multi-service business platform 510. There may be several ways to improve quality of the transcripts. For example, hints may relate to preprocessing where the hints may be sent with the audio to the transcription service. There may be some post processing as well at some point to improve transcription results from the transcription service. Automation Triggering with Keywords

In example embodiments, there may be keywords that may be interconnected with workflow, workflow triggering, and/or action triggering. This may be set up by users (e.g., customers) or may be predetermined in setup of the CI system 528 and the multi-service business platform 510 (e.g., framework). For example, there may be alerts setup or workflow actions setup based on the automation platform of the framework (e.g., as described in the disclosure).

In an example embodiment, keywords may be added and then the CI system 528 communicates with portions of the multi-service business platform 510 to set a workflow action (e.g., may send an email). For example, a competitor may be referred to in a deal that may have a workflow action that sends an email out to customers (e.g., saying "we've noticed you spoke about the competitor, here's a discount"). In another example, the workflow action may be a notification sent to an internal team that a particular speaker on the transcript is discussing a competitive deal such that the system (e.g., multi-service business platform 510) may recommend extra attention on this contact speaker from the conversation. This may also be used to change the score of a deal such that the deal may need more attention, the deal may need less attention, or the deal is more or less likely to close with manual scoring. In terms of the automated approach, the user may enter keywords then the system (e.g., multi-service business platform 510) may send out an email digest of the relevant calls that have that keyword in them. In examples, the CI system 528 may allow for selection of a keyword by the user to trigger an action. The automation of the multi-service business platform 510 (e.g., framework) may allow for filtering by any of the properties of the objects in the framework.

The automated framework (e.g., automated components of the multi-service business platform 510) may be con-

nected to the CI system 528 such that keywords may be created as unified events in the framework and unified events may flow through this. The unified events may be used to trigger workflow actions. As a unified event, connecting to the framework (e.g., platform 510) may be easily completed such that conversation objects (e.g., conversation object 5000) may interact with current and future aspects of the framework.

Examples of Keywords with Events

In example embodiments, every time a particular keyword is mentioned, it may trigger a unified event. These unified events may refer to relatively large event records that apply to a particular type of object (e.g., user may have an event log for the user that may log several different types of events). In examples, keywords may be triggered by keyword events (e.g., may be unified events) which may be predefined by the multi-service business platform 510 (e.g., framework as related to core objects) and/or some keywords may be customizable as being created by each user. The keyword events may be integrated into the broader platform which may then be aggregated in reports, in workflows, list segmentation, and other functionalities/features of the system framework or platform. In some examples, each keyword may relate to a separate unique event where each event may have different attributes that have different values. In other examples, multiple keywords may relate to the same event. A category may be supported on top of these events. In some examples, this may be used with a JavaScript Object Notation (JSON) schema where a keyword may be detected and then the property may be the word (e.g., “sell” and the category may be “pricing”).

In example embodiments, it may be preferred that each keyword has its own event record which may be only one type of event. For example, every time the word occurs in conversation, this may refer to one type of event with a different attribute of what the word refers to in the event. For example, out of the keyword mention event type, there may be two events for two different words (e.g., “buy” and “sell”) such that these words refer to individual events that may trigger any individual automation. Keywords may be grouped with a particular event type (e.g., keyword detected) as one event type or multiple event types. For example, there may be event types such as pay queue where a page view form submission or integration event may be triggered. This event may be triggered by a next event keyword mention that may be defined with one or multiple keywords (e.g., based on developer and/or user preferences) with each word providing a trigger independently and flowing independently. In some examples, properties of keywords and/or properties of events may trigger workflows. For example, if a competitor’s name is mentioned in the conversation and this attribute may be in an event, it may trigger an action such as sending an email.

In example embodiments, keywords may be grouped together such that multiple keyword events may be searched together per call (e.g., a list of keywords being tracked as mentions). This may refer to a list of all competitors which may be grouped under one category of keywords (e.g., category keyword event may include multiple keywords as being under same category of event for keyword searches or mentions). In some examples, each user may create their own keyword events. In other examples, keywords may already be created as default keyword events or as generally recommended keywords for most users (e.g., use model that recommends keywords for users to add). For example, there may be categories of keywords that may be suggested to users such as competitors, products, services which may

vary based on the user. In this example, the model may be used to recommend categories of keywords and/or keywords themselves based on the transcript (e.g., the CI system 528 may communicate that it detects that Company X is a competitor to user’s company and ask user whether they would like to add “Company X” in the form of a keyword suggestion). In other examples, the CI system 528 may recommend keywords based on data in other parts of the platform (e.g., the CMS and support articles). For example, if a user has a blog discussing their products, the platform may recommend the products discussed on the user’s blogs. With these examples, the CI system 528 may include a suggestion phase where the CI system 528 suggests keywords or keywords events (e.g., based on category of keywords and/or keywords themselves) to users to be considered for creation. The CI system 528 may mine the transcript, email, and/or other conversation communications to identify these suggested keyword events.

For example, the word “next steps” may be set up by the CI system 528 using keyword technology or may be created through a call topic tagging model or a next steps model (e.g., analysis and creation of keywords based on models). The call topic tagging may involve monitoring of generic topics which may be set by the user. This may be tuned over time such that some of these topics may be defaults within the CI system 528. In general, keywords may be either defined by the user or extracted through the conversation. In some specific examples, tf-idf scores may be used to identify keywords from the transcript (e.g., a word is mentioned 10-12 times in a call but is usually mentioned a couple times or less in most conversations such that this word may be suggested as a keyword).

In example embodiments, during post processing, the call topic tagging (e.g., tagging conversation as a pricing call) may be a unified event such that the CI system 528 may trigger a workflow. The automatic models may also be used with these events. For example, topics may be unified events that may trigger one or more workflows. The keyword events may trigger these post-conversation workflows. Unified events may provide interesting information with respect to a user’s journey analytics. For example, when considering a lifecycle of a customer from a user’s website to pricing call to buying a product, use of unified events may be particularly important and useful with respect to this lifecycle. Transcript Reporting and Analytics

In example embodiments, the CI system 528 may pull all relevant information to be added as events which may be reportable in the report builder. The CI system 528 may allow for cross object reports with any other object (e.g., core object and/or custom object) in the platform 510 (e.g., framework) which may be reported as preferred by the user. These events may be aggregated in the coaching view (e.g., in the coaching app) of the GUI 7200 as shown in FIG. 72. This means that these events may be placed in the coaching GUI so that a user (e.g., manager) may select a preferred position in the call without having to listen to the entire call or search around. Related metadata may also be used by the CI system 528 such that the user may find desired information.

In some examples, the CI system 528 may process all transcripts in bulk to find item suggestions that may be unaware to the user (e.g., a user may not necessarily know to make these suggested events, but the CI system 528 may monitor/analyze across calls and correlate to build suggestions). For example, this may involve an insights dashboard such as a sales AI dashboard that may provide common topics mentioned that the CI system 528 may determine are

relevant to each user. The CI system 528 may drill down to the list of conversations (e.g., calls) to observe and analyze what may be going on and/or identify list of relevant emails. For example, the CI system 528 may identify some deals that may be in danger because conversations may include mentions (e.g., a few words) that may be detected as being relevant to the user with respect to deal concerns. The CI system 528 may monitor across all conversations and provide analysis outside of the event frameworks. In some examples, attribution reporting may be used with the CI system 528. For example, the events may be used when keywords are mentioned that may have a monetary value correlated to the revenue that a user's company/business earned. In general, the CI system 528 may utilize searching, keyword integration, trigger actions, models, transcript analytics, etc. to provide these capabilities. In example embodiments, the CI system 528 may provide general call analysis & reporting that may include searching for keywords and phrases across a user or a team's entire history of calls and may build detailed reports tracking call metrics and keyword mentions. In some examples, there may be keyword reporting where reporting on keyword mentions in transcripts may be provided in the custom report builder as described in the disclosure. For example, the "custom report builder" may be selected, then calls, keywords, and categories may be selected from data sources. The keywords may be adjusted by the user such that there may be different categories of keywords (e.g., "competitors" may be one category and a "finance" may be a second category where "finance" includes keywords such as "budget" and "funds").

Predictions and Forecasts

In example embodiments, the CI system 528 may provide a host of models to make forecasts and predictions (e.g., predict deal scores) as well as improving accuracy of these forecasts and predictions. For example, the keywords may trigger a deal score to change which may be used for forecasting. The CI system 528 may determine from one or more conversations whether a deal appears it will close or whether the deal may not close (e.g., based on a number of different factors such as tone, words that were used, and/or a combination of these factors). The CI system 528 may also monitor frequency of touch points in making these determinations (e.g., how many emails are being sent by customer, how many phone calls from customer, and/or a combination of these actions may lead to a relatively accurate prediction). In some examples, there may be a correlation functionality (e.g., correlation engine) as may be used with correlating/correlations described in the disclosure. This correlation function/engine may monitor conversation data (e.g., in the conversation datastore 6720) to find correlations between touch points in one or more conversations (e.g., frequency of words, tones, frequency of contacting, etc.) for indicating one or more predictions based on models. In examples, the models may be setup to monitor preselected correlations and/or these models may be used to identify new correlations specific to each user's conversations (e.g., properties of conversation objects). These models may be trained generally on the entire Internet but may learn to identify nuances of each particular user-based training with the user data over time.

Legal

The background description is presented simply for context, and is not necessarily well-understood, routine, or conventional. Further, the background description is not an admission of what does or does not qualify as prior art. In

fact, some or all of the background description may be work attributable to the named inventors that is otherwise unknown in the art.

Physical (such as spatial and/or electrical) and functional relationships between elements (for example, between modules, circuit elements, semiconductor layers, etc.) are described using various terms. Unless explicitly described as being "direct," when a relationship between first and second elements is described, that relationship encompasses both (i) a direct relationship where no other intervening elements are present between the first and second elements and (ii) an indirect relationship where one or more intervening elements are present between the first and second elements.

Example relationship terms include "adjoining," "transmitting," "receiving," "connected," "engaged," "coupled," "adjacent," "next to," "on top of," "above," "below," "abutting," and "disposed."

The detailed description includes specific examples for illustration only, and not to limit the disclosure or its applicability. The examples are not intended to be an exhaustive list, but instead simply demonstrate possession by the inventors of the full scope of the currently presented and envisioned future claims. Variations, combinations, and equivalents of the examples are within the scope of the disclosure.

No language in the specification should be construed as indicating that any non-claimed element is essential or critical to the practice of the disclosure.

The term "exemplary" simply means "example" and does not indicate a best or preferred example.

The term "set" does not necessarily exclude the empty set—in other words, in some circumstances a "set" may have zero elements. The term "non-empty set" may be used to indicate exclusion of the empty set—that is, a non-empty set must have one or more elements.

The term "subset" does not necessarily require a proper subset. In other words, a "subset" of a first set may be coextensive with (equal to) the first set. Further, the term "subset" does not necessarily exclude the empty set—in some circumstances a "subset" may have zero elements.

The phrase "at least one of A, B, and C" should be construed to mean a logical (A OR B OR C), using a non-exclusive logical OR, and should not be construed to mean "at least one of A, at least one of B, and at least one of C."

The use of the terms "a," "an," "the," and similar referents in the context of describing the disclosure and claims encompasses both the singular and the plural, unless contradicted explicitly or by context.

Unless otherwise specified, the terms "comprising," "having," "with," "including," and "containing," and their variants, are open-ended terms, meaning "including, but not limited to."

Each publication referenced in this disclosure, including foreign and domestic patent applications and patents, is hereby incorporated by reference in its entirety.

Although each of the embodiments is described above as having certain features, any one or more of those features described with respect to any embodiment of the disclosure can be implemented in and/or combined with features of any of the other embodiments, even if that combination is not explicitly described. In other words, the described embodiments are not mutually exclusive, and permutations of multiple embodiments remain within the scope of this disclosure.

One or more elements (for example, steps within a method, instructions, actions, or operations) may be

executed in a different order (and/or concurrently) without altering the principles of the present disclosure.

Unless technically infeasible, elements described as being in series may be implemented partially or fully in parallel. Similarly, unless technically infeasible, elements described as being in parallel may be implemented partially or fully in series.

While the disclosure describes structures corresponding to claimed elements, those elements do not necessarily invoke a means plus function interpretation unless they explicitly use the signifier “means for.”

While the drawings divide elements of the disclosure into different functional blocks or action blocks, these divisions are for illustration only. According to the principles of the present disclosure, functionality can be combined in other ways such that some or all functionality from multiple separately-depicted blocks can be implemented in a single functional block; similarly, functionality depicted in a single block may be separated into multiple blocks.

Unless explicitly stated as mutually exclusive, features depicted in different drawings can be combined consistent with the principles of the present disclosure.

In the drawings, reference numbers may be reused to identify identical elements or may simply identify elements that implement similar functionality.

Numbering or other labeling of instructions or method steps is done for convenient reference, not to indicate a fixed order.

In the drawings, the direction of an arrow, as indicated by the arrowhead, generally demonstrates the flow of information (such as data or instructions) that is of interest to the illustration. For example, when element A and element B exchange a variety of information but information transmitted from element A to element B is relevant to the illustration, the arrow may point from element A to element B. This unidirectional arrow does not imply that no other information is transmitted from element B to element A. As just one example, for information sent from element A to element B, element B may send requests and/or acknowledgements to element A.

Unless otherwise indicated, recitations of ranges of values are merely intended to serve as a shorthand way of referring individually to each separate value falling within the range, and each separate value is hereby incorporated into the specification as if it were individually recited.

Special-Purpose System

A special-purpose system includes hardware and/or software and may be described in terms of an apparatus, a method, or a computer-readable medium. In various embodiments, functionality may be apportioned differently between software and hardware. For example, some functionality may be implemented by hardware in one embodiment and by software in another embodiment. Further, software may be encoded by hardware structures, and hardware may be defined by software, such as in software-defined networking or software-defined radio.

In this application, including the claims, the term module refers to a special-purpose system. The module may be implemented by one or more special-purpose systems. The one or more special-purpose systems may also implement some or all of the other modules.

In this application, including the claims, the term module may be replaced with the terms controller or circuit.

In this application, including the claims, the term platform refers to one or more modules that offer a set of functions.

In this application, including the claims, the term system may be used interchangeably with module or with the term special-purpose system.

The special-purpose system may be directed or controlled by an operator. The special-purpose system may be hosted by one or more of assets owned by the operator, assets leased by the operator, and third-party assets. The assets may be referred to as a private, community, or hybrid cloud computing network or cloud computing environment.

For example, the special-purpose system may be partially or fully hosted by a third party offering software as a service (SaaS), platform as a service (PaaS), and/or infrastructure as a service (IaaS).

The special-purpose system may be implemented using agile development and operations (DevOps) principles. In embodiments, some or all of the special-purpose system may be implemented in a multiple-environment architecture. For example, the multiple environments may include one or more production environments, one or more integration environments, one or more development environments, etc.

Device Examples

A special-purpose system may be partially or fully implemented using or by a mobile device. Examples of mobile devices include navigation devices, cell phones, smart phones, mobile phones, mobile personal digital assistants, palmtops, netbooks, pagers, electronic book readers, tablets, music players, etc.

A special-purpose system may be partially or fully implemented using or by a network device. Examples of network devices include switches, routers, firewalls, gateways, hubs, base stations, access points, repeaters, head-ends, user equipment, cell sites, antennas, towers, etc.

A special-purpose system may be partially or fully implemented using a computer having a variety of form factors and other characteristics. For example, the computer may be characterized as a personal computer, as a server, etc. The computer may be portable, as in the case of a laptop, netbook, etc. The computer may or may not have any output device, such as a monitor, line printer, liquid crystal display (LCD), light emitting diodes (LEDs), etc. The computer may or may not have any input device, such as a keyboard, mouse, touchpad, trackpad, computer vision system, barcode scanner, button array, etc. The computer may run a general-purpose operating system, such as the WINDOWS operating system from Microsoft Corporation, the MACOS operating system from Apple, Inc., or a variant of the LINUX operating system.

Examples of servers include a file server, print server, domain server, internet server, intranet server, cloud server, infrastructure-as-a-service server, platform-as-a-service server, web server, secondary server, host server, distributed server, failover server, and backup server.

Hardware

The term hardware encompasses components such as processing hardware, storage hardware, networking hardware, and other general-purpose and special-purpose components. Note that these are not mutually-exclusive categories. For example, processing hardware may integrate storage hardware and vice versa.

Examples of a component are integrated circuits (ICs), application specific integrated circuit (ASICs), digital circuit elements, analog circuit elements, combinational logic circuits, gate arrays such as field programmable gate arrays (FPGAs), digital signal processors (DSPs), complex programmable logic devices (CPLDs), etc.

Multiple components of the hardware may be integrated, such as on a single die, in a single package, or on a single

printed circuit board or logic board. For example, multiple components of the hardware may be implemented as a system-on-chip. A component, or a set of integrated components, may be referred to as a chip, chipset, chiplet, or chip stack.

Examples of a system-on-chip include a radio frequency (RF) system-on-chip, an artificial intelligence (AI) system-on-chip, a video processing system-on-chip, an organ-on-chip, a quantum algorithm system-on-chip, etc.

The hardware may integrate and/or receive signals from sensors. The sensors may allow observation and measurement of conditions including temperature, pressure, wear, light, humidity, deformation, expansion, contraction, deflection, bending, stress, strain, load-bearing, shrinkage, power, energy, mass, location, temperature, humidity, pressure, viscosity, liquid flow, chemical/gas presence, sound, and air quality. A sensor may include image and/or video capture in visible and/or non-visible (such as thermal) wavelengths, such as a charge-coupled device (CCD) or complementary metal-oxide semiconductor (CMOS) sensor.

Processing Hardware

Examples of processing hardware include a central processing unit (CPU), a graphics processing unit (GPU), an approximate computing processor, a quantum computing processor, a parallel computing processor, a neural network processor, a signal processor, a digital processor, a data processor, an embedded processor, a microprocessor, and a co-processor. The co-processor may provide additional processing functions and/or optimizations, such as for speed or power consumption. Examples of a co-processor include a math co-processor, a graphics co-processor, a communication co-processor, a video co-processor, and an artificial intelligence (AI) co-processor.

Processor Architecture

The processor may enable execution of multiple threads. These multiple threads may correspond to different programs. In various embodiments, a single program may be implemented as multiple threads by the programmer or may be decomposed into multiple threads by the processing hardware. The threads may be executed simultaneously to enhance the performance of the processor and to facilitate simultaneous operations of the application.

A processor may be implemented as a packaged semiconductor die. The die includes one or more processing cores and may include additional functional blocks, such as cache. In various embodiments, the processor may be implemented by multiple dies, which may be combined in a single package or packaged separately.

Networking Hardware

The networking hardware may include one or more interface circuits. In some examples, the interface circuit(s) may implement wired or wireless interfaces that connect, directly or indirectly, to one or more networks. Examples of networks include a cellular network, a local area network (LAN), a wireless personal area network (WPAN), a metropolitan area network (MAN), and/or a wide area network (WAN). The networks may include one or more of point-to-point and mesh technologies. Data transmitted or received by the networking components may traverse the same or different networks. Networks may be connected to each other over a WAN or point-to-point leased lines using technologies such as Multiprotocol Label Switching (MPLS) and virtual private networks (VPNs).

Examples of cellular networks include GSM, GPRS, 3G, 4G, 5G, LTE, and EVDO. The cellular network may be

implemented using frequency division multiple access (FDMA) network or code division multiple access (CDMA) network.

Examples of a LAN are Institute of Electrical and Electronics Engineers (IEEE) Standard 802.11-2020 (also known as the WIFI wireless networking standard) and IEEE Standard 802.3-2018 (also known as the ETHERNET wired networking standard).

Examples of a WPAN include IEEE Standard 802.15.4, including the ZIGBEE standard from the ZigBee Alliance. Further examples of a WPAN include the BLUETOOTH wireless networking standard, including Core Specification versions 3.0, 4.0, 4.1, 4.2, 5.0, and 5.1 from the Bluetooth Special Interest Group (SIG).

A WAN may also be referred to as a distributed communications system (DCS). One example of a WAN is the internet.

Storage Hardware

Storage hardware is or includes a computer-readable medium. The term computer-readable medium, as used in this disclosure, encompasses both nonvolatile storage and volatile storage, such as dynamic random access memory (DRAM). The term computer-readable medium only excludes transitory electrical or electromagnetic signals propagating through a medium (such as on a carrier wave). A computer-readable medium in this disclosure is therefore non-transitory, and may also be considered to be tangible.

EXAMPLES

Examples of storage implemented by the storage hardware include a database (such as a relational database or a NoSQL database), a data store, a data lake, a column store, a data warehouse.

Example of storage hardware include nonvolatile memory devices, volatile memory devices, magnetic storage media, a storage area network (SAN), network-attached storage (NAS), optical storage media, printed media (such as bar codes and magnetic ink), and paper media (such as punch cards and paper tape). The storage hardware may include cache memory, which may be collocated with or integrated with processing hardware.

Storage hardware may have read-only, write-once, or read/write properties. Storage hardware may be random access or sequential access. Storage hardware may be location-addressable, file-addressable, and/or content-addressable.

Example of nonvolatile memory devices include flash memory (including NAND and NOR technologies), solid state drives (SSDs), an erasable programmable read-only memory device such as an electrically erasable programmable read-only memory (EEPROM) device, and a mask read-only memory device (ROM).

Example of volatile memory devices include processor registers and random access memory (RAM), such as static RAM (SRAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), synchronous graphics RAM (SGRAM), and video RAM (VRAM).

Example of magnetic storage media include analog magnetic tape, digital magnetic tape, and rotating hard disk drive (HDDs).

Examples of optical storage media include a CD (such as a CD-R, CD-RW, or CD-ROM), a DVD, a Blu-ray disc, and an Ultra HD Blu-ray disc.

Examples of storage implemented by the storage hardware include a distributed ledger, such as a permissioned or permissionless blockchain.

Entities recording transactions, such as in a blockchain, may reach consensus using an algorithm such as proof-of-stake, proof-of-work, and proof-of-storage.

Elements of the present disclosure may be represented by or encoded as non-fungible tokens (NFTs). Ownership rights related to the non-fungible tokens may be recorded in or referenced by a distributed ledger.

Transactions initiated by or relevant to the present disclosure may use one or both of fiat currency and cryptocurrencies, examples of which include bitcoin and ether.

Some or all features of hardware may be defined using a language for hardware description, such as IEEE Standard 1364-2005 (commonly called “Verilog”) and IEEE Standard 1076-2008 (commonly called “VHDL”). The hardware description language may be used to manufacture and/or program hardware.

A special-purpose system may be distributed across multiple different software and hardware entities. Communication within a special-purpose system and between special-purpose systems may be performed using networking hardware. The distribution may vary across embodiments and may vary over time. For example, the distribution may vary based on demand, with additional hardware and/or software entities invoked to handle higher demand. In various embodiments, a load balancer may direct requests to one of multiple instantiations of the special purpose system. The hardware and/or software entities may be physically distinct and/or may share some hardware and/or software, such as in a virtualized environment. Multiple hardware entities may be referred to as a server rack, server farm, data center, etc. Software

Software includes instructions that are machine-readable and/or executable. Instructions may be logically grouped into programs, codes, methods, steps, actions, routines, functions, libraries, objects, classes, etc. Software may be stored by storage hardware or encoded in other hardware. Software encompasses (i) descriptive text to be parsed, such as HTML (hypertext markup language), XML (extensible markup language), and JSON (JavaScript Object Notation), (ii) assembly code, (iii) object code generated from source code by a compiler, (iv) source code for execution by an interpreter, (v) bytecode, (vi) source code for compilation and execution by a just-in-time compiler, etc. As examples only, source code may be written using syntax from languages including C, C++, JavaScript, Java, Python, R, etc.

Software also includes data. However, data and instructions are not mutually-exclusive categories. In various embodiments, the instructions may be used as data in one or more operations. As another example, instructions may be derived from data.

The functional blocks and flowchart elements in this disclosure serve as software specifications, which can be translated into software by the routine work of a skilled technician or programmer.

Software may include and/or rely on firmware, processor microcode, an operating system (OS), a basic input/output system (BIOS), application programming interfaces (APIs), libraries such as dynamic-link libraries (DLLs), device drivers, hypervisors, user applications, background services, background applications, etc. Software includes native applications and web applications. For example, a web application may be served to a device through a browser using hypertext markup language 5th revision (HTML5).

Software may include artificial intelligence systems, which may include machine learning or other computational intelligence. For example, artificial intelligence may include one or more models used for one or more problem domains.

When presented with many data features, identification of a subset of features that are relevant to a problem domain may improve prediction accuracy, reduce storage space, and increase processing speed. This identification may be referred to as feature engineering. Feature engineering may be performed by users or may only be guided by users. In various implementations, a machine learning system may computationally identify relevant features, such as by performing singular value decomposition on the contributions of different features to outputs.

Examples of the models include recurrent neural networks (RNNs) such as long short-term memory (LSTM), deep learning models such as transformers, decision trees, support-vector machines, genetic algorithms, Bayesian networks, and regression analysis. Examples of systems based on a transformer model include bidirectional encoder representations from transformers (BERT) and generative pre-trained transformer (GPT).

Training a machine-learning model may include supervised learning (for example, based on labelled input data), unsupervised learning, and reinforcement learning. In various embodiments, a machine-learning model may be pre-trained by their operator or by a third party.

Problem domains include nearly any situation where structured data can be collected, and includes natural language processing (NLP), computer vision (CV), classification, image recognition, etc.

Architectures

Some or all of the software may run in a virtual environment rather than directly on hardware. The virtual environment may include a hypervisor, emulator, sandbox, container engine, etc. The software may be built as a virtual machine, a container, etc. Virtualized resources may be controlled using, for example, a DOCKER container platform, a pivotal cloud foundry (PCF) platform, etc.

In a client-server model, some of the software executes on first hardware identified functionally as a server, while other of the software executes on second hardware identified functionally as a client. The identity of the client and server is not fixed: for some functionality, the first hardware may act as the server while for other functionality, the first hardware may act as the client. In different embodiments and in different scenarios, functionality may be shifted between the client and the server. In one dynamic example, some functionality normally performed by the second hardware is shifted to the first hardware when the second hardware has less capability. In various embodiments, the term “local” may be used in place of “client,” and the term “remote” may be used in place of “server.”

Some or all of the software may be logically partitioned into microservices. Each microservice offers a reduced subset of functionality. In various embodiments, each microservice may be scaled independently depending on load, either by devoting more resources to the microservice or by instantiating more instances of the microservice. In various embodiments, functionality offered by one or more microservices may be combined with each other and/or with other software not adhering to a microservices model.

Some or all of the software may be arranged logically into layers. In a layered architecture, a second layer may be logically placed between a first layer and a third layer. The first layer and the third layer would then generally interact with the second layer and not with each other. In various embodiments, this is not strictly enforced—that is, some direct communication may occur between the first and third layers.

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What is claimed is:

1. A method, comprising:
receiving a selection of a custom object within a customer relationship management (CRM) system;
receiving a definition of an event type definition for the custom object;
storing an ontology that defines a set of objects and respective associations between the set of objects, wherein the set of objects includes the custom object;
wherein the ontology is a client-specific ontology of a client of a multi-service platform that includes custom objects that are specific to the client;
constructing a custom event record for the custom object;
configuring the custom event record to track an event type defined by the event type definition;
monitoring for an occurrence of an event instance of the event type; and
in response to an occurrence of the event instance associating the event instance with an event log of the custom event record.
2. The method of claim 1, comprising:
creating the event log to provide a time-sequenced log of event instances recorded for the custom object.
3. The method of claim 1, comprising:
storing event record instances of different types of event records, wherein an event record includes an association to a primary object and defines event types tracked with respect to an object instance of the primary object.
4. The method of claim 1, comprising: receiving a user selection of the custom object and the event type definition to associate with the custom event record, wherein the event type definition indicates a set of event types tracked with respect to an instance of the custom object and a set of event properties.
5. The method of claim 4, comprising:
updating a client-specific ontology to associate the set of event types with the custom object.
6. The method of claim 1, comprising:
receiving user input indicating a new event type, a set of new event properties of the new event type, and, for each new event property of the set of new event properties, a data type of the new event property.
7. The method of claim 6, comprising:
receiving a data source selection indicating a third-party data source to monitor for the new event type.
8. The method of claim 6, comprising:
configure a custom listening thread that monitors a third-party data source for event instances of the new event type.
9. A computing system including a processor and memory configured to perform operations comprising:
receiving a selection of a custom object within a customer relationship management (CRM) system;
receiving a definition of an event type definition for the custom object;
storing an ontology that defines a set of objects and respective associations between the set of objects, wherein the set of objects includes the custom object;

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- wherein the ontology is a client-specific ontology of a client of a multi-service platform that includes custom objects that are specific to the client;
constructing a custom event record for the custom object;
configuring the custom event record to track an event type defined by the event type definition;
monitoring for an occurrence of an event instance of the event type; and
in response to an occurrence of the event instance associating the event instance with an event log of the custom event record.
10. The computing system of claim 9, wherein the operations comprise: creating the event log to provide a time-sequenced log of event instances recorded for the custom object.
 11. The computing system of claim 9, wherein the operations comprise:
storing event record instances of different types of event records, wherein an event record includes an association to a primary object and defines event types tracked with respect to an object instance of the primary object.
 12. The computing system of claim 9, wherein the operations comprise:
receiving a user selection of the custom object and the event type definition to associate with the custom event record, wherein the event type definition indicates a set of event types tracked with respect to an instance of the custom object and a set of event properties.
 13. The computing system of claim 12, wherein the operations comprise:
updating a client-specific ontology to associate the set of event types with the custom object.
 14. A non-transitory machine readable medium comprising instructions for performing a method, which when executed by a machine, causes the machine to perform operations comprising:
receiving a selection of a custom object within a customer relationship management (CRM) system;
receiving a definition of an event type definition for the custom object;
storing an ontology that defines a set of objects and respective associations between the set of objects, wherein the set of objects includes the custom object;
wherein the ontology is a client-specific ontology of a client of a multi-service platform that includes custom objects that are specific to the client;
constructing a custom event record for the custom object;
configuring the custom event record to track an event type defined by the event type definition;
monitoring for an occurrence of an event instance of the event type; and
in response to an occurrence of the event instance associating the event instance with an event log of the custom event record.
 15. The non-transitory machine readable medium of claim 14, wherein the operations comprise: creating the event log to provide a time-sequenced log of event instances recorded for the custom object.

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